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Chapter 4. Manuscript frames 237–334. Catalogue snapshot: 5 October 2026.

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Chapter 4

GELFAND-P000237



Uncertain reading. The upper right has a single upright stroke. It may be the numeral 1.

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Gelfand Seminar. 1968-69.

1. Mumford. Moduli spaces. (Geronimus Sasha)

2. ~~Kr.~~ Kirillov. Characters. (lectures)



Manuscript notation. The short crossed-out Kp. (Kr.) precedes Kirillov on the same line.

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3. Kazhdan Dwork ?

4. Griffiths Mixture of diff. eqs., geom., top., complex analysis

5. Spencer resolution. (Palamodov)

6. **Gromov**

7. Buslaev on eigenvalues (incl. Maslov's work)

8. Grauert and Remmert, p-adic analysis.

(old idea of L. G. Shnirelman about the Cauchy $\int$ and the like over an arbitrary field)

9. Nilson

10. Vershik on factors $\overline{\text{II}}$ and $\overline{\text{III}}$

or Sinai (and K°?)



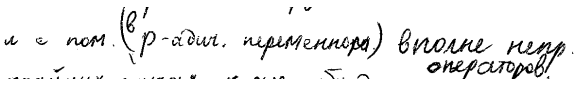
Uncertain reading. The small superscript on K is open at the top. Provisional $\circ$, possibly 0.

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11. Hörmander work on Lie groups and diff. eqs. (hypoell.).

Idea of **Dwork** (Kazhdan)

If some series converges near 0 in a complex variable and at every p-adic one, then it is a rational function.
idea: represent the ζ-function using ([Above the line: in] p-adic variables) completely continuous operators.



Manuscript notation. The word в (in) is written above "p-adic variables" in the parenthesis before "completely continuous operators".

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On Gromov: 2 extreme cases which are unified.

Bundle techniques are transferred to foliations (sometimes). For foliations of dimension 1 — the question of a vector field.

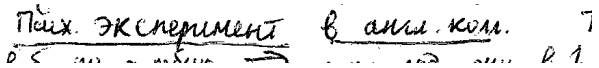
A foliation of codim. ≥ 1 always exists if a family of planes exists (!)



Uncertain reading. The sign before 1 is heavily inked. The lower strokes favour $\geq$.

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The sheaf of germs and jets — a weak homotopy equiv.
(If there is a set of coeff. glob., then one can construct a function)
Psych. experiment in an Engl. coll.



Manuscript notation. The underlined words read в англ. колл. (in an Engl. coll.).

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Tests $\Rightarrow$ teachers secretly told about the giftedness of 20% of the pupils, selected by lot $\Rightarrow$ after a year they are 1.5 times better than avg.! (not by marks, but by tests)

GELFAND-P000238

Geronimus on Mumford.
Object of study — compact Riem. surfaces.
$g = 0$ — all isomorphic (sphere).
$g = 1$ — complex elliptic curves.

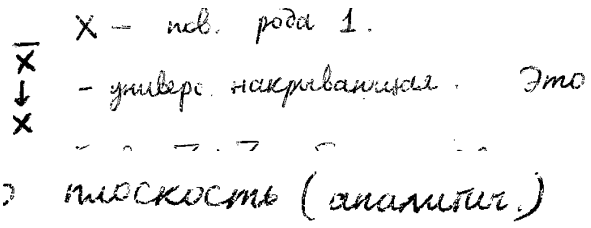


Diagram and typed description.
X — surface of genus 1.

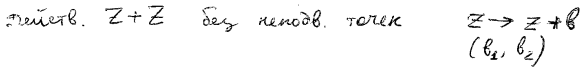
$$\begin{array}{c} \overline{X} \\ \downarrow \\ X \end{array}$$

— universal cover. It is the plane (analytically)

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$Z + Z$ acts without fixed points

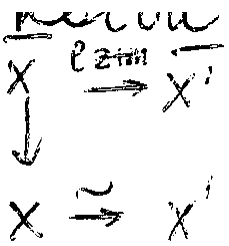
$$z \underset{(b_1,b_2)}{\to} z + b$$



Manuscript notation. The pair (b_1, b_2) is written below the translation formula.

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When do we obtain different surfaces?



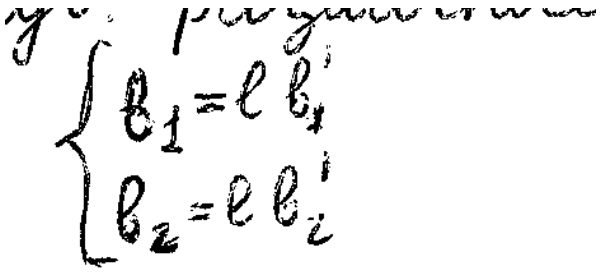
Uncertain reading.

$$\begin{array}{ccc} & \ell z + m & \\ X & \longrightarrow & X' \\ \downarrow & & \\ X & \underset{\sim}{\longrightarrow} & X' \end{array}$$

Rules are visible above X and X'. The left rule is close to the underlined heading.

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$$\begin{cases} b_1 = \ell b_1', \\ b_2 = \ell b_i'. \end{cases}$$



Manuscript notation. The final subscript has a dot and a lower hook and reads i. The left of the second line clearly reads b_2.

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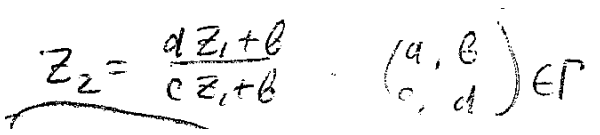
One can assume ($b_i \neq 0$) that $b_1 = 1$.

$(1, z_2)$ an automorphism of $Z + Z$ also changes nothing.

$$(1, z) \qquad \operatorname{Im} z > 0$$

$$\mathrm{SL}(2, Z) = \Gamma$$

$$z_2 = \frac{dz_1 + b}{cz_1 + b} \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$$



Manuscript notation. The numerator reads $dz_1 + b$, and the denominator $cz_1 + b$. The first and last entries of the matrix are a and d.

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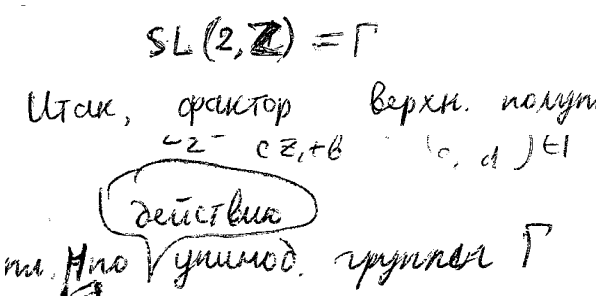
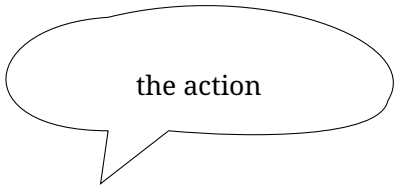


Diagram and typed description.

Thus, the quotient of the upper half-plane H by

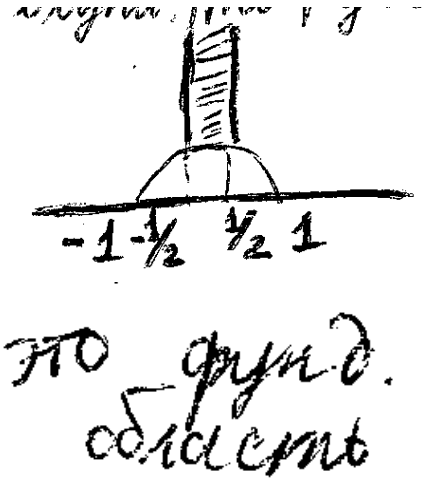


of the unimodular group Γ

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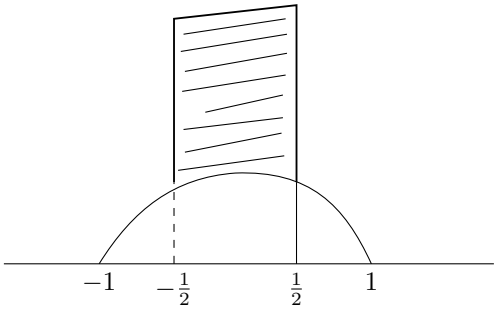
$$H \longrightarrow H/\Gamma = \Pi$$

On H there is a fund. domain



Original source details: GELFAND-P000238

Diagram and typed description.



this is a fund. domain

The plane — the analytic set of all elliptic curves.

General def. 1

There is a collection of analytic sets K.

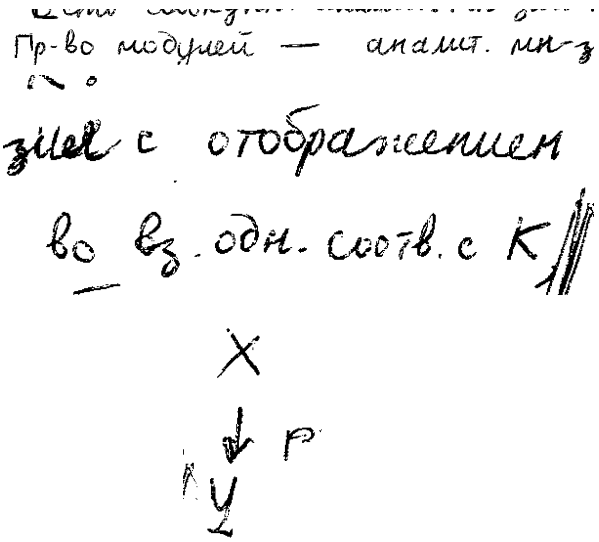


Diagram and typed description.

Moduli space — an analytic set with a map

$$\begin{array}{c} X \\ \downarrow p \\ \underline{Y} \end{array}$$

Original source details: GELFAND-P000238

($X, \underline{Y}$ — manifolds and an analytic map). Points of $\underline{Y}$ in one-to-one corresp. with K and, in addition, $p^{-1}(y) \cong$ the corresponding object of K

(objects of K cannot be analytically isom.)

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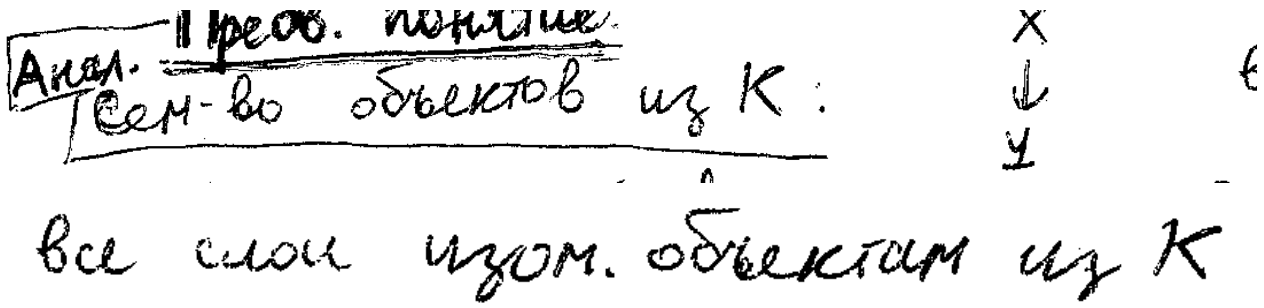
[source number: 2]

$\underline{Y} = K$ with the structure of an analytic set

If the moduli space exists, it is unique up to analytic isom. (easy to prove).

Another def. takes account of possible automorphisms of objects of K.

Prelim. notion.



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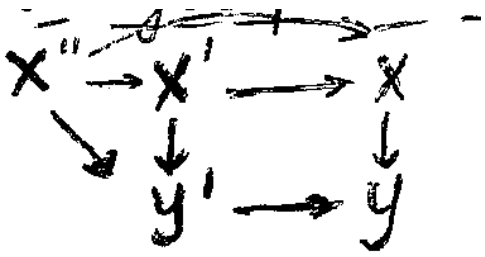
Diagram and typed description.

Anal. Family of objects of K:

$$\begin{array}{c} X \\ \downarrow \\ \underline{Y} \end{array}$$

all fibers isom. to objects of K

Induced family — analogous to an induced bundle (it always exists)



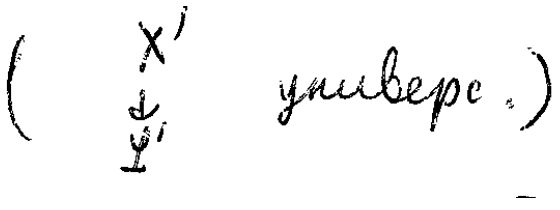
Original source details: GELFAND-P000239

Diagram and typed description.

$$\begin{array}{ccccc} X'' & \longrightarrow & X' & \longrightarrow & X \\ & \searrow & \downarrow & & \downarrow \\ & & Y' & \longrightarrow & Y \end{array}$$

This is a commut. diagram + univers.

Universality



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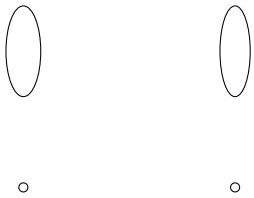
Diagram and typed description.

$$\begin{array}{ccc} & X' & \\ & \downarrow & \\ (& & \text{univers.}) \\ & \underline{Y'} & \end{array}$$

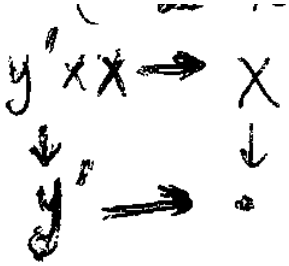


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Diagram and typed description.



A merely commut. diagram is insufficient (2 points and any one where a map $\exists$).



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Diagram and typed description.

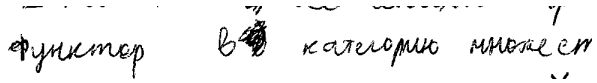
$$\begin{array}{ccc} Y' \times X & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & \bullet \end{array}$$

Kirillov (conjecture)

Representable functor. The representing object is the family of moduli

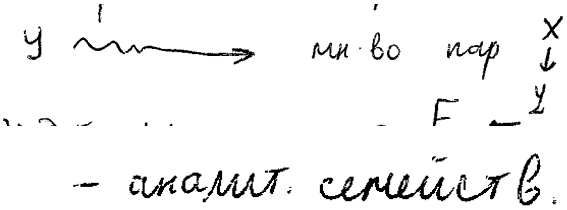
2 categ. — 1) all analytic spaces.

functor into the [Crossed out: 2)?] category of sets



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Uncertain reading. There is a heavily crossed-out mark just after “into”. Provisional 2), possibly 2.



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Diagram and typed description.

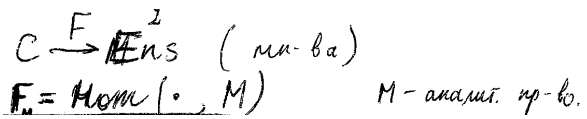
$$\begin{array}{ccc} Y & \rightsquigarrow & \text{set of pairs} \\ & & X \\ & & \downarrow \\ & & \underline{Y} \end{array}$$

— analytic families.

Repr. of the functor

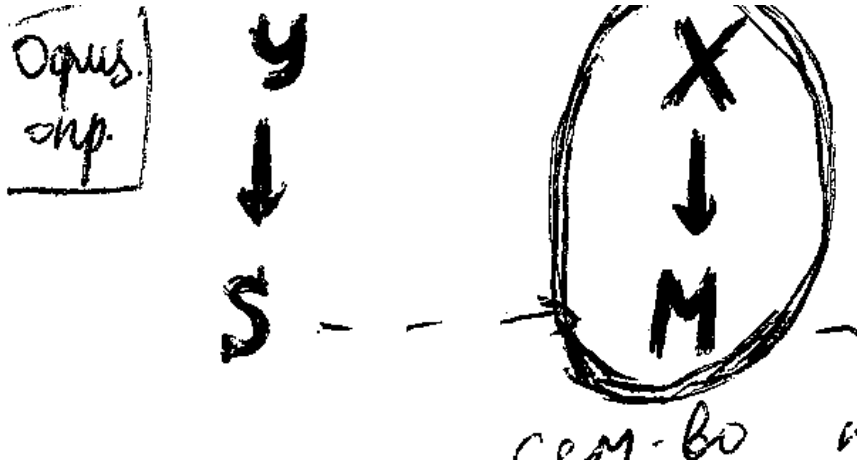
$$C \xrightarrow{F} \text{Ens} \quad (\text{sets})$$

$$F_M = \text{Hom}(\bullet, M) \qquad M \text{ — analytic space.}$$



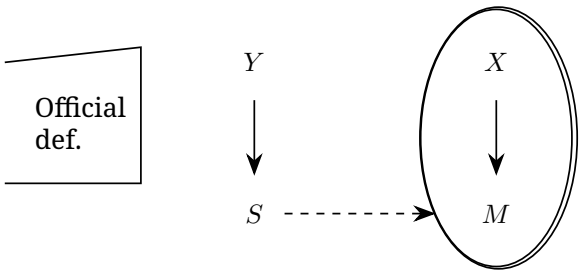
Manuscript notation. The upper line has $C \to \text{Ens}$ with F above the arrow. The first argument of Hom is a dot.

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Original source details: GELFAND-P000239

Diagram and typed description.



— the family is called a moduli space if $\forall$ other family is obtained as an induced one.

GELFAND-P000240

A universal family for curves of a given genus does not exist (even for elliptic ones)

Exists for polarized ones (with a chosen homology basis)

Modification — final object of the category of families

If it does not exist, adjoin it formally.

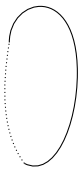
An additional structure is introduced, allowing cohom. with coeff. in a sheaf to be considered, and so on.

There is no universal object for elliptic curves



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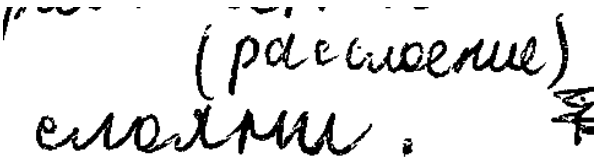
Diagram and typed description.



M

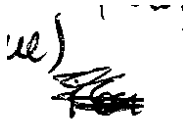
all points map unambiguously $\Rightarrow$ as a set — elliptic curves do not repeat.

Suppose there is a nontrivial family (base not a point) with identical fibers. [Above “fibers”: (bundle).]



Original source details: GELFAND-P000240

Manuscript notation. The parenthetical word расслоение (bundle) is written above слоями (fibers).

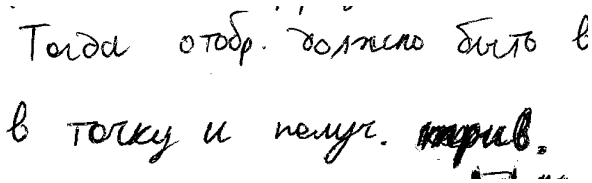


Original source details: GELFAND-P000240

Undeciphered passage. [Cancelled fragment to the right of “fibers”: reading not established.]

(because there are automorphisms)

Then the map must be into a point and one obtains triv.?



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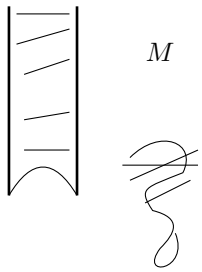
Uncertain reading. The ending reads трив. (triv.). Heavy strokes precede it. A possible не (non-) is indistinct.

Exists in a weaker sense — the plane



Original source details: GELFAND-P000240

Uncertain reading.



M

The upper right mark resembles M. The crossed-out lower mark is illegible.

Most have one autom. ($x \to -x$) (with a marked point), and some have additional rotations.

Thus, the work consists in studying the properties of a nonexistent universal object.

Start precisely $6^{\underline{15}}$



Manuscript notation. The 15 is raised and underlined.

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GELFAND-P000241

[source number: 3]

$9/\underline{\text{IX}}$. **Continuation of the talk by Geronimus.**

Nonexistence of the moduli space is overcome by fixing a homology basis (a more rigid object). This is not considered.

For elliptic curves only a point is fixed (and even that not immediately).

A way (a different one) to overcome this difficulty will be indicated.

Def. of a Grothendieck topology.

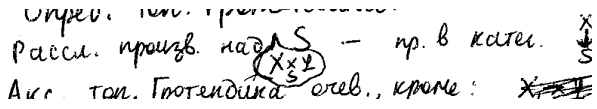
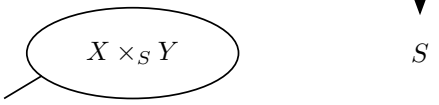


Diagram and typed description.

Fiber product over S — product in the categ.

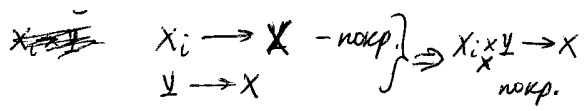




Original source details: GELFAND-P000241

The axioms of a Grothendieck topology are obvious except: [Cancelled: $X_i \times Y$]

$$\left.\begin{array}{c} X_i \longrightarrow X \text{ — cover} \\ \underline{Y} \longrightarrow X \end{array}\right\} \Rightarrow \begin{array}{c} X_i \times_X \underline{Y} \longrightarrow X \\ \text{cover} \end{array}$$



Manuscript notation. The conclusion points to X. The two occurrences of Y are underlined. The crossed-out writing is to the left of the hypotheses.

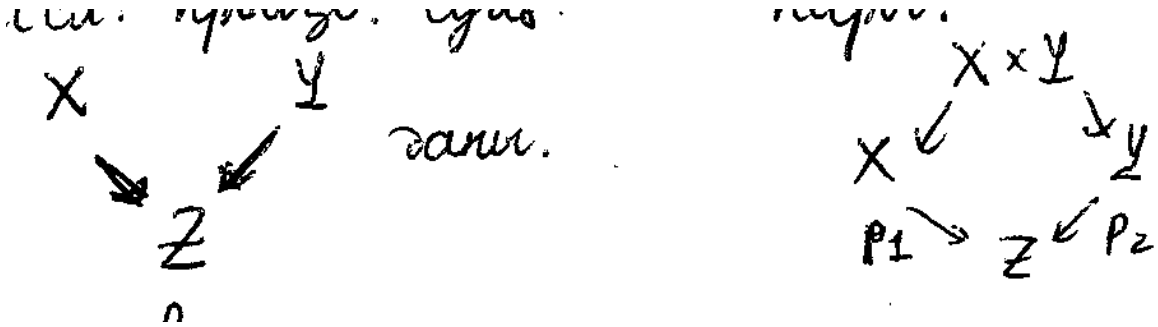
Original source details: GELFAND-P000241

A morphism of Grothendieck topologies —

— a functor $T_1 \to T_2$ preserving all structures.

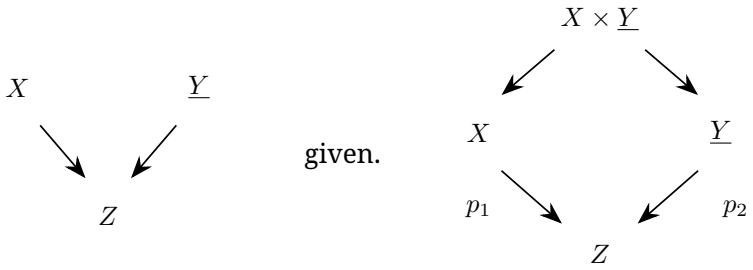
Example 1. T_Γ — Γ-sets (Γ — a group).

Fiber products exist — pairs.



Original source details: GELFAND-P000241

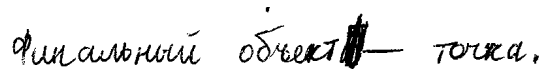
Diagram and typed description.



(in $X \times Y$ take pairs (x, y) such that $p_1(x) = p_2(y)$).

$X_i \xrightarrow{f} X$ — a covering if $\bigcup f_i(X_i) = X$.

Final object — a point.



Original source details: GELFAND-P000241

Uncertain reading. Dense looped strokes follow "object". Any underlying reading is unresolved.

Cohomology of a point with coeff. in various sheaves. $H^q(\cdot, F)$

(F — an arbitrary sheaf)

~~On~~ All sheaves are constructed as follows.

Manuscript notation. The word Ha (On) is crossed out before "All sheaves".



Original source details: GELFAND-P000241

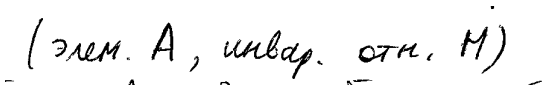
If the value $F(\Gamma) = A$ is specified (Γ acts on Γ on the left), then on all the others it is uniquely determined by the group A. (if this really is a sheaf and not a presheaf).

$\Gamma \longrightarrow \Gamma/H$ a covering.

Apply the sheaf property to the covering of the open set Γ/H. We obtain

$$F(\Gamma/H) = A^H$$

(elements of A invariant under H?)



Original source details: GELFAND-P000241

Uncertain reading. The letter in the parenthesis has a curved middle stroke. Nearby A^H supports provisional H. An alternative is M.

(Γ acts on A, because maps of sets induce maps of sections by the definition of a presheaf).

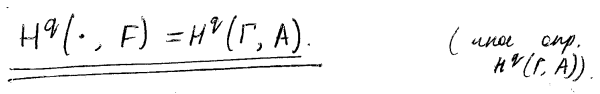
GELFAND-P000242

For an arbitrary Γ-set that is a union of disjoint orbits, one must consider the embeddings of all orbits, and there will be a direct sum.

Thus, everything is uniquely determined by the Γ-module A.
Here

$$\underline{\underline{H^q(\cdot, F) = H^q(\Gamma, A)}}.$$

(another def. of $H^q(\Gamma, A)$).

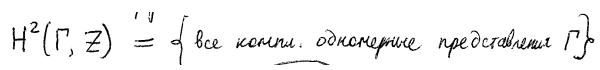


Original source details: GELFAND-P000242

Manuscript notation. The equality has two underlines. At right is the note “another def.” with $H^q(\Gamma, A)$.

A fact about group cohom.

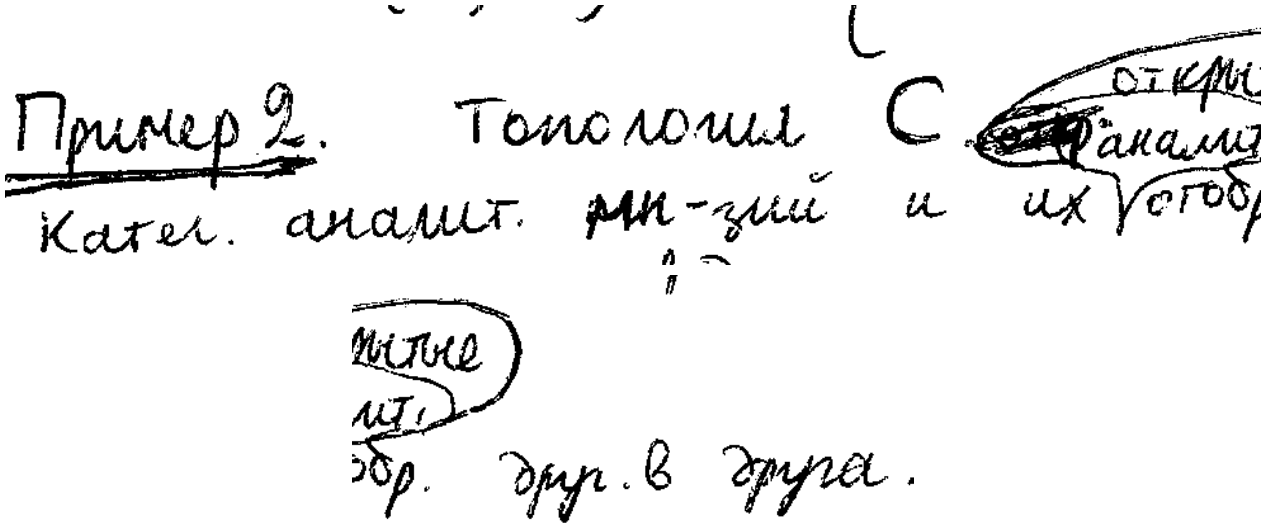
$$H^2(\Gamma, Z) = \{\text{all complex one-dimensional representations of } \Gamma\}$$



Original source details: GELFAND-P000242

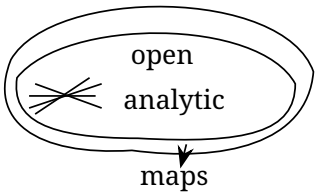
Manuscript notation. The left side is $H^2(\Gamma, Z)$. The right side gives all complex one-dimensional representations of Γ.

Example 2. Topology C



Original source details: GELFAND-P000242

Diagram and typed description.



The categ. of analytic sets and their maps to one another.

Undeciphered passage. The cancelled fragment to the left of “analytic” inside the enclosing stroke could not be deciphered.



Original source details: GELFAND-P000242

There is a fiber product.

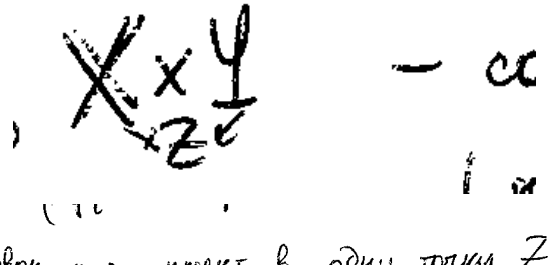
Coverings (!) X — an analytic set

$X_i \xrightarrow{f_i} X$ — a covering if

1) $\bigcup f_i(X_i) = X$

2) each f_i — an unramified covering of X_i over an open set of X

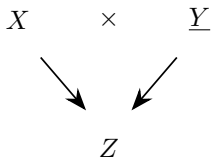
(f_i^{-1} a discrete set of points).



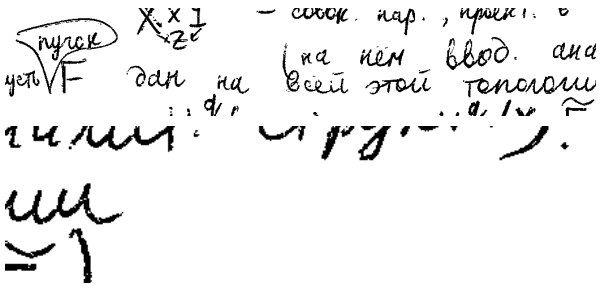
Original source details: GELFAND-P000242

(an analytic structure is introduced on it)

Diagram and typed description.

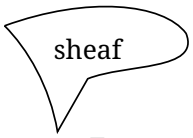


— the collection of pairs projecting to the same point of Z



Original source details: GELFAND-P000242

Diagram and typed description.



let F be given on this entire topology

$$H^q(X, F) = H^q(X, \widetilde{F})$$

$\widetilde{F}$ — a sheaf on the usual topology (restriction)

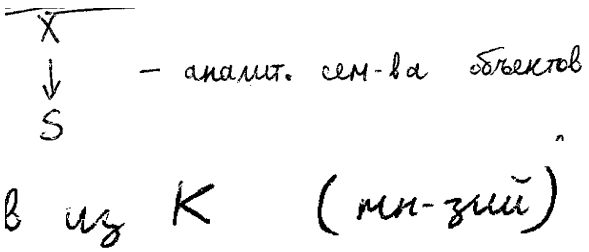
(ordinary cohomology)

(over a sufficiently small open set all coverings split).

GELFAND-P000243

[source number: 4]

Example 3. (the main one) Given K.



Original source details: GELFAND-P000243

Diagram and typed description.

$$\begin{array}{c} X \\ \downarrow \\ S \end{array}$$

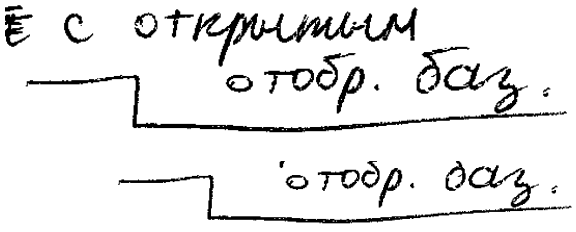
— analytic families of objects of K (sets)

A map — induction,



Original source details: GELFAND-P000243

Uncertain reading. Several strokes cross out a short word. Some loops remain visible. A possible reading is это (this).

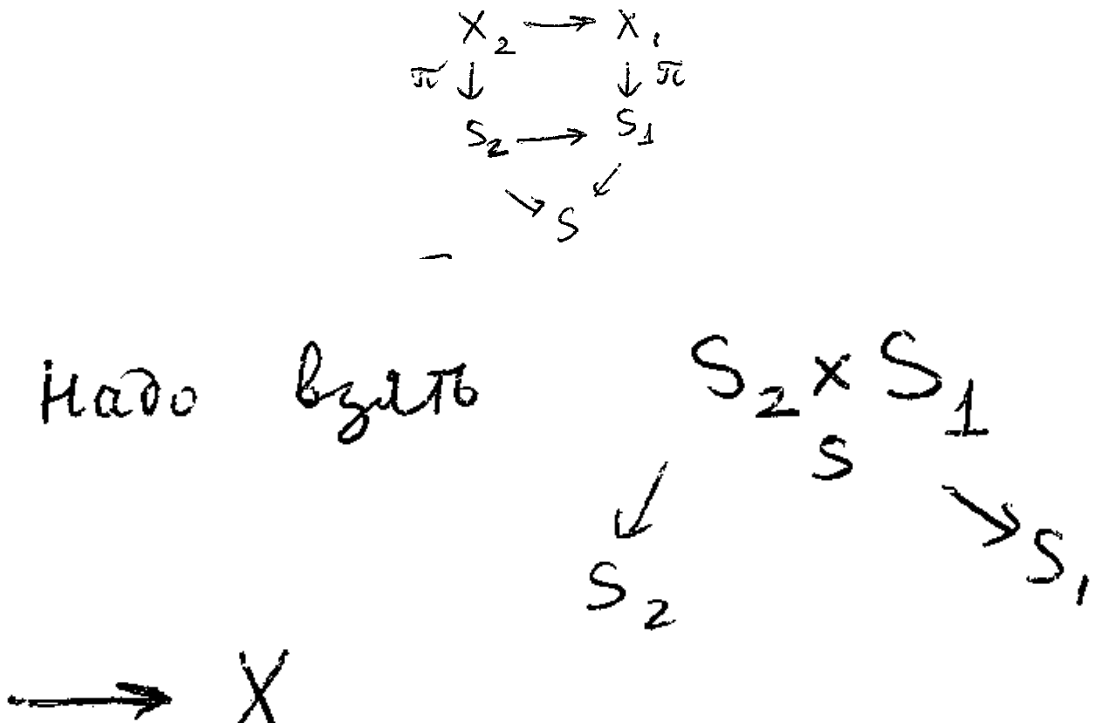


Original source details: GELFAND-P000243

Diagram and typed description.

with an open map of bases

Fiber products exist



Original source details: GELFAND-P000243

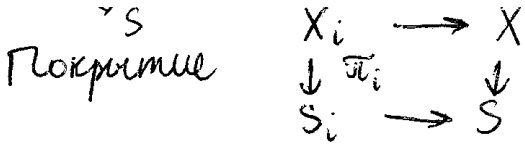
Diagram and typed description.

$$\begin{array}{ccc} X_2 & \longrightarrow & X_1 \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ S_2 & \longrightarrow & S_1 \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

One must take

$$\begin{array}{ccc} & S_2 \times_S S_1 & \\ \swarrow & & \searrow \\ S_2 & & S_1 \end{array}$$

and induce a family over it (from S_2 and S_1 one obtains the same thing)



Original source details: GELFAND-P000243

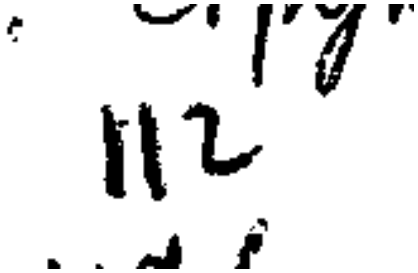
Diagram and typed description.
A covering

$$\begin{array}{ccc} X_i & \longrightarrow & X \\ \downarrow & \pi_i & \downarrow \\ S_i & \longrightarrow & S \end{array}$$

if the images cover all of S and each map —
— a covering (in the sense of Ex.2).

$H^q(\pi, F)$ equiv. to the preceding example $\cong$ cohom. of S in the preceding example.

(since the structures of open sets are the same and the topologies are isomorphic).



Original source details: GELFAND-P000243

Uncertain reading.

112

Two upright strokes and a small 2-like mark stand between the cohomology lines. The meaning of the smaller mark is uncertain.

$H^q(S, \widetilde{F})$ — ordinary cohomology.

If there is a terminal object.

Suppose there is no terminal object.

Example 4.

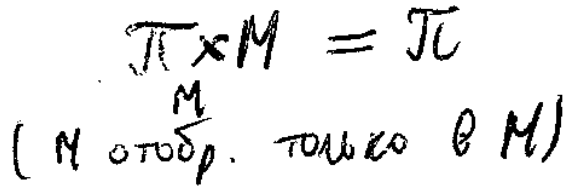
T_M — categ. — the old one + a formally adjoined terminal object M.

Its cohomology is computed as if it existed.

A fiber product exists.

$$\pi \times_M M = \pi$$

(M maps only to M)



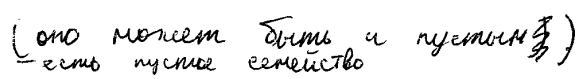
Original source details: GELFAND-P000243

Manuscript notation. The product has M below the $\times$ sign. The equation repeats π.

The fiber product $X \times_M Y$ remains

In any category $\boldsymbol{X} \times_{\boldsymbol{M}} \boldsymbol{Y}$ — the direct product.

(a tautological fact).



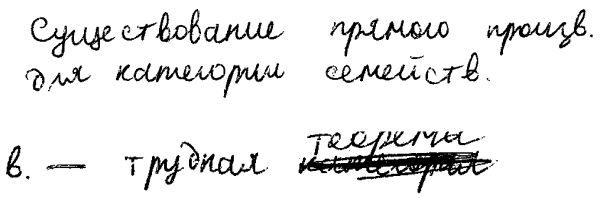
Original source details: GELFAND-P000243

Uncertain reading. (it may also be empty [a sign resembling ξ, possibly cancelled])

— there is an empty family

GELFAND-P000244

Existence of a direct product — a difficult [crossed out: category?] [inserted above: theorem] for the category of families.



Uncertain reading. The insertion above the crossed-out word reads теорема (theorem). Under the strokes, категория (category) is the likelier reading.

Original source details: GELFAND-P000244

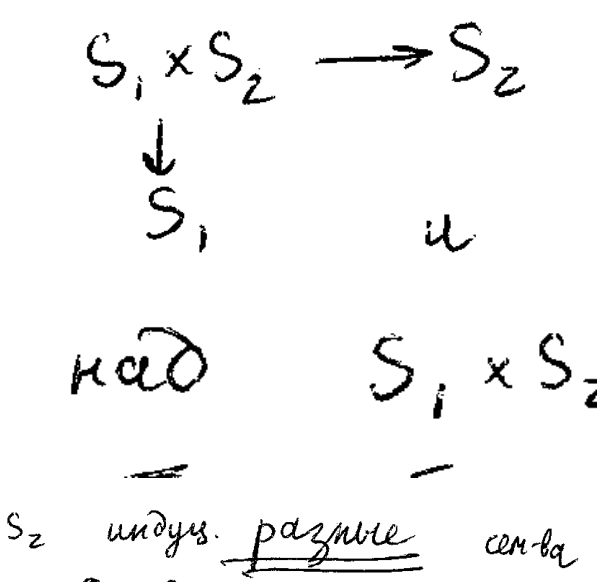


Diagram and typed description.

$$\begin{array}{ccc} S_1 \times S_2 & \longrightarrow & S_2 \\ \downarrow & & \\ S_1 & & \end{array}$$

over $S_1 \times S_2$ different families are induced

Original source details: GELFAND-P000244

and one cannot take $S_1 \times S_2$ as the base.

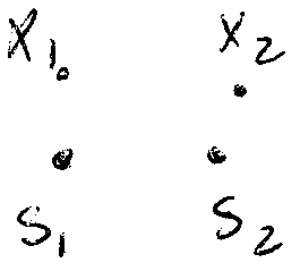
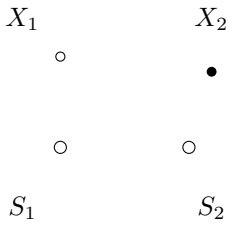


Diagram and typed description.



Original source details: GELFAND-P000244

If $X_1 = X_2$ then the product — itself.
If $X_1 \neq X_2$, then empty.



Manuscript notation. The crossed-out word before “If a point” reads Но (But).

Original source details: GELFAND-P000244

If a point, then as a rule empty.

If close to universal, then usually nonempty.

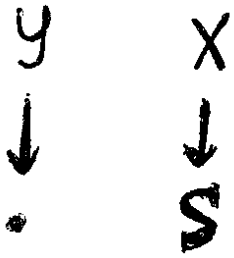
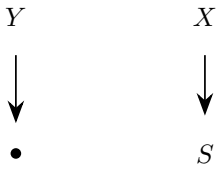


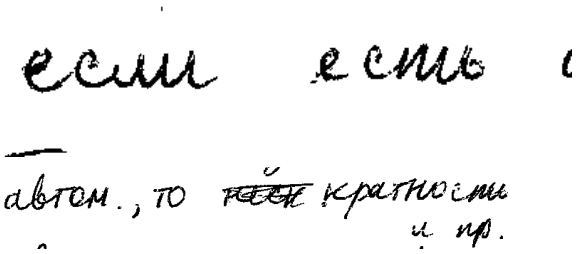
Diagram and typed description.



Original source details: GELFAND-P000244

The base — a subset of S, with fibers isom. to Y.

if there are autom., then [crossed out: there is?] multiplicity etc.



Uncertain reading. A short word before “multiplicity” is crossed out, possibly есть (there is).

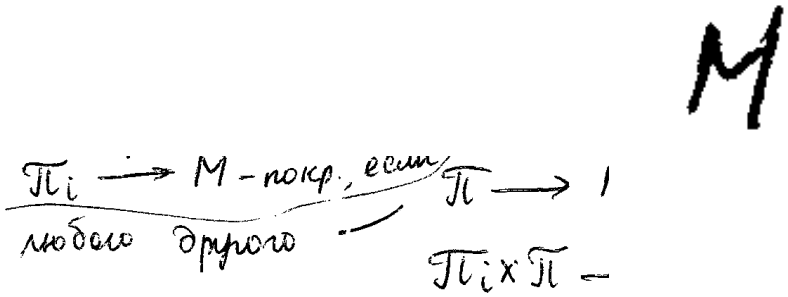
Original source details: GELFAND-P000244

Set-theoretically — the base — points in S with fibers Y, multiplicity — the number of automorphisms.

This — an analytic set! (a difficult theorem)

(the number of ways of mapping Y into this family)

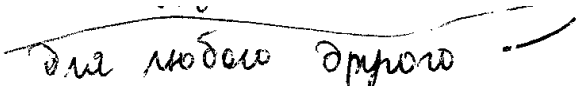
Def. of a covering of M:



Original source details: GELFAND-P000244

Diagram and typed description.

$\pi_i \longrightarrow M$ — a cover if $\pi \longrightarrow M$

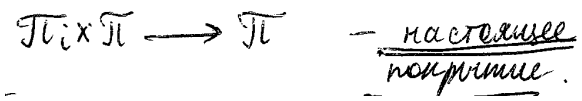


Original source details: GELFAND-P000244

Diagram and typed description.

for any other

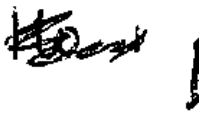
$\pi_i \times \pi \longrightarrow \pi$ — a genuine covering.



Manuscript notation. There is no index under $\times$. The words “genuine covering” are underlined twice.

Original source details: GELFAND-P000244

(one axiom is ensured by this def., the others — automatically)
All π_i — not a covering!



Uncertain reading. [Cancelled: But / To us?]

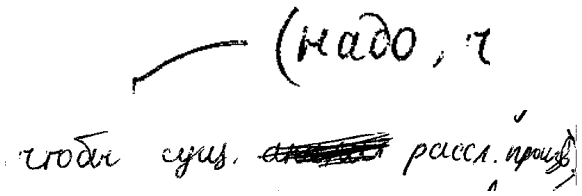
Original source details: GELFAND-P000244

Next time, continuation and or the Spencer resolvent or Dwork.

GELFAND-P000245

[source number: 5]

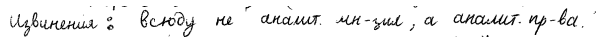
16/IX Geronimus.



Undeciphered passage. (it is necessary that there exist [a cancelled word cannot be read beneath dense strokes] fiber products)

Original source details: GELFAND-P000245

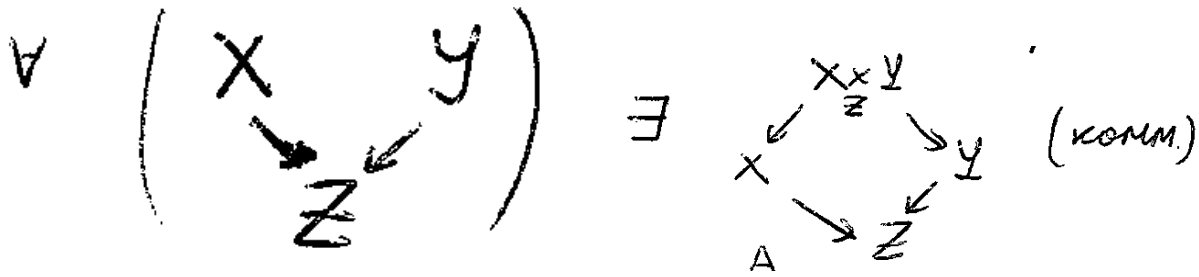
Apologies: everywhere, not analytic sets, but analytic spaces.



Manuscript notation. The two abbreviations mean “analytic sets” and “analytic spaces”.

Original source details: GELFAND-P000245

Fiber product.

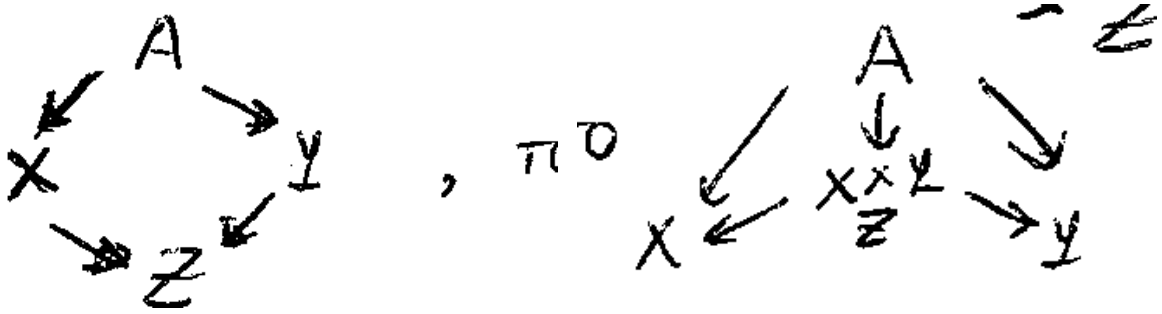


Original source details: GELFAND-P000245

Diagram and typed description.

$$\forall \left(\begin{array}{ccc} X & & Y \\ & \searrow \quad \swarrow & \\ & Z & \end{array} \right) \quad \exists \quad \begin{array}{ccccc} & & X \times_Z \underline{Y} & & \\ & \swarrow & & \searrow & \\ X & & & & \underline{Y} \\ & \searrow & & \swarrow & \\ & & Z & & \end{array} \qquad \text{(comm.)}$$

and universally, i.e. if



Original source details: GELFAND-P000245

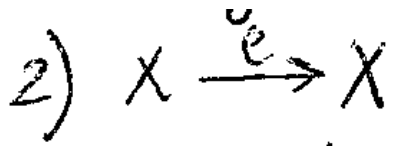
Diagram and typed description.

$$\begin{array}{ccc} & A & \\ \swarrow & & \searrow \\ X & & \underline{Y} \\ \searrow & & \swarrow \\ & Z & \end{array} \quad \text{, then} \quad \begin{array}{ccc} & A & \\ & \downarrow & \\ \swarrow & & \searrow \\ X & & \underline{Y} \\ \searrow & & \swarrow \\ & Z & \end{array}$$

Covering axioms

1) $X_{ij} \longrightarrow X_i \longrightarrow X$

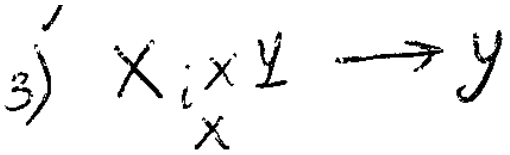
2) $X \xrightarrow{e} X$



Original source details: GELFAND-P000245

Manuscript notation. The label above the arrow reads e, with a short upper loop.

3) $X_i \times_X \underline{Y} \longrightarrow Y$



Manuscript notation. The arrow points to Y. The earlier Y is underlined and X stands below $\times$.

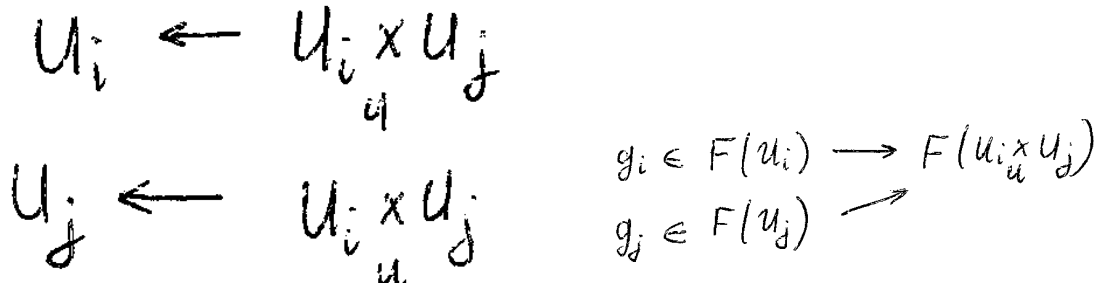
Original source details: GELFAND-P000245

Presheaf: a contravariant functor F to the categ. of groups.

Sheaf.

$U_i \longrightarrow U$ — a covering

$g_i \in F(U_i)$



Original source details: GELFAND-P000245

Diagram and typed description.

$$\begin{array}{ccccc} U_i & \longleftarrow & U_i \times_U U_j & & \\ & & & & \\ U_j & \longleftarrow & U_i \times_U U_j & & \end{array} \qquad \begin{array}{ccc} g_i \in F(U_i) & \longrightarrow & F(U_i \times_U U_j) \\ & \nearrow & \\ g_j \in F(U_j) & & \end{array}$$

Compatible if the g_i map to the same element.

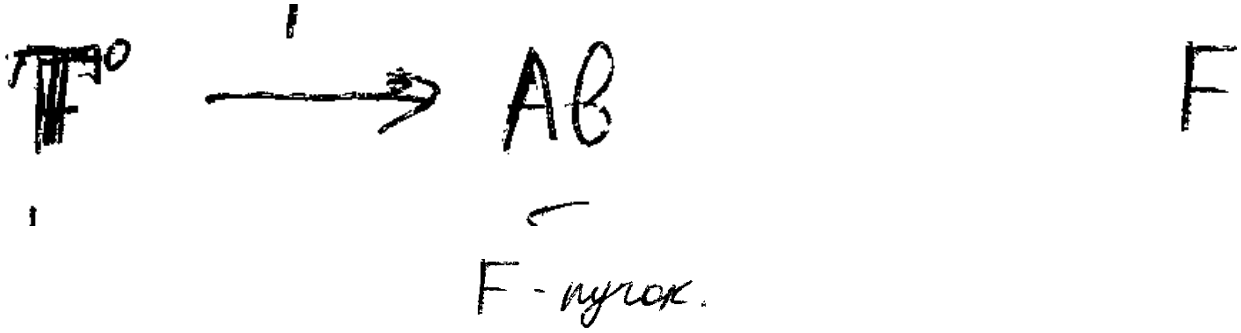
$U_i \longrightarrow U \qquad F(U) \longrightarrow F(U_i)$ (over g_i)



Uncertain reading. [Cancelled: def. / exists / this?]

Original source details: GELFAND-P000245

a sheaf: if $g \in F(U)$ exists and is unique, mapping to the g_i.



Original source details: GELFAND-P000245

Uncertain reading.

$T^\circ \xrightarrow{F}$ Ab $\qquad F$ — a sheaf.

categ. with reversed arrows. abel. groups

The base letter is overwritten and provisionally reads T. The superscript is round and is set as $\circ$.

GELFAND-P000246

Cohomology — these are groups
$H^\nu(U, F)$,



Original source details: GELFAND-P000246

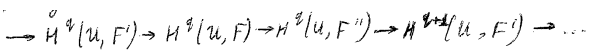
Manuscript notation. The opening degree is angular and open, ν. The exact sequence uses a looped q.

such that $H^0(U, F) = F(U)$. and if

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0, \quad \text{then}$$

(there is a procedure allowing one to construct a sheaf canonically from any presheaf — for any topology),

$$\longrightarrow H^q(U, F') \longrightarrow H^q(U, F) \longrightarrow H^q(U, F'') \longrightarrow H^{q+1}(U, F') \longrightarrow \dots$$



Original source details: GELFAND-P000246

Manuscript notation. The first three terms have q, followed by $q+1$. The final written term has one prime on F.

(they exist).

And for any injective sheaf cohom. 0

(and there are sufficiently many inj. sheaves)

Suppose there is a map compatible with all structures (a map of topologies)

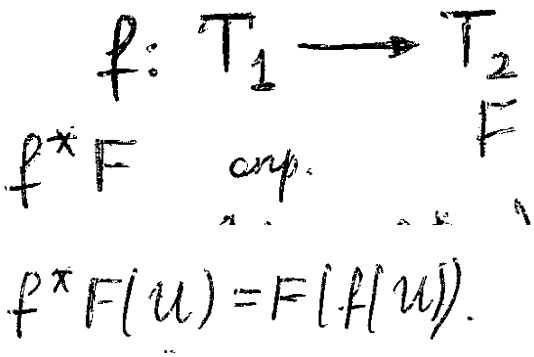


Diagram and typed description.

$$f: \quad T_1 \longrightarrow T_2$$

f^*F def. F

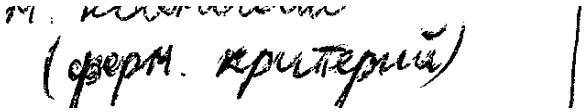
$$f^*F(U) = F(f(U)).$$

Original source details: GELFAND-P000246

$$H^q(U, f^*F) \longrightarrow H^q(f(U), F).$$

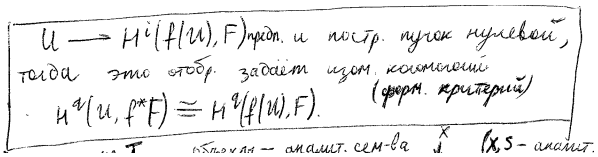
$U \longrightarrow H^i(f(U), F)$ a presheaf and the constructed sheaf is zero,
then this map gives an isom. of cohomology (form. criterion)

$$H^q(U, f^*F) \cong H^q(f(U), F).$$



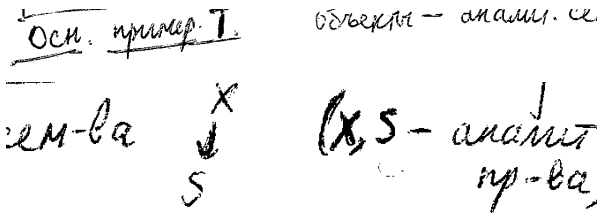
Original source details: GELFAND-P000246

Manuscript notation. The parenthesis reads форм. критерий (form. criterion). The words above it read изом. когомологий (isom. of cohomology).



Original source details: GELFAND-P000246

Manuscript notation. The box ends with $H^q(U, f^*F) \cong H^q(f(U), F)$.



Original source details: GELFAND-P000246

Diagram and typed description.

Main example T. objects — analytic families

$$\begin{array}{c} X \\ \downarrow \\ S \end{array}$$

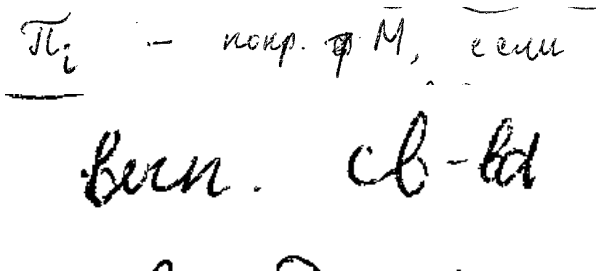
(X, S — analytic spaces)

It is necessary that

New category T_M with M. (a terminal object)

Existence of fiber products.

(topol. of the moduli space)



Uncertain reading. π_i — a covering [cancelled sign: φ / o?] of M if the properties hold

Original source details: GELFAND-P000246

1) each K occurs as a fiber in one of the families π_i

2) each K occurs only a discrete number of times

GELFAND-P000247

[source number: 6]

3) Extension axiom. $\forall X$.

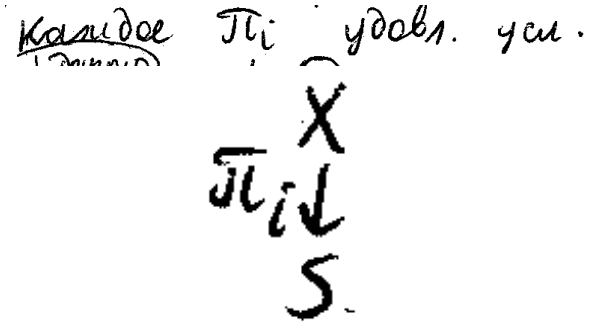


Diagram and typed description.

Each Π_i satisfies the cond.

$$\begin{array}{cc} & X \\ \pi_i & \downarrow \\ & S \end{array}$$

Original source details: GELFAND-P000247

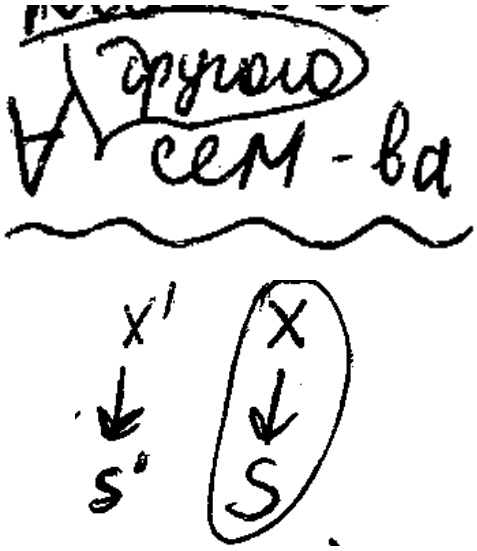
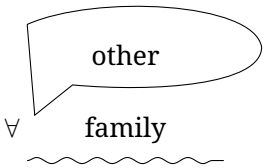


Diagram and typed description.



$$\begin{array}{cc} X' & X \\ \downarrow & \downarrow \\ S' & S \end{array}$$

Original source details: GELFAND-P000247

and a subfamily (base — a submanifold and [crossed out: i.e.?] induced restr.)



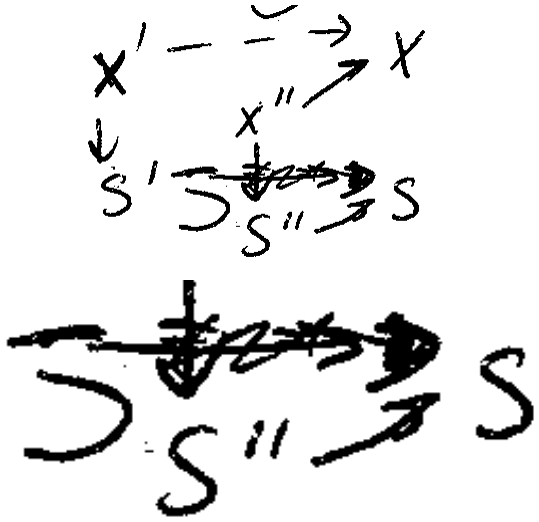
Original source details: GELFAND-P000247

Uncertain reading. A point and a low loop are visible under the strike. Provisional т.е. (i.e.), possibly т.д. (etc.).

and ∀ map of the subfamily

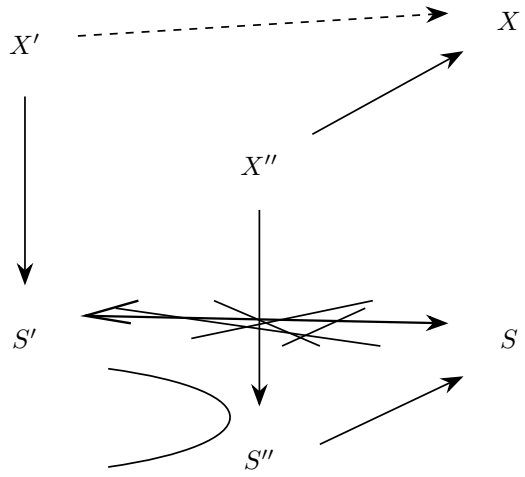
∃ its extension (local)-

— to a tubular neighborhood S''.



Original source details: GELFAND-P000247

Uncertain reading.



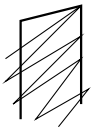
A right arrowhead and a dense left wedge survive on the crossed-over line $S' \to S$. The meaning of the curved stroke below is uncertain.

Next time — example with ellipt. curves.



Original source details: GELFAND-P000247

Diagram and typed description. Isolated hatched figure in the lower margin.



GELFAND-P000248

Palamodov.

Spencer resolution.

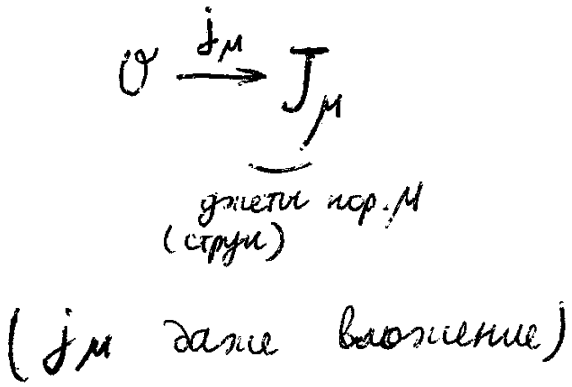


Original source details: GELFAND-P000248

Diagram and typed description.

Working on this	Sweeny Holdschmidt Guillemin Quillen Sternberg Singer	presented by Sweeny in Acta math.

A method for solving the system $p(x, D)u = w$, where p is a matrix, u, w are vectors.



Original source details: GELFAND-P000248

Diagram and typed description.

M — a smooth manifold. $\mathcal{O}$ — the sheaf of germs of diff. fns.

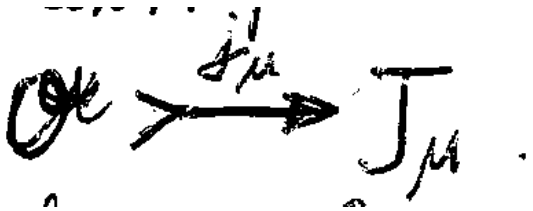
$$\mathcal{O} \xrightarrow{j_\mu} J_\mu$$

jets of ord. μ

(jets)

(j_μ is even an embedding)

J_μ — vect. bundle (or the sheaf of its germs)



Original source details: GELFAND-P000248

Uncertain reading.

$$\mathcal{O} \xrightarrow{j_\mu} J_\mu \quad .$$

Retraced $\mathcal{O}$ / $\mathcal{O}$ with an upper sign; overlapping strokes not deciphered.

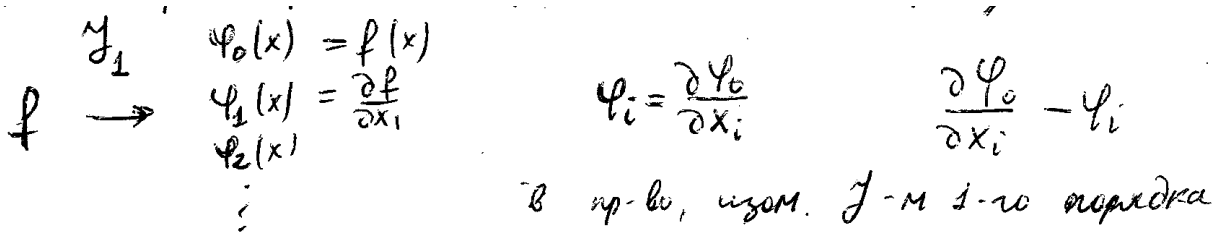
The Spencer resolution of the sheaf $\mathcal{O}$ begins with j_μ and continues so that the seq. of sheaves is exact.

$$0 \longrightarrow \mathcal{O} \xrightarrow{j_\mu} J_\mu \longrightarrow$$

$\mu = 0$ — map to 0.

A collection is given (I. M. G.'s project)

$$\varphi_i \qquad i \text{ — multi-index}, \qquad |i| \le \mu.$$



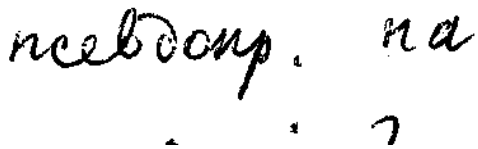
Original source details: GELFAND-P000248

Diagram and typed description.

$$f \xrightarrow{j_1} \begin{matrix} \varphi_0(x) = f(x) \\ \varphi_1(x) = \dfrac{\partial f}{\partial x_1} \\ \varphi_2(x) \\ \vdots \end{matrix} \qquad \varphi_i = \frac{\partial \varphi_0}{\partial x_i} \qquad \frac{\partial \varphi_0}{\partial x_i} - \varphi_i$$

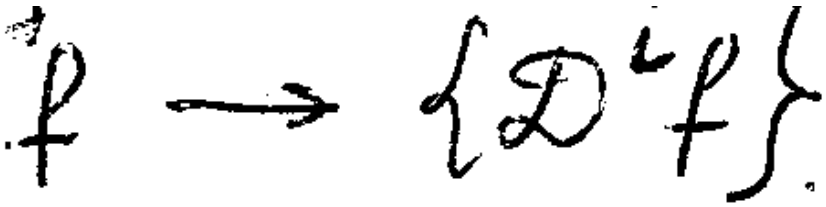
to a sp., isom. to the J's of 1st order

Cf. псевдопр.? one order higher.



Uncertain reading. Provisional reading of the abbreviation псевдопр. Its expansion is uncertain.

Original source details: GELFAND-P000248



Original source details: GELFAND-P000248

Uncertain reading.

$$f \longrightarrow \{D^i f\}$$

Above the first f are short strokes; their meaning is not established.

GELFAND-P000249

[source number: 7]

notation $1_j = (0, \ldots, \underset{j}{1}, \ldots, 0)$

cf. φ_i with φ_{i+1_j}.

Need to chk. that

$$\frac{\partial \varphi_i}{\partial x_j} = \varphi_{i+1_j}, \qquad \underset{|i| \le \mu - 1}{\forall i}, \qquad j = 1, \ldots, n.$$

Map to



Uncertain reading. Crossed out:



The first mark is heavily crossed out, possibly J or j. The start of «джеты» (jets) is rewritten.

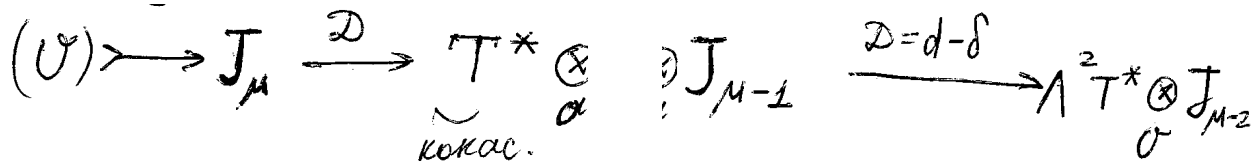
Original source details: GELFAND-P000249

jets of order lower by 1.

$$\{\varphi_i\} \longrightarrow \left\{ \frac{\partial}{\partial x_j} \varphi_i - \varphi_{i+1_j} \right\}$$

$$|i| \le \mu - 1, \qquad j = 1, 2, \ldots, n.$$

$$\sum \left(\frac{\partial}{\partial x_j} \varphi_i - \varphi_{i+1_j} \right) dx_j \qquad \text{(added).}$$



Original source details: GELFAND-P000249

Diagram and typed description.

$$(\mathcal{O}) \rightarrowtail J_\mu \xrightarrow{\mathcal{D}} T^* \otimes_{\mathcal{O}} J_{\mu-1} \xrightarrow{\mathcal{D} = d - \delta} \Lambda^2 T^* \otimes_{\mathcal{O}} J_{\mu-2}$$

cotang.

$$\left(\frac{\partial}{\partial x_j} \varphi_i - \varphi_{i+1_j} \right) = (d - \delta)\{\varphi_i\}$$

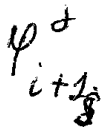
(difference of the operators of actual and formal diff.)

On $T^* \otimes J_{\mu-1}$ both ops. are also def.

d — exterior diff. in the 1st coord.

δ — formal diff.

$$\{\varphi_i^j(x) dx_j\} \overset{\mathcal{D}=d-\delta}{\longrightarrow} d(\varphi_i^j(x) dx_j) - \varphi^j_{i+1_s}\, dx_j \wedge dx_s$$



Uncertain reading. Provisional $i + 1_s$. The lower s is overwritten, possibly over an earlier j.

Original source details: GELFAND-P000249

and so on.

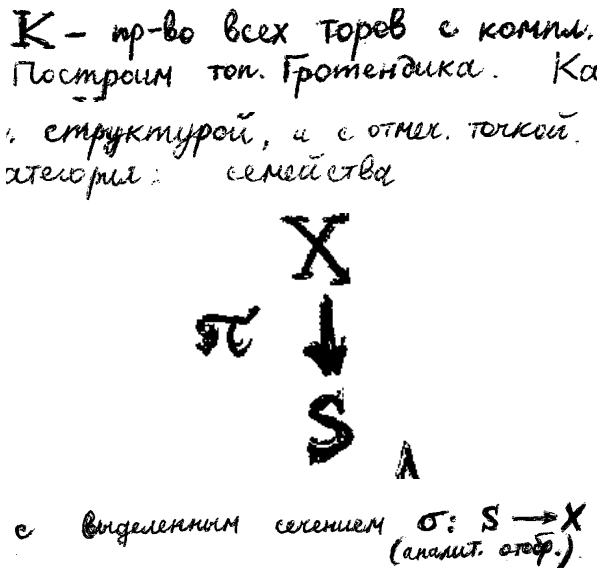
The resolution will terminate (and it is exact)

Next time Geronimus, Palamodov and Kazhdan

GELFAND-P000250

$23/\overline{\text{IX}}$

Geronimus.



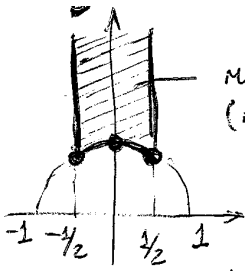
Original source details: GELFAND-P000250

Diagram and typed description.

K — sp. of all tori with a compl. structure, and with a marked point.
Let us construct Grothendieck's top.
Category: families

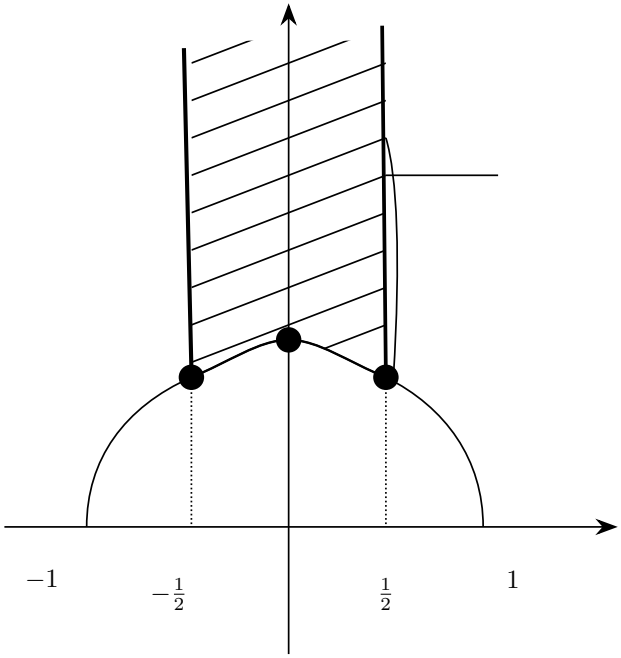
$$\begin{array}{ccc} & & X \\ & \pi & \downarrow \\ & & S \end{array}$$

with a distinguished section $\sigma\colon S \to X$ (analyt. map)



Original source details: GELFAND-P000250

Uncertain reading.



Two close strokes run along the right boundary. Their exact junction is uncertain.

set of all lattices S'

(moduli sp. in the coarse sense of the word)

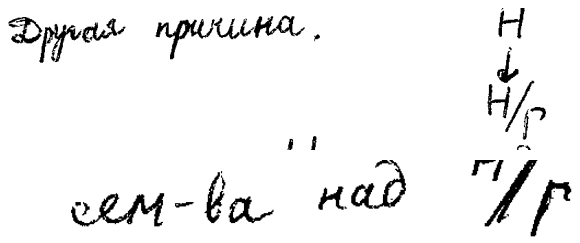
(*coarse*)

Grothendieck top. — mod. sp. in the fine (*fine*) sense of the word.

Quotient H/Γ — H — upper half-plane

There is a family with base S', but this is not a fine moduli sp.

(does not induce nontriv. bundles with const. fiber).



Original source details: GELFAND-P000250

Diagram and typed description.

Another reason.



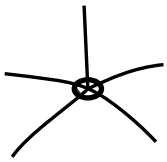
The family over H is not induced from a family over H/Γ.

(Under the action on H there are fixed points (3 of them)
there is branching there



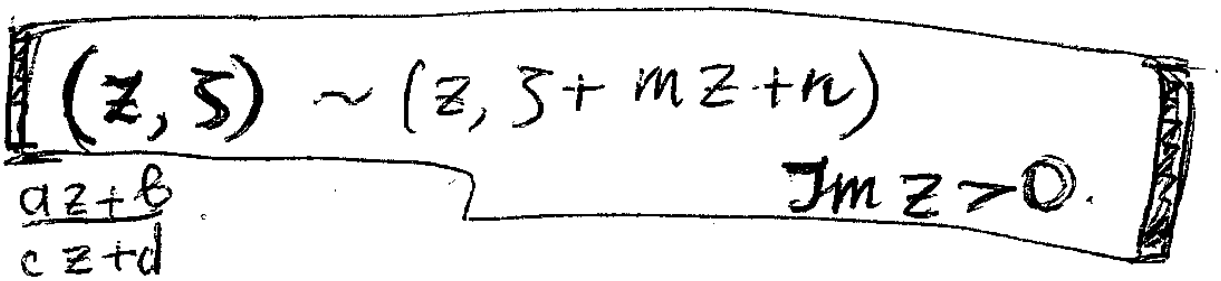
Original source details: GELFAND-P000250

Uncertain reading.



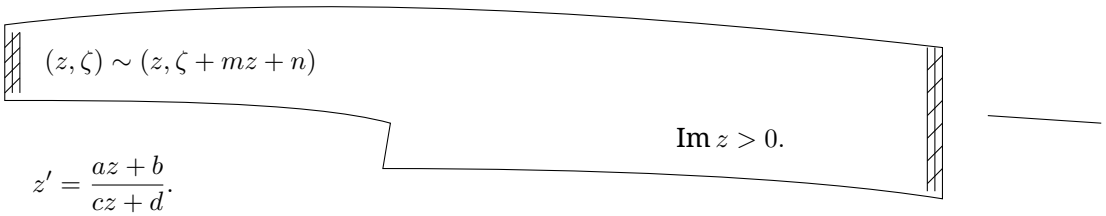
Dense retracing at the node and stroke ends; arrowheads are not established.

and different struct. of the ellipt. curve are obtained)
Family over H $\quad H \times \mathbb{C} \quad z \in H \quad (z, 1)$ —
— a lattice.



Original source details: GELFAND-P000250

Diagram and typed description.



Problem: this family,

GELFAND-P000251

[source number: 8]

and there the group acts

$$z' = \frac{az+b}{cz+d}$$
$$\zeta' = \zeta(cz+d)^{-1}$$



Uncertain reading. Crossed out in the upper-right margin. Provisional reading там (there).

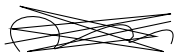
Original source details: GELFAND-P000251

Calculation of cohomology in this case.

The topology and everything else were defined earlier.



Uncertain reading. A separate word is cancelled:



Individual strokes survive under the crossing. The word is uncertain.

Original source details: GELFAND-P000251

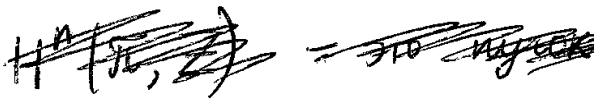


Diagram and typed description. Crossed-out formula and words.

~~$H^n(\pi, Z)$~~ ~~this is a sheaf~~

Original source details: GELFAND-P000251

$$F(\pi) = \underbrace{Z + Z + Z}_{\text{by the number of conn. comp.}}$$

(so to speak, a const. sheaf).

It is easy to see that

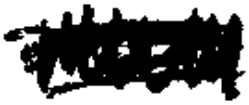
$$\underset{(\pi\text{ — ordinary})}{H^n(\pi, Z)} = \underset{\text{(Topological).}}{H^n(S, Z)}$$

Cohom. of the final object.

1) T_Γ — another top.; Γ — a group of matrices with det 1.

2) T'_Γ — those Γ-sets where Γ does not act too triv.

(if the stabilizer of $\forall$ point — integral matr. with



Original source details: GELFAND-P000251

Uncertain reading. The cancelled word has not been deciphered.

1 mod 3.

$$\sigma x = x \to \sigma \in \Gamma_3.)$$

Map $T'_\Gamma \to T$

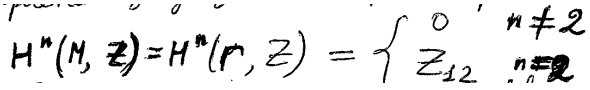
$$S \to (H \times S)/\Gamma \qquad (S \text{ — a discr. set}).$$

$$M' \to M.$$

It induces an isomorphism of cohomology.

(which is checked using the criterion stated last time).

Hence



Original source details: GELFAND-P000251

Uncertain reading.

$$H^n(M, Z) = H^n(\Gamma, Z) = \begin{cases} 0 & n \neq 2 \\ Z_{12} & n = 2 \end{cases}$$

The sign and digit in the second condition are retraced. Provisional $n = 2$.

(which, if Γ acted without fixed points, would coincide with the cohom. of the ordinary moduli sp.).

The answer is valid over any field (Char $\neq 2$).

Families have an invar.

mod 12.

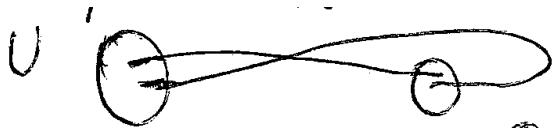
GELFAND-P000252

Shafarevich: an example of Grothendieck top.

A syst. of diff. eqns. (ordinary) is given on a set

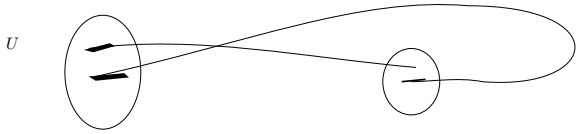
U — transv. patch

Map



Original source details: GELFAND-P000252

Uncertain reading.



The directions of all parts and exact joins are not established; the left-hand thickenings are retained by their shape.

$U \to V$

if from points of U

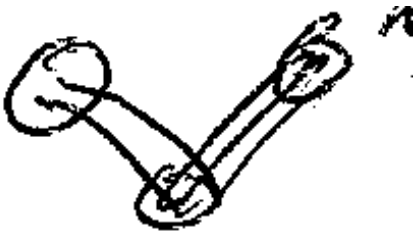
draw [crossed out: ...] traject. into V?.



Original source details: GELFAND-P000252

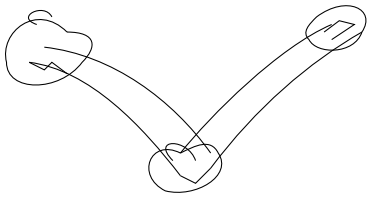
Uncertain reading. The crossed-out fragment is uncertain. The overwritten final sign is provisionally V.

there is a fiber product



Original source details: GELFAND-P000252

Uncertain reading.



Additional marks are visible within the outlines and at the upper right. Their meaning is uncertain.

If a fibration, then the cohom. of the base and the syst. are triv.

(then there is a final object).

On the plane diff. eqn. and const. sheaf of Groth. top.,

then its Euler char. — the difference [inserted above: of numbers] of lim. cycles and sing. points.



Original source details: GELFAND-P000252

Diagram and typed description. Insertion of “of numbers”.

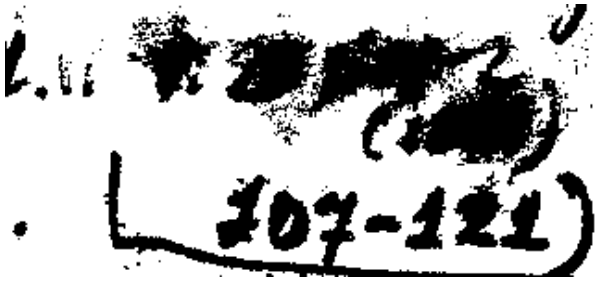
Next time Kazhdan. and Palamodov.

GELFAND-P000253

[source number: 9]

30/IX. Kazhdan.

Work of Ihara. (Jap.). (Ihara, Journ. Math. Soc. of J.



Undeciphered passage. Two short strokes follow "J." A dense cancellation follows. The range 107–121 is legible.

Original source details: GELFAND-P000253

107-121)

One-dimensional surf. over C are connected with discr. groups.

An attempt to transfer this to curves over finite fields.

$X = \Gamma\backslash M$ $\quad$ M — Lobach. pl.

Γ describes both X and unramif. coverings.

(The congruence monodromy problem).

In [crossed out: ...] $k[X, Y]$ consider the principal ideal (f)



Uncertain reading. The sign before the ring is heavily crossed out. Its reading is uncertain.

Original source details: GELFAND-P000253

$A = k[x, y]/(f)$ — ring of fns. on the curve, max. ideals — geom. points of the curve

A map $Y \longrightarrow X$ is a map of rings $A \longrightarrow B$

$$B \longleftarrow A$$

An unramif. map — unramif. at $\forall$ point.

$\underset{\text{max. ideal.}}{m} \subset A$ $\quad$ Consider the image and B/mB

if it does not contain nilpot. elems., then it is unramified at m.

There will be no seminar next Monday.

The time after next there will be...

GELFAND-P000254

14/X Palamodov.

X — set

O — sheaf of germs of diff. fns.

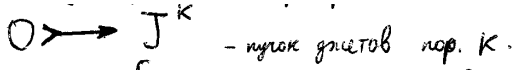


Diagram and typed description.

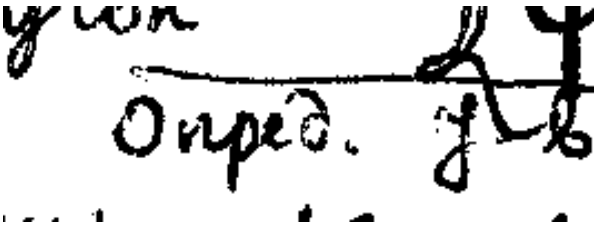
$O \rightarrowtail J^k$

— sheaf of jets of order k.

Original source details: GELFAND-P000254

presheaf $\{\varphi_i, \quad i = (i_1, \ldots, i_n), \quad |i| \leq k\}$ $\quad$ transition — as for derivatives.

Gelfand. Def. j?

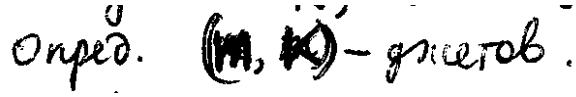


Original source details: GELFAND-P000254

Uncertain reading. After "Def." there is a small sign, provisionally j.

Let k, m be integers.

Def. of (m, k)-jets.



Original source details: GELFAND-P000254

Uncertain reading. The first letter is retraced. Provisional (m, k).

X — diff. set.
Consider fns. of the $(m+1)$-st variable.



Original source details: GELFAND-P000254

Uncertain reading. Provisional $m+1$. The letter is retraced.

(ordinary — $(1, k)$-jets).

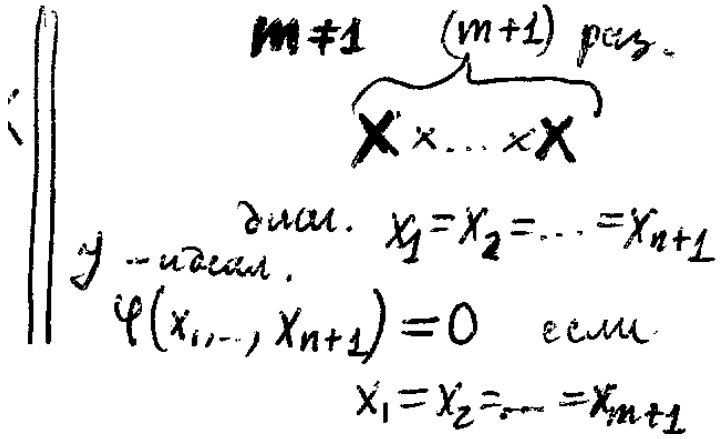
$(1, k)$-jets.

$\varphi(x, y)$ — fns. on the square $X \times X$

Consider the diagonal $x = y$

Consider the collection of fns. $= 0$ on the diagonal

This is an ideal.



Original source details: GELFAND-P000254

Diagram and typed description. Addition to the right of the two vertical strokes.

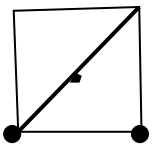
$m \neq 1 \qquad \overbrace{X \times \cdots \times X}^{(m+1) \text{ times}}$
diag. $x_1 = x_2 = \cdots = x_{n+1}$
$\mathcal{J}$ — ideal.
$\varphi(x_1, \ldots, x_{n+1}) = 0$ if
$x_1 = x_2 = \cdots = x_{m+1}$

Consider $\underline{\mathcal{J}^{k+1}}$ (fns. of k-th order of smallness on the diag.)

We regard $\varphi_1 \sim \varphi_2$ if $\varphi_1 - \varphi_2 \in \mathcal{J}^{k+1}$



Diagram and typed description. Unlabelled square, thick rising diagonal and two dark lower corners; central protrusion: point / small arrowhead.



Original source details: GELFAND-P000254

$\varphi(x, y)$ — [crossed out: ...]

Uncertain reading. Cancelled single sign: 0 / e.



Original source details: GELFAND-P000254

$(1, 0)$-jet. — fn. on the diagonal.

Undeciphered passage. Short cancelled entry at the left; possible beginning $(1, \ldots)$, remainder unreadable.



Original source details: GELFAND-P000254

$(1, 1)$ — jet

Uncertain reading. 0 / O

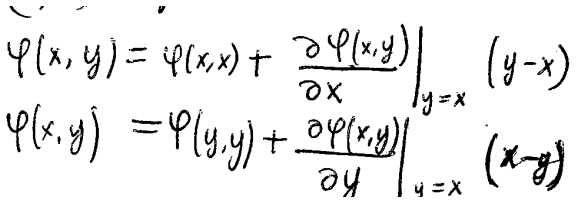


Original source details: GELFAND-P000254

on the diag. with 1-st derivatives.

$$\varphi(x, y) = \varphi(x, x) + \left.\frac{\partial \varphi(x, y)}{\partial x}\right|_{y=x} (y - x)$$
$$\varphi(x, y) = \varphi(y, y) + \left.\frac{\partial \varphi(x, y)}{\partial y}\right|_{y=x} (x - y)$$

Diagram and typed description. The formulas in the manuscript.



Original source details: GELFAND-P000254

There are 2 ways to define jets

$$a(x) + b(x)(y - x)$$
$$a(y) + b(y)(x - y)$$

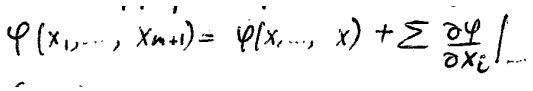
These are different ways.

GELFAND-P000255

[source number: 10]

One can differentiate — this gives a jet of lower order. 10

$$\varphi(x_1,\dots,x_{n+1}) = \varphi(x,\dots,x) + \sum \frac{\partial\varphi}{\partial x_i}\Big|_{\dots}$$



Uncertain reading. Short dots follow the lower end of the evaluation bar. The entry stops there.

Original source details: GELFAND-P000255

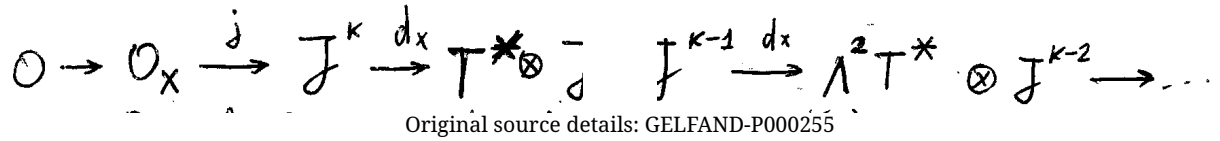
$(1,m)$-jet — scalar fn. + diff. form

Palamodov.

The case of general m reduces to $m=1$.

$$\varphi(x,y) \sim \sum_{|i|\le m} \varphi_i(x)\frac{\xi^i}{i!}$$

$$O_Y \longrightarrow O_{X\times Y}/\mathcal{J}^{k+1}$$



Original source details: GELFAND-P000255

Diagram and typed description.

$$0 \longrightarrow O_X \xrightarrow{j} J^k \xrightarrow{d_x} T^*\otimes J^{k-1} \xrightarrow{d_x} \Lambda^2 T^*\otimes J^{k-2} \longrightarrow \dots$$

Spencer resol. (triv. case)

In the gen. case — O_X is replaced by the sheaf of germs of solns. of an arbitrary lin. syst. of eqns.

Sweeny

Acta Math., 120 $N=3-4$.

and development by Malgrang

Quillen — a large article in proceedings...

Next time: Palamodov again.

GELFAND-P000256

[source number: 17]

$27/\overline{\text{X}}$. Prof. Weinstein (USA).

on operators.

4/XI?



Original source details: GELFAND-P000256

Uncertain reading. Provisional XI. The Roman numeral and the 4 are retraced.

Kirillov — Traces of operators !!

11/XI — skipped.

18/XI Kazhdan ζ-fn. following Dwork

25/XI ——— || ——— || ——— || ———

Kirillov.

$\mathcal{A}$ — operator.

M — comp. set, $\mathcal{L}_2(M,H)$ H — Hilbert.

μ — measure on it

(or a Hermitian bundle)

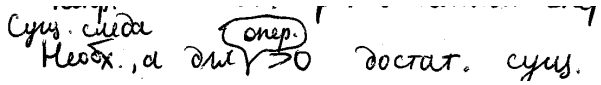
$$\mathcal{K}f(x) = \int_M K(x,y)f(y)\,d\mu(y)$$

(K — operator fn., $K(x,y)\colon H \longrightarrow H$)

jointly cont. in the weak oper. topology.

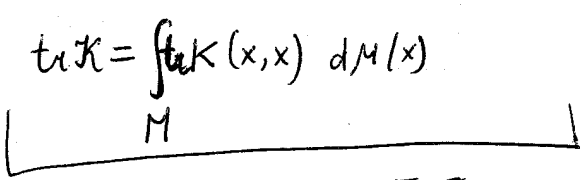
Existence of the trace

Necessary, and for [inserted above: oper.] > 0 sufficient existence.



Original source details: GELFAND-P000256

Diagram and typed description. Insertion of "oper." above the > 0 sign.



Original source details: GELFAND-P000256

Diagram and typed description. The trace formula, underlined in the manuscript.

$\operatorname{tr}\mathcal{K} = \int_M \operatorname{tr} K(x,x)\,d\mu(x)$ (tr in the usual sense).

Scalar fns.: see Gohberg and Krein (Carleman theorem)

Nontriv.: if $\mathcal{K}$ has a trace, then $\operatorname{tr} K(x,x)$ exists almost everywhere.

Weinstein.

A — self-adj. oper. with discr. spectrum.

$$\mathbf{A}+\mathbf{B} \qquad \min R(u) \qquad R(u) = \frac{(Au,u)}{(u,u)}$$

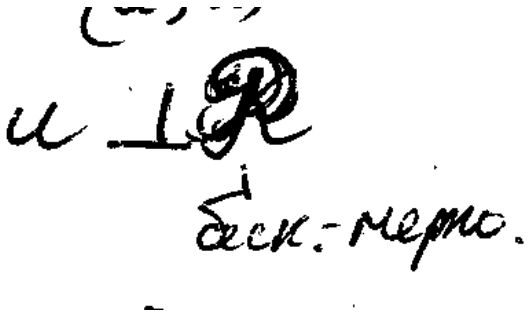
The Euler eqn. gives $Au = \lambda u = 0$



Original source details: GELFAND-P000256

Uncertain reading. The first sign after Au is retraced. Provisional $Au = \lambda u = 0$.

$u \perp \mathcal{R}$ infinite-dim.



Uncertain reading. The retraced letter has a lower-right stroke. Provisional $\mathcal{R}$, possibly $\mathcal{P}$.

Original source details: GELFAND-P000256

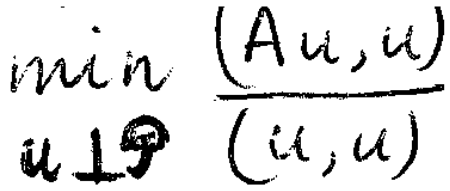
$$\mathcal{P} = \bigcup \mathcal{P}_n$$

$\min\limits_{u \perp \mathcal{P}_n} R(u)$ soln. exists: $\det \ldots = 0$

17

GELFAND-P000257

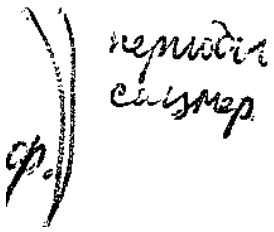
$$\min_{u \perp \mathcal{P}} \frac{(Au, u)}{(u, u)}$$



Uncertain reading. The retraced letter after $u \perp$ under the minimum is provisionally $\mathcal{P}$.

Original source details: GELFAND-P000257

exists for the Schrödinger oper.
(not only completely cont.).
In general. classical theory — estimates of eigenvals., from one side.
Weinstein and Co. — from the other side.
Someone from Poland. Rolevich.
Periodic solns. of diff. eqns. with delay.
Reduces to solving a syst. of ordinary eqns. with periodic coeffs.



Uncertain reading. Beside the double bracket, “periodic” and provisionally “case”.

periodic case?

Original source details: GELFAND-P000257

A few more applications, etc.

Appl. to control problems.

Next will be Kirillov.

$4/\overline{\text{XI}}$ Kirillov.

G — simply conn. Lie group

T — irred. representation

$$T(\varphi) = \int_G \varphi(g) T(g)\, dg \quad \text{— operator}$$

tr as a generalized fn. sometimes exists.

$$\operatorname{tr} T(\varphi) = \int_G \varphi(g)\chi(g)\, dg$$

$\chi(g)$ — generalized fn.

In the examples it is no more complicated than a measure, but there is no general Theorem.

There are examples with no trace for any φ and even narrower than C_0^∞.

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$

for it one must narrow C_0^∞ to fns. with

$$\int \varphi(a,b)\, db \underset{a}{\equiv} 0$$

(over a conjugacy class of elems. — 0)

GELFAND-P000258

[source number: 18]

$\mathfrak{g}$ — Lie alg. (real) $\quad \mathfrak{g}^*$ — dual sp. (real).

G acts in $\mathfrak{g}$ and $\mathfrak{g}^*$.

$\chi(g)$ resembles another generalized fn.

$\mathfrak{g}^* \supset \Omega$, $\quad \Omega$ — orbit of G.

Ω — symplectic sp. (there is a nondegen. 2-form there).

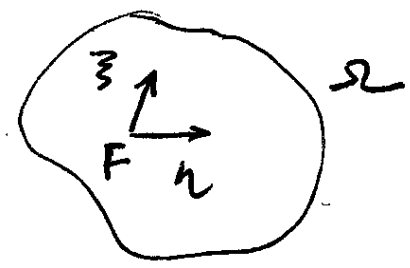
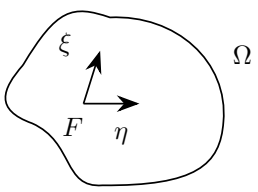


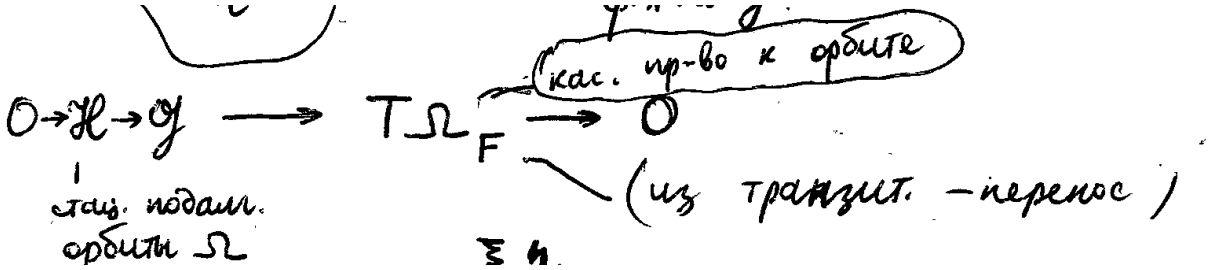
Diagram and typed description.



Original source details: GELFAND-P000258

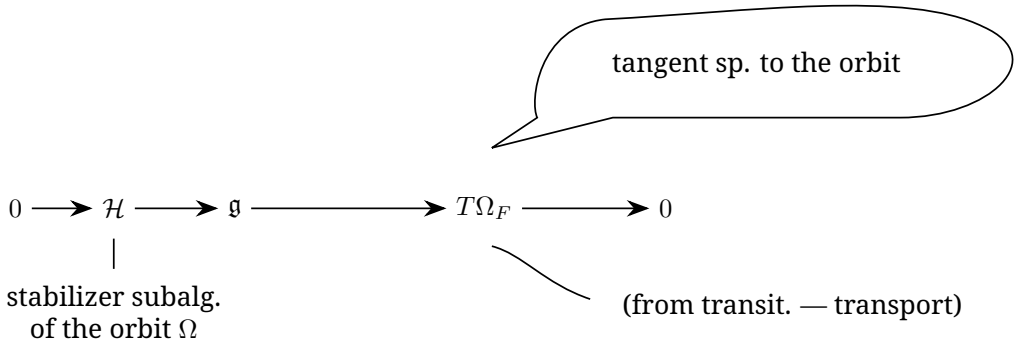
$B(\xi, \eta)$ $F \in \Omega$

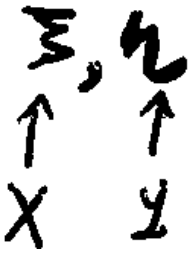
functional on $\mathfrak{g}$.



Original source details: GELFAND-P000258

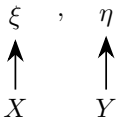
Diagram and typed description.





Original source details: GELFAND-P000258

Diagram and typed description.



One must take

$$\frac{1}{2\pi}\langle F, [X, Y]\rangle = B(\xi, \eta)$$

skew-symm.
nondegen.
form

(well-defined — independent of the choice [two signs crossed out] of representatives mod$\mathcal{H}$)



Original source details: GELFAND-P000258

Uncertain reading. Two signs after “choice” are crossed out. Their reading is uncertain.

Inhomog. form.

$$\beta = e^B = 1 + B + \frac{B \wedge B}{2!} + \cdots + \frac{B \wedge \cdots \wedge B}{k!}$$

(it terminates).

In fact, only one term is needed for integr.

Thus, a measure is defined on each orbit.

generalized fns. $\begin{cases} \mathcal{J}_\Omega(x) = \int_\Omega e^{i\langle F,X\rangle}\beta(F) & \text{(Fourier transform of the measure on the orbit)} \\ \chi(\exp X) \end{cases}$

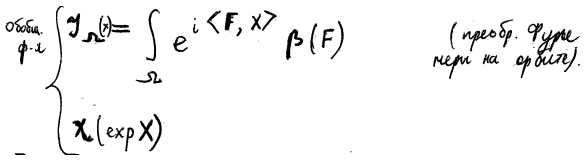


Diagram and typed description. The formulas and annotation in the manuscript.

Original source details: GELFAND-P000258

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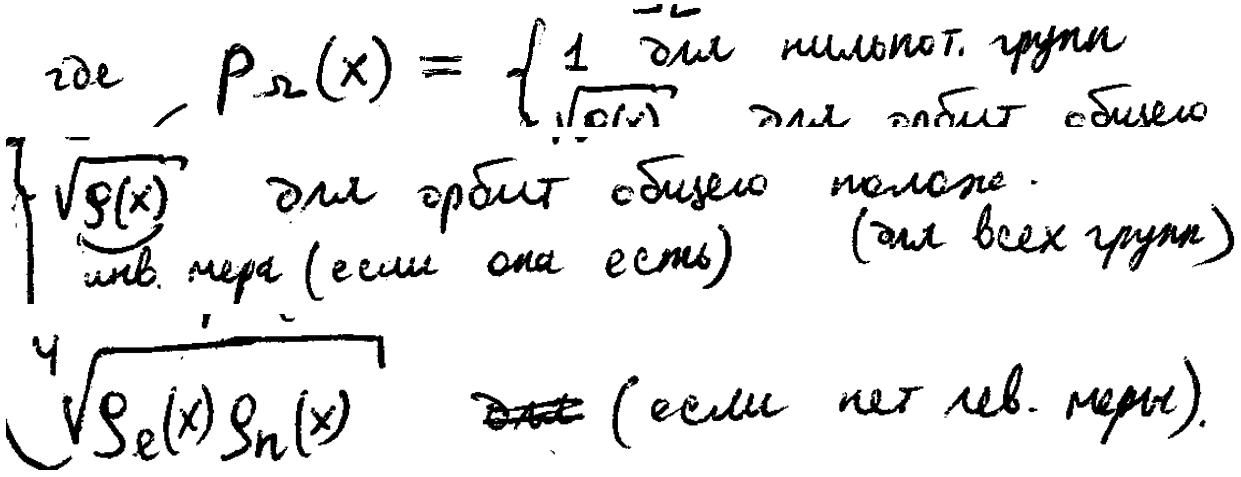
resemble one another.

the hypothesis that they coincide is false. But:

GELFAND-P000259

Their ratio $\in C^\infty$ and one can write a formula.
$\forall$ irred. repr.

$$\chi(\exp X) = p_\Omega^{-1}(x) \int_\Omega e^{i\langle F,X\rangle}\beta(F),$$

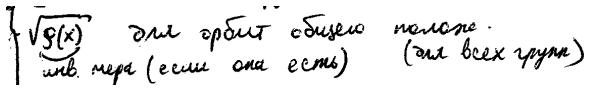


Original source details: GELFAND-P000259

Diagram and typed description.

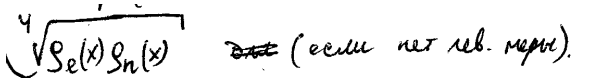
where

$$p_\Omega(x) = \begin{cases} 1 & \text{for nilpot. groups} \\ \underbrace{\sqrt{\rho(x)}}_{\text{inv. measure (if there is one)}} & \text{for orbits in general position (for all groups)} \\ \sqrt[4]{\rho_л(x)\rho_п(x)} & \text{~~for~~ (if there is no left measure)} \end{cases}$$



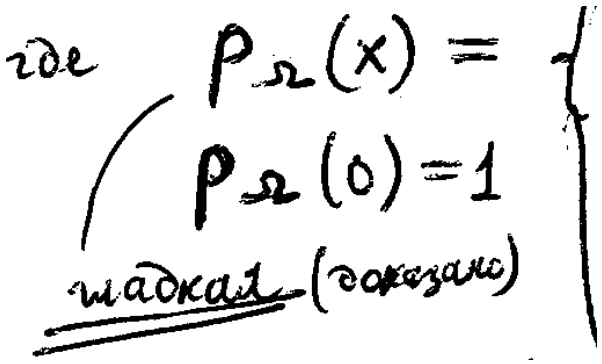
Original source details: GELFAND-P000259

Uncertain reading. Provisional $\rho(x)$ under the square root. Below it reads “inv. measure (if there is one)”.



Original source details: GELFAND-P000259

Uncertain reading. Provisional $\rho_л(x)\rho_п(x)$ under the fourth root. “For” is crossed out on the right.

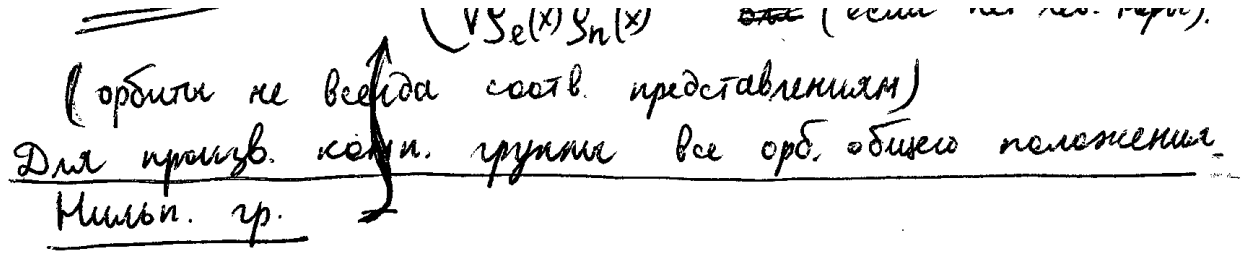


Original source details: GELFAND-P000259

Diagram and typed description.

$p_\Omega(0) = 1$

smooth (proved)



Original source details: GELFAND-P000259

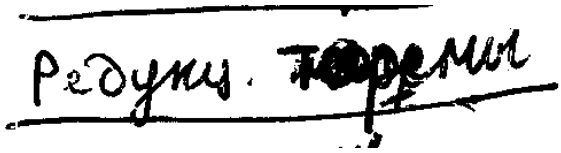
Uncertain reading. The arrow rises from beside “Nilpot. gr.” through the “comp.” line towards “always”; its reference to the formula or the orbit remark is not established.

(orbits do not always correspond to representations)

For an arbitrary comp. group all orbits are in general position.

Nilpot. gr.

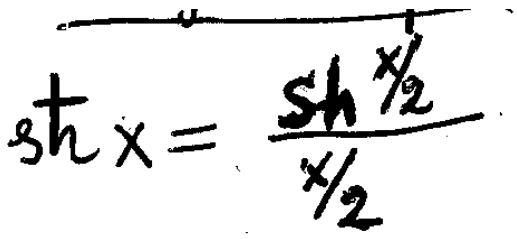
Reduction theorems



Original source details: GELFAND-P000259

Uncertain reading. Provisional reading of the retraced word: theorems (теоремы); initial strokes remain undeciphered.

$$\operatorname{sh} x = \frac{\operatorname{sh} \frac{x}{2}}{x/2}$$

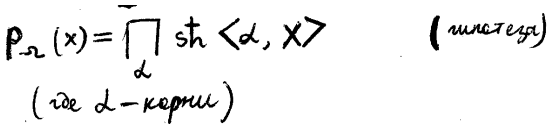


Original source details: GELFAND-P000259

Uncertain reading. A horizontal stroke crosses the ascender of the h.

$$p_\Omega(x) = \prod_\alpha \operatorname{sh}\langle \alpha, X\rangle \qquad \text{(conjecture)}$$

(where α — roots)



Original source details: GELFAND-P000259

Uncertain reading. A horizontal stroke crosses the ascender of the h.

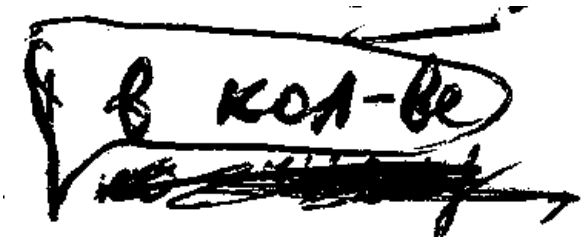
Unit. irred. reprs. of all Lie groups.

in $\mathfrak{g}^*$, orbits Ω.

Choose those for which $B \in H^2(\Omega, \mathbb{Z})$

($\int$ over $\forall$ cycle — an integer)

An integral orbit corresponds to a repr. [crossed-out entry]



Original source details: GELFAND-P000259

Undeciphered passage. The crossed-out writing is uncertain. “In a number” is inserted above it.

in a number

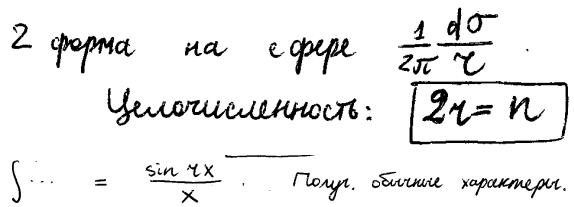
equal to the number of characters of the fund. group of the orbit.
(the fund. gr. is commutative in this case).

GELFAND-P000260

[source number: 19]

19

~~On~~ For $SU(2)$ (almost $SO(3)$)
2-form on the sphere $\frac{1}{2\pi}\frac{d\sigma}{r}$.
Integrality: $\boxed{2r = n}$ — such spheres.
$\int \ldots = \frac{\sin rx}{x}$. Obtain the usual characters.



Original source details: GELFAND-P000260

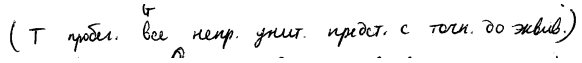
Uncertain reading. The same sign occurs in all three formulas. Provisional r, with a form that could be τ.

Universal Plancherel formula for all groups.

$$\int_G |f(g)|^2\, dg = \int \mathrm{tr}[T(f)T^*(f)]\, d\mu(T)$$

$$T(f) = \int_G f(g)T(g)\, dg$$

(T ranges over all irred.? unit. reprs. up to equiv.)

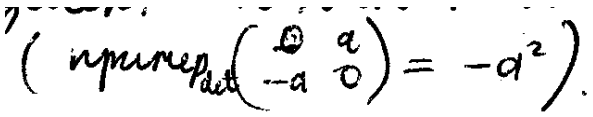


Original source details: GELFAND-P000260

Uncertain reading. The manuscript has непр., provisionally “irred.”.

Abstr. theorem — true, the question is the calculation of the measure $d\mu$.

Problem. For a skew-symm. matrix, extract a polynomial sq. root from its det.

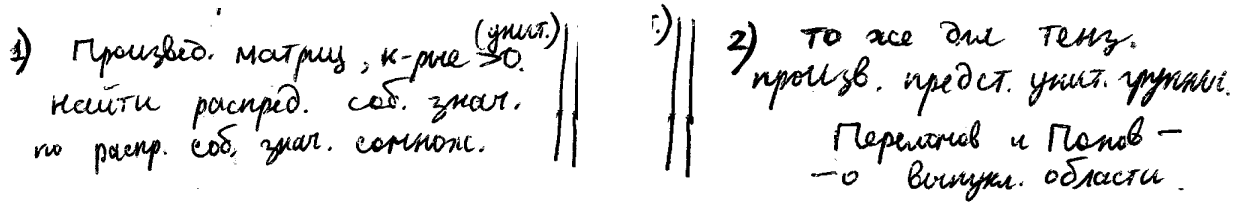


Original source details: GELFAND-P000260

Diagram and typed description. The sign before a^2 in the manuscript.

(example $\det \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} = -a^2$).

One nontriv. duality:



Original source details: GELFAND-P000260

Diagram and typed description.

1) Product of matrices which $\overset{\text{(unit.)}}{> 0}$.
Find distrib. of eigenvalues
from distrib. of eigenvalues of factors.

2) the same for tensor
product of reprs. of the unit. group.

Perelomov and Popov —
— on a convex region.

$[(x^2-1)^n]^{(n)}$ — Legendre

$[(x-1)^{m_1}(x+1)^{m_2}]^{(m_3)}$ — Jacobi polynomials

m_3 — fractional — hypergeom. function

(or $x(x-1)$)

GELFAND-P000261

$18/\overline{\text{XI}}$.



Undeciphered passage. Heavily cancelled inscription to the right of the date not deciphered; two strokes below it.

Original source details: GELFAND-P000261

Lit. on representation theory.

Minimum (0th level)

1. Kirillov lectures “Elem. theory of reprs.”, $\overline{\text{I}}, \overline{\text{II}}$
2. Gelfand, Minlos, Shapiro “Reprs. of the rotation group”
3. Zhelobenko, Lectures on theory of reprs.
4. Jacob Schwartz “Lie alg. and Lie groups” (Engl)
5. Serre — "_____
6. Vinberg “Compact Lie groups”.

7. Sophus Lie Seminar “Reprs. of Lie algs.* reprs. of Lie groups”

and

1. general theorems on fin. and comp. grs.

2. detailed analysis for rotation gr. and matrices of 2nd order — illustr. and concr. for 1

3. useful for physicists

4. general theory of Lie groups and class. comp. and semisimple Groups

6. ... comp. groups and Lie algebras (Schwartz, but shorter)

7. modern exposition of theory of semisimple groups

1st level.

1. Gelfand, Naimark ~~Reprs.~~ „Unit. reprs. of class. groups “

(Acad. Sci. 1950)

(do not read the chapter on the Plancherel theorem,

read the introd. and article of Gelfand and Graev in Trans. Math. Soc. vol. 4.

(Plancherel theorem).

2. Godement Matem. 5:5

Simple proof of the Harish-Chandra theorem on existence of char.

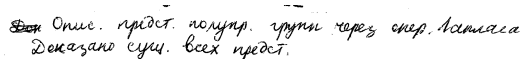
Also there, description of fin.-dim. reprs.

3 Berezin Trans. + letter to ed.

vol. 6 vol. 12

[Cancelled: For?] Description of reprs. of semisimple groups through the Laplace oper.

Existence of all reprs. proved.



Original source details: GELFAND-P000261

Uncertain reading. Provisional «Для» (for). The word is crossed out.

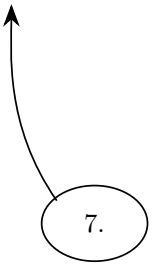
4. Kirillov Reprs. of nilpot. groups

Mackey Addendum to Segal’s book (induc. reprs.)

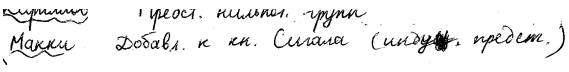


Original source details: GELFAND-P000261

Uncertain reading. Beside “Mackey” is a circled numeral, provisionally 7. An arrow points upward at the left. Below the circle is “(in the 0th)”.



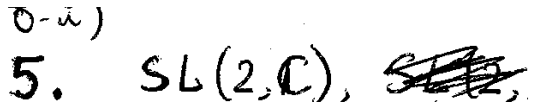
(in the 0th)



Original source details: GELFAND-P000261

Uncertain reading. The parenthesis reads инду... предст. (induc... reprs.). The end of the first word is heavily overwritten.

5. $SL(2,C)$, ~~SL_2~~,



Uncertain reading. Cancelled notation: SL_2 / $SL(2)$; position of 2 and parentheses unresolved.

Original source details: GELFAND-P000261

(5th issue of generalized functions, Naimark, Unitary repr. of the Lorentz gr.)

GELFAND-P000262

[source number: 20]

20

$SL(2,\mathbb{R})$ (5th issue of generalized functions)

$SL(2,Q_p)$ (6th issue of gen. functions) ~~or Gelfand's article)~~

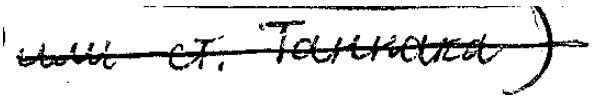


Diagram and typed description. Crossed-out reference to an article.

Original source details: GELFAND-P000262

to the next level.

1st level: no final theorems, except Berezin's work + Harish-Chandra

6. J. Schwartz, C^*-algebras *; Naimark, UMN (book)

2nd level.

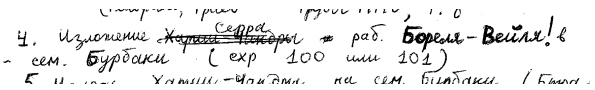
1. $SL(2,Q_p)$. Gen. functions 6.

2. Karpelevich, Gindikin (Izv. AN) ~~Method~~ Plancher. theorem for symm. spaces.

3. Method of horospheres $\rightarrow$ 1st level (Gelfand, Graev — Trans. MMO, vol. 8

Serre

4. Exposition by ~~Harish-Chandra~~ ~~o~~ of **Borel-Weil!** work in — Bourbaki sem. (exp 100 or 101)

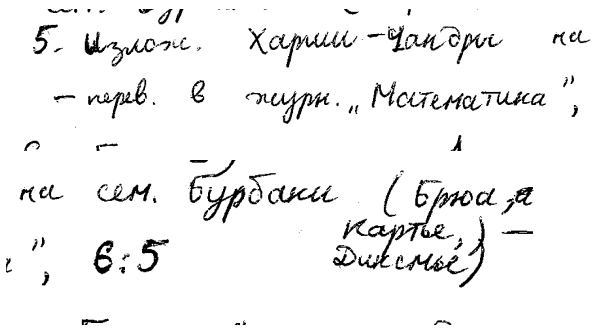


Uncertain reading. A single sign after the insertion "Serre" is heavily crossed out, possibly o.

Original source details: GELFAND-P000262

5. Exposition of Harish-Chandra at the Bourbaki sem. (Bruhat, ~~a~~ Cartier, Dixmier) —

— transl. in the journal „Matematika “, 6:5



Uncertain reading. A letter after “Bruhat” is crossed out, provisionally a.

Original source details: GELFAND-P000262

6. Groups and algs. of type 1, Glimm, Ferr, etc. —

— in Dixmier’s book „C^*-algebras and their reprs. “

7. Lie algebras and enveloping algebras.



Uncertain reading. The first numeral before the heading is overwritten. Provisional 7.

Original source details: GELFAND-P000262

7. Enveloping algebras:

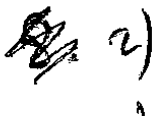
а) Sophus Lie; ch. I.

б) Harish-Chandra from 5)

в) Gelf., Kirillov; Fr. work and journal FA, vol. 3, issue 1

(Publ. Math.)

~~в)~~



Uncertain reading. A separate crossed-out sign before г) has two loops and resembles 8. Its reading is uncertain.

Original source details: GELFAND-P000262

г) Kostant, Amer. Journ., 85, pp. 327–404

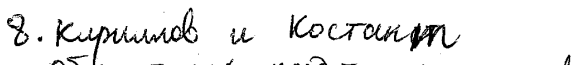
д) Zhelobenko „Enveloping algs. “

е) Nelson „Anal. vectors “ (there is a transl. in the journal Matem.)

6:3

GELFAND-P000263

8. Kirillov and Kostant

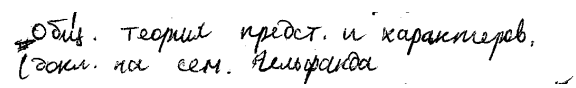


Uncertain reading. Provisional “Kostant”. The end of the surname is retraced.

Original source details: GELFAND-P000263

Gen. theory of reprs. and characters.

(talk at Gelfand’s sem.



Uncertain reading. A dense mark appears to the left of “Gen.”. Its reading is uncertain.

Original source details: GELFAND-P000263

9. Harish-Chandra — work of the latest series. (from 1963)

10. Kazhdan — func. analysis and Fr. translation at the Bourbaki sem.

vol. Ī, issue 1.

(Kirillov's talk)

11. G/Γ — extensive lit. (conference Boulder)
Sommer Sicool?

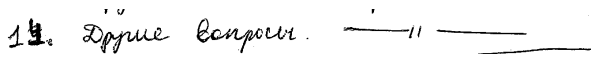
Proceed. — translation in preparation.



Original source details: GELFAND-P000263

Uncertain reading. Below "Boulder", provisionally "Sommer Sicool". The first strokes of the second word are uncertain.

12. Other fields. Extensive lit.,
14? Other questions. — // —



Original source details: GELFAND-P000263

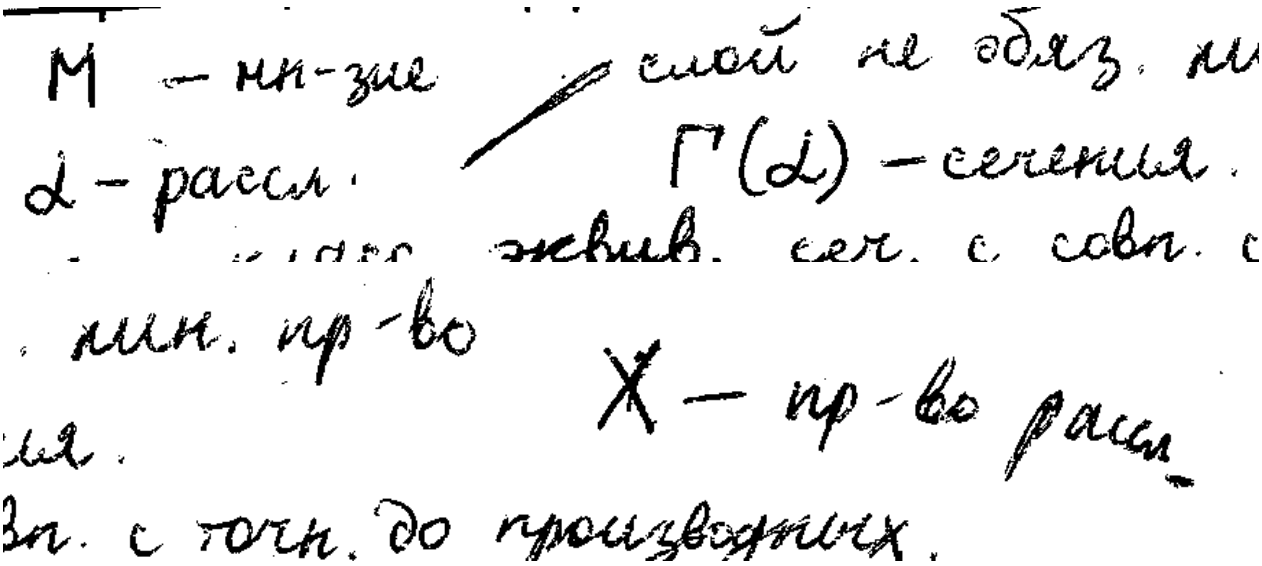
Uncertain reading. The final numeral is overwritten. Provisional 14, possibly 13.

13. List of questions.

Everything will be in "Uspekhi"

Gromov.

1. Invar. def. of diff. oper.



Original source details: GELFAND-P000263

Diagram and typed description.

M — set

fibre not necessarily a linear space

$\mathcal{L}$ — bundle

$\Gamma(\mathcal{L})$ — sections.

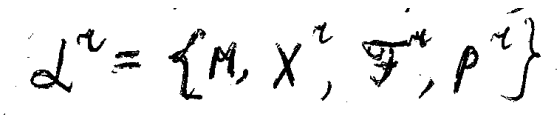
X — bundle space

Jets — equiv. class of sects. coinciding up to derivatives.
$\mathcal{L}^r$ — jet extension.

X^r — its space

Partial diff. equation —
— subset $A \subset \boldsymbol{X}^r$.
In appl. — a subset (not necessarily closed),
linear equation — a subbundle.
(add. to X of vect. spaces)

$$\mathcal{L}^r = \{M, X^r, \mathcal{F}^r, \rho^r\}$$



Uncertain reading. Last symbol of the quadruple: ρ^r / p^r; superscript r retained, the letter unresolved.

Original source details: GELFAND-P000263

Subset A in class. theory is usually closed.
But the equation $Lu \neq 0$ — an open set
$L(u) = 0$ — closed (if the func. does not satisfy, then

GELFAND-P000264

[source number: 21]

21

Or example.

$$f : \underset{g_1}{M_1} \longrightarrow \underset{g_2}{M_2}$$ with quad. forms.

smooth map, inducing forms $f^*(g_2) = g_1$.

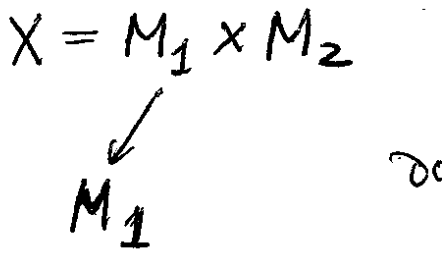


Diagram and typed description.

$X = M_1 \times M_2$

M_1

Original source details: GELFAND-P000264

map $M_1 \to M_2 \Leftrightarrow$ section
add an open cond. — so that it is an embedding.
General formulation:
solution exists or not.

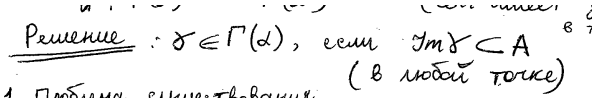
$$d^r : \Gamma(\mathcal{L}) \longrightarrow \Gamma(\mathcal{L}^r)$$

(sect. has a jet at a point).

Solution:

$$\gamma \in \Gamma(\mathcal{L}), \quad \text{if} \quad \operatorname{Im}\gamma \subset A$$

(at any point)



Uncertain reading. The same looped sign occurs twice, provisionally γ. It follows Im directly.

Original source details: GELFAND-P000264

1. Problem of existence.

Connection with others

2. Uniqueness. never occurs (almost never)
open sets

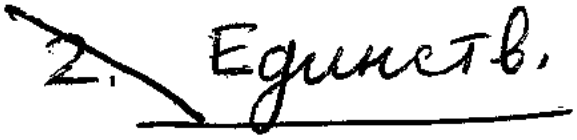


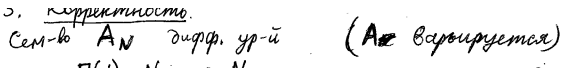
Diagram and typed description. Slanting stroke across the numeral 2.

Original source details: GELFAND-P000264

But one can speak of well-posedness.

3. Well-posedness.

A family A_N of diff. equations ($A_?$ varies)



Uncertain reading. The left occurrence is A_N. The right subscript is overwritten and uncertain.

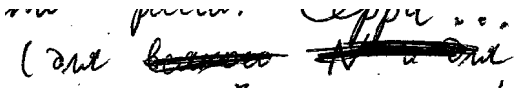
Original source details: GELFAND-P000264

$$\Gamma(\mathcal{L}) \times N \longrightarrow N$$

Well-posedness — this is a Serre fibration!!!

of a family of problems.

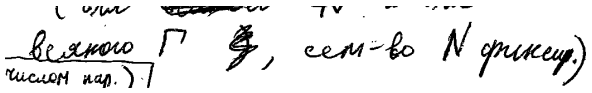
(for ~~every~~



Undeciphered passage. Following cancelled continuation unreadable; only elements resembling N and “и” are distinguishable. Complete words and number of hidden signs unknown.

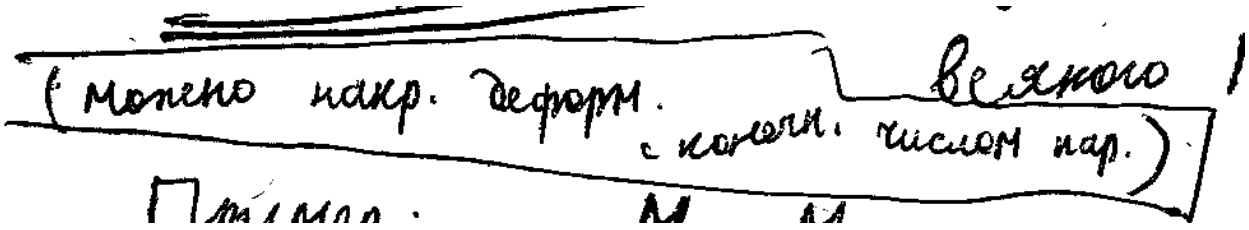
Original source details: GELFAND-P000264

every Γ, family N fixed)



Uncertain reading. After Γ, a separate cancelled looped sign is undeciphered; lower stroke and following comma visible in the complete fragment.

Original source details: GELFAND-P000264



Original source details: GELFAND-P000264

Diagram and typed description.

(one may, e.g., deform

with a finite number of param.)

Example:

M_1, M_2 a pair of forms defines an eq.
$g_1 \quad g_2$

All pairs of forms — N,

Idea: as a rule, one assumes that locally everything is known,

and what is global — unknown.

Local part: Nash's implicit-function theorems.

in ex.

GELFAND-P000265

Stab. well-posedness allows solving eqs.

Theorem C^1-Nash.

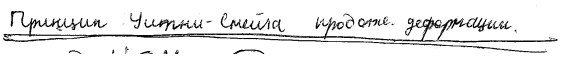
problem of C^1-imm. is posed correctly:

with a deform. of the metric one can deform the immersion.

With some metric an imm. exists — top. methods,

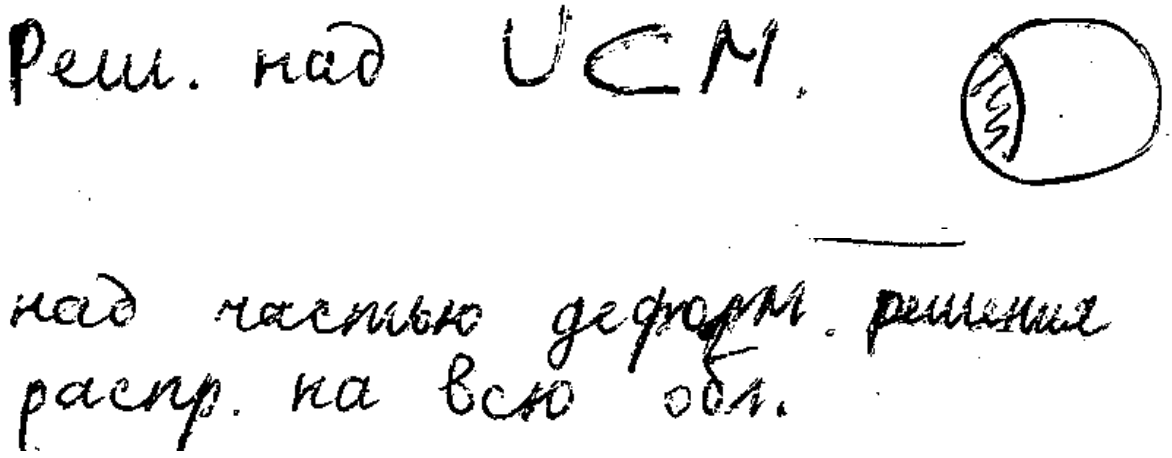
then deform it into any — obtain an emb. !!

Whitney–Smale princ. of contin. deformation.



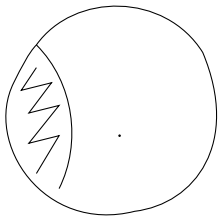
Original source details: GELFAND-P000265

Uncertain reading. The abbreviation is provisionally продолж. The upper stroke ends in a point on the right.



Original source details: GELFAND-P000265

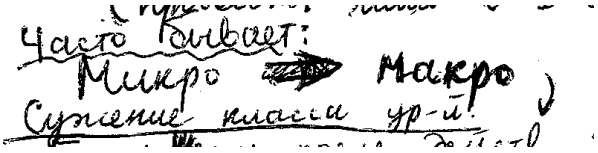
Diagram and typed description.



Sol. over $U \subset M$, deform the solution over a part

extend to the whole region

(if the boundary effect is ignored)
They proved it in a number of cases.
~~From corr. and this princ. one can~~
Microscopic realizability of these principles —
— this is the implicit-function theorem.
(contin. only at the 1st instant).



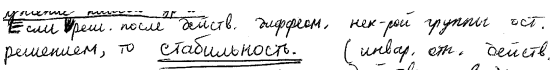
Original source details: GELFAND-P000265

Diagram and typed description.

Often:

Micro ⟶ Macro ↘

Narrowing the class of eqs.
If **[?]** a sol. after the action of diffeom. of some group remains
a solution, then Stability.



Original source details: GELFAND-P000265

Uncertain reading. A heavily overwritten sign precedes “sol.”. Its reading is uncertain.

(invar. under the action
of the group of all diff., which acts there in X —
— if this is given)
example — tang. bundle.
(maps the eq. to itself)



Original source details: GELFAND-P000265

Diagram and typed description. Crossed-out title of the problem.

~~Plateau prob~~

Example: Plateau prob.

$$R^2 \longrightarrow R^3$$ minim. films invar. under diffeom.

Then (in the stable case) micro ⇒ macro.

Another principle.

(Theorem ^{2}.)

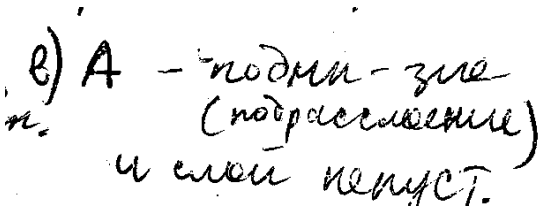
a) M parall.

noncomp. conn.

в) A — subset

(subbundle)

and fibres nonempty.



Uncertain reading. The letter before A has two loops: “в)” / “b)”. The order of the conditions does not determine its alphabet.

Original source details: GELFAND-P000265

⇒

13-04 from 16-18 Tue.; sem. Arnold

GELFAND-P000266

[source number: 22]

c) there is micro

d) there is stability

Then the eq. has a solution.

It contains Smale’s immersion theory and so on.

Noncomp.: micro ⇒ macro.

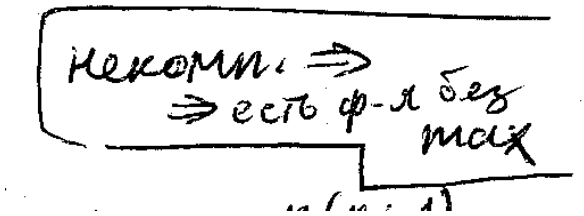


Diagram and typed description.

noncomp. ⇒

⇒ there is a fn. without max

Original source details: GELFAND-P000266

Geometric example

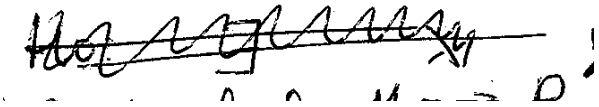
M^n — Riemannian

$M^n \subset R^s$ loc. in $s = \frac{n(n+1)}{2}$

comp. of metr. tensor

one cannot, generally speaking, fit into a lower one.

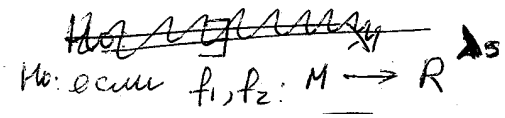
~~But, [?]~~



Original source details: GELFAND-P000266

Undeciphered passage. After “But,” the cancelled continuation is unreadable; looped and horizontal strokes are visible.

But: if $f_1, f_2 : M \to R^{?s}$.



Original source details: GELFAND-P000266

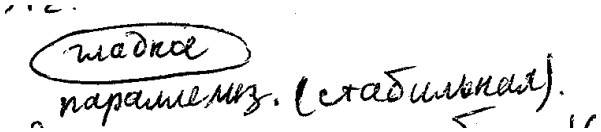
Uncertain reading. The sign before the superscript s is heavily overwritten. Its final form is uncertain.

$\underline{\underline{0 < \lambda < 1}}$



Original source details: GELFAND-P000266

Undeciphered passage. Cancelled formula at the right; lower loop resembles g_0. Left-hand part and relation sign are unreadable.



Original source details: GELFAND-P000266

Diagram and typed description.

smooth

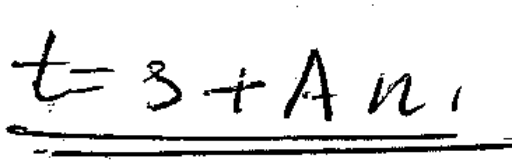
parametriz. (stable).

they are C^0-approxim. by maps φ_1, φ_2, such that the induced metrics are equal.

(Essentially there is no unique determination)

In large dim. $t \geq s$

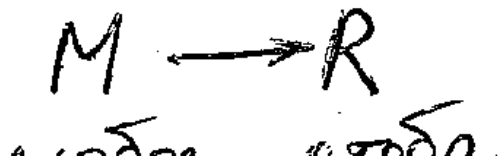
$\underline{\underline{t = 3 + An}}$



Original source details: GELFAND-P000266

Manuscript notation. The formula $t = 3 + An$ is doubly underlined.

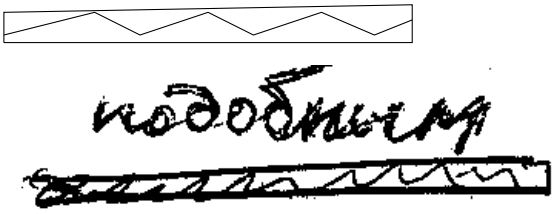
$$M \to R$$



Original source details: GELFAND-P000266

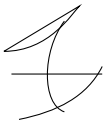
Manuscript notation. The manuscript gives $M \to R$ without an exponent.

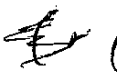
any map is approxim. by a similar one



Original source details: GELFAND-P000266

Uncertain reading. Narrow hatched outline below “similar”. No letters could be identified.





Original source details: GELFAND-P000266

Uncertain reading. Separate overwritten sign: f / t. Order of the superimposed strokes undetermined.

(with induced metric λg)

All this follows from theorem 2.

Next time: Gromov

GELFAND-P000267

25/XI

~~I?I~~

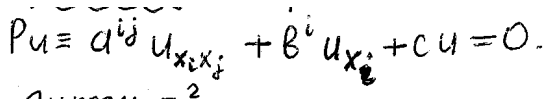


Original source details: GELFAND-P000267

Uncertain reading. Dense cancelled sign at the right of the date is unreadable.

Oleinik Radkevich’s results on eqs.

$$Pu \equiv a^{ij} u_{x_i x_j} + b^i u_{x_i} + cu = 0.$$



Original source details: GELFAND-P000267

Uncertain reading. The subscript on x is heavily overwritten. The dot and terminal curve support a provisional i.

hypoell. — ?

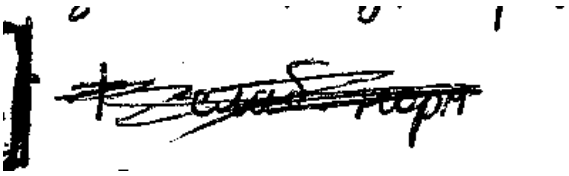
Hörmander:

$$\sum_{i=1}^{n} X_i^2 u + X_0 u + cu = f$$

if at every point there are n lin. indep. commutators, then hypoell., which is obtained from the est.:

$$\|u\|_\varepsilon \leq C\big[\|u\|_0 + \|Pu\|_0\big]$$

~~+ weak norm[?]~~

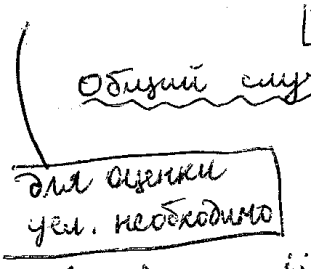


Original source details: GELFAND-P000267

Uncertain reading. After +, provisionally "weak norm". The ending of the last word is crossed out.

$\boxed{\varepsilon > 0}$ (necessity not proved)

General case: without resorting to the concrete form



Original source details: GELFAND-P000267

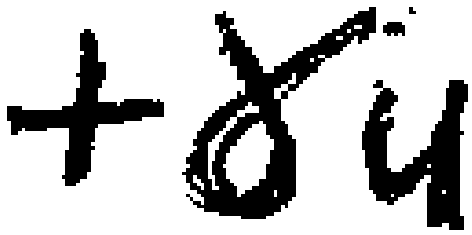
Diagram and typed description.

for the estimate

cond. necessary

Radkevich:

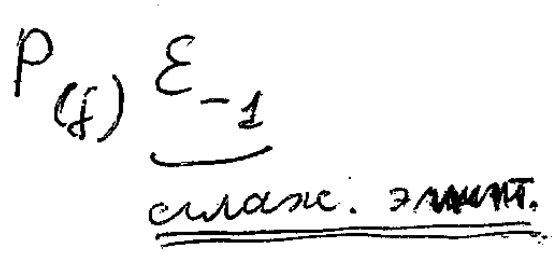
$$Lu = a^{ij} u_{x_i x_j} + \delta u$$



Original source details: GELFAND-P000267

Uncertain reading. The looped sign with crossed upper strokes before u is provisionally δ.

$$P^0 = a^{ij}u_{ij}, \qquad P^{(j)} = a^{ij}(x)D_i,$$

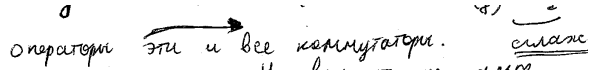


Original source details: GELFAND-P000267

Diagram and typed description.

$P_{(j)}$ $\mathcal{E}_{-1}$

smooth. ellipt.



Original source details: GELFAND-P000267

Diagram and typed description.

Operators these and all commutators.

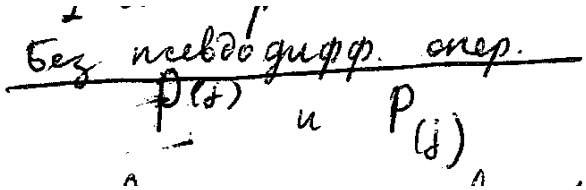
The condition is the same

($2n$ first-order oper.)

$$P_{(j)} = \sum_{i,k} \frac{\partial a_{ik}}{\partial x_j} \frac{\partial^2}{\partial x_i \partial x_k}(\mathcal{E}_{(-1)})$$

Rank n — one can select an elliptic system.

First order — the same



Original source details: GELFAND-P000267

Diagram and typed description. Short stroke below the notation on the left.

without pseudodiff. oper.

$P^{(j)}$ and $P_{(j)}$

everywhere we keep the principal parts

It is necessary to be able to obtain an ellipt. system.

There is as yet no formulation incl. Hörmander's theorem.

GELFAND-P000268

[source number: 25]

Gelfand: for applications it is important to know: when is every solution a summable function?

25

Kazhdan

Work of Dwork: rationality of the ζ-functions of all alg. varieties over a fin. field.

k — a field.

$A = k[x_1, \ldots, x_n]/\mathfrak{J}$ — an aff. scheme



Original source details: GELFAND-P000268

Uncertain reading. The ideal sign has a long descending hook. Provisionally $\mathfrak{J}$.

Points — closed points (max ideals)

Hilbert's thm.:

$$K = A/I \quad \text{— a fin. ext. of } k.$$

Prime ideals — points. $K \supset k$.

Geom. point with values in K: a homomorphism $A \to K$.

(a max ideal corresp. to it)

$$A = \frac{R[x]}{x^2 + 1}$$

2 geom. points, although 1 scheme point

i and $-i$

Alg. variety — if there are no nilpot. elements

(this is the difference from a scheme)

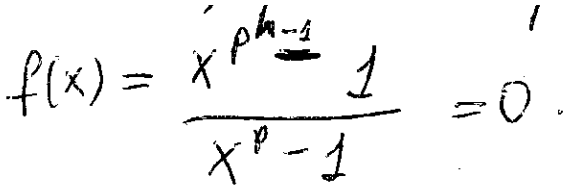
[?]



Original source details: GELFAND-P000268

Uncertain reading. At the left before F_{p^n} is a separate densely cancelled unreadable sign.

F_{p^n} — a fin. field, ext. of Z_p by the eq.



Original source details: GELFAND-P000268

Uncertain reading. In the numerator $n-1$ sits above p. A heavy horizontal stroke below $n-1$ precedes a separate 1. The function of the heavy stroke is uncertain.

$$f(x) = \frac{x^{p^{n-1}} - 1}{x^p - 1} = 0.$$



Original source details: GELFAND-P000268

Diagram and typed description.

K

$\uparrow$

k

The autom. group of K is cyclic and has a canon. gen. — the Frobenius autom.

$$\sigma : x \to x^p.$$

$k = Z_p$.

A — an alg. variety. ζ-func. of it — ?

[?].



Uncertain reading. Interlinear mark above the following line: 3. / §.; reading and role undetermined.

Original source details: GELFAND-P000268

Consider a closed point of A — $x = I \subset A$

A/I — a finite field — an ext. of some degree.

Degree — degree of the closed point.

GELFAND-P000269

$$\zeta(t, A) = \prod_{x \in X} \frac{1}{1 - t^{\deg x}}$$

(t — a formal variable)

Theorem. (Dwork)

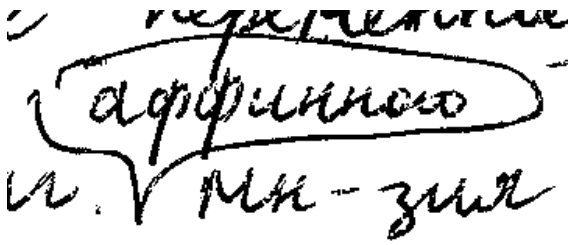


Diagram and typed description.

affine

$\forall$ alg. variety — this is a rat. func. of t

Original source details: GELFAND-P000269

(was Weil's conjecture, proved earlier for curves).

There are usually infinitely many points.

If A — the ring

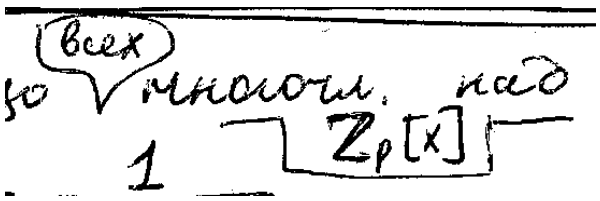


Diagram and typed description.

all

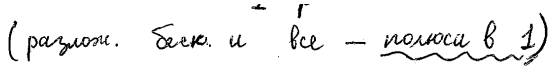
polynom. over Z_p, then

$Z_p[x]$

Original source details: GELFAND-P000269

$$\zeta(t) = \frac{1}{1 - pt}$$

(expansion inf. and all — poles at 1)



Manuscript notation. The manuscript says "and all — poles at 1", with a wavy underline under "poles at 1".

Original source details: GELFAND-P000269

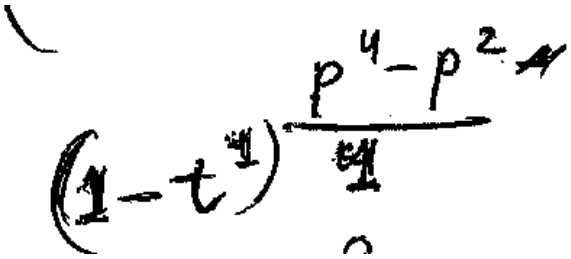
Degree 1 — exactly p points (x — Z_p).

Degree 2 — in $(p^2 - p)$ elem.

(so as not to fall among those counted
$\exists 1$ nontriv. autom.

$$\frac{1}{(1-t)^p} \quad \frac{1}{(1-t^2)^{\frac{p^2-p}{2}}} \quad \frac{1}{(1-t^3)^{\frac{p^3-p}{3}}} \quad \dots$$

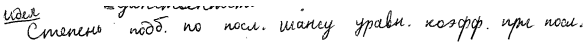
$$(1-t^4)^{\frac{p^4-p^2}{4}}$$



Uncertain reading. The fourth entry is on its own line. Two occurrences of 4 are overwritten. A short sign after p^2 is crossed out.

Original source details: GELFAND-P000269

Uniqueness — ?
Idea
Degree chosen by the last chance to equate the coeff. at the last ...



Uncertain reading. Provisionally по посл. шансу (“by the last chance”). The final abbreviation посл. is left unexpanded.

Original source details: GELFAND-P000269

Hence everything can be expanded.

GELFAND-P000270

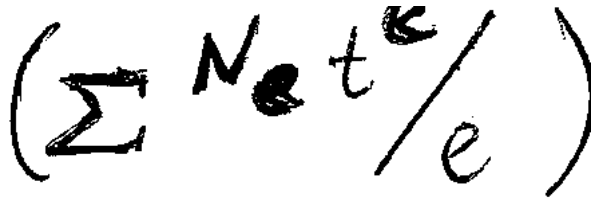
[source number: 26]

26

$$\zeta(t, A) = \sum c_k t^k \qquad c_k \text{ — integers} \qquad (\text{obv.}).$$

The equality holds.

$$\zeta(t, A) = \exp\left(\sum N_e t^e / e\right),$$



Manuscript notation. The looped letter e occurs in the indices on N and t and after the slash.

Original source details: GELFAND-P000270

where N_e — the number of geom. points with values in a field with deg $= e$.
(number of homom. from A to F_{p^e}).
This is a general equality.

But [?] it does not immediately follow that c_k are integers



Original source details: GELFAND-P000270

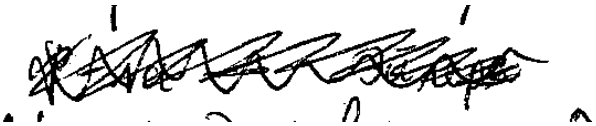
Uncertain reading. After “But” a short entry is crossed out and boxed. Its reading is uncertain.

Compute for $Z_p[x]$. $\quad N_e = p^e$.

$$\sum (pt)^e/e = -\ln(1-pt) \qquad \text{and the rest is clear.}$$

Apply at $A = Z$. The usual ζ-func. results.

[?...?]



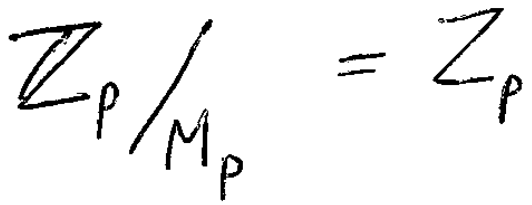
Original source details: GELFAND-P000270

Undeciphered passage. Cancelled line not fully read. Possibly Russian “Идея ... Дворк” / “... Дворк” (“Idea ... Dwork” / “... Dwork”); content beneath the hatching undetermined.

Idea of proof. The series $\zeta(t, A)$ in a compl. variable — conv. in a disc.
Consider it in a p-adic variable. It will also conv. for a small norm.
Merom. p-adic (ratio of holom. p-adic) $\Longrightarrow$
$\Longrightarrow$ autom. rational (general theorem) —
— a simple theorem

$$\mathbb{Z}_p \text{ — ring} = \{x \in Q_p;\ |x| \le 1\}$$
$$M_p = \{x \in Q_p :\ |x| < 1\} \quad \text{— max. ideal}$$

$$\mathbb{Z}_p / M_p = Z_p \quad \text{— a field of } p \text{ elem.}$$



Manuscript notation. The left Z has a doubled diagonal. The Z on the right has a single diagonal.

Original source details: GELFAND-P000270

$$x_n \in Q_p. \qquad \sum_{\text{conv.}} x_n \iff x_n \to 0.$$

Banach sp. — the usual def. carries over.
Completely cont. oper.

GELFAND-P000271

Thm. In a Ban. sp. a compl. cont. oper. is nuclear,

[?]



Original source details: GELFAND-P000271

Uncertain reading. After "nuclear" there is a comma and a short crossed-out entry. Its reading is uncertain.

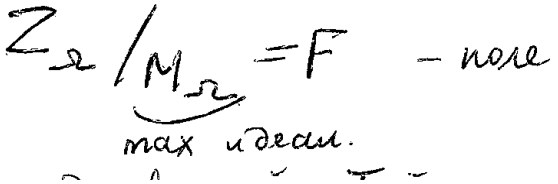
and therefore there exists $\det(1 - tA)$

— an entire p-adic func. of a p-adic variable t.

Alg. extensions of Q_p.

~~Closure~~

Ω — alg. completion of the alg. closure of Q_p



Original source details: GELFAND-P000271

Diagram and typed description.

$Z_\Omega / M_\Omega = F$ — field

max ideal

Teichmüller representatives exist.

Every elem. $f \in F$ has a canon. representative in Z_Ω.

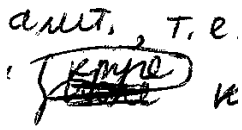
$F^* \longrightarrow Z_\Omega^*$ there is a canon. homomorphism.

Structure of analytic funcs. on the field of p-adic numbers.

$$f(t) = \sum c_k t^k, \qquad c_k \in \Omega.$$

Entire analytic, i.e. c_k sufficiently rapidly $\to 0$.

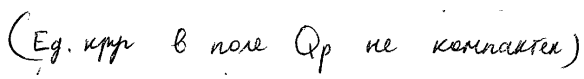
Then in every disc [?] of finite radius there are finitely many zeros.



Original source details: GELFAND-P000271

Uncertain reading. "disc" is inserted above a cancelled word and outlined with a lower point. Initial word undetermined.

(Unit disc in the field Q_p is not compact)



Original source details: GELFAND-P000271

Manuscript notation. The parenthesis reads "Unit disc in the field Q_p is not compact".

(Shnirelman)

Proof: the max principle, Schwarz lemma $\Rightarrow$ finitely many zeros.

$$f(z) = \frac{\prod(1-\alpha_i z)}{\prod(1-\beta_j z)} = 1 + \sum c_k z^k$$

c_k — integers

α_i, β_j — alg. $\Longrightarrow$ they are alg. integers.

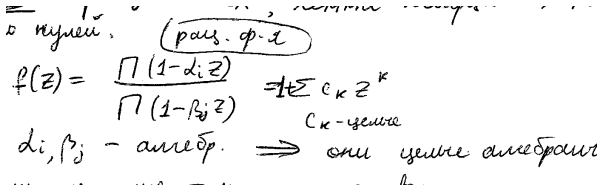
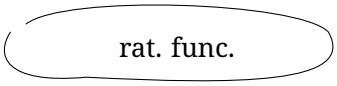


Diagram and typed description.



Original source details: GELFAND-P000271

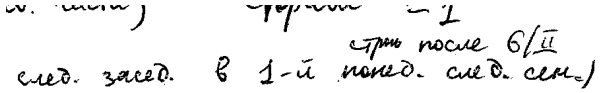
If not integers, a contradiction with the norms is obtained

$|\ | < 1$ (on the left) on the right $= 1$

(This is an example)

str[?] after $6/\overline{\text{II}}$

next mtg. on the 1st Mon. of next sem.)



Uncertain reading. A joined entry above "Mon." and before "after 6/II", provisionally стр.

Original source details: GELFAND-P000271

GELFAND-P000272

$10/\overline{\text{II}} - 1969$ **yr.**

Margulis. Integer points in a disc on the Lobachevsky plane.

Int. points — preimages on the univers. covering.

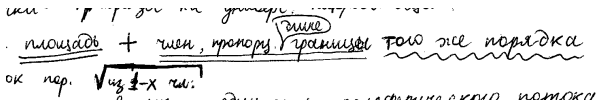
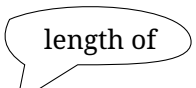


Diagram and typed description.

Obtain area +



term, proport. to boundary of the same order

Remainder of order $\sqrt{\text{of the 1st terms}}$

Original source details: GELFAND-P000272

The ergodicity of the horospherical flow is used in full.

There are works of Linnik: how many points on a sphere.

What will there be?

1. Kazhdan. 2. Margulis. 3. Kirillov.

Gelfand:

3-adic numbers (for schoolchildren)

Inf. number of digits to the left, rather than to the right.

The rules and top. are introduced analogously.

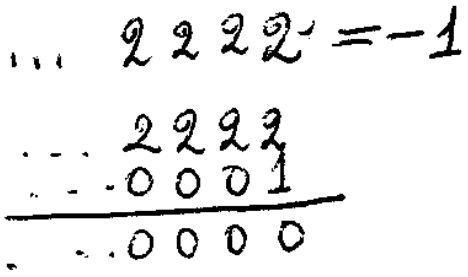
Ordinary numbers are worse because of the identification of



Diagram and typed description.

$0{,}999\ldots$ and 1 disconnected becomes connected

Original source details: GELFAND-P000272



Original source details: GELFAND-P000272

Diagram and typed description.

$$\ldots 2222 = -1$$

$$\begin{array}{r} \ldots 2222 \\ \ldots 0001 \\ \hline \ldots 0000 \end{array}$$

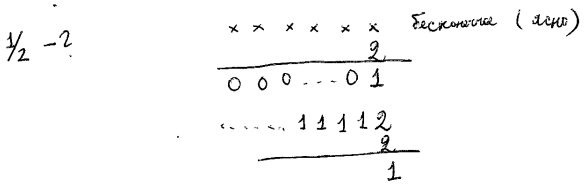


Diagram and typed description.

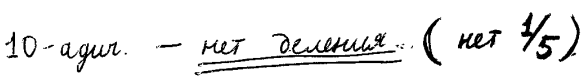
$\frac{1}{2}$ — ?

$$\begin{array}{r} \times\ \times\ \times\ \times\ \times\ \times \\ 2 \\ \hline 000\ldots01 \end{array}$$ infinite (clear)

$$\begin{array}{r} \ldots 11112 \\ 2 \\ \hline 1 \end{array}$$

Original source details: GELFAND-P000272

10-adic — no division (no $1/5$)



Manuscript notation. The last entry reads “no $1/5$”.

Original source details: GELFAND-P000272

GELFAND-P000273

$31/\overline{\text{III}}$ Bleher’s talk. On the spectr. func.

Acta Math. 121

$N = 3 - 4$, 1968 (L. Hörmander)

$$(Pu, u) \geq c(u, u),$$

$$\hat{p} = \int \lambda \, dE_\lambda, \qquad e(x, y, \lambda) \in C^\infty(\Omega \times \Omega)$$

— the spectral function of the operator.

Asymptotics in λ — ?

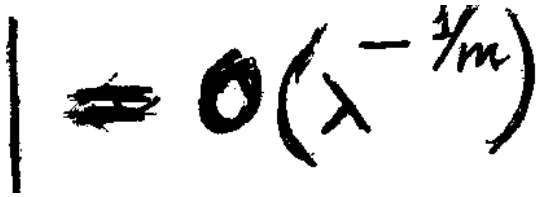
Theorem.

$$\left| \lambda^{-n/m} e(x, x, \lambda) - (2\pi)^{-n} \int_{p^0(x,\xi)<1} d\xi \right| = O(\lambda^{-1/m})$$



Original source details: GELFAND-P000273

Uncertain reading. Second argument: final x over y / an earlier x; initial letter undetermined.



Original source details: GELFAND-P000273

Uncertain reading. Before O, two horizontal strokes lie over slanted strokes. Provisionally $=$.

Off the diagonal 0.

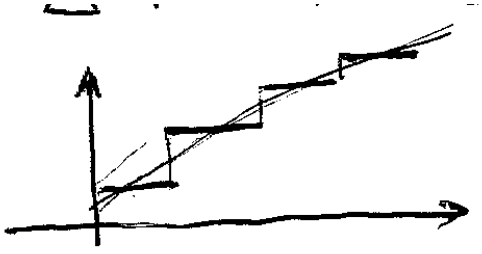


Original source details: GELFAND-P000273

Uncertain reading. After “diagonal”, a round sign provisionally read as 0, possibly the letter O.

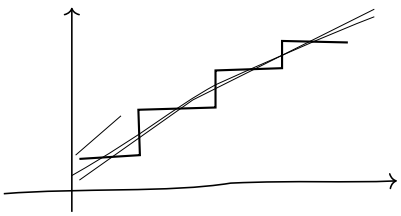
Avakumovich’s example.

Δ on $S^n \subset \mathbb{R}^{n+1}$



Original source details: GELFAND-P000273

Uncertain reading. Axes with upward and rightward arrows, four horizontal steps and three rises. Sloping strokes have no assigned formulas; their number as distinct comparison lines and their role are undetermined. No axis labels.



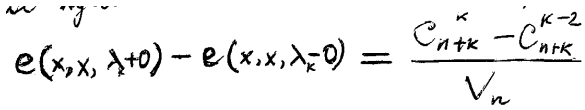
λ of large multiplicity.

the jump magnitude is of just this order.

Large multiplicities due to invariance of the sphere under rotations.

The worst situation for asymptotics is a **multiple spectrum.**

$$e(x,x,\lambda_k+0)-e(x,x,\lambda_k-0)=\frac{C^{k}_{n+k}-C^{k-2}_{n+k}}{V_n}.$$



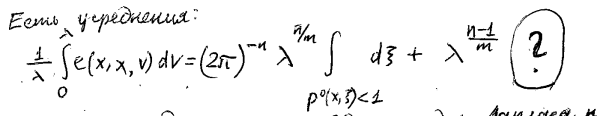
Uncertain reading. Both lower indices read $n+k$ and the second upper index reads $k-2$. The first subscript on λ is retraced, provisionally k.

Original source details: GELFAND-P000273

$$\frac{(2\pi)^{-n}}{V_n}\lambda^{n/2}$$

Jump $\sim\lambda^{(n-1)/2}$ and $C\lambda^{(n-1)/2}$ cannot be written.
There are averages:

$$\frac{1}{\lambda}\int_0^{\lambda} e(x,x,\nu)\,d\nu=(2\pi)^{-n}\lambda^{n/m}\int_{p^0(x,\xi)<1}d\xi+\lambda^{(n-1)/m}$$



Uncertain reading. The leading coefficient is written $(2\pi)^{-n}$; a question mark is outlined after the second power term.

Original source details: GELFAND-P000273

This has not been done. It has been done for the Laplacian on a square.

GELFAND-P000274

[source number: 38]

See Dikii, UMN, 1957 on traces.

and earlier in 1954 Gelfand.

38

There are 2 problems.
1) obtain the subsequent terms of the asymptotics.
2) obtain a **formal asympt. expansion** (a subtle problem).

$$A=\hat{P}^{1/m}\quad\text{— a pseudodiff. operator.}$$

$$A=a_1+a_0+a_{-1}+\cdots$$

$$a_1=p^{1/m}(x,\xi).$$

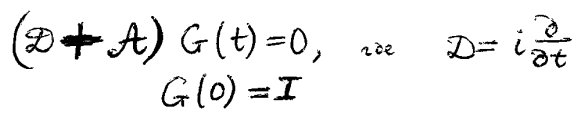
It suffices to do everything for the operator A. (pseudodiff., first order)

$$G(t)=\int e^{it\lambda}\,dE_\lambda=\exp(itA).$$

E_λ — spectr. decomp. of A: $A=\int\lambda\,dE_\lambda$.

$$(D + A)G(t) = 0, \qquad \text{where} \quad D = i\frac{\partial}{\partial t}$$

$$G(0) = I$$



Manuscript notation. A bold plus appears between D and A. At the right, $D = i\,\partial/\partial t$.

Original source details: GELFAND-P000274

This gives a hyperbolic problem, which can therefore be solved.

Generalized Huygens principle: singularities propagate at finite speed.

$$G(t)u = \iint e^{i\varphi(x,t,y,\xi)} q(x,t,y,\xi) u(y)\, dy\, d\xi + Ku,$$

$\varphi(x,t,y,\xi) = \psi(x,y,\xi) - ta_1(y,\xi).$

$$q \sim q_0 + q_1 + \cdots$$

Only this is important for the asymptotics.

We need small t. ($|t| < \varepsilon$).

$$\psi(x,y,\xi) = (x-y,\xi) + O(|x-y|^2|\xi|).$$

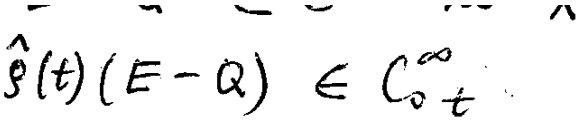
One can then refer to Marchenko's Tauberian theorem.

$E(x,y,t)$ — kernel of $G(t)$

$Q(x,y,t)$ — approx. kernel (regulariz.)

$$E - Q \in C^\infty \quad \text{in } x \text{ and } y \text{ for } |t| < \varepsilon$$

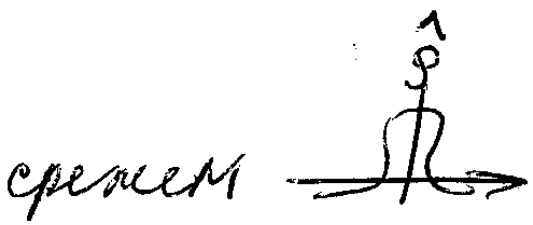
$$\hat{\varrho}(t)(E-Q) \in C^\infty_{0t}$$



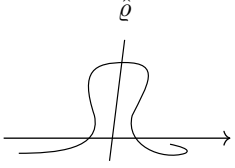
Uncertain reading. The signs 0 and t are below and to the right of C^∞; t is lower and farther right than 0. Strict index hierarchy undetermined.

Original source details: GELFAND-P000274

cut off



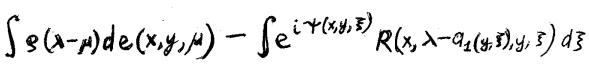
Uncertain reading. A slanted stroke marked $\hat{\varrho}$ crosses the rounded rise. Strokes continue below the horizontal axis on both sides. Their function is uncertain.



Original source details: GELFAND-P000274

GELFAND-P000275

$$\int \varrho(\lambda-\mu)\,de(x,y,\mu) - \int e^{i\psi(x,y,\xi)} R(x,\lambda-a_1(y,\xi),y,\xi)\,d\xi$$



Original source details: GELFAND-P000275

Manuscript notation. The first line is the difference of two integrals. The next line separately defines R.

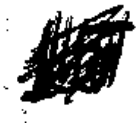
$$R(x,\lambda,y,\xi) = F^{-1}_{t\to\lambda}\big(q(x,t,y,\xi)\hat{\varrho}(t)\big)$$

Original source details: GELFAND-P000275

Diagram and typed description.

GELFAND-P000276

$14/\underline{\text{IV}} - 69.$



Original source details: GELFAND-P000276

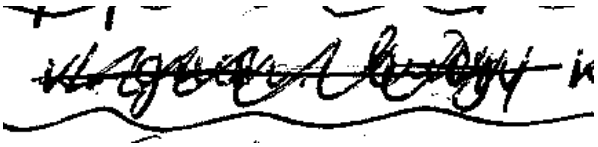
Uncertain reading. An illegible short mark is crossed out before the date.

I. M. Gelfand.

On the work of Yik-Hoi Au-Yeung (Proceed. of Amer. M. S. February 1969)

If two quadr. forms are not simult. 0, then they can be

simult. reduced [Crossed out: to ... form.]



Original source details: GELFAND-P000276

Uncertain reading. The crossed-out phrase reads “to ... form”. The middle word is uncertain.

to a sum of squares.

(Greg[?] “Lin. algebra”)



Original source details: GELFAND-P000276

Uncertain reading. The surname in the “Lin. algebra” reference is provisionally Greg. The last letter is uncertain.

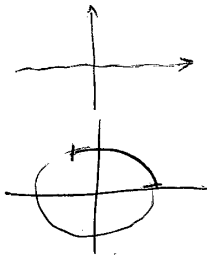
$$\xi^* A\xi, \qquad \xi^* B\xi \quad \text{— corresp. quadr. forms}$$

a pair of numbers — a vector of the plane.

[Above the line: Normalize.]

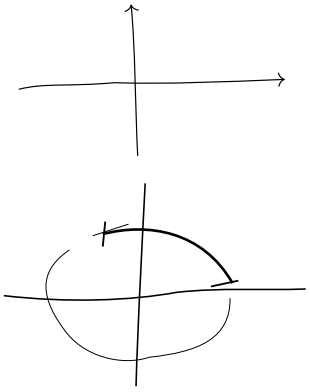
If $\forall \xi$ this is $\neq 0$, then it runs through no more than a semicirc.

After this rotate and one will be pos. definite.



Original source details: GELFAND-P000276

Uncertain reading. The upper axes have upward and rightward arrows. Below are axes without arrows and an open, nearly circular curve, heavier at the upper right. Short strokes occur at the ends of the heavy arc; their role as arrows or boundary marks is undetermined. There are no labels.



[Crossed out: in R / in R^3.]



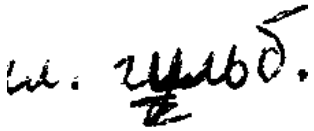
Original source details: GELFAND-P000276

Uncertain reading. A possible superscript at R is obscured by hatching; the digit 3 is undetermined.

Davis (Toronto)

Davis, Acta Sci. Math. (1968)

H — compl. Hilb.



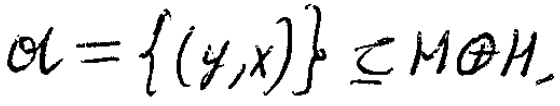
Original source details: GELFAND-P000276

Uncertain reading. Under the beginning of "Hilb." is a small correction group resembling Cyrillic "и"; its role is undetermined.

y^*x — scalar product, $(x, y) \in H \oplus H$.

$$A : H \to H, \qquad \underset{y=Ax}{(y,x)} \quad \text{— graph}$$

More gen. def. — a relation — a set $\alpha = \{(y, x)\} \subseteq M \oplus H$.



Original source details: GELFAND-P000276

Manuscript notation. The right-hand side reads $M \oplus H$.

1°

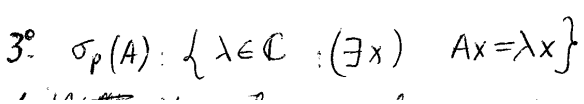
$$\|A\| \le a \quad \Rightarrow \quad \left\{ \frac{\|Ax\|^2}{\|x\|^2} : x \ne 0 \right\} \subseteq [0, a^2]$$

2°. Range of values of the form.

$$w(A) = \left\{ \frac{x^* Ax}{\|x\|^2} : x \neq 0 \right\}.$$

3°.

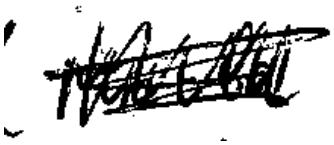
$$\sigma_p(A) : \{\lambda \in \mathbb{C} : (\exists x) \quad Ax = \lambda x\}$$



Manuscript notation. The parenthesis reads $(\exists x)$.

Original source details: GELFAND-P000276

([Crossed out: Numerical?] Numerical sets of the operator)



Uncertain reading. The cancelled word: Numerical?; some letters are obscured.

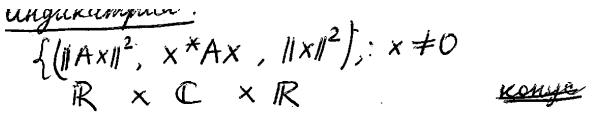
Original source details: GELFAND-P000276

Univers. inform. about the operator.
Indicatrix.

4°.

$$\{(\|Ax\|^2,\ x^* Ax,\ \|x\|^2) : x \neq 0$$

$$\mathbb{R} \times \mathbb{C} \times \mathbb{R}$$



Manuscript notation. The set is written with one opening brace. “Cone” is doubly underlined below the formula.

Original source details: GELFAND-P000276

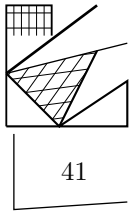
cone

GELFAND-P000277

[source number: 41]



Diagram and typed description.

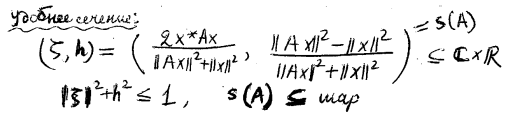


Original source details: GELFAND-P000277

A convenient section:

$$(\xi, h) = \left(\frac{2x^* Ax}{\|Ax\|^2 + \|x\|^2}, \frac{\|Ax\|^2 - \|x\|^2}{\|Ax\|^2 + \|x\|^2} \right)^{=s(A)} \subseteq \mathbb{C} \times \mathbb{R}$$

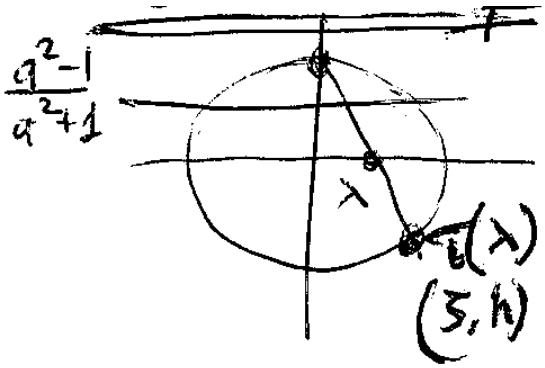
$$\|\xi\|^2 + h^2 \le 1, \qquad s(A) \subseteq \text{ball}.$$



Original source details: GELFAND-P000277

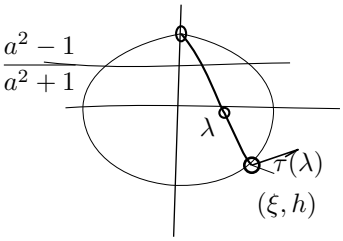
Manuscript notation. The insertion $= s(A)$ is above the right parenthesis. The main line gives $\subseteq \mathbb{C} \times \mathbb{R}$.

Eigenvector — when it belongs to the sphere.
eigenval. — stereogr. projection



Original source details: GELFAND-P000277

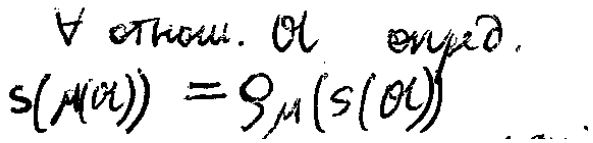
Uncertain reading. A circle, a vertical line, two horizontal lines and a sloping stroke with three marked points. Labels $\frac{a^2-1}{a^2+1}$, λ, $\tau(\lambda)$, (ξ, h). Short strokes at the lower-right point have no established arrow direction; the exact boundaries of the densely marked points are undetermined.



Norm — if it does not go above the line $\frac{a^2-1}{a^2+1}$, then a — the norm.

Th. 1. Let $\mu(\lambda) = \frac{a\lambda+b}{c\lambda+d}$. $ad - bc \neq 0$.
For each relation α, $\mu(\alpha)$ is defined,

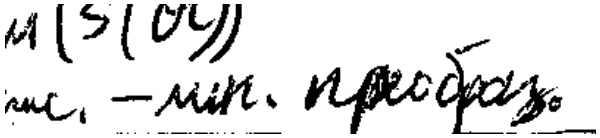
$$s(\mu(\alpha)) = \varrho_\mu(s(\alpha))$$



Original source details: GELFAND-P000277

Uncertain reading. Before the subscript μ is a loop with a curved tail. Provisionally ϱ_μ.

[Under the transformation symbol:]
[?]-lin. transf.

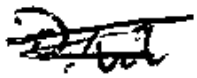


Original source details: GELFAND-P000277

Uncertain reading. The first abbreviation is uncertain. It is followed by "-lin. transf.".



[Crossed out: **[?]**]

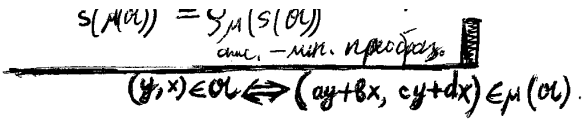


Original source details: GELFAND-P000277

Uncertain reading. A short cancelled word before “If”: Для? (For?); reading undetermined.

If A — an oper., then as usual
(if there exists $\mu(A) = (aA + b)(cA + d)^{-1}$).

$$(y, x) \in \alpha \iff (ay + bx, cy + dx) \in \mu(\alpha).$$



Original source details: GELFAND-P000277

Manuscript notation. Transformation rule below the long line.

def.:

$$X \text{ — a spectr. set of } A \iff$$

(**[Cancelled illegible mark]** $\forall$ rat. funcs. f: $f(X) \subseteq D \Rightarrow \|f(A)\| \leq 1$.)

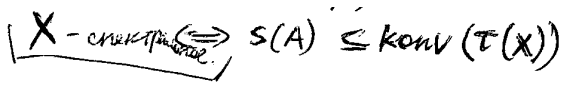


Original source details: GELFAND-P000277

Uncertain reading. After the opening parenthesis, before $\forall$, a single mark is densely crossed out and unread.

$$D = \{\lambda : |\lambda| \leq 1\}.$$

Th. 2 description of spectr. sets
provided that $\tau(X)$ (on the sph.) — a segment (spherical).



Original source details: GELFAND-P000277

Diagram and typed description. Criterion marked by a bracket.

X — spectral $\iff$ $s(A) \subseteq \text{konv}(\tau(X))$.

GELFAND-P000278

V. Buchstaber.

$$T^n = \underbrace{S^1 \times \ldots \times S^1}_{n}.$$

Thm. There are finitely many smooth manifolds of homotopy type torus.
(up to diffeom.)
For simply conn., it is proved: everything is determined by Pontryagin classes.
and the hom. type
(up to finitely many possibilities)

Milnor: for spheres (except $\underset{\text{unknown}}{4}$ and 3)



Manuscript notation. The note “unknown” is written below the line just to the left of 4.

Original source details: GELFAND-P000278

Novikov: For simply conn. manifolds: they are determined by the homot. type and charact. classes of the tangent bundle — then finitely many possibilities remain.

M^n — a manifold.

Smale:

Let $\partial W^{n+1} = M_1^n \cup M_2^n$

$M_1^n \subset W^{n+1}$ — a deform. retract

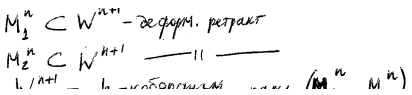


Diagram and typed description.

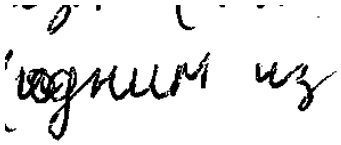
$M_2^n \subset W^{n+1}$ ——————||——————

Original source details: GELFAND-P000278

Then W^{n+1} — h-cobordism of the pair (M_1^n, M_2^n).

Smale’s theorem.

h-cobordism (film) is completely (up to diffeom.) determined by a boundary (one of M_i^n) and an invar.



Uncertain reading. The beginning of the final word “одним” (one) is overwritten over an illegible earlier entry.

Original source details: GELFAND-P000278

$$\alpha \in \underset{\text{(Whitehead gr.)}}{\mathrm{Wh}}(\pi_1(M_1^n)).$$

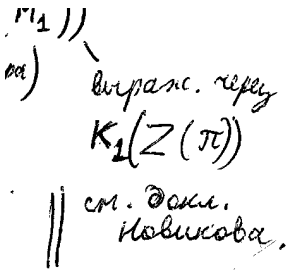


Diagram and typed description.

\

expressed through

$K_1(Z(\pi))$

||

see the talk of

Novikov.

Original source details: GELFAND-P000278

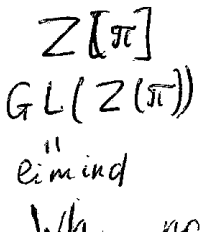


Diagram and typed description.

$Z[\pi]$

$GL(Z(\pi))$

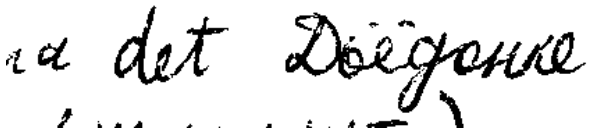
| |

lim ind

Original source details: GELFAND-P000278

group det Dieudonné $K_1(Z(\pi))$.

Uncertain reading. The name is Dieudonné. The sequence of middle strokes in the handwritten Russian spelling is uncertain.

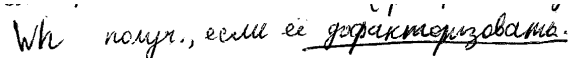


Original source details: GELFAND-P000278

(noncommut.)

(quotient by the commutator subgroup).

Wh is obtained by taking a further quotient.



Original source details: GELFAND-P000278

Manuscript notation. The underlined verb means “take a further quotient”.

The element α is obtained through Smale’s func. (nice-function).

In the Smale-Novikov case [crossed out: this?] h cobordisms $\Longleftrightarrow$ diffeom., and a cobord. is already a homot. [cancelled illegible mark] concept. (There $W^{n+1} = M_1^n \times I$)

Uncertain reading. A short word between “Smale-Novikov” and “h cobordisms” is cancelled: это? (this?); reading undetermined.



Original source details: GELFAND-P000278

Uncertain reading. Between “homot.” and “concept”, a mark or beginning of a word is cancelled; reading undetermined.



Original source details: GELFAND-P000278

GELFAND-P000279

[source number: 42]

42

Tori could be classif. because of Bass’s theorem:

$$K_1(Z + Z + \ldots + Z) = 0.$$

(of the group ring of the group).

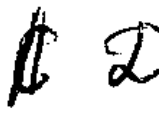
(algebr. fact)



Original source details: GELFAND-P000279

Manuscript notation. The line uses K_1, ordinary plus signs and $= 0$.

For Z it is obvious.



Uncertain reading. Left of “For Z”, a narrow loop-like mark with a stroke; reading undetermined.

Original source details: GELFAND-P000279

Matrices of polynomials ~~det =~~ invertibility $\Longleftrightarrow$ det $= \pm 1$.



Uncertain reading. Cancelled: det and a following sign: $det =$ / $det \neq$; the sign is undetermined.

Original source details: GELFAND-P000279

(see Artin’s book “Geometric Algebra” — it has the Dieudonné determinant).

Whitehead det. 1 — if reducible by elem. transformations to 1

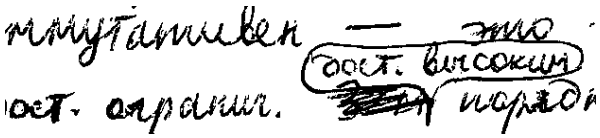
(and increase of order)

The quotient is commutative — this is a remarkable theorem! (Whitehead).

Bass — it suffices to restrict to

sufficiently high

~~[?]~~ order.

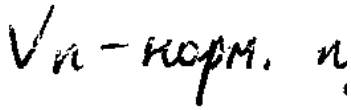


Uncertain reading. An illegible word is crossed out below the outlined “sufficiently high”.

Original source details: GELFAND-P000279

$$M^n \overset{\nu_n}{\subset} \mathbb{R}^N$$

ν_n — normal bundle



Uncertain reading. Initial letter: ν_n / V_n; the first lower index is n, and M is used below.

Original source details: GELFAND-P000279

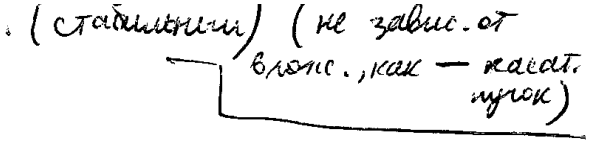


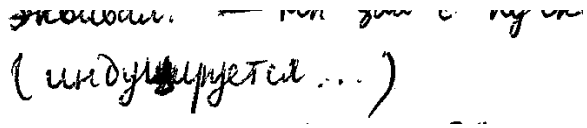
Diagram and typed description. Continuation at the right of the line.

(stable) (does not depend on embedding, like the tangent bundle)

Original source details: GELFAND-P000279

Equivalence — of manifolds with a bundle.

(is induced...)



Uncertain reading. In “индуцируется” (is induced), the middle letters are heavily overwritten; the earlier entry is undetermined.

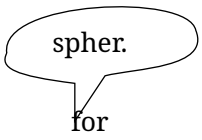
Original source details: GELFAND-P000279

Let us calculate for the torus.

From Adams's theorem on the J-funct.



Diagram and typed description. The insertion “spher.” above “for”.



Original source details: GELFAND-P000279

$M^n \sim T^n$ everything is completely determined by the Pontryagin classes of M^n.

Pontryagin classes — an infinitely valued invar.

Novikov: for tori, Pontryagin classes are a homot. invariant.

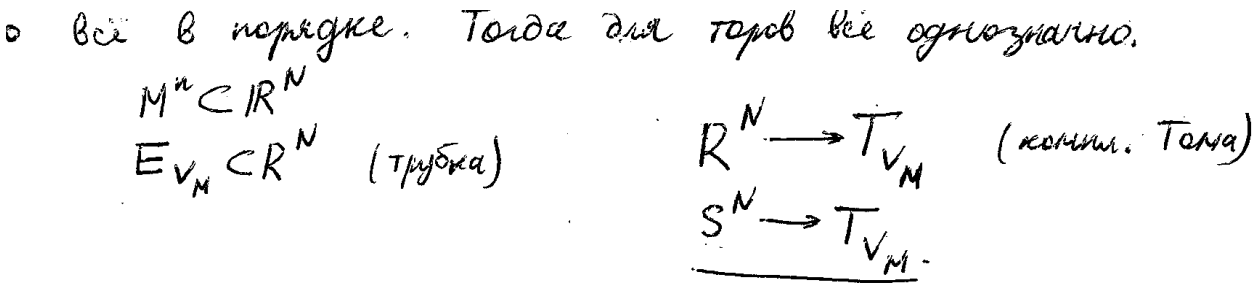
There $\nu_M = 0$.

But: on $S^2 \times S^4$ there are infinitely many smooth structures.

Again everything works for tori, since the torus is parallelizable and $\nu_M = 0$.

$$\{M_1^n\} \qquad f : M_1^n \longrightarrow M_2^n, \quad f^*\nu_{M_2} = \nu_{M_1}.$$

How many such classes? If there are finitely many here, then everything is in order. Then for tori everything is unique.



Original source details: GELFAND-P000279

Diagram and typed description.

$M^n \subset \mathbb{R}^N$

$R^N \longrightarrow T_{\nu_M}$ (Thom compl.)

$E_{\nu_M} \subset R^N$ (tube)

$S^N \longrightarrow T_{\nu_M}$.

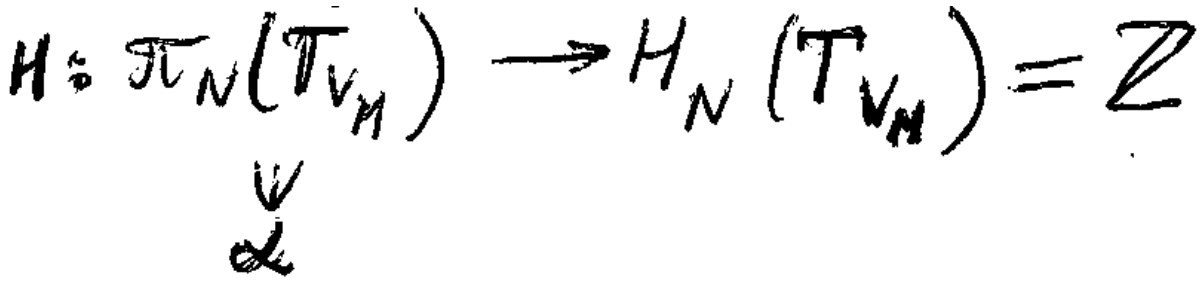
GELFAND-P000280

An elem. of $\pi_N(T_{\nu_{M_1}})$ is obtained.

The space T_{ν_M} does not depend on the manifold (!) — they are all homot. equiv., since on the base they are homot. equiv.

We obtain an element of $\pi_N(T_{\nu_M})$.

We need to consider the Hurewicz homom.



Original source details: GELFAND-P000280

Diagram and typed description.

$$H: \quad \pi_N(T_{\nu_M}) \longrightarrow H_N(T_{\nu_M}) = \mathbb{Z}$$

$$\pi_N(T_{\nu_M}) \downarrow \alpha$$

If α is realized in this way, then $H(\alpha) \longrightarrow +1$.

(on the interior of the tube it is the identity).

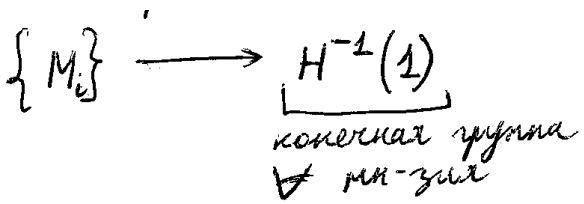
(by the Excision theorem).

There is a map

$$\{M_i\} \longrightarrow H^{-1}(1)$$

finite group

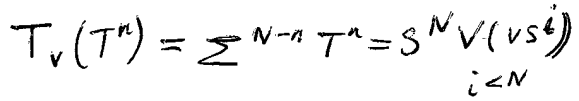
$\forall$ manifolds



Manuscript notation. The rectangular bracket is labelled “finite group $\forall$ manifolds”.

Original source details: GELFAND-P000280

$$T_\nu(T^n) = \Sigma^{N-n} T^n = S^N \vee \left(\vee_{i<N} S^i\right).$$

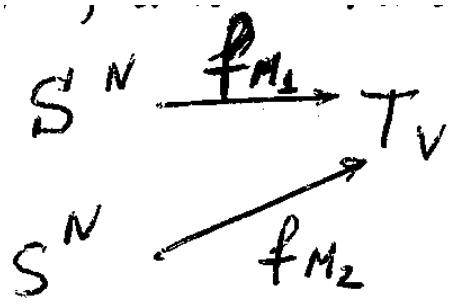


Manuscript notation. Both signs in the line are equality signs. The last parenthesis has $i < N$ below it.

Original source details: GELFAND-P000280

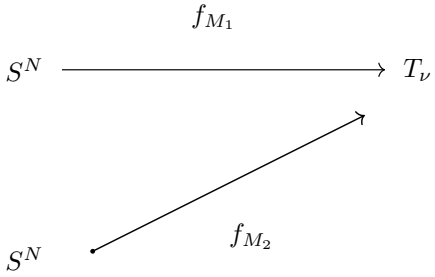
Everything follows from Serre’s finiteness theorem.

It remains to show that Ker is finite.



Original source details: GELFAND-P000280

Uncertain reading. The f_{M_2} stroke has an arrowhead on the right. The left end is thickened. A second arrowhead is indistinct.

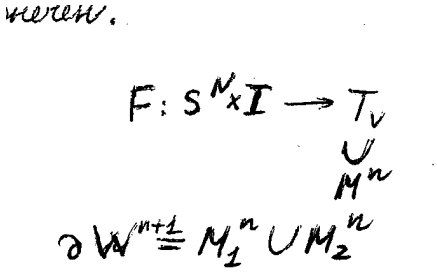


Suppose they are homotopic.

$$\partial W^{n+1} = M_1^n \cup M_2^n$$

and there exist retractions and with bundles

Everything is in order.

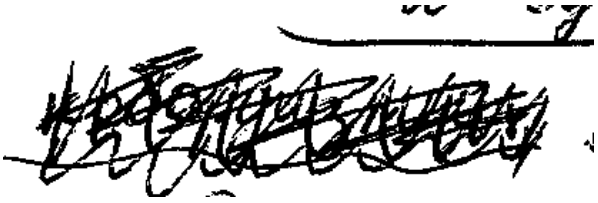


Original source details: GELFAND-P000280

Diagram and typed description.

$$\begin{array}{ccc} F : S^N \times I & \longrightarrow & T_\nu \\ & & \cup \\ & & M^n \end{array}$$

We need to classify ~~[?]~~ films which deform. retract.



Original source details: GELFAND-P000280

Uncertain reading. A possible reading under the heavy cancellation is кобордизмы (cobordisms).

A class of finite index, for which a film exists.
This is the same as saying that $i : M_1 \subset W$ induces an isom. of all homot. groups.
Obstruction theory for the modification of the film.

GELFAND-P000281

[source number: 43]

21/IV Kirillov. Borel–Weil theorem. (in a new form).

43

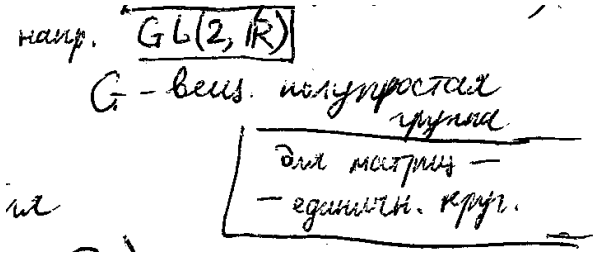
$G \supset K$ K — compact subgr. (max).
It is assumed that rank G = rank K (dimension of a max. torus).
$K \supset H$ — Cartan subgr. (common to K and G).

e.g. $GL(2,\mathbb{R})$

G — a real semisimple group.

for matrices —

— unit circle.



Original source details: GELFAND-P000281

Manuscript notation. The frame around $GL(2,\mathbb{R})$ is open on the left. The frame of the matrix note is open on the right.

$D = G/H$ — complex manifold (always)

The number of complex structures — the number of elements of the Weyl group.

$\mathcal{L}_\lambda$ — 1-dimensional bundles over D(fiber $\mathbb{C}$)

$$\mathcal{L}_\lambda \xrightarrow{\mathbb{C}} D$$

1-dimensional representation — a character of the Cartan subgroup

$A_0^i(\mathcal{L}_\lambda)$ — smooth $(0,i)$-forms with values in $\mathcal{L}_\lambda$.

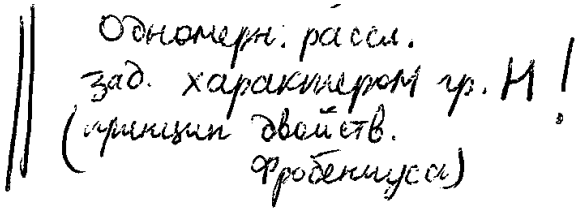
(and with compact support)

$$\bar\partial : A_0^i \longrightarrow A_0^{i+1}$$

$$\Box = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial.$$

(Laplace–Beltrami)

The space of L^2-cohomology. (with coefficients in the sheaf)



Original source details: GELFAND-P000281

Diagram and typed description. Note in the right margin.

A one-dimensional bundle
is given by a character of the group H!
(the principle of Frobenius
reciprocity)

G — compact $\Rightarrow$ this is the Borel–Weil theorem.

On D there is an invariant measure $\Rightarrow$ there is [?] a Hilbert space L^2. (completion of smooth ones)



Original source details: GELFAND-P000281

Uncertain reading. Before "a Hilbert space L^2", a densely crossed-out entry in a rectangle; its reading is undetermined.

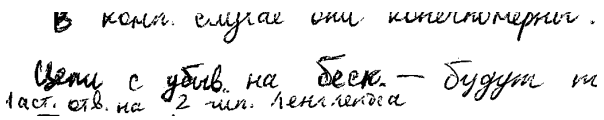
$\bar{\partial}$ on a dense set.

$$\mathrm{Ker}\,\Box = \mathcal{H}^i(\mathcal{L}_\lambda)$$

— i-dimensional L^2-cohomology.

In the compact case they are finite-dimensional. In the general case they are not.

If with decay at infinity — there will be the same cohomology.



Original source details: GELFAND-P000281

Uncertain reading. The beginning of "Если" (If) is heavily overwritten; the earlier letters are undetermined.

A partial answer to 2 conjectures of Langlands

Theorem 1. λ — a weight, $|(\lambda, \alpha)| > b \quad \forall$ simple root α.

sufficiently nondegenerate.

Then

$$\mathcal{H}^i(\mathcal{L}_\lambda) = \begin{cases} 0, & i \neq k(\lambda + \rho), \\ \pi_{\lambda+\rho}, & i = k(\lambda + \rho). \end{cases}$$

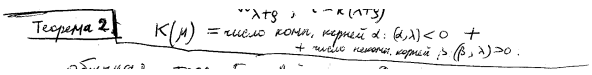


Original source details: GELFAND-P000281

Uncertain reading. A looped mark after λ in the first row is struck out. It is followed by $+\rho$.

Theorem 2.

$$K(\mu) = \text{number of compact roots } \alpha : (\alpha, \lambda) < 0 \quad + \\ + \text{number of noncompact roots } \beta : (\beta, \lambda) > 0.$$



Original source details: GELFAND-P000281

Manuscript notation. The left side is $K(\mu)$ and both pairings on the right use λ. The plus sign is repeated at the line break.

For the usual Bor.–Weil theorem see Demazur



Original source details: GELFAND-P000281

Manuscript notation. The surname is written “Demazur”.

GELFAND-P000282





Original source details: GELFAND-P000282

Uncertain reading. Two loops before the surname are struck out. The letter is undetermined.

Buchstaber.

Manifolds of the homot. type of a torus are divided by h-cobordisms into a finite number of classes

Problem: to show that there are finitely many in each class

Next, the simplest films are classified by Morse modifications.



Original source details: GELFAND-P000282

Diagram and typed description. Rule at the end of the entry.





Original source details: GELFAND-P000282

Uncertain reading. A curved stroke with a dot above it is visible at the very edge of the scan. Its meaning is uncertain.

GELFAND-P000283

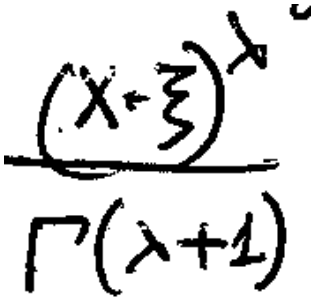
15.9.69 yr.

B. Vainberg on results of Hörmander. (see Vishik seminar). on singularities of fund. sol.
General sol. of homog. eqn.: (homog. polynomial)

$$\int_{P(\xi)=0} f[(x,\xi)]A(\xi)\,d\xi \qquad \text{(Borovikov)}$$

f, A — arbitrary functions.
Singularities in the homog. case are described by Borovikov.
For example:

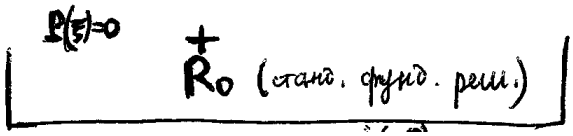
$$\int_{P(\xi)=0} \frac{(x+\xi)^\lambda}{\Gamma(\lambda+1)} A(\xi)\,d\xi$$



Original source details: GELFAND-P000283

Uncertain reading. The short crossed stroke is provisionally read as $+$. The preceding formula has (x,ξ).

— proposal of Borovikov.



Original source details: GELFAND-P000283

Diagram and typed description. Plus above R_0 within the angular rule.

$\overset{+}{R}_0$ (stand. fund. sol.)

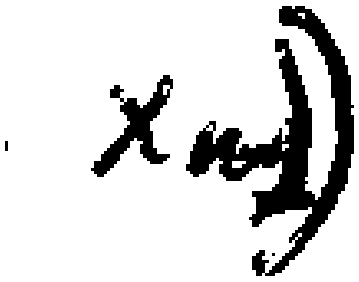
Grushin:

$$\sum_j \int \frac{\alpha_j(\xi)e^{i(x,\xi)}}{P(\xi)\pm i0}\,d\xi$$

α_j — partition of unity on the sphere.

The Bernstein problem is solved.

$$\left\{\begin{array}{l} z = f(x_1, \ldots, x_{n-1}) \text{ — min. surf.} \\ (x\text{-arbitrary}) \implies \text{it is a plane.} \end{array}\right\}$$



Uncertain reading. The final x has the provisional index $n-1$. Its end overlaps the closing parenthesis.

Original source details: GELFAND-P000283

true for $n \leq 8$; for $n \geq 9$ false.

De Giorgi, Simons and Bombieri.

Kazhdan and Bernstein.

Convergence of the spectrum as the domains expand.

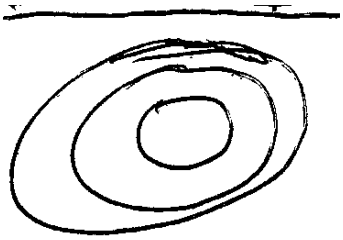
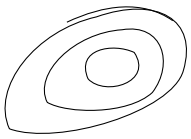


Diagram and typed description.



Original source details: GELFAND-P000283

A — self-adj. op. on the manifold X.

$\Lambda \subset \mathbb{R}$.

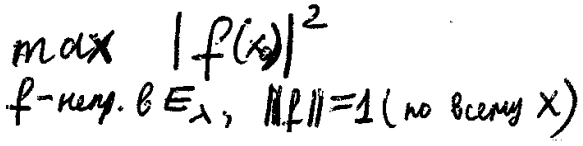
[Below Λ:] interval.

The continuous functions contained in E_Λ are dense in it.

(required) — so for elliptic op.

$x_0 \longrightarrow$ a Plancherel measure on the spectrum of A.

$$M_{x_0}(\Lambda) = \max_{\substack{f\text{-continuous in } E_\lambda,\ \|f\|=1 \\ \text{(over all of } X\text{)}}} |f(x_0)|^2.$$



Manuscript notation. Below max the manuscript has E_λ. Above it has E_Λ.

Original source details: GELFAND-P000283

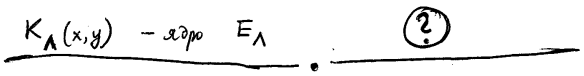
[At right:] This is a measure (possibly infinite).

GELFAND-P000284

[source number: 53]

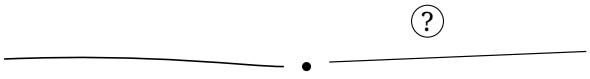
$K_\Lambda(x,y)$ — the kernel of E_Λ.

53



Original source details: GELFAND-P000284

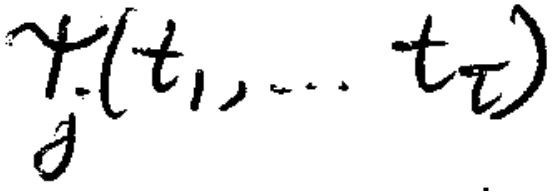
Diagram and typed description. Question mark over the right part of the rule.



Kazhdan. M. Artin in the Bourbaki seminar.

$$f_i(x_1,\dots,x_n)=0, \qquad 1\le i\le k.$$

f_i — convergent power series.
When does a sol. exist $\psi_j(t_1,\dots,t_\tau)$



Manuscript notation. The forms read ψ_j and t_τ.

Original source details: GELFAND-P000284

Answer: if and only if there is a sol. in formal power series.
(arbitrarily close to the formal one)
[Below the parenthesis:] can be completed

22.9.69 yr.
Fichera on unilateral bound. problems.

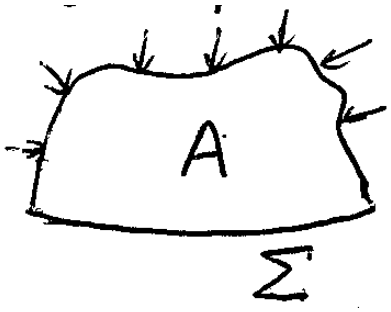
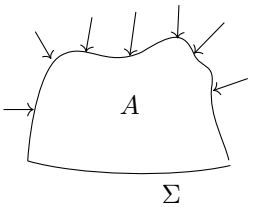


Diagram and typed description.



Original source details: GELFAND-P000284

An elastic body on a rigid, frictionless surface Σ.
And the stress tensor $\equiv 0$.
A — natural volume of the body.

On the free part of the boundary, a force $\underbrace{\varphi}_{\text{density}}$ on $\partial A - \Sigma = \Sigma^*$.

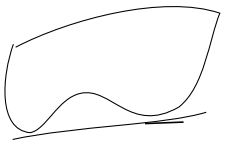
Inside there are also certain forces f — volume forces.

Let the body move to a position of equil.

It may separate from Σ.



Uncertain reading. Contact is visible at the left. At the right, overlapping strokes and a narrow light strip; exact contact or separation and the length of contact are undetermined. There are no labels or arrows.



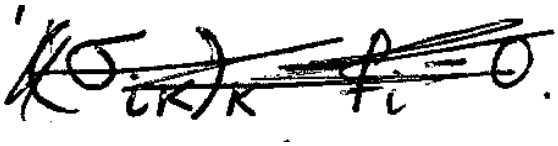
Original source details: GELFAND-P000284

This is an analytic problem (formulation A. Signorini, 1959)

σ_{ik} — the stress tensor.

Inside the body

[Crossed out:] ~~$(\sigma_{ik})_k \quad f_i = 0$~~



Manuscript notation. Earlier formula under the cancellation.

Original source details: GELFAND-P000284

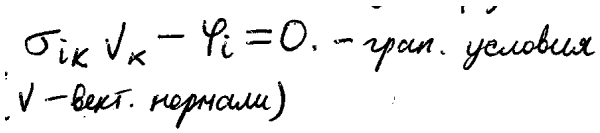
[Written at the right:]

$$\sigma_{ik/k} - f_i = 0.$$

$(\operatorname{div}\sigma = f)$.

$$\sigma_{ik}\nu_k - \varphi_i = 0.$$

— bound. conditions.
(ν — normal vector)



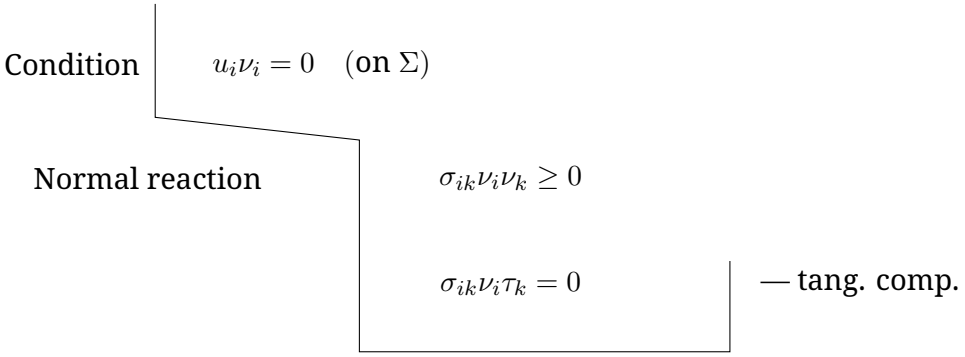
Uncertain reading. The sign looks like V in both lines. Provisionally ν.

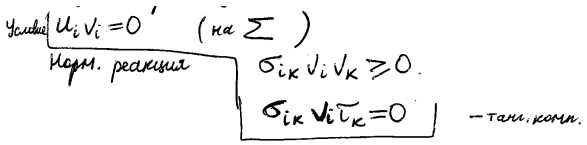
Original source details: GELFAND-P000284

GELFAND-P000285

On Σ.

u_i — the displacement vector.



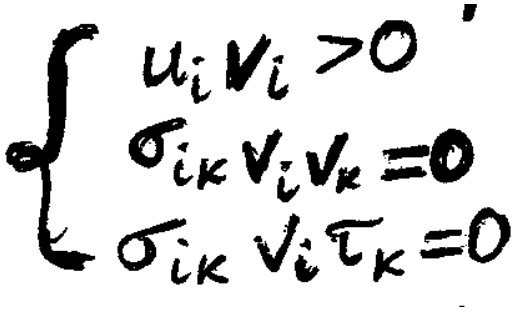


Uncertain reading. The conditions are joined by a stepped rule. The normal sign looks like V and is provisionally read as ν.

Original source details: GELFAND-P000285

At a point of the free surface:

$$\begin{cases} u_i\nu_i > 0, \\ \sigma_{ik}\nu_i\nu_k = 0, \\ \sigma_{ik}\nu_i\tau_k = 0. \end{cases}$$



Manuscript notation. The first line starts with $u_i\nu_i$.

Original source details: GELFAND-P000285

But it is unknown where which occurs.

We have not yet taken into account that the body is elastic.

$$\underline{\underline{W(x,\varepsilon)}}$$
potential

$$\varepsilon = \{\varepsilon_{ik}\} \qquad \varepsilon_{ik} = 2^{-1}(u_{i/k} + u_{k/i})$$

Strain tensor.

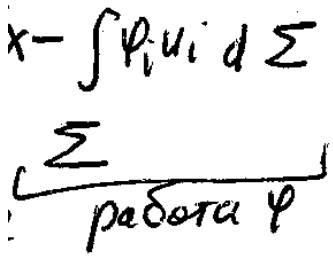
Relation:

$$\sigma_{ik} + W_{\varepsilon_{ik}} = 0 \qquad \text{on } A.$$

Problem of exist. and uniqueness.

$$I(u) = \int_A W[x, \varepsilon(u)]\, dx - \underbrace{\int_A f_i u_i\, dx}_{\text{work of } f} - \underbrace{\int_\Sigma \varphi_i u_i\, d\Sigma}_{\text{work of } \varphi}.$$

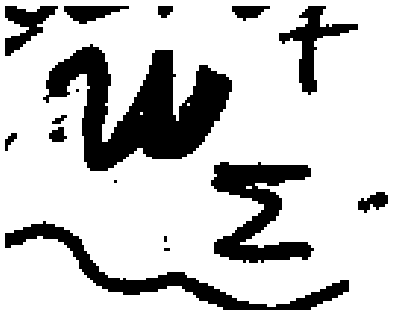
(Total energy)



Original source details: GELFAND-P000285

Manuscript notation. This integral has domain Σ. The model problem below has Σ^*.

In the class of admissible displacements $\mathcal{W}_\Sigma$.

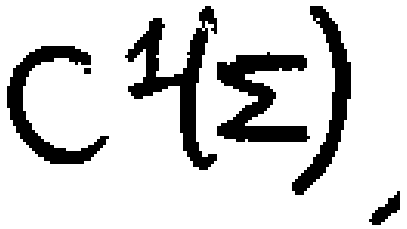


Original source details: GELFAND-P000285

Manuscript notation. The letter with two lower bends reads $\mathcal{W}_\Sigma$. The model below uses $\mathcal{U}_\Sigma$.

If a minimum exists in $C^1(\Sigma)$, then it is a solution.

conversely, a solution is a min.



Original source details: GELFAND-P000285

Manuscript notation. The line reads $C^1(\Sigma)$.

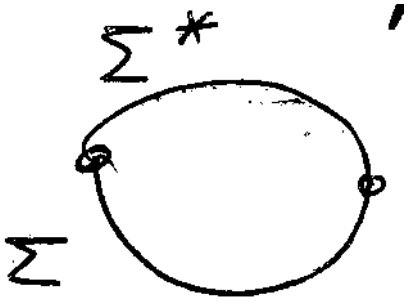
But: in this class min does not exist.

$$I(u) = \frac{1}{2}\int_A a_{nk}(x) u_{/n} u_{/k}\, dx - \int_A f u\, dx - \int_{\Sigma^*} \varphi u\, d\sigma.$$

$$a_{nk}(x)\mu_n\mu_k \geq \rho_0 \mu_n \mu_n.$$

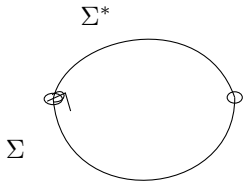
$$\mathcal{U}_\Sigma : \quad u|_\Sigma \geq 0.$$

(Model problem)



Original source details: GELFAND-P000285

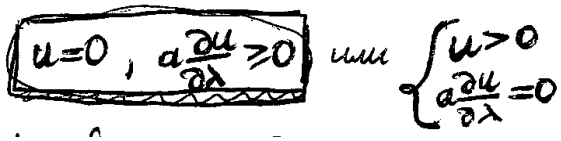
Uncertain reading. The upper arc: Σ^*; at lower left: Σ. Two ringed junction marks; the left one is retraced, with an oblique stroke. Its role and an orientation of traversal are undetermined.



GELFAND-P000286

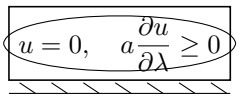
[source number: 54]

From this we obtain the problem.



Original source details: GELFAND-P000286

Diagram and typed description. Frame and brace at the two alternatives.

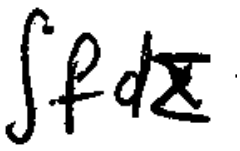


$$u = 0, \quad a\frac{\partial u}{\partial \lambda} \geq 0 \quad \text{or} \quad \begin{cases} u > 0, \\ a\dfrac{\partial u}{\partial \lambda} = 0. \end{cases}$$

$$(a_{nk}u_{/k})_{/n} = f. \qquad 54$$

Necessary:

$$\int_A f\,dx + \int_{\Sigma^*} \varphi\,d\Sigma \leq 0.$$



Original source details: GELFAND-P000286

Uncertain reading. The diagonal strokes after d overlap upper and lower horizontal lines. Provisionally dx.



Original source details: GELFAND-P000286

Uncertain reading. Below the first integral is a retraced A. The earlier strokes are illegible.

Let $f \leq 0$ on A and $\varphi < 0$ on Σ^*.

By the min principle, u cannot have a min. in A.

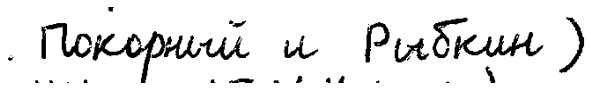
There is no sol. in $\underline{\underline{C^1(\bar{A})}}$.



(deduced from Green's formula)
an immediate contradiction — a discontinuity of the derivative is obtained.

29.9.69

Kazhdan Formulation of results of M. Artin. (1968).
(in the USSR independently Pokornii and Rybkin)



Original source details: GELFAND-P000286

Uncertain reading. The ending of the first surname provisionally reads "-ний". The second is written "Рыбкин" (Rybkin).

UMN, vol. 15, $N = 4$ (1960) p. 169
and Melamed
Sib. math. journal

$$f_i(x_1, \dots, x_k) = 0, \qquad 1 \le i \le n.$$

a collection of convergent power series (in a nbd. of 0
We try to find

$$\boxed{x_j = x_j(y_1, \dots, y_\ell)} \quad \boxed{j = 1, \dots, k.}$$

so that the equalities hold upon substitution.
It is sufficient to find such functions in formal power series.

Example.

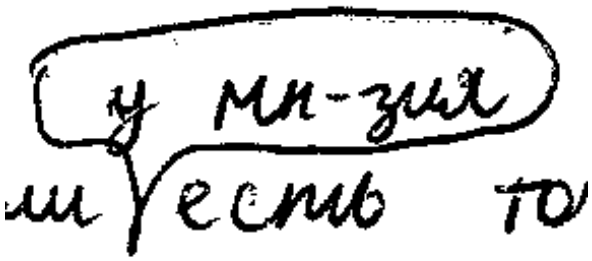
$$x_1^2 - x_2^3 = 0 \qquad \begin{aligned} x_1 &= y^3, \\ x_2 &= y^2. \end{aligned}$$

Then a sol. in analytic functions will be found, arbitrarily close. (with any number of derivatives)

the problem is handled by induction on the number of x_j.
and a more complicated ind.
Reformulation.
These are equations of an analytic set and 2 rings

$$\begin{aligned} A &= \mathbb{C}\{y_1, \dots, y_\ell\} && \text{convergent series.} \\ B &= \mathbb{C}[[y_1, \dots, y_\ell]] && \text{formal series.} \end{aligned}$$

If the manif. has a point with values in B, then there is also a point with values in A.



Manuscript notation. Insertion between “If” and “has”.

Original source details: GELFAND-P000286

GELFAND-P000287

Induction also on conn. components. (their number)

Irreduc. — a subpiece has smaller dimension.

1) If for irreduc. varieties (of the given dim.) proved, then also always proved. (given dim.)

Choose eqns. lying in the ideal, **[crossed out: ?]** corresp. to this component.



Uncertain reading. The initial Cyrillic letters appear to be ко…. The rest is obscured by the cancellation.

Original source details: GELFAND-P000287

Adding eqns. Consider a prime ideal.

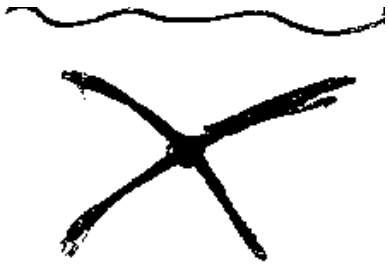
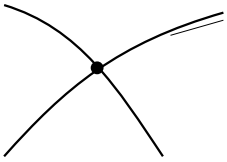


Diagram and typed description.



Original source details: GELFAND-P000287

$\mathcal{J} =$ (Intersect. of primes).

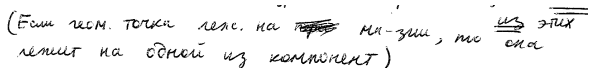


Uncertain reading. The sign before the equals sign has a long lower hook. Provisionally $\mathcal{J}$.

Original source details: GELFAND-P000287

We obtain 2 systs. of eqns. Every solution satisfies one of these systems.

(If a geom. point lies on **[crossed out: ?]** a variety, then it lies on one of the components.)



Original source details: GELFAND-P000287

Uncertain reading. A short word or word beginning between “on” and “a variety” is covered by several lines; no reliable sequence of letters is established.

[Cancelled mark in the left margin: ?]



Original source details: GELFAND-P000287

Uncertain reading. A compact repeatedly crossed-out mark with loops. Whether it is a letter, number or nontextual mark is undetermined.

The rest — next time.
L. Faddeev.

On the results of a computational experiment, **[crossed out: ?]** rel. to Kirillov’s programme

descr. of reprs.



Original source details: GELFAND-P000287

Uncertain reading. The cancelled word appears to begin with the Cyrillic letters ре.... The rest is obscured.

Representation — quantization of class. mechanics.

Examples and probls. on quantization rules.

M_{2n} — a sympl. manifold, equipp. Ω — a 2-form (nondeg.). $\Omega = d\omega$.

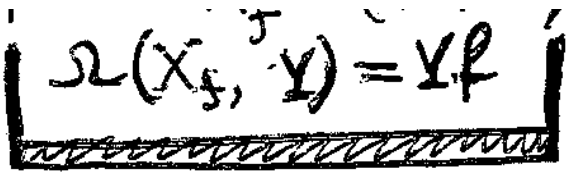


Original source details: GELFAND-P000287

Manuscript notation. The abbreviation before the form reads обесп. (“equipp.”).

$d\mu = \Omega^n$ — a measure.

$f \longrightarrow X_f$ (a field from a fn.)



Original source details: GELFAND-P000287

Diagram and typed description.

$\Omega(X_f, Y) = Yf$

Hamiltonian fields X_f form a Lie algebra

$$[X_f, X_g] = X_{\{f,g\}},$$

where $\{f, g\}$ — the Poisson bracket (it is defined by this).
Oper. X_f is antisymm. in L^2 with measure $d\mu = \Omega^n$. (It is preserved by Liouville's theorem.)
Lie-algebra elem. $X_f \longrightarrow$ corresponding oper. in L^2.
1951 Van Hove found a 1-param. family of representations.

$$Z_f = X_f + \alpha\big(\omega(X_f) - f\big)$$

GELFAND-P000288

[source number: 55]

55

Then the representation condition holds

$$\omega(X_f) - f = \ell_f$$

Van Hove's assertion: This is again a repr., not equiv. for any α

$$[Z_f, Z_g] = Z_{\{f,g\}}$$

But this is not yet a passage to quantum mechanics, since everything is still in the space of all variables, rather than half of them.
Consider now $SO(3)$ with its nat. repr. in $\mathbb{R}^3$. On the orbits (S^2) there is a natural compl. structure.
Result of computing Z_f, when X_f — a gener. of the rotation group.



Diagram and typed description.

~~X_1, X_2, X_3~~

Original source details: GELFAND-P000288

$$x_1^2 + x_2^2 + x_3^2 = a^2 \quad \text{— orbit}$$

Functions x_1, x_2, x_3. Corresp. Van Hove opers.:

Quantization problem.

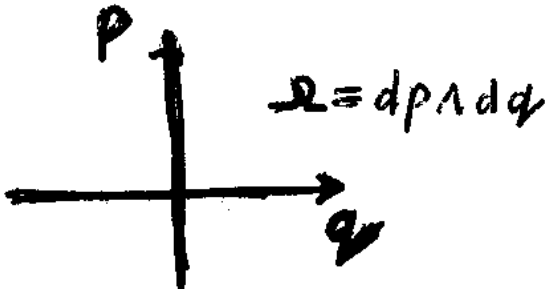
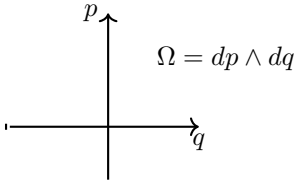


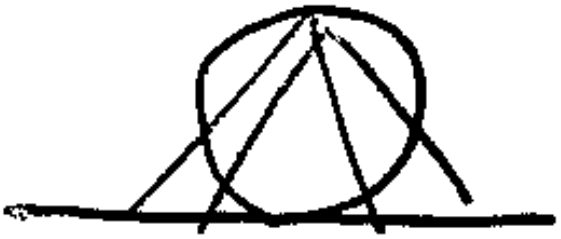
Diagram and typed description.



Original source details: GELFAND-P000288

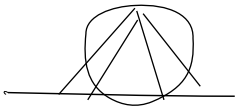
Hamiltonian $f = \dfrac{p^2}{2m}$ (free particle).
An operator must be assigned to it.

$f = x_1$.



Original source details: GELFAND-P000288

Uncertain reading. A rounded contour touches a nearly horizontal line; four sloping strokes extend from its upper region. The rightmost ends above the lower line. A small left hook may be a leftward arrow or a line irregularity; its direction and the exact upper-stroke junctions are undetermined. There are no labels.



After stereogr. projection (coord. z) — ω is not single-valued.

$$\Omega = a\frac{dz \wedge d\bar{z}}{(1+|z|^2)^2}$$

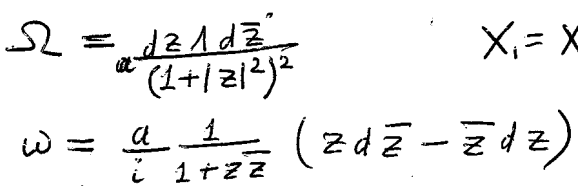
$$X_i = X_f \quad \text{for } f = x_i$$



Uncertain reading. The subscript on X_i has a dot. The short stroke under x_i has no distinct dot.

Original source details: GELFAND-P000288

$$\omega = \frac{a}{i}\frac{1}{1+z\bar{z}}(z\,d\bar{z} - \bar{z}\,dz)$$



Manuscript notation. The factor before the fraction in Ω is a. The entry for ω has a/i.

Original source details: GELFAND-P000288

No sing. on the plane, but a sing. at ∞ (north pole before projection).

$$X_1 = \frac{1}{2i}\left((1-z^2)\frac{\partial}{\partial z} - (1-\bar{z}^2)\frac{\partial}{\partial \bar{z}}\right)$$

$$Z_1 = X_1 + \frac{\alpha a}{2}(z+\bar{z}).$$

The analyt. and antianal. parts commute!
Therefore we take

$$Y_1 = \frac{1}{2i}(1-z^2)\frac{d}{dz} + \frac{\alpha a}{2}z$$

Y_2, Y_3 analogously. This is a basis of the Lie alg., satisfying commutation rels.

GELFAND-P000289

They give a repr. of the rotation gr.

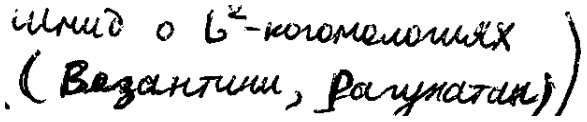
A measure where they are self-adj. exists only for integral a!

$$\frac{1}{(1+\bar{z}z)^a}$$

Square-integrable ones — a finite number — polynomials. Therefore a finite-dimensional representation.

Next time: Kazhdan;

(Schmid on L^2-cohomology (Vezantini, Raghunathan)).



Original source details: GELFAND-P000289

Uncertain reading. The first surname in parentheses provisionally reads Везантини (Vezantini).

Starts at 18^{30}. (end 20^{30})

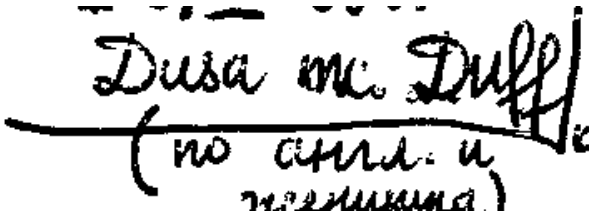
6/X-69 yr.



Original source details: GELFAND-P000289

Uncertain reading. A narrow mark before the digit 6 is heavily struck through. Its identity is uncertain.

Dusa mc Duff



Original source details: GELFAND-P000289

Manuscript notation. The name is written “Dusa mc Duff”, with lowercase “mc”.

On factors of type II_1.

(in English and a woman).



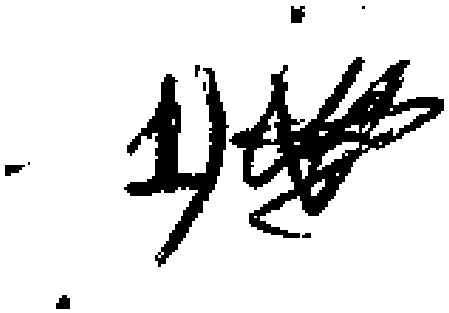
Original source details: GELFAND-P000289

Diagram and typed description. Three rules under the reference in the left margin.

see Naimark.

A subring of the ring of all opers. in Hilb. space is called a factor, if

1) [crossed out: ∀?] $X \in M \Longrightarrow X^* \in M$.



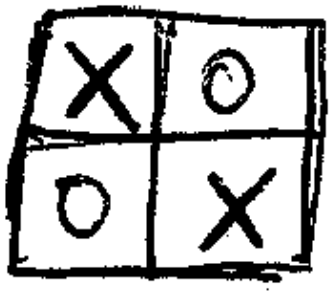
Original source details: GELFAND-P000289

Uncertain reading. After "1)" the sign beneath slanting strokes may be the quantifier ∀; it has not been recovered.

2) M is weakly closed (for ∀ seq. X_n the weak limit belongs).

3) The centre consists of opers. proportional to I.

Example. In the fin.-dimensional case:



Original source details: GELFAND-P000289

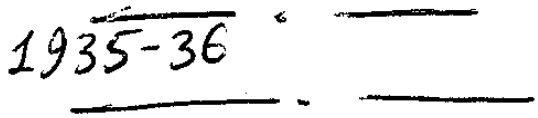
Diagram and typed description.

X	0
0	X

Every factor in the fin.-dimensional case is like this: in $H_1 \otimes H_2$ consider opers. $X \otimes 1$ (X — arbitrary).

A vect. — a matrix ξ_{ik}. An oper. — acts on the 1st coordinate.

Every weakly closed ring (with involution) decomposes into a direct integral of factors.



Original source details: GELFAND-P000289

Diagram and typed description. The date with rules above and below it.

Von Neumann

1935–36

Let M — a factor; then $\frac{X+X^*}{2}, \frac{X-X^*}{2i}$ generate them.

(self-adj.) $a = \int \lambda \, de_\lambda$, $e_\lambda \in$ the factor, and generate the factor.

A factor is generated by projections with $e^2 = e$, $e^* = e$.

GELFAND-P000290

[source number: 56]

56

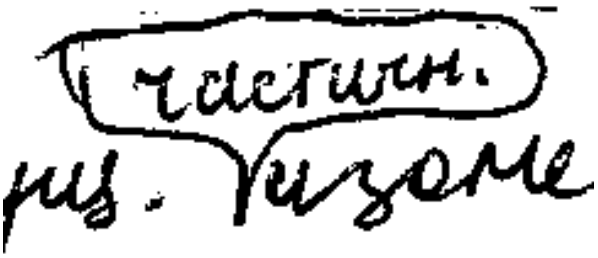
partial

If [crossed out: illegible] there exists an isometry from M, joining the subsp. corresponding to the projections, then the 2 opers. are called equivalent.
(subsp.)



Original source details: GELFAND-P000290

Uncertain reading. After "If" and before "there exists", tall signs are densely crossed by diagonals; the underlying signs have not been separated or read.



Original source details: GELFAND-P000290

Manuscript notation. "Partial" is inserted above "isometry".

If equiv. ones are identified, then an ordering results (by inclusion).

Equiv. classes of projection opers. can be added.

Hence subsp. can be measured (assign numbers to a subsp.).

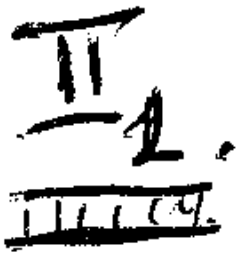
(No Archim. axiom — ∞ is present).

Thm. If there is a min subsp., then it is isom. to the ring of all opers. in Hilb. space.

$$R = M \otimes M' \qquad (M' \text{ — commutant factor}).$$

No min $\Longrightarrow$ either II_1, II_∞ or III.

II_1 — there is never a min, but everything is meas. by numbers in $(0, 1]$ (there is a max) and the Archim. axiom.



Original source details: GELFAND-P000290

Diagram and typed description.

II_1.

[cancelled sign: ?] $\forall$ oper. has a trace and det; the minimax theory of eigenvalues holds (everything that holds for matrices).



Uncertain reading. Two nearly vertical strokes are visible under the slanted cancellation before the legible $\forall$. The cancelled sign is uncertain.

Original source details: GELFAND-P000290

$$\mathrm{Tr}(a) = \int \lambda \, d(\dim e_\lambda) \qquad (\dim \text{ from } 0 \text{ to } 1)$$

The space itself has not yet been discovered.

Which factors of type II_1?

1) hyperfinite Murray and Neumann — there is only one

2) J. Schwartz

3) Ching

4) Sakai

5–6) Dixmier, Lance

7–8) Zeller-Meier

$7, 8'$) Stepin, Golodets

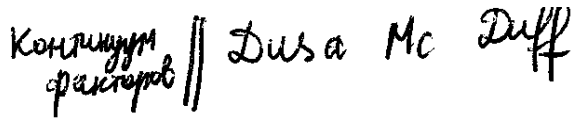


Diagram and typed description.

Continuum of factors // Dusa Mc Duff

Original source details: GELFAND-P000290

GELFAND-P000291

13/X-69 yr. Next time Minlos and Sinai.

Gelfand.

On Araki's work.

Question: Let

$$\begin{pmatrix} \alpha(t) & \beta(t) \\ -\bar{\beta}(t) & \overline{\alpha(t)} \end{pmatrix} = u(t)$$

a unit. matrix with det 1,

$$|\alpha(t)|^2 + |\beta(t)|^2 = 1, \qquad t \in [0, 1].$$

Find all irred. unit. representations of this group. This has not been done.

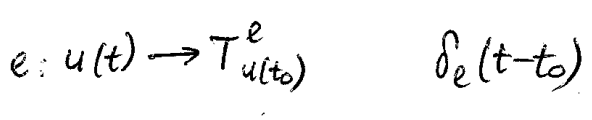
supp of repr. F — the smallest closed set for which the condition holds

$$u(t) \equiv 1 \quad \forall t \in U(F) \Longrightarrow T_{u(t)} = I$$

Problem: does an irred. unit. representation exist with supp — an interval.

supp — a point — the operator $u(t_0)$ (2-dim. repr.).
This is the repr. $\delta(t-t_0)$.

$$e:\ u(t) \longrightarrow T^e_{u(t_0)} \qquad \delta_e(t-t_0)$$

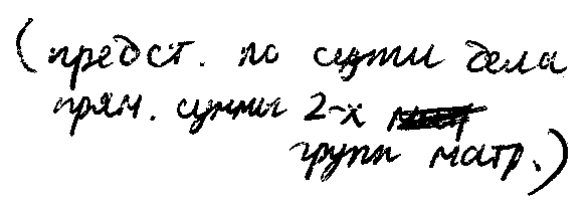


Manuscript notation. The three matching looped letters read e.

Original source details: GELFAND-P000291

$$u(t) \longrightarrow u(t_1) \otimes u(t_2)$$

an irred. repr. with supp — a pair of points.
(a repr., in effect, of a direct sum of 2 [crossed out: ?] groups of matrices).



Uncertain reading. A word after "2" is crossed out. "Groups of matrices" is written separately below.

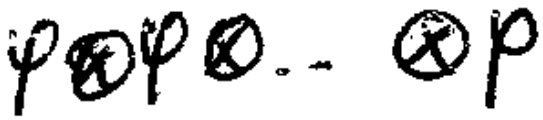
Original source details: GELFAND-P000291

One can take n points.
For a countable set this can no longer be done (it can be done for a convergent seq.).
Araki.
Idea:

$$e^K = \bigoplus_{n=0}^{\infty} S^n(K)$$

(K — Hilb. space)

$$e^{\varphi} = \bigoplus_{n=0}^{\infty} \frac{1}{\sqrt{n!}} \varphi \otimes \varphi \otimes \cdots \otimes p$$



Uncertain reading. The last factor has a right-hand loop and straight descender. Provisionally p.

Original source details: GELFAND-P000291

$\sqrt{n!}$, so that $(e^{\varphi}, e^{\psi}) = e^{(\varphi,\psi)}$.
$K \to e^K$ — a functorial passage

$$K_1 \oplus K_2 \longrightarrow e^{K_1} \otimes e^{K_2}$$

At $\forall$ point t we specify a representation in the space $K(t)$ (e.g. 2-dimensional)

$$\int_0^1 K(t)\,dt$$

GELFAND-P000292

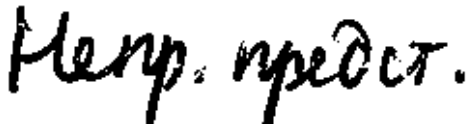
[source number: 57]

57

Consider $e^{\int_0^1 K(t)\,dt}$.
A repr. is obtained in it (from funct.) which is reducible.

———— • ————

The gr. of matrices of order 1 $\alpha(t) = e^{x(t)}$.
"Nepr." repr.



Uncertain reading. The line reads "Непр. предст." ("Nepr. repr."). The first abbreviation is unresolved.

Original source details: GELFAND-P000292

$$e^{x(t)} \longrightarrow C, \qquad e^{x_1(t)}e^{x_2(t)} \longrightarrow C_1C_2, \qquad C = e^a$$
$$x(t) \longrightarrow a$$

a — a lin. funct. of $X(t)$ — a generalized func. or distribution.



Uncertain reading. The letter before (t) is taller than the nearby x signs. Provisionally $X(t)$.

Original source details: GELFAND-P000292

$$\delta(t - t_0): \qquad e^{x(t)} \longrightarrow e^{x(t_0)}$$

$$x(t) \longrightarrow e^{\int_0^1 a(t)x(t)\,dt},$$

On the circle an integrality condition is added.

—— • ——

A repr. here is a generalization of a distribution.

GELFAND-P000293

[source number: 60]

17/XI-69 yr. Hewitt.
One result (sharp for all comp. groups)
as for the circle.
[crossed out: illegible]



Original source details: GELFAND-P000293

Uncertain reading. Dense crossing strokes precede the heading “On s-numbers”; the underlying letters or signs have not been read.

On s-numbers:

Gantmakher;
Wielandt
Berezin, Gelfand Trudy M. M. O. —
— $\forall$ semisimple Lie groups.
from repr. theory for a compl. func.

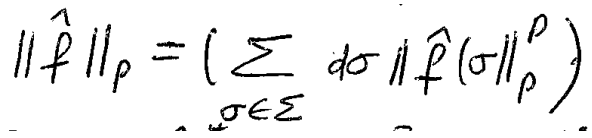
H. Weyl and Ky Fan.
[crossed out: illegible] All this is for an arbitrary group and a class of representations:



Original source details: GELFAND-P000293

Uncertain reading. Before “All this”, joined vertical and slanting strokes are cancelled; the underlying sequence is undetermined.

$$\|\hat{f}\|_p = (\sum_{\sigma\in\Sigma} d\sigma \|\hat{f}(\sigma\|_p^p)$$



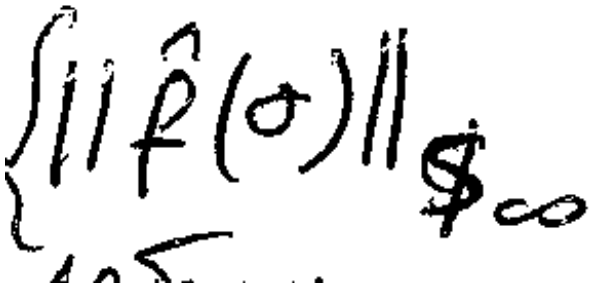
Original source details: GELFAND-P000293

Manuscript notation. The σ next to d sits on the line. The argument σ is followed immediately by the norm bars. No exponent is legible after the outer parenthesis.

Peter–Weyl–Plancherel thm.

$$\|\hat{f}\|_2^2 = \int_G |f|^2 \, d\lambda;$$

$$\|\hat{f}\|_\infty = \sup\{\|\hat{f}(\sigma)\|_{\mathfrak{S}_\infty} : \ \sigma \in \Sigma\}.$$



Uncertain reading. A vertical stroke crosses the S-shaped sign before ∞. Provisionally $\mathfrak{S}$.

Original source details: GELFAND-P000293

Riem.–Lebesgue lemma:

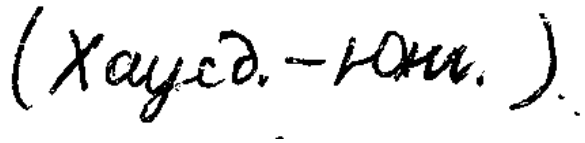
$$\forall f \in L_1(G), \quad \forall \varepsilon > 0 \quad \{\sigma : \ \|\hat{f}(\sigma)\|_{S_\infty} > \varepsilon\}$$

— a finite set.

Kunze's thm. $1 < p < 2$

$$\|\hat{f}\|_{p'} \le \|f\|_p.$$

(Hausd.–Youn.).



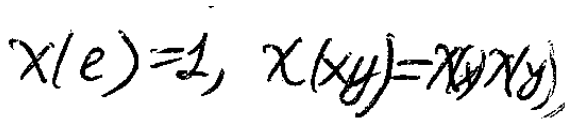
Uncertain reading. The abbreviation after "Хаусд.-" provisionally reads "Юн.". The final strokes are close together.

Original source details: GELFAND-P000293

Thm. of Hardy–Littlewood — equality only for $f(x) = \alpha e^{inx}$;

For a general group things are not so simple: there, for funcs. of the form

χ — a cont. func. on an open subgroup G_0, $\chi(e) = 1$, $\chi(xy) = \chi(x)\chi(y)$,



Uncertain reading. The first argument on the right has been retraced. Provisionally x.

Original source details: GELFAND-P000293

$\chi = 0$ outside G_0.

There are analogues of the Riesz–Thorin theorems.

GELFAND-P000294

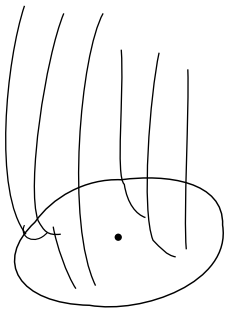
23.2.70 yr.

Kazhdan on Griffiths's work.



Original source details: GELFAND-P000294

Uncertain reading. On the left is an oval contour with an interior point and six long curves above it. A small loop and nearly parallel lower strokes occur at the left edge. The short-stroke connections and curve continuations are undetermined; there are no labels or definite arrows.

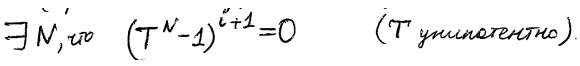


$$T:\ H^i(V_s) \longrightarrow H^i(V_s)$$

(on going around a puncture)

$$\exists N, \text{ such that } \quad (T^N - 1)^{i+1} = 0$$

(T is unipotent).



Original source details: GELFAND-P000294

Manuscript notation. To the right of $(T^N - 1)^{i+1} = 0$ the manuscript separately says "(T is unipotent)".



Original source details: GELFAND-P000294

Diagram and typed description.

W

$\downarrow$

S

an algebraic bundle

base — a 1-dim. alg. variety (a Riem. surface with punctures).

$$\pi_1(S):\ H^K(V_s) \longrightarrow H^K(V_s) \qquad (\text{coeffs. } \mathbb{C}).$$

and defines a vect. bundle [crossed out: an illegible sign] (analytic)



Uncertain reading. A narrow cancelled sign occurs between “bundle” and “(analytic)”; the underlying letter is undetermined.

Original source details: GELFAND-P000294



Diagram and typed description.

$$\begin{array}{c} E \\ \downarrow \\ S \end{array}$$

Original source details: GELFAND-P000294

$O(E)$ — the sheaf of germs of analyt. sects.

There is a canon. connection from parallel transports on the universal cover $\tilde{S}$. (There it is a direct product.)

We shall study it near a puncture.

Write the connection in the form of an eqn.

$$\begin{array}{l} f_1(z) \text{ — sections} \\ \dots \\ f_n(z) \end{array} \qquad \text{(a basis near } z\text{)}.$$

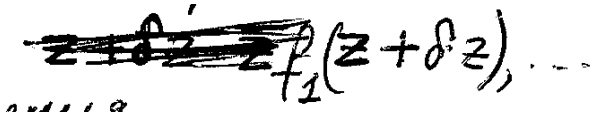
Picard–Fuchs eqn. $\dfrac{\partial}{\partial z} = A(z)$



Manuscript notation. This line reads $\partial/\partial z = A(z)$. The later equation separately includes $f(z)$.

Original source details: GELFAND-P000294

Transport $f_1(z)$ horizontally to $z + \delta z$ and expand in the basis [crossed out: $z + \delta z$, followed by illegible signs] $f_1(z + \delta z), \dots$



Uncertain reading. After “basis”, the cancelled sequence $z + \delta z$ has an illegible ending; there may be another z on the right. The uncancelled $f_1(z + \delta z), \dots$ is read separately.

Original source details: GELFAND-P000294

Consider any flat sect. of the bundle. it satisfies the eqn.

$$\frac{\partial}{\partial z} f(z) = A(z) f(z)$$

and all flat sects. are solutions of this eqn.

Example of flat sections.

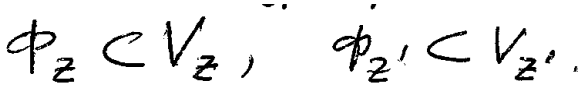
Let Φ_z — a cycle in the fibre (K-dimensional).



Original source details: GELFAND-P000294

Uncertain reading. After "Let" a loop overlaps a heavy vertical stroke. Provisionally Φ_z.

$$\phi_z \subset V_z, \qquad \phi_{z'} \subset V_{z'}.$$



Original source details: GELFAND-P000294

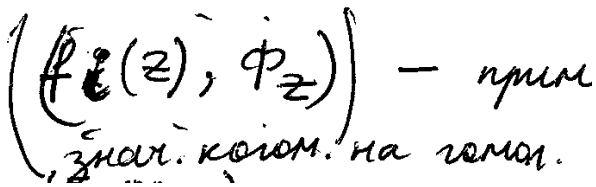
Manuscript notation. Both letters on the lower line have a thin loop.

GELFAND-P000295

[source number: 62]

62

— an example of a flat section. —



Original source details: GELFAND-P000295

Uncertain reading. The index on f is heavily retraced. Provisionally a subscript i. The outer parentheses descend toward the explanation.

$$\Big((f_i(z), \phi_z) \quad \text{value of cohom. on hom.} \Big)$$

— a vector func.

Thm. (Griffiths)

$\exists$ a basis $f_i(z)$, such that the eqn. has a reg. singular point,

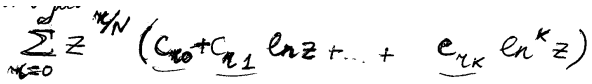
i.e. $zA(z)$ extends holomorphically to 0. (regular).

(Allows one to infer the behaviour of periods on approaching a singular point).

Corollary

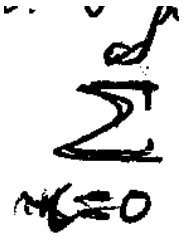
Consider any flat section — a sol. of the eqn. Then it expands in a series

$$\sum_{n\le 0} z^{n/N}\left(\underline{C_{n0}} + \underline{C_{n1}}\ln z + \ldots + \underline{c_{nk}}\ln^k z\right)$$



Original source details: GELFAND-P000295

Uncertain reading. The lower limit is provisionally $n \le 0$. The final initial resembles lowercase c or e. Three coefficients are underlined.

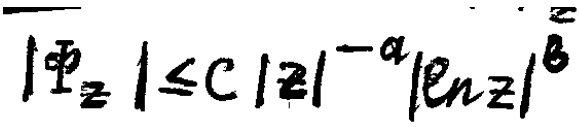


Original source details: GELFAND-P000295

Uncertain reading. A loop above the sum merges with the descender in “ряд” (“series”). The upper limit is unresolved.

Lemma $\forall$ flat sect. ϕ_z

$$|\Phi_z| \le C|z|^{-a}|\ln z|^b$$



Original source details: GELFAND-P000295

Uncertain reading. The exponent on $|\ln z|$ has two close loops. Provisionally b. The bound uses a larger Φ_z.

Original source details: GELFAND-P000295

Diagram and typed description.

. . . . . .

GELFAND-P000296

7/IX-1970 yr. Seminar of Gelfand.

Seminar programme.

1. Kazhdan on the relation between formal and algebraic geometry.
2. Arnold: Griffiths’s works.
3. Mityagin and Henkin on their works.

[cancelled number: 4.?] $1'$. Moishezon on Artin’s works.



Original source details: GELFAND-P000296

Uncertain reading. Before the separate $1'$ is a cancelled number, provisionally 4. A separate “?.” appears below.

?. K-theory of finite groups

Brown, Tate, Tomson, Swan, Katz.

(Kazhdan, Bernstein, etc.)

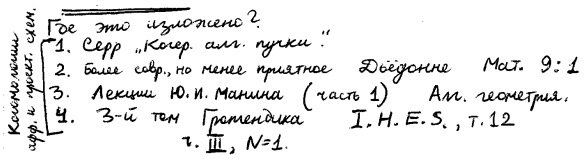
4. Hironaka.
For next semester.

ζ-func. (p-adic harm. analysis).

Finite groups.

Kazhdan.

Where is this expounded?



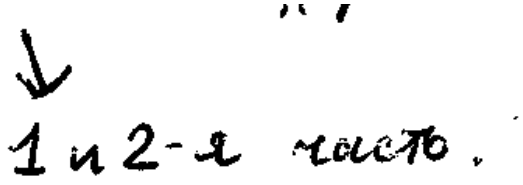
Original source details: GELFAND-P000296

Diagram and typed description.

Cohomology of aff. and proj. schemes.

1. Serre, “Coherent alg. sheaves”.
2. More modern, but less pleasant Dieudonné Mat. 9:1
3. Lectures of Yu. I. Manin (part 1) Alg. geometry.
4. 3rd volume of Grothendieck I. H. E. S., vol. 12 part III, $N = 1$.

Gelfand’s remark: learn Serre’s books “Finite groups” and “Lie gr. and Lie alg.”.
(Part of the minimum in lin. algebra)



Original source details: GELFAND-P000296

Diagram and typed description.

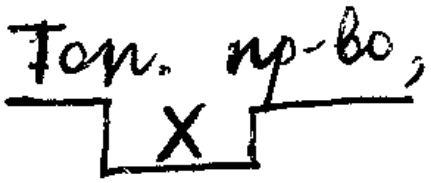


1 and 2nd part.

Part 1: Cohomology of coherent sheaves.

§1. Definition of a scheme.

A scheme — a ringed space of a special kind



Original source details: GELFAND-P000296

Diagram and typed description. Top. space, X equipped with a sheaf of rings. O (structure sheaf).

Affine scheme. A — a comm. ring with 1.

Spec A Points — prime ideals $\wp \subset A$.

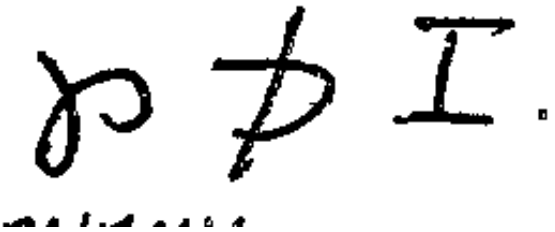


Uncertain reading. The looped ideal sign admits $\wp$ / $\mathfrak{p}$ / p. Both occurrences show the nearest available glyph $\wp$; the letter's identity is undetermined.

Original source details: GELFAND-P000296

Topology: $\forall \underset{\text{ideal}}{I} \subset A$ assign the set of prime ideals $U_I \subset \operatorname{Spec} A$,
consisting of prime ideals

$$\wp \not\supset I.$$



Uncertain reading. A looped ideal glyph and a crossed inclusion sign open to the left. No lower equality bar is discernible.

Original source details: GELFAND-P000296

The sets U_I are called open.

GELFAND-P000297

[source number: 67]

67

Problem: check that a topology results.
A basis of open sets:

$$f \in A \rightsquigarrow U_f \subset \operatorname{Spec} A \quad \text{from} \quad \wp \not\ni f.$$



Uncertain reading. The looped ideal sign admits $\wp$ / $\mathfrak{p}$ / p; the candidate $\wp$ is shown. The following sign has a rounded outline, a middle crossbar and a crossing stroke; $\not\ni$ is shown.

Original source details: GELFAND-P000297

Problem: U_f — a basis.
Problem: the weakest separation ax. holds (of 2 points, 1 can be separated by a nbhd.)

Localization.

Let S — a multipl. set in A, not containing 0 (closed under mult.)

The ring A_S consists of pairs $\frac{a}{s}$:

$$\frac{a}{s} = \frac{a'}{s'} \iff \exists s'' \in S \qquad s''(s'a - sa') = 0.$$

Problem: this is an equivalence rel.

Later this will be needed for modules.

Struct. sheaf on Spec A:

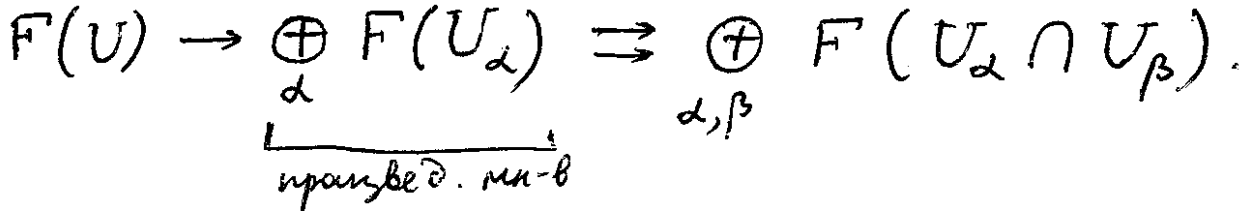
$$\Gamma(U_f, O) = A_{\substack{\{f^n\},\\ n\geq 0}}, \qquad S = \{f^n\}.$$

Reminder.

Def. of a sheaf.

A presheaf and $\forall U \subset X$ and a cover $\{U_\alpha^*\}$ of the set U the sequence is exact.

$$\{U_\alpha^*\} \subset U \qquad \left(\bigcup U_\alpha^* = U\right).$$



Original source details: GELFAND-P000297

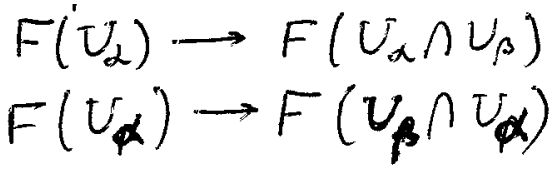
Diagram and typed description. Both signs are circled crosses. Below the first expression the note reads “product of sets”.

$$F(U) \longrightarrow \bigoplus_{\alpha} F(U_\alpha) \rightrightarrows \bigoplus_{\alpha,\beta} F(U_\alpha \cap U_\beta).$$

product of sets

$$F(U_\alpha) \longrightarrow F(U_\alpha \cap U_\beta)$$

$$F(U_\alpha) \longrightarrow F(U_\beta \cap U_\alpha)$$



Uncertain reading. The indices on the second line are overwritten. Provisionally α on the left and β, α on the right. The underlying strokes cannot be separated.

Original source details: GELFAND-P000297

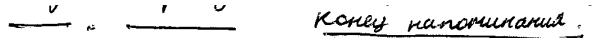
Question: why are these different arrows?

Exactness:

$$\operatorname{Im} i = \{\cdot \mid \varphi_1 = \varphi_2\}.$$

The stalk of a sheaf (presheaf) at pt. x etc.

Problem: prove that ∀ presheaf ∃ a sheaf with the same stalks (and unique)
Remark: to define a sheaf, it suffices to know the presheaf on a basis.

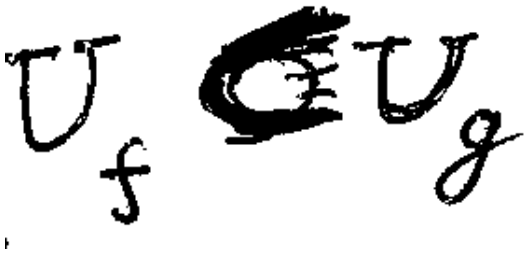


Original source details: GELFAND-P000297

Diagram and typed description.

End of reminder.

$$U_f \subset U_g \iff \exists n: \quad f^m = gh.$$



Uncertain reading. The sign between U_f and U_g is heavily overwritten; the outer contour $\subset$ is shown. The original sign is undetermined.

Original source details: GELFAND-P000297



Uncertain reading. After ∃ the letter reads n. The exponent on f provisionally reads m, with the final arches joined.

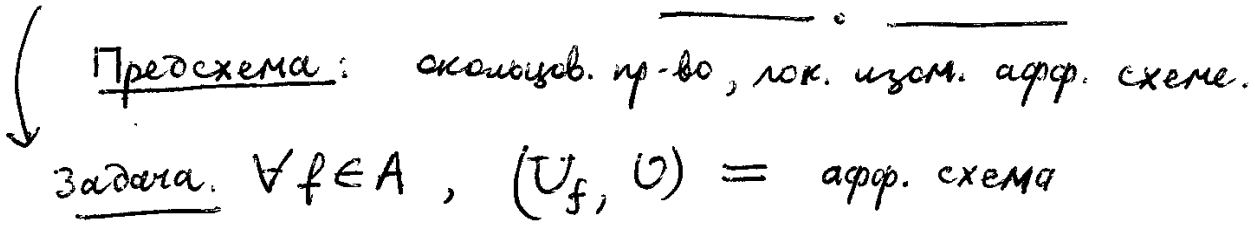
Original source details: GELFAND-P000297

(Equiv. to the thm. that ⋂ of all prime ideals is the radical)

GELFAND-P000298

$$A_{\{g^n\}} \longrightarrow A_{\{f^n\}}.$$

$$\frac{a}{g} \longrightarrow \frac{ah}{f^m}, \qquad \frac{a}{g^k} \longrightarrow \frac{ah^k}{f^{mk}}.$$



Original source details: GELFAND-P000298

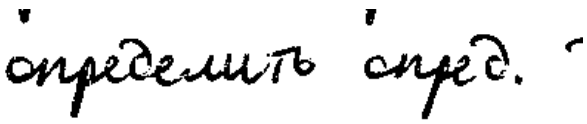
Diagram and typed description.

Prescheme: a ringed space, loc. isom. to an aff. scheme.
Problem. $\forall f \in A$, $(U_f, O) =$ an aff. scheme

Product of preschemes. Spec A, Spec B

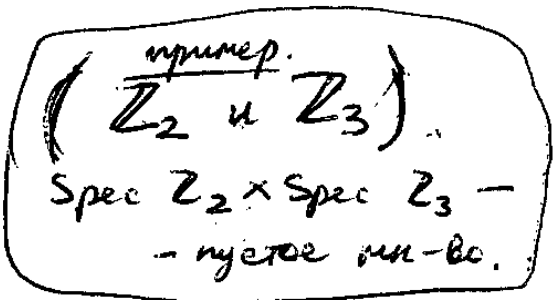
$$\text{Spec}\, A \times \text{Spec}\, B = \text{Spec}(A \otimes B)$$

Problem: determine def. of two preschemes.



Original source details: GELFAND-P000298

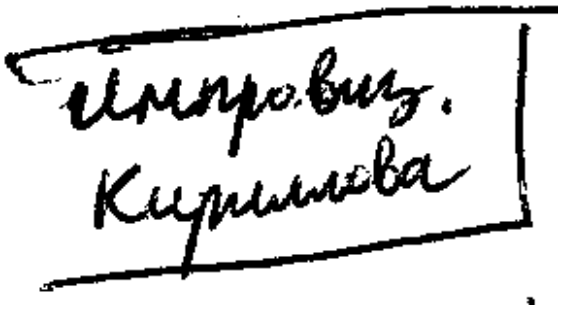
Manuscript notation. The manuscript reads “определить опред.” (“determine def.”).



Original source details: GELFAND-P000298

Diagram and typed description.

example.
(Z_2 and Z_3).
Spec Z_2 × Spec Z_3 —
— the empty set.



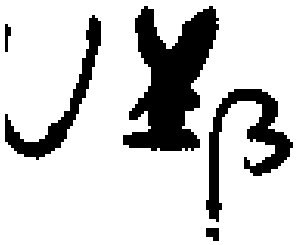
Original source details: GELFAND-P000298

Diagram and typed description.

Improvis.
of Kirillov

$$\begin{array}{cc} X, & Y \\ \| & \| \\ \bigcup X_\alpha & \bigcup Y_\beta \end{array}$$

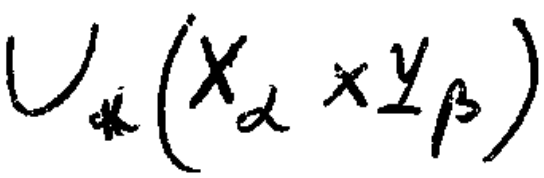
— preschemes



Original source details: GELFAND-P000298

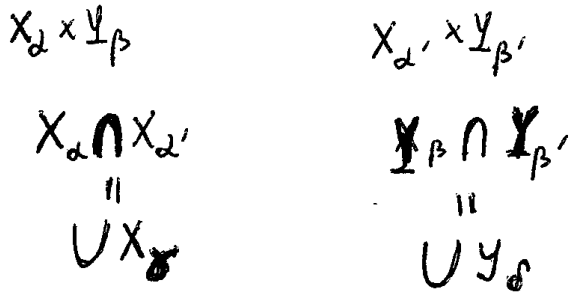
Uncertain reading. The Y above β is overwritten. The overwritten Y's in the right-hand intersection appear in the figure below.

in $\bigcup_{*}(X_\alpha \times Y_\beta)$



Original source details: GELFAND-P000298

Uncertain reading. The dark index under the union has radiating strokes, provisionally an asterisk.



Original source details: GELFAND-P000298

Diagram and typed description.

$$\begin{array}{cc} X_\alpha \times Y_\beta & X_{\alpha'} \times Y_{\beta'} \\ X_\alpha \cap X_{\alpha'} & Y_\beta \cap Y_{\beta'} \\ \| & \| \\ \bigcup X_\gamma & \bigcup Y_\delta \end{array}$$

Uncertain reading. The index in the lower left union is heavily overwritten. Provisionally γ.



Original source details: GELFAND-P000298

Lemma: $X_\alpha \subset X_{\alpha'}, Y_\beta \subset Y_{\beta'} \Rightarrow X_\alpha \times Y_\beta \subset X_{\alpha'} \times Y_{\beta'}$.

See Serre.

A prescheme X is called a scheme if $\Delta \subset X * X$ is closed.



Manuscript notation. An asterisk stands between the two X's.

Original source details: GELFAND-P000298

Problem: prove that an aff. scheme is a scheme.

Let X — an arbitr. prescheme, $\mathcal{F}$ — a sheaf of $\mathcal{O}$-modules over [cancelled sign: x?] X.



Uncertain reading. Before X is a cancelled sign with several short radiating strokes.

Original source details: GELFAND-P000298

It is called quasicoherent if it is loc. isom.

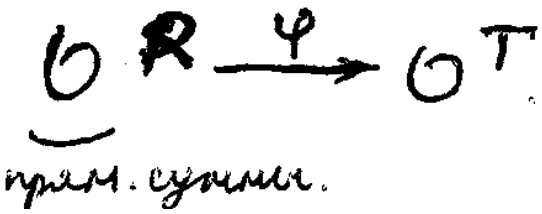
GELFAND-P000299

[source number: 68]

to the cokernel of the map:

$$\underbrace{O^R}_{\text{direct sums.}} \xrightarrow{\varphi} O^T.$$

S, T — arbitrary sets.



Uncertain reading. The first exponent resembles R, with dense strokes on the left. A separate line below reads S, T.

Original source details: GELFAND-P000299

Problem: give an example of a nonquasicoherent sheaf.

Theorem. Let A — a comm. ring and Spec A — an aff. scheme.

The map from the categ. of quasicoherent sheaves of modules

$$\mathcal{F} \longrightarrow \Gamma(\text{Spec } A, \mathcal{F})$$

to the categ. of modules is an isomorphism.

I.e. the categ. of q.-c. sheaves of mod. over the aff. scheme A is isom. to the categ. of modules over A.

(The module of glob. sections.)

(This is equivalent to saying that for aff. spaces the cohom. are zero)

(это равносильно тому, что для афф. пр-в когом. нулевые)

Original source details: GELFAND-P000299

Manuscript notation. The parenthesis is legible throughout. Degrees and coefficients are unspecified.

A quasicoherent sheaf of modules is determined by its glob. sections.

28/IX.

Direct and inv. image of sheaves.

$$X \xrightarrow{f} Y, \qquad X, Y \text{ — top. spaces,}$$

$\mathcal{F}$ — a sheaf on X,

$$\Gamma(U, f_*\mathcal{F}) := \Gamma(f^{-1}(U), \mathcal{F}), \qquad U \subset Y$$

a presheaf on Y results.

Exercise (easy) — this presheaf is a sheaf

Example: if Y — a point, then $f_*\mathcal{F}$ — the group of sections of the orig. sheaf,
($f^{-1}(U)$ may also be empty, then the set of sections is empty).



Original source details: GELFAND-P000299

Manuscript notation. The parenthesis reads "empty" twice.

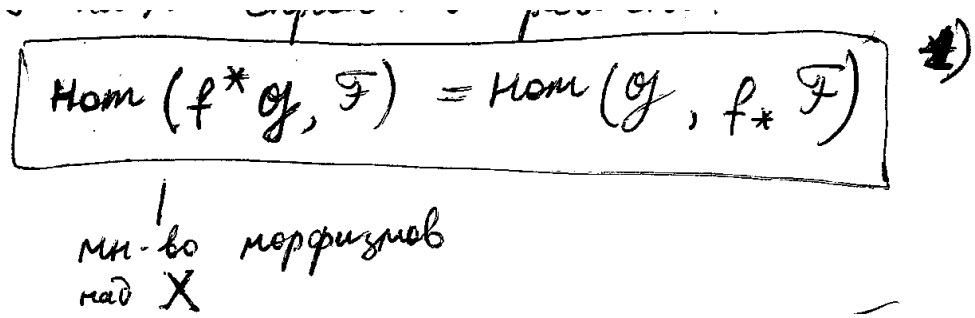
A very important concept.
Inverse image.

$$X \xrightarrow{f} Y$$

$\mathcal{G}$ — a sheaf on Y

GELFAND-P000300

The sheaf $f^*\mathcal{G}$ is defined as follows:
$\forall \mathcal{F}$ on X the equality holds:



Original source details: GELFAND-P000300

Uncertain reading. To the right of the box is a heavily overwritten number with a closing parenthesis. Provisionally 1).

$$\boxed{\mathrm{Hom}(f^*\mathcal{G}, \mathcal{F}) = \mathrm{Hom}(\mathcal{G}, f_*\mathcal{F})}^{1)}$$

set of morphisms over X

I.e. $\forall$ sheaf $\mathcal{F}$ a one-to-one corr. between these sets is given, compatible under morphisms of $\mathcal{F}$ into one another.

(An isomorphism of functors on the cat. of sheaves).

Construction, proving the existence of $f^*\mathcal{G}$

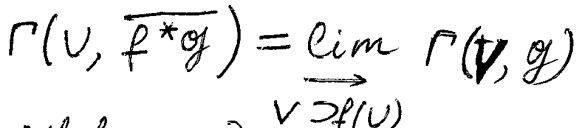
[A short expression is cancelled: ?]



Original source details: GELFAND-P000300

Uncertain reading. Beneath the diagonal strokes are an initial shaped like Γ or F and several loops. The formula is illegible.

$$\Gamma(U, \overline{f^*\mathcal{G}}) = \varinjlim_{V \supset f(U)} \Gamma(V, \mathcal{G})$$

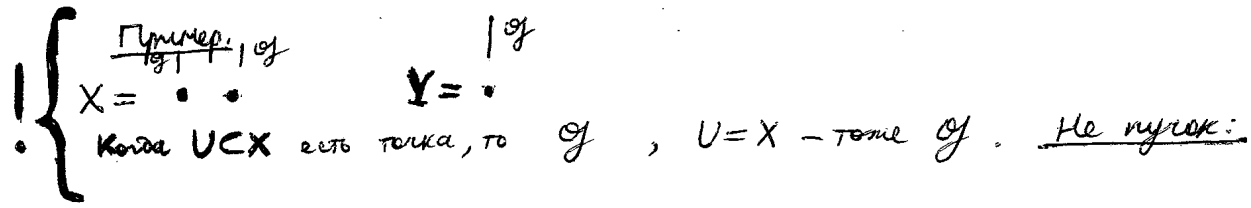


Uncertain reading. The first argument of the last Γ is heavily overwritten. The upper strokes read as V.

Original source details: GELFAND-P000300

this equality defines a presheaf $(V \subset Y)$

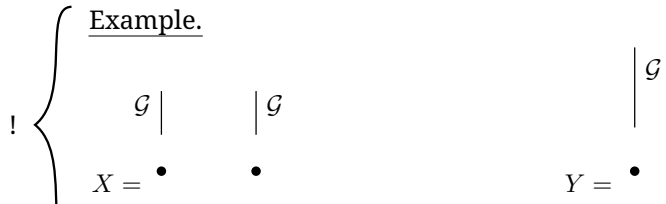
Exercise: the sheaf assoc. to it is the inverse image



Original source details: GELFAND-P000300

Diagram and typed description.

Example.



When $U \subset X$ is a point, then $\mathcal{G}$, $U = X$ — also $\mathcal{G}$. Not a sheaf.

---

1) Definition of Hom:

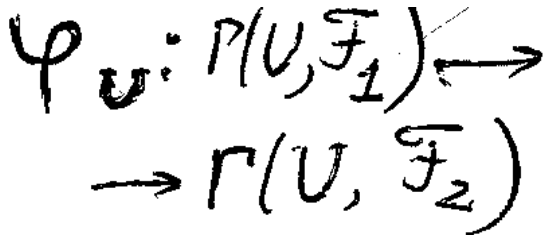
A map $\mathcal{F}_1 \longrightarrow \mathcal{F}_2$ over X is called [cancelled word: ?] a collection of maps



Uncertain reading. After "is called" and before "a collection", a short word is cancelled; a lower loop and right-hand vertical strokes are visible, but no reliable letters are established.

Original source details: GELFAND-P000300

$$\varphi_U : \Gamma(U, \mathcal{F}_1) \longleftrightarrow$$
$$\longrightarrow \Gamma(U, \mathcal{F}_2)$$
$$\forall U \subset X.$$



Uncertain reading. A long arrow stands at the right of the first line. The short slanted strokes at its left end may be a second head. A separate arrow points right on the following line.

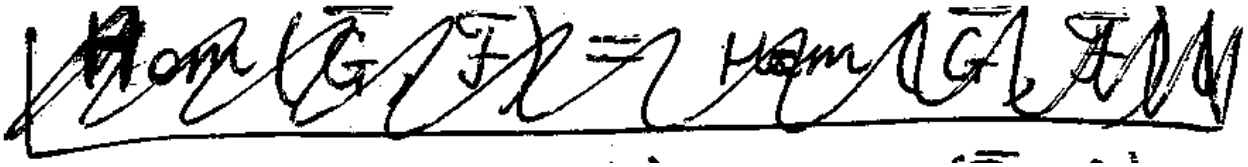
Original source details: GELFAND-P000300

and commute with the restriction opers. (A morphism of functors)

GELFAND-P000301

[source number: 69]

69



Original source details: GELFAND-P000301

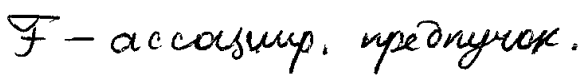
Uncertain reading. Separate bars are visible above both G signs. The upper strokes above $\mathcal{F}$ merge with the letter loops and cancellation. The bar above the second $\mathcal{F}$ is provisional.

~~$\mathrm{Hom}(\overline{G}, \mathcal{F}) = \mathrm{Hom}(\overline{G}, \mathcal{F})$~~

— def. of the sheaf corr. to a presheaf

$$\mathrm{Hom}(\mathcal{F}, G) = \mathrm{Hom}(\overline{\mathcal{F}}, G)$$

$\overline{\mathcal{F}}$ — a presheaf, $\mathcal{F}$ — the assoc. presheaf.



Manuscript notation. The entry on the right reads “the assoc. presheaf.”

Original source details: GELFAND-P000301

$\overline{\mathcal{F}} \longrightarrow \mathcal{F}$ — a map (of presheaves) must be given

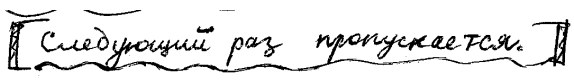


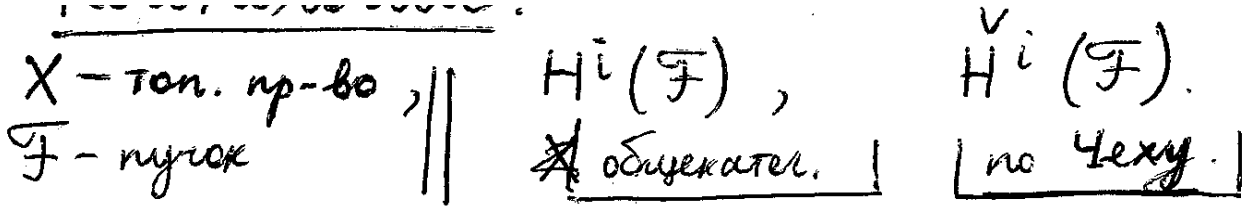
Diagram and typed description.

Next time is skipped.

Original source details: GELFAND-P000301

Reminder, § 2.

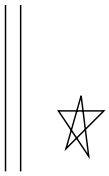
Cohomology.



Original source details: GELFAND-P000301

Uncertain reading. Two vertical rules stand at the left. The labels sit above the lower strokes of the frames.

X — a top. space, $\mathcal{F}$ — a sheaf



$H^i(\mathcal{F})$, general-categ.

$\check{H}^i(\mathcal{F})$. Čech.



Uncertain reading. Before “general-categ.” is a mark with a crossbar and a long right stroke. Whether it is a letter or an asterisk is unresolved.

Original source details: GELFAND-P000301

$H^i(\mathcal{F})$ — a deriv. functor of the sections functor.

Consider the category C of sheaves of ab. groups over the space X

Lemma 1. — this is an abelian category.

Inject. sheaves are important.

Lemma 2. There are enough inject. objects (any sheaf embeds in an inject. sheaf).

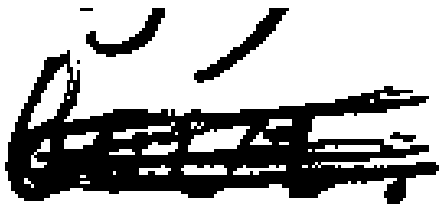
Functor to a sheaf $\longrightarrow$ sections over all of X.

$$0 \longrightarrow \Gamma(\mathcal{F}') \longrightarrow \Gamma(\mathcal{F}) \longrightarrow \Gamma(\mathcal{F}'')$$

Γ — a semiexact functor (on the left), and since there are enough inject. objects, the derived functors are defined.
Let X — a top. space $\{U\}$ — a basis of the topology in X, such that

$$\check{H}^i(\mathcal{F}, \{u_i\}) = 0, \qquad i > 0.$$

Then the actual cohom. [cancelled: calc.?] agree with the Čech calc.



Uncertain reading. After “actual” the first stroke is a tall loop. The cancelled reading is provisionally “выч.” (“calc.”).

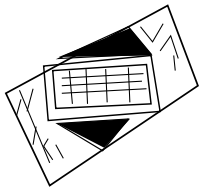
Original source details: GELFAND-P000301

And one can calc. them in acycl. covers.



Original source details: GELFAND-P000301

Uncertain reading. Slanted quadrilateral outlines, inner frames, dark triangular regions and short edge strokes. The reconstruction follows the visible contours; labels, exact internal-stroke junctions and mathematical purpose are undetermined.



GELFAND-P000302

An ellipt. syst. of order 2 — on a manif. without lower-order terms. Does it have solutions?

For the Laplace oper. it is known that there are none

The number of solns. $(\text{mod } 2)$.

In principle max is topology

(The Dirichlet prob. has a 0-th solution)

$\longleftarrow \quad \cdot \quad$

Kazhdan on one specific work.

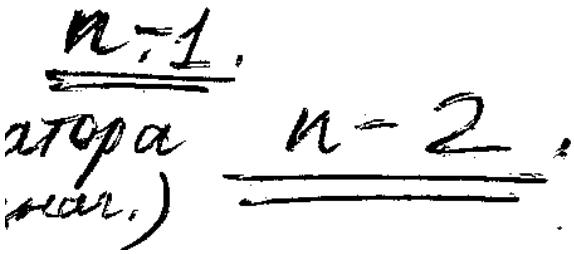
G — a compl. semis. group $(A_n$ or $D_n)$.

A reg. elem. — if the centralizer has rank $n-1$.

Subreg. — if dim of the centralizer is $n-2$,

(two coincident eigenvals.)

Uncertain reading. $n-1$ and $n-2$ are double underlined. A separate short stroke lies below the first minus.



Original source details: GELFAND-P000302

Consider

$$G \xrightarrow{\varphi} H/W \qquad \begin{array}{l}(W \text{ — the Weyl gr.} \\ H \text{ — a Cartan subgroup})\end{array}$$

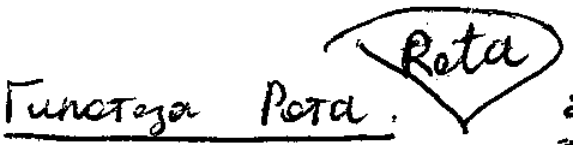
An elem. [cancelled: sub / not?] is reg. $\Longleftrightarrow$ the map is reg. there.



Original source details: GELFAND-P000302

Uncertain reading. Heavily cancelled loops precede “reg.” The readings “sub” and “not” are unresolved.

12/X.



Original source details: GELFAND-P000302

Diagram and typed description.



Rota’s conjecture.

Fix a finite field F_q.

Let there be given ℓ and $k > \ell$. Then $\exists N$ [above the line: such?], that if one partitions the ℓ-dimensional subspaces in an N-dimensional subspace,



Original source details: GELFAND-P000302

Uncertain reading. Above “that” are the first letters “та” and overlapping strokes. Provisionally “такое” (“such”).

[cancelled: that?]



Original source details: GELFAND-P000302

Uncertain reading. Before “then” a short word is heavily struck through. The last letter “о” is visible.

then $\exists$ k-dimensional one, such that all subspaces of this k-dimensional one fall into one of the classes.

$$F^N \supset E \qquad \dim E = k$$

and all its ℓ-dim. subsp. are in one class.

For $\ell = 1$ it is proved.

One can use proj. terms (there a line is a point).

If P^N is partitioned into 2 classes, there is a line in one class.

GELFAND-P000303

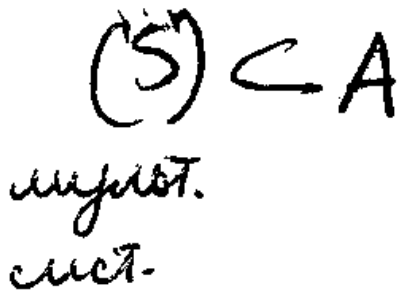
[source number: 70]

Kazhdan.

A quasicoherent sheaf — a sheaf loc. isom. to the cokernel of a map of free sheaves.

Example. A — a ring, M — a module, $\mathcal{F}_M$ — a sheaf of modules on Spec A.

$$\underset{\text{multipl. syst.}}{(\dot{S})} \subset A \qquad M_S \qquad \frac{m}{s} = \frac{m'}{s'} \quad \text{if } s''(s'm - sm') = 0.$$



Uncertain reading. A short separate stroke stands above S. Two other strokes touch the tops of the parentheses.

Original source details: GELFAND-P000303

$$M_f = M_{(f^n)}.$$

Sect. on U_f are M_f.

$$\Gamma(U_f, \mathcal{F}_M) = M_f.$$

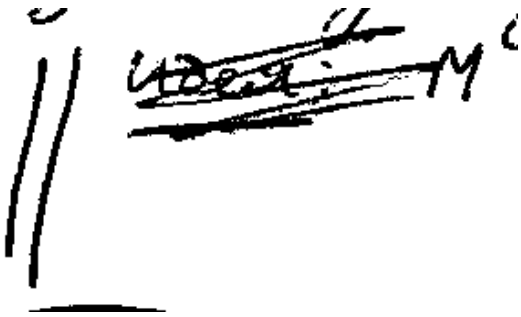
The check is left to the listeners.

Lemma: this is a sheaf.

Another def. of a quasicoherent sheaf.

Def. 2. Quasicoherent sheaves — sheaves loc. isom. to a sheaf of the form $\mathcal{F}_M$.

Problem: prove equiv. of the definitions.



Uncertain reading. Beyond two slanted rules a short word is cancelled, provisionally “idea:”. It is followed by M.

// [cancelled: idea:?] M

Original source details: GELFAND-P000303

Main theorem.

Spec A

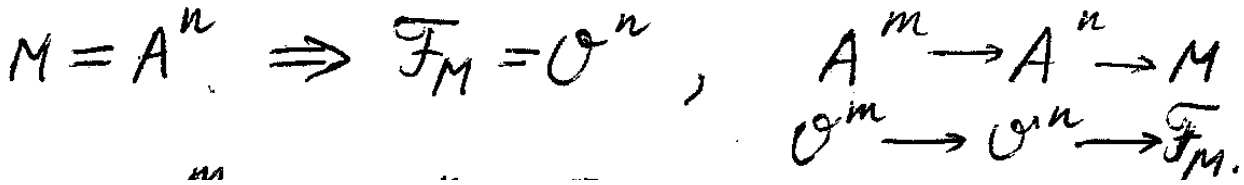
$\forall$ quasicoherent sheaf $\mathcal{F}$ on an aff. scheme has the form $\mathcal{F}_M$, where M is an A-module.



Manuscript notation. Spec A is inserted above "aff. scheme".

Original source details: GELFAND-P000303

Problem (sol.)



Original source details: GELFAND-P000303

Diagram and typed description.

$$M = A^n \Rightarrow \mathcal{F}_M = \mathcal{O}^n, \qquad \begin{array}{c} A^m \longrightarrow A^n \longrightarrow M \\ \mathcal{O}^m \longrightarrow \mathcal{O}^n \longrightarrow \mathcal{F}_M. \end{array}$$

In the reverse direction:

$$\mathcal{O}^m \longrightarrow \mathcal{O}^n \longrightarrow \mathcal{F} \quad \text{loc.}$$

[overwritten: and?] then sect. $A^m \longrightarrow A^n \longrightarrow M$.



Uncertain reading. A densely inked, looped sign precedes "then". Provisionally "and".

Original source details: GELFAND-P000303

Proof of the theorem.

$$\mathcal{F} \qquad M = \Gamma(X, \mathcal{F}).$$

$$\forall f \in A \qquad \Gamma(U_f, \mathcal{F}) = M_f.$$

$\forall$ aff. scheme is quasicompact (no separation). From $\forall$ open cover select a finite one.

$$\mathcal{F}|_{U_{f_i}} = \mathcal{F}_{M_i}, \qquad M_i \quad A_{f_i}\text{-module.}$$

There is a nat. map

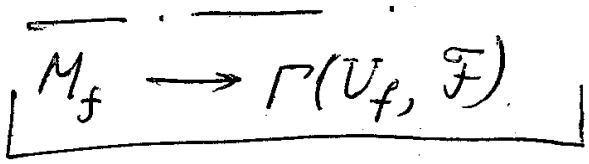


Diagram and typed description.

$M_f \longrightarrow \Gamma(U_f, \mathcal{F})$.

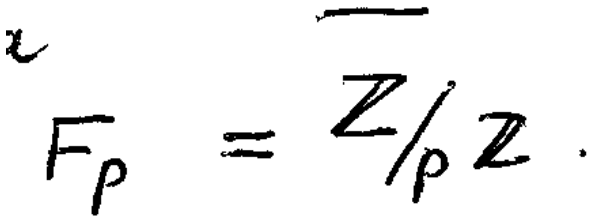
Original source details: GELFAND-P000304

$1/f \in \Gamma(U_f, \mathcal{O}_X)$, and $\mathcal{F}$ — a sheaf of modules.

Thompson as expounded by Gelfand I. M.

(G, M) — a pair.

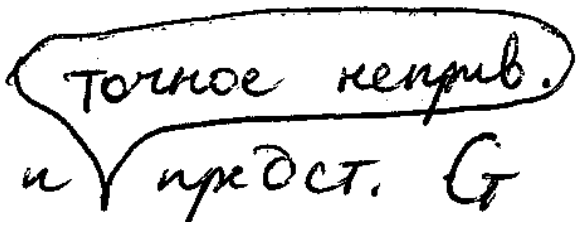
G — a group, $F_p = \overline{Z}/pZ$.



Uncertain reading. A separate horizontal stroke stands above the first Z.

Original source details: GELFAND-P000304

M — a module over F_p, and faithful irred. a repr. of G in M.



Manuscript notation. Insertion “faithful irred.” before “repr.”.

Original source details: GELFAND-P000304

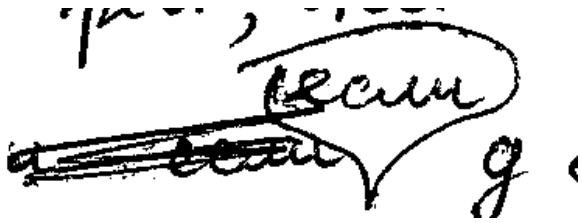
Require that [cancelled sign: M?] G be generated by a collection $\langle Q \rangle$, of such elems.:



Uncertain reading. Before the clean G, a tall sign with several nearly vertical strokes is cancelled, possibly M; its letter is undetermined.

Original source details: GELFAND-P000304

~~if~~ if $g \in Q$, then $M(g-1)^2 = 0$.

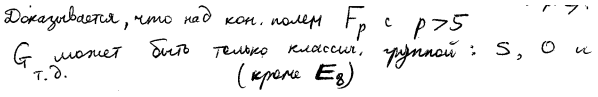


Original source details: GELFAND-P000304

Manuscript notation. "If" is inserted after the cancelled entry.

(Example: a matrix group generated by elems. with Jordan blocks of order 2).

It is proved that over the finite field F_p with $p > 5$ G can only be a class. group: S, O etc. (except E_8).



Original source details: GELFAND-P000304

Uncertain reading. The manuscript reads "$p > 5$", "class. group", and separately "(except E_8)".

Characteristics 2 and 3 remain.

Kirillov from Thompson's account.

Another finite simp. group (Conway) —

— belonging to no series.

The automorphism gr.

[margin: в]

[Cancelled: $R^{24} \supset Z^{24}$?; then an illegible cancelled word.]



Original source details: GELFAND-P000304

Uncertain reading. The cancelled line has R^{24}, an inclusion sign and Z^{24} / L^{24}, then an illegible word; the candidate Z^{24} is shown. Later loops prevent separating the original letter.

$$\mathbb{R}^{24} \supset \underset{\text{lattice}}{L}.$$

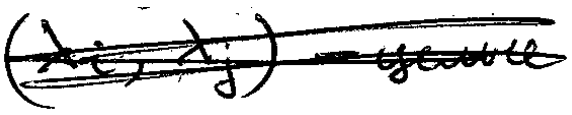


Original source details: GELFAND-P000304

Manuscript notation. The note below L reads "lattice".

1) unimodular — the fund. domain has unit volume.

2) ~~(λ_i, λ_j) — integers.~~



Original source details: GELFAND-P000304

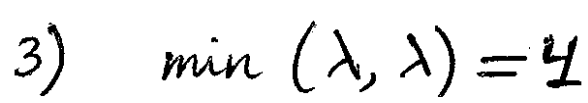
Uncertain reading. The first subscript provisionally reads i, with a loop below the cancellation. The second reads j.

(λ, μ) is always an integer, and (λ, λ) is even.

GELFAND-P000305

[source number: 71]

3) $\min(\lambda, \lambda) = 4$.



Original source details: GELFAND-P000305

Manuscript notation. The manuscript has $\min(\lambda, \lambda) = 4$ without a specified range.

[Cancelled: 4)?]

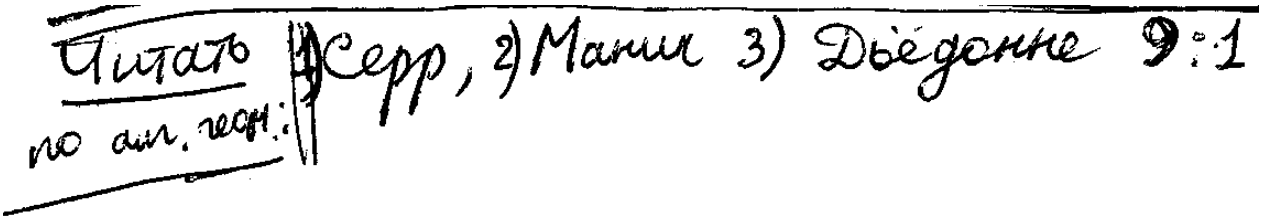


Original source details: GELFAND-P000305

Uncertain reading. Before "Such a lattice", a short sign with a parenthesis is cancelled, possibly 4); its original and cancelling strokes cannot be separated.

Such a lattice is uniquely det. up to autom.

The signature is div. by 8 for pos. def. forms.



Original source details: GELFAND-P000305

Diagram and typed description.

To read on alg. geom. ‖ 1) Serre, 2) Manin 3) Dieudonné 9:1



Manuscript notation. After "Dieudonné" the entry reads 9:1.

Original source details: GELFAND-P000305

19/X. I. M. Gelfand.

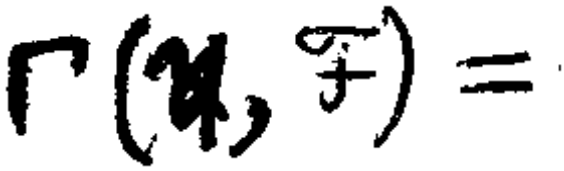
Let a sheaf of modules be given on X.

A, $X = \operatorname{Spec} A$, M — a module, $\mathcal{F}_M$ — the corresp. sheaf (sheafification of the module M).

Example of a non-quasi-coh. sheaf over Spec A:

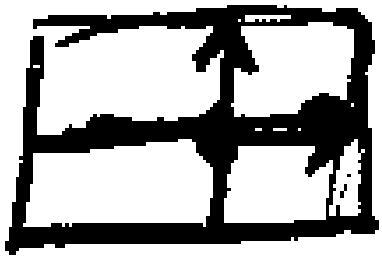
stalk A at the point x and 0 at the other points.

$$\Gamma(U,\mathcal{F}) = \begin{cases} A & \text{if } x \in U, \\ 0 & \text{if } x \notin U. \end{cases}$$



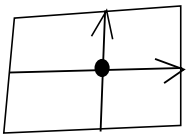
Original source details: GELFAND-P000305

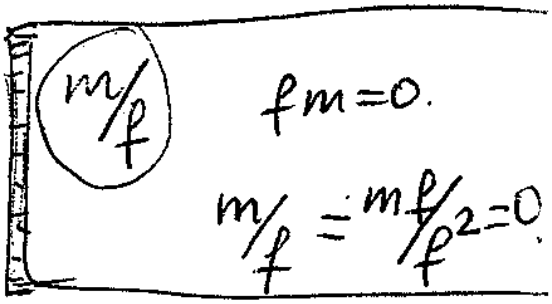
Uncertain reading. The first argument of Γ has heavily overlapping strokes. Provisionally U, from the upper strokes and the entry at right.



Original source details: GELFAND-P000305

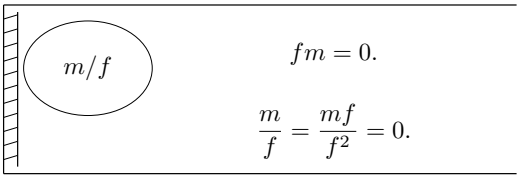
Uncertain reading. The frame contains crossing strokes with heads at the top and right and a thickened junction. The purpose of the drawing is undetermined.





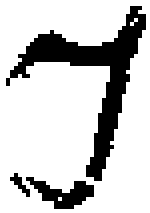
Original source details: GELFAND-P000305

Diagram and typed description.



Example (I. Bernstein).

$\mathbb{C}^n$, J — an ideal in the polynomial ring; to it there corresp. a subset, $\underset{M}{\frac{P}{Q}}$ is def. at every point of the subset.



Original source details: GELFAND-P000305

Manuscript notation. The ideal letter hooks to the left at the foot. It reads J.

Then to it there corresp. a polynomial R such that

$$\frac{P}{Q} = R \pmod{J} \quad \text{(on it)}$$



Manuscript notation. The parenthesis reads "mod J".

Original source details: GELFAND-P000305

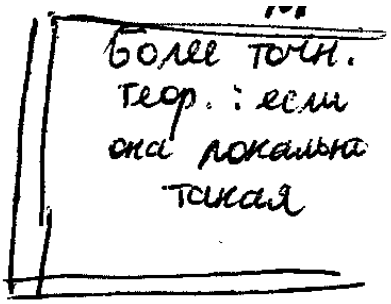


Diagram and typed description.

More prec. thm.: if it is locally such.

Original source details: GELFAND-P000305

$A = \mathbb{C}[x_1, \ldots, x_n]$, $\mathcal{O}_X$ — the struct. sheaf on $X = \operatorname{Spec} A$.

$$\operatorname{Spec} A \supset \mathbb{C}^n.$$

(not only on $\mathbb{C}^n$).

$$\underset{\text{subsheaf}}{J \subset \mathcal{O}}$$

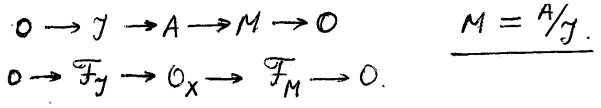
J — the ideal $(f_1, \ldots, f_n)$, a module over A.
$\mathcal{F}_J$ — the corresp. sheaf, $\mathcal{F}_J \subset \mathcal{O}_X$.

$$(J \subset A \Rightarrow \mathcal{F}_J \subset \mathcal{F}_A).$$

GELFAND-P000306

$$0 \longrightarrow J \longrightarrow A \longrightarrow M \longrightarrow 0, \qquad M = A/J.$$

$$0 \longrightarrow \mathcal{F}_J \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{F}_M \longrightarrow 0.$$



Manuscript notation. The letter J recurs in the exact sequences and in A/J.

Original source details: GELFAND-P000306

[Cancelled: $\mathcal{F}_M$ — rat. f-n?]



Original source details: GELFAND-P000306

Uncertain reading. After the cancelled $\mathcal{F}_M$, “рац. ф-” is legible; the final letter admits а / я / и. The candidate “ф-а?” is shown as “rat. f-n?”; the later clean “rat. f-n” does not resolve the cancelled entry.

Sections of the sheaf $\mathcal{F}_M$:

1) constr. from M

2) from the exact seq.

Let Y — the set of zeros of the ideal J.

P/Q — a rat. f-n, $Q \neq 0$ on Y.

$$P/Q \in \Gamma(X, \mathcal{F}_M) \qquad \text{from exactness of the sequence.}$$



Original source details: GELFAND-P000306

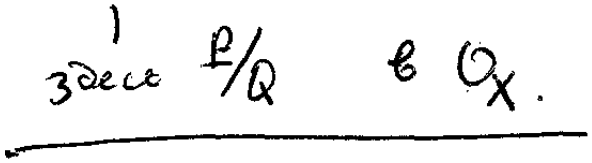
Uncertain reading. The sign before Γ is heavily overwritten. Provisionally $\in$.

Consider X_Q [a small sign is cancelled] and $X \setminus Y$.

Uncertain reading. Between X_Q and “and”, a small sign with crossing strokes is cancelled; its original letter is undetermined.



Original source details: GELFAND-P000306



Original source details: GELFAND-P000306

Diagram and typed description.

here P/Q in $\mathcal{O}_X$.

Over $X \setminus Y$ consider the section $\equiv 0$.

$$0 \in \Gamma(X \setminus Y, \mathcal{F}_M).$$

$$i\left(\frac{P}{Q}\right) \in \Gamma(X_Q, \mathcal{F}_M).$$

Construct a section of the sheaf on X.

On the intersection they agree, since sections of $\mathcal{F}_M$ on $X \setminus Y$ are ident. 0, since $\mathcal{F}_M$ is supported on Y.

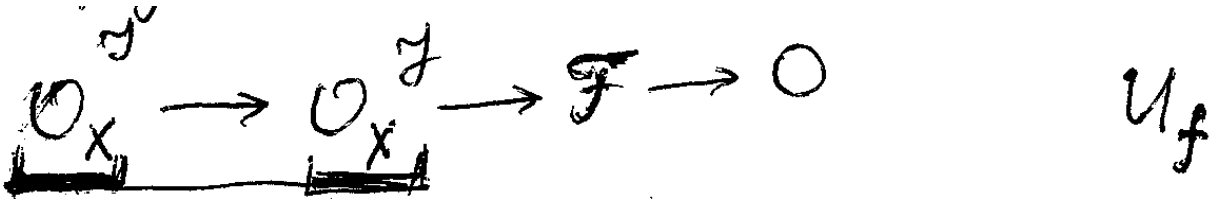
But! a section of this sheaf is an elem. of M — a polynomial.

[A sign is cancelled at the left: $\mathcal{O}$?]



Uncertain reading. Before the first $\mathcal{O}_X$ is a dark corrected sign with a long loop and a rising right stroke, possibly $\mathcal{O}$; the upper-right stroke cannot be separated from the cancellation.

Original source details: GELFAND-P000306

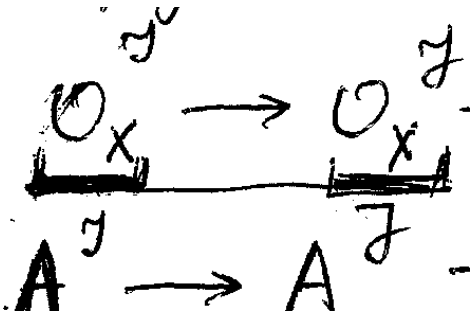


Original source details: GELFAND-P000306

Diagram and typed description.

$$\mathcal{O}_X^J \longrightarrow \mathcal{O}_X^{\mathcal{J}} \longrightarrow \mathcal{F} \longrightarrow 0 \qquad U_f$$

$$A^J \longrightarrow A^{\mathcal{J}} \longrightarrow M \longrightarrow 0.$$



Uncertain reading. In both rows the first superscript resembles a hooked J. The second has a lower loop and a cross-stroke. Provisionally J and $\mathcal{J}$.

Original source details: GELFAND-P000306

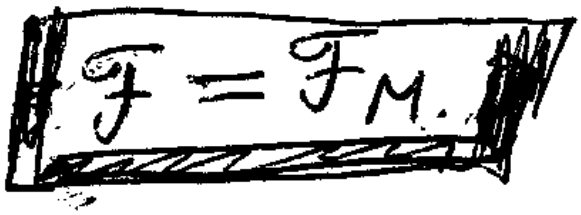
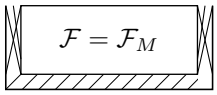


Diagram and typed description.



Original source details: GELFAND-P000306

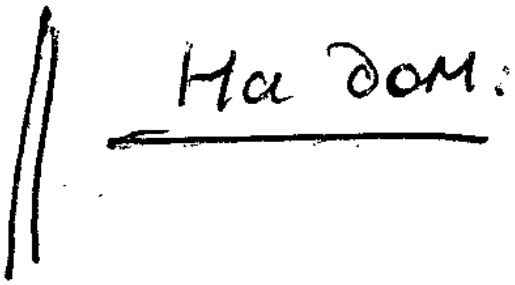


Diagram and typed description.

Homework:

Original source details: GELFAND-P000306

GELFAND-P000307

[source number: 72]

72

Ya. Sinai. Noncommut. probability theory.

H — finite-dimensional Hilbert. $\dim H = 2$ already nontriv.

$\ldots, H_{-1}, H_0, H_1, H_2, \ldots$ with a fixed isom. onto H (basis).

I — a finite subset,

$$H_I = \bigotimes_{i \in I} \text{[a sign is cancelled]}\, H_i. \qquad \dim H_I = 2^k, \quad \text{where } k = |I|.$$



Uncertain reading. Between $\otimes$ and the clean H_i is a dark cancelled sign with several vertical strokes; an earlier H is possible, but the original entry is undetermined.

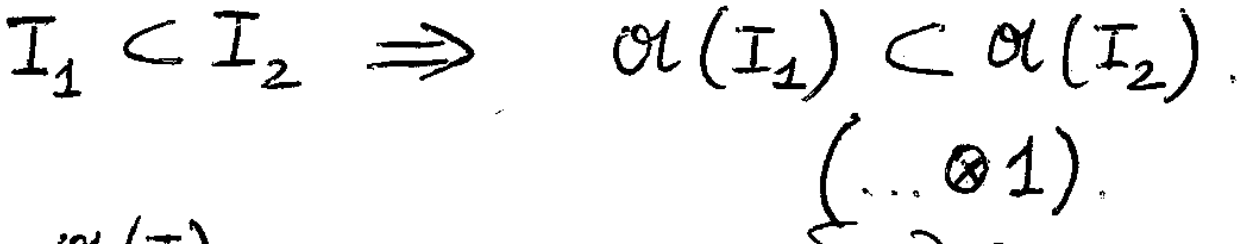
Original source details: GELFAND-P000307

$\alpha(I)$ — the algebra of matrices over H_I. [Cancelled: “Всем” (to all)?]



Uncertain reading. After the full stop following H_I, a short looped word is cancelled, resembling “Всем” / “всем” (to all); crossing strokes prevent the complete word from being established.

Original source details: GELFAND-P000307



Original source details: GELFAND-P000307

Diagram and typed description.

$I_1 \subset I_2 \implies \alpha(I_1) \subset \alpha(I_2).$

$(\ldots \otimes 1).$

Elems. of $\alpha(I)$ — local observables.

Close $\bigcup \alpha(I)$ in the uniform metric.

$$[-n, n].$$

$$\alpha([-n, n]) \subset \alpha([-n-1, n+1]).$$

closed in norm (completion) (uniform).

Obtain the algebra $\underline{\alpha}$ quasilocal observables. (with invol.).

Probability th. — Ω — a space (comp. metric), on $C(\Omega)$ positive functionals (probability distrs.).

α plays the role $C(\Omega)$.

A state φ on α [a sign is cancelled: φ?] — a positive normalized functional (probability distr.).



Uncertain reading. After α, a looped sign with a long lower stroke and several oblique strokes is cancelled; φ is possible, but original and cancelling lines cannot be separated.

Original source details: GELFAND-P000307

$$\varphi(a) \geq 0 \quad \text{when } a \geq 0; \qquad \varphi(e) = 1.$$

$\varphi(a^*)$ — real when $a^* = a$.



Uncertain reading. After a, dark crossing strokes overlap the closing parenthesis. The asterisk is provisional.

Original source details: GELFAND-P000307

1. State of measure type.
Consider the values of the functional on $\alpha(I)$.

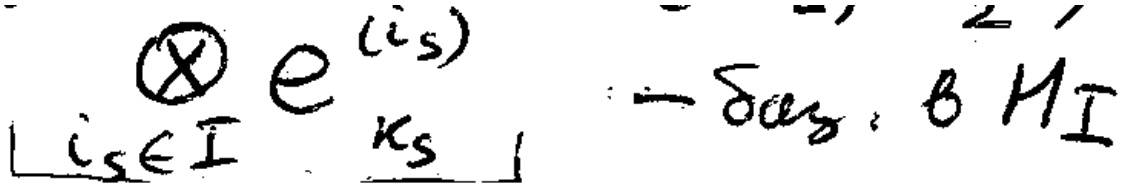
$$\varphi(a) = \operatorname{tr}(\rho_I \cdot a), \qquad \text{where } \rho_I \geq 0, \quad \operatorname{tr} \rho_I = 1.$$

State of measure type — the eigenv. of ρ_I agree in the following sense:
in H_i there exists a basis $(e_1^{(i)}, e_2^{(i)})$ — a basis in H_i (depends on φ).



Uncertain reading. The first subscript has heavy upper and lower strokes and a separate dot. Provisionally i.

Original source details: GELFAND-P000307



Original source details: GELFAND-P000307

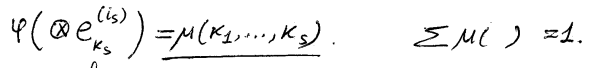
Diagram and typed description.

$\bigotimes_{i_s \in I} e_{k_s}^{(i_s)}$ — a basis in H_I

They — eigen.
for ρ_I

GELFAND-P000308

$$\varphi\left(\bigotimes e_{k_s}^{(i_s)}\right) = \mu(k_1, \ldots, k_s). \qquad \sum \mu(\,) = 1.$$

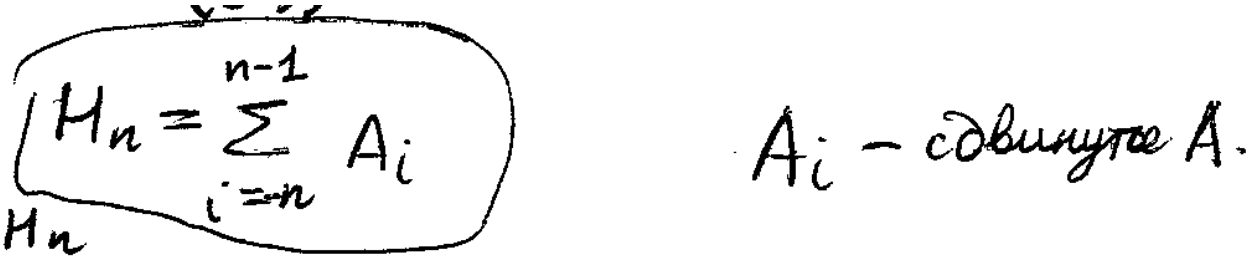


Original source details: GELFAND-P000308

Manuscript notation. The argument of φ is written as a tensor product. After $\sum \mu$ the parentheses are empty.

probability distr.
A measure on the space of seqs. taking the values 1 and 2.
2. States generated by a Hamiltonian.
$A \in \alpha((0,1)), \qquad H((I))$.

On $[-n, n]$ consider



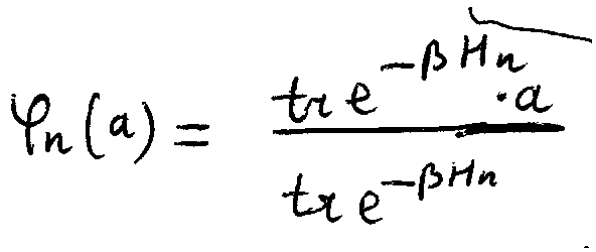
Original source details: GELFAND-P000308

Diagram and typed description.

$$H_n = \sum_{i=-n}^{n-1} A_i$$

A_i — shifted A.

$$\varphi_n(a) = \frac{\operatorname{tr} e^{-\beta H_n} \cdot a}{\operatorname{tr} e^{-\beta H_n}}.$$



Manuscript notation. The numerator has a dot before a. The expression after tr has no enclosing parentheses.

Original source details: GELFAND-P000308

$$\underline{\varphi(a) = \lim_{n\to\infty} \varphi_n(a) \text{ exists}} \quad \forall A. !$$

This is called a state, generated by the Hamiltonian A.
(from the quantum Gibbs distr.)
A noncommut. analogue of Markov chains.
Let φ — a state. A weakly closed ring of opers. in Hilbert space is constructed.
In the alg.

$$\langle a_1, a_2 \rangle = \varphi(a_2^* \cdot a_1).$$

Quotient by the ideal . . .
H_φ — Hilbert space.
[Crossed out: $h \in H_\varphi$]



Uncertain reading. Beneath the cancellation one can read $h \in H_\varphi$. The subscript extends right from the long stroke.

Original source details: GELFAND-P000308

$$h \in H_\varphi \implies L_a = ah.$$

Weak closure — M_φ.
when is it a factor?
Mainly of type 3.

GELFAND-P000309

[source number: 73]

26/X. Kazhdan.

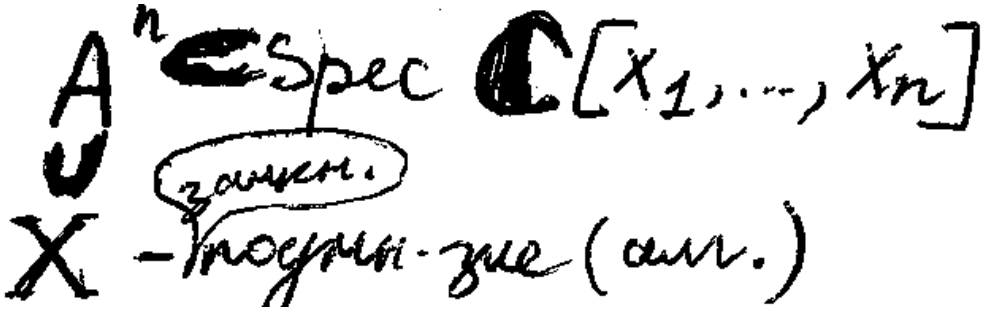
73

Triviality of the cohom. of a quasicoher. sheaf on an aff. scheme.



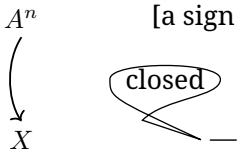
Uncertain reading. Between A^n and Spec is a heavily overwritten rounded sign open to the right. No middle stroke is discernible.

Original source details: GELFAND-P000309



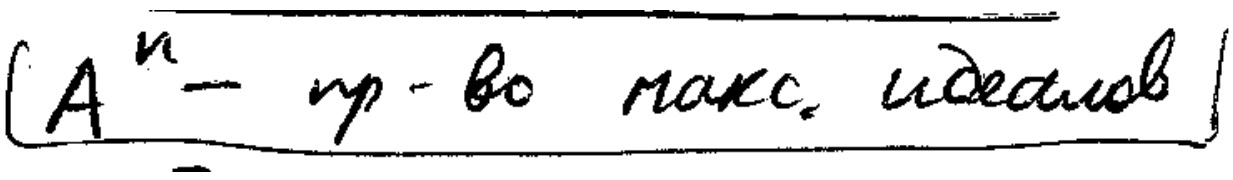
Original source details: GELFAND-P000309

Diagram and typed description.



[a sign is cancelled] Spec $\mathbb{C}[x_1, \ldots, x_n]$

— a subset (alg.).



Original source details: GELFAND-P000309

Diagram and typed description.

A^n — the space of max. ideals

$Q|_X \neq 0$, then P/Q — a func. on X, $\forall P$.
Then $\exists R$ — a polynomial, $(\forall P \exists R)$.

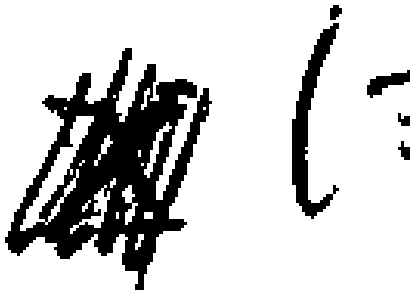
$$\left.\frac{P}{Q}\right|_X = R|_X.$$

Consider the ideal corresp. to X (0 on X),

$F_1, \ldots, F_k$ — a basis of this ideal.

Then $(F_1, \ldots, F_k, Q) = k[x_1, \ldots, x_n]$.

[A sign is cancelled: H / R?]



Original source details: GELFAND-P000309

Uncertain reading. Before the opening parenthesis is a cancelled short sign with several upright and oblique strokes; H / R are possible, but its original letter is undetermined. No closing parenthesis is visible in the following phrase.

(this ideal is not contained in any maximal

$\exists$ polynomials $L_0, \ldots, L_k$.

$$L_0 Q + L_1 F_1 + \ldots + L_k F_k = P;$$



Original source details: GELFAND-P000309

Manuscript notation. The right-hand side reads P, with an overlapping loop.

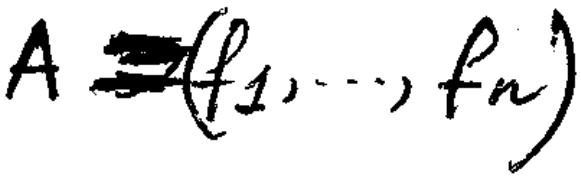
Hence one can take $R = L_0$.

(Hilbert's thm. — description of max. ideals)

This proof is analogous to the proof of triviality of cohomology

(this is triviality of 1-dimensional cohomology).

$X = \operatorname{Spec} A$, A [a sign is cancelled: $=$?] $(f_1, \ldots, f_n)$.



Original source details: GELFAND-P000309

Uncertain reading. Between A and $(f_1, \ldots, f_n)$, horizontal strokes are heavily overwritten, with an additional oblique line below. A cancelled $=$ is shown as a candidate; the exact earlier sign is undetermined.

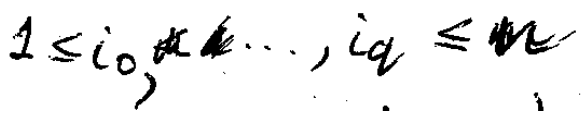
Consider $X_i = X_{f_i}$.

$$A = (f_1, \ldots, f_n) \Longleftrightarrow X = \bigcup X_i.$$

Let M — A-module, $\mathcal{F}_M$ — the corresp. sheaf.

Čech complex:

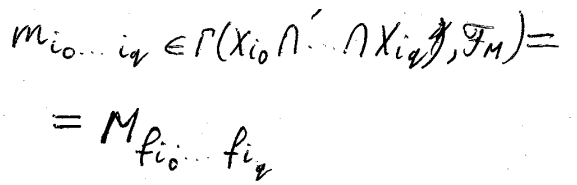
q-chain: $\forall$ index collection $1 \leq i_0$, [two overwritten signs: <?] $\ldots, i_q \leq n$ is given



Original source details: GELFAND-P000309

Uncertain reading. After i_0 are a comma and two dense marks. The final letter after $\leq$ is overwritten, provisionally n.

$$m_{i_0 \ldots i_q} \in \Gamma(X_{i_0} \cap \ldots \cap X_{i_q}), \mathcal{F}_M) = \\ = M_{f_{i_0} \ldots f_{i_q}}.$$

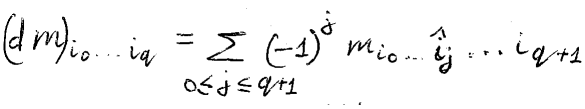


Original source details: GELFAND-P000309

Uncertain reading. After X_{i_q} is a crossed curved stroke resembling a parenthesis, followed by a comma and $\mathcal{F}_M) =$. The equals sign is repeated on the next line.

GELFAND-P000310

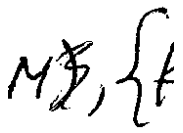
$$(dm)_{i_0 \ldots i_q} = \sum_{0 \leq j \leq q+1} (-1)^j m_{i_0 \ldots \widehat{i_j} \ldots i_{q+1}}.$$



Original source details: GELFAND-P000310

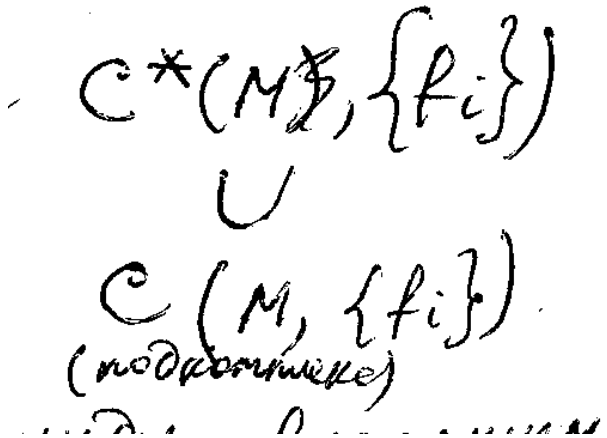
Manuscript notation. The left side has i_q, the right side i_{q+1}; the upper bound of summation is $q+1$.

$d : C^q \longrightarrow C^{q+1}$.
$d^2 = 0$ and the cochains form the Čech complex



Original source details: GELFAND-P000310

Uncertain reading. After M, before the comma, a tall looped letter is cancelled. The formula provisionally shows the candidates F/φ; the original sign is undetermined.



Original source details: GELFAND-P000310

Diagram and typed description.

$$C^*(M\,[F/\varphi?], \{f_i\})$$

$$\cup$$

antisymm. cochains — $C(M, \{f_i\})$

(subspace)

Prop.: the cohomologies coincide (the isom. is induced by inclusion).

For convenience — sometimes C^*, sometimes C.

Theorem: $H^q(\{X_i\}; \mathcal{F}_M) = 0$ for $q \neq 0$.

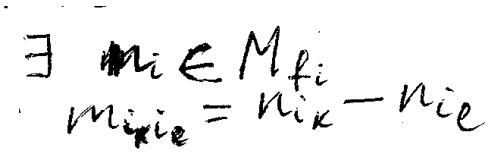
Proof: Let $m_{i_0 \dots i_q} \in M_{f_{i_0} \dots f_{i_q}}$ — cycles.

There we had $A = (f_1, \dots, f_m, Q)$,

$$m_{i_0 i_1} \in M_{f_{i_0} f_{i_1}},$$
$$(dm)_{i_0 i_1 i_2} = m_{i_0 i_1} - m_{i_0 i_2} + m_{i_1 i_2} = 0.$$

Need to prove:

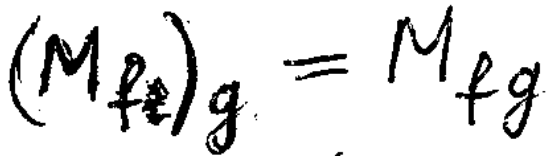
$$\exists n_i \in M_{f_i}$$
$$m_{i_k i_\ell} = n_{i_k} - n_{i_\ell}.$$



Original source details: GELFAND-P000310

Uncertain reading. The letter before i after $\exists$ is heavily overwritten; the candidate n_i is provisional. The small subscripts on m are also overwritten; the reading $i_k i_\ell$ is not final.

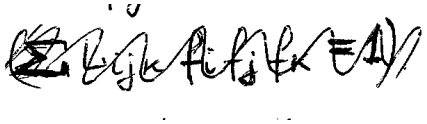
Ex.: $(M_f)_g = M_{fg}$.



Original source details: GELFAND-P000310

Manuscript notation. Inside the first parenthesis, a mark after f is heavily overwritten. The original reading is undetermined.

[Cancelled; provisional reading: $(\sum L_{ijk} f_i f_j f_k = 1)$.]



Original source details: GELFAND-P000310

Uncertain reading. Oblique strokes cancel the sum, coefficient, and product. The candidate L_{ijk}, the small subscripts on f, and the final $= 1$ remain uncertain; the right parenthesis is visible.

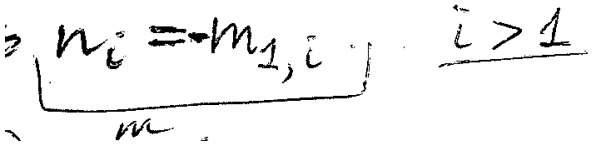
$m_{1,i} = n_1 - n_i$.

Put $n_1 = 0$, [a sign is cancelled: m?]



Original source details: GELFAND-P000310

Uncertain reading. Before $n_i = -m_{1,i}$, a short sign is densely crossed out; m is possible, but the original reading is undetermined.



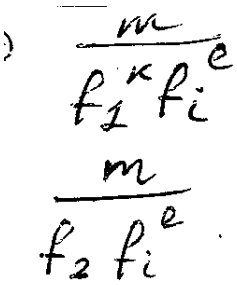
Original source details: GELFAND-P000310

Diagram and typed description.

$n_i = -m_{1,i}$ $\quad i > 1$

False, since $m_{1,i}$ has the form

$$\frac{m}{f_1^k f_i^\ell},$$
$$\frac{m}{f_2 f_i^\ell},$$
$$\vdots$$



Original source details: GELFAND-P000310

Manuscript notation. In both fractions f_i has a looped exponent, provisionally ℓ. No exponent is visible on f_2.

and combine, using the fact that this is a cover.

GELFAND-P000311

[source number: 74]

74

[Cancelled: “Consider.”?] For simplicity, let $q = 1$.



Original source details: GELFAND-P000311

Uncertain reading. After the initial R-shaped letter, several horizontal strokes obscure the middle letters. The candidate “Consider.” remains provisional.

Given $m_{i_0 i_1} - m_{i_0 i_2} + m_{i_1 i_2} = 0$. $\quad 1 \le i_k \le n$.



Original source details: GELFAND-P000311

Uncertain reading. Before $m_{i_0 i_2}$ is a dark overwritten sign with two horizontal regions and a short lower extension. A minus is shown as a candidate; the earlier sign is unrecovered.

Need to find $n_i \in M_{f_i}$: $\quad m_{i_0 i_1} = n_{i_0} - n_{i_1}$.

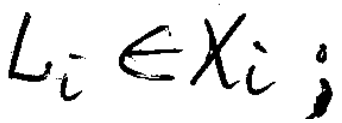
$m_{i_0 i_1} =$ [a cancelled fraction: $\bar{m}/f_{i_0}$?] $\dfrac{\bar{m}_{i_0 i_1}}{f_{i_0}^N f_{i_1}^N}$, $\quad \bar{m} \in M$.



Original source details: GELFAND-P000311

Uncertain reading. Oblique strokes cross the old numerator and fraction bar. The reading $\bar{m}/f_{i_0}$ is provisional; a second denominator factor in the old expression is unrecovered.

Find $L_i \in X_i$;



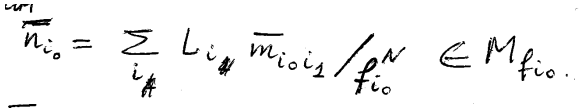
Original source details: GELFAND-P000311

Manuscript notation. After the membership sign the manuscript reads X_i.

$$\sum L_i f_i^N = 1.$$

Define

$$\bar{n}_{i_0} = \sum_{i_1} L_{i_1} \bar{m}_{i_0 i_1} / f_{i_0}^N \in M_{f_{i_0}}.$$

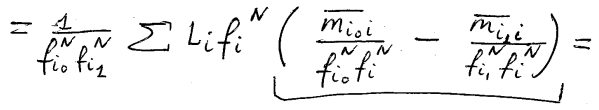


Original source details: GELFAND-P000311

Uncertain reading. The digits at the sum and on L are heavily overwritten, provisionally i_1. The second subscript on $\bar{m}$ is distinctly i_1.

$$\bar{n}_{i_0} - \bar{n}_{i_1} = \sum_i \left(\frac{L_i}{f_{i_0}^N} \bar{m}_{i_0 i} - \frac{L_i}{f_{i_1}^N} \bar{m}_{i_1 i} \right) =$$

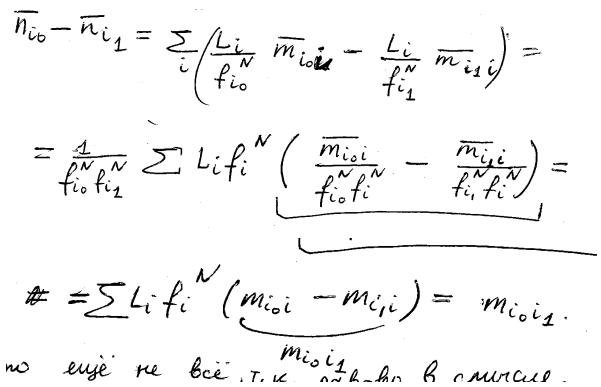
$$= \frac{1}{f^N_{i_0} f^N_{i_1}} \sum L_i f_i^N \left(\frac{\bar m_{i_0 i}}{f^N_{i_0} f^N_i} - \frac{\bar m_{i_1 i}}{f^N_{i_1} f^N_i} \right) =$$



Original source details: GELFAND-P000311

Manuscript notation. The factor $\frac{1}{f^N_{i_0} f^N_{i_1}}$ is written before the sum. The denominators of both fractions above the angular bracket are also written out.

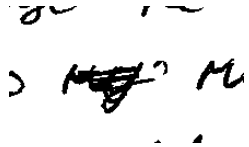
$$\text{[a sign is cancelled]} \quad = \sum L_i f_i^N \underbrace{(m_{i_0 i} - m_{i_1 i})}_{m_{i_0 i_1}} = m_{i_0 i_1}.$$



Original source details: GELFAND-P000311

Uncertain reading. In the first line, the last subscript on the first $\bar m$ is overwritten; the candidate i is provisional. A small expression before the last line's equality is cancelled. The difference of fractions has a long unlabelled angular bracket below it; the last difference has a curve labelled $m_{i_0 i_1}$.

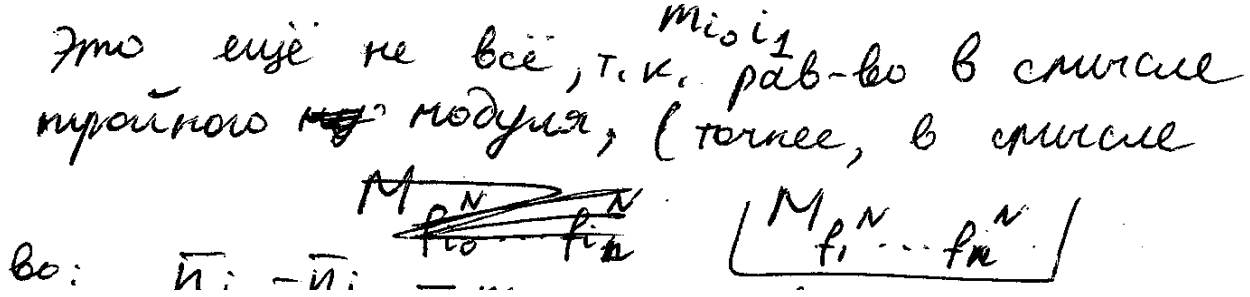
This is not all, since the eq. is in the sense of a twisted



Original source details: GELFAND-P000311

Undeciphered passage. Before "module", a short word is cancelled. An initial n-shaped stroke and a small final curve are visible; the text is unrecovered.

module, (more precisely, in the sense of [Cancelled; candidate: $M_{f^N_{i_0} \dots f^N_{i_n}}$?]



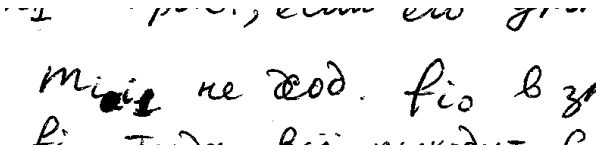
Original source details: GELFAND-P000311

Uncertain reading. Several long strokes cancel the old localization expression. The final subscript in the candidate $f^N_{i_n}$ remains uncertain; i_1 is also possible. At the right, the new expression $M_{f^N_1 \dots f^N_n}$ has lower and side strokes. No closing parenthesis is visible after it.

$M_{f^N_1 \dots f^N_n}$

The eq.: $\bar n_{i_0} - \bar n_{i_1} = m_{i_0 i_1}$ holds if it is multiplied by some power.

Bernstein's proof: let all $m_{i_0 i_1}$ not contain f_{i_0} in the denom.



Uncertain reading. The two small subscripts on m are heavily overwritten. The candidate $i_0 i_1$ is provisional.

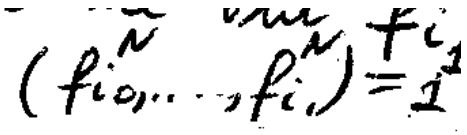
Original source details: GELFAND-P000311

and eqs. to 0 do not require multiplication by f_{i_0}. Then everything goes through.
Consequence:

$f_{i_0}^N m_{i_0 i_1}$ — a boundary. The same for $f_{i_1}^N$ and so on.

Cohom. is a module.

Since $(f_{i_0}^N, \dots, f_{i_n}^N) = 1$, everything is all right.

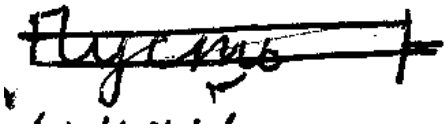


Manuscript notation. The end of the last subscript is unclear, provisionally i_n.

Original source details: GELFAND-P000311

GELFAND-P000312

[Cancelled: "Let" and a sign ┼]



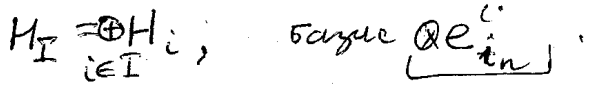
Uncertain reading. After "Let" is a tall vertical stroke with a short cross-stroke. The candidate φ is undetermined: no closed loop is visible.

Original source details: GELFAND-P000312

Sinai. H_i — 2-dimensional, $-\infty < i < +\infty$.
$I =$ any finite set.

$$H_I = \bigoplus_{i \in I} H_i;$$

basis $\bigotimes e^i_{t_n}$



Uncertain reading. The manuscript has a circled plus before H_i and a circled cross before e. The subscript on e is provisionally t_n; its first letter is heavily overwritten.

Original source details: GELFAND-P000312

$\alpha(I)$ — all matr. in H_I.
In $\alpha(I)$ introduce the operator norm in Hilb. space.

$$I \subset I_1 \Rightarrow \alpha(I) \subset \alpha(I_1). \qquad (\otimes 1).$$

$$\alpha(I_1) = \alpha(I) \otimes \alpha(I_1 \setminus I).$$

This gives a normed ring (completion) α.
(algebra of quasiloc. observables).
New operators added:

$\varphi(a)$ — a functional (lin.), $\varphi(a) \geq 0$ $\forall$ positive matr. a.
In $\alpha(I)$ we have $\varphi(a) = \mathrm{Sp}(a\rho) = \mathrm{Tr}(a\rho)$.

$$\underline{\underline{\varphi(a) \geq 0}} \Longleftrightarrow \rho \geq 0.$$

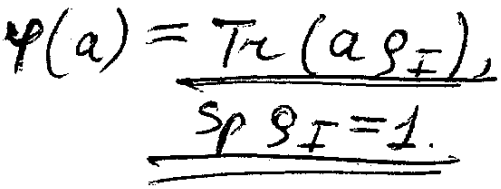
$\underline{\underline{\varphi(e) = 1}}$ — a noncommut. analogue of a probability distrib.; a state in QFT.

$\alpha(I)$ — a ring of matrices [cancelled: “on”?], $\varphi(a) = \underline{\underline{\mathrm{Tr}(a\rho_I)}}$, $\underline{\underline{\mathrm{Sp}\,\rho_I = 1}}$.



Original source details: GELFAND-P000312

Uncertain reading. After “matrices”, a short word is densely crossed out. The candidate “on” remains provisional.



Original source details: GELFAND-P000312

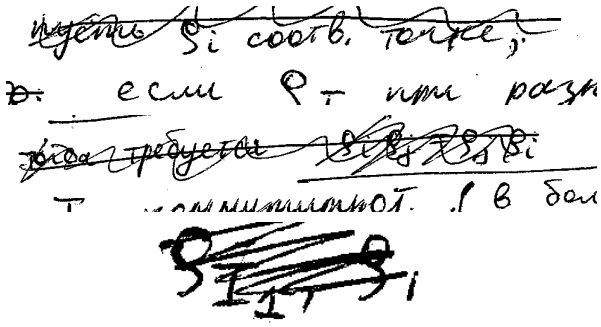
Manuscript notation. There are two strokes below $\mathrm{Tr}(a\rho_I)$ and two below $\mathrm{Sp}\,\rho_I = 1$.

On ρ_I the compatibility cond.:

$$\mathrm{Tr}_{I_1 - I}(\rho_{I_1}) = \rho_I.$$

$$\alpha(I_1) = \alpha(I) \otimes \alpha(I_1 - I).$$

$\underline{\varphi$ — a state of measure type.$}$ $\longrightarrow$ [Cancelled: let ρ_i corresp. to a point;]
[Cancelled: then require $\underline{\rho_i\rho_j = \rho_j\rho_i}$?]
if ρ_I for different I commute (in a larger space).
[At the left, two signs are cancelled; candidates $\rho_{I_1}\rho_i$?]
(one-point ρ_i and ρ_j always commute.)



Original source details: GELFAND-P000312

Uncertain reading. In the upper cancelled line, “let” is followed by a rho-shaped letter with a small subscript, then “corresp. to a point;”. In the next line, a condition with four rho-shaped letters is crossed out; the subscripts i, j are provisional. Below at the left are two separately cancelled letters whose small subscripts also remain uncertain.

GELFAND-P000313

[source number: 75]

75.

ρ_i — a matrix in H_i,

Let e_1^i, e_2^i — a basis of eigenvectors of ρ_i,

$$e_{\alpha_1}^{i_1} \otimes e_{\alpha_2}^{i_2} \otimes \ldots \otimes e_{\alpha_n}^{i_n}$$

Then $\rho(I)$ is diagonal in it, since it commutes with all ρ_i (obvious if the spectrum is simple) ($\exists$ such a basis).

$2^{|I|}$ eigenvectors. $\mu(\alpha_1, \ldots, \alpha_n)$ — an eigenvalue.

It characterizes a distrib. of measure type. Its compatibility cond.:

$$\sum_{\alpha_n} \mu(\alpha_1, \ldots, \alpha_n) = \mu(\alpha_1, \ldots, \alpha_{n-1})$$

and so on.

Consider an infinite seq. of 0 and 1.

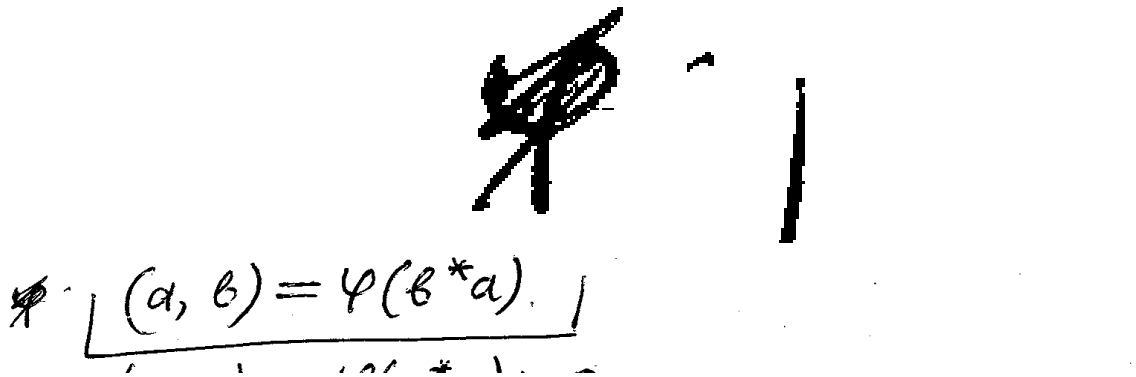
$$\ldots, \varepsilon_{-n}, \varepsilon_{-n+1}, \ldots, \varepsilon_0, \ldots$$

a group (addition mod 2). A measure is defined on cylinder sets $\alpha_1, \ldots, \alpha_n$ — fixed $\mu(\alpha_1, \ldots, \alpha_n)$.

The condition is compatibility of these measures.

$\prod \mathbb{Z}_2^i$. One can take Haar measure.

[A sign is cancelled; candidate φ?]



Original source details: GELFAND-P000313

Uncertain reading. Before (a, b), oblique strokes cross a looped letter; a lower vertical extension is visible. The candidate φ remains provisional. The uncancelled formula has a bracket with a straight lower stroke and short sides.

$(a, b) = \varphi(b^* a).$

$$(a, a) = \varphi(a^* a) \geq 0.$$

Now quotient and complete — a Hilb. space.

$xa = X_a$ — a lin. operator — continuous in this inner product.



Original source details: GELFAND-P000313

Manuscript notation. The manuscript has xa on the left and X_a on the right.

Our ring consists of lin. operators.

(see the supplement to the book by Gelfand, Shilov, Raikov)

(Naimark et al.)

The ring generated by the state φ.

Factors of type 3. — they are important — these are shifts.



Original source details: GELFAND-P000313

Manuscript notation. The manuscript has the numeral 3 and the abbreviation «сдвиг.».

GELFAND-P000314

2/XI I. M. Gelfand.

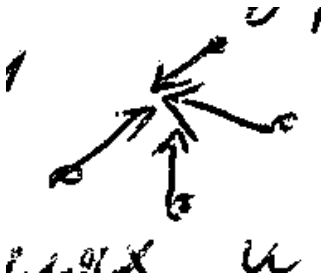
Results of the speaker and V. Ponomaryov (not the one from the Faculty of Mechanics and Mathematics).

P — a finite-dimensional space, $E_1, \dots, E_4$ — 4 subspaces (the order matters).

$$S = \{P; E_1, E_2, E_3, E_4\}.$$

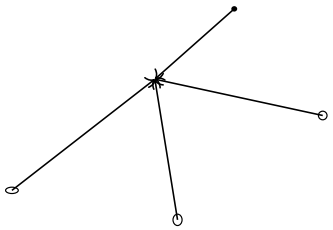
This is a category (morphisms map into).

Equivalent to studying the category of diagrams



Original source details: GELFAND-P000314

Diagram and typed description.



in the category of finite-dimensional spaces.

The methods give information about modules and submodules.

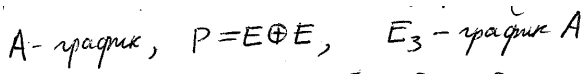
$S \sim S'$, if $\exists$ a lin. isom.

Direct sum is naturally introduced.

The permutation group acts.

Example. E; $A : E \to E$. Graph $\{(x, Ax)\} \subset E \oplus E$.

A — graph, $P = E \oplus E$, E_3 — the graph of A.



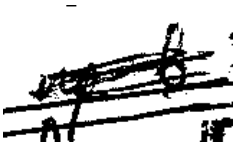
Original source details: GELFAND-P000314

Manuscript notation. This line reads "A — graph" and later "E_3 — the graph of A".

To describe the system $E \oplus E$ one must specify

$E_1 = E \oplus 0$, $E_2 = 0 \oplus E$ and the diagonal $E_4 = \{(x, x)\}$.

Pairs of maps



Original source details: GELFAND-P000314

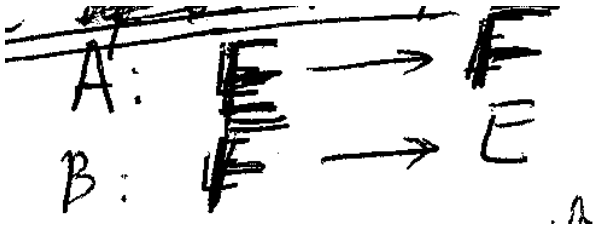
Undeciphered passage. After “Pairs”, several letters before the final tall looped sign are crossed out. Their text is unrecovered.



Original source details: GELFAND-P000314

Uncertain reading. Horizontal and oblique strokes cross the final tall sign; the candidates φ / b remain unresolved.

$$A : E \longrightarrow F,$$
$$B : F \longrightarrow E.$$



Original source details: GELFAND-P000314

Uncertain reading. The starting letter of each arrow is overwritten, as is the ending letter of the first. The reading shown is $A : E \to F$, $B : F \to E$.

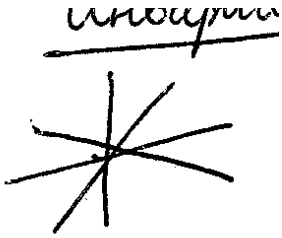
(problem are also described.

Kronecker’s problem — in the same direction.

Relation — any graph.

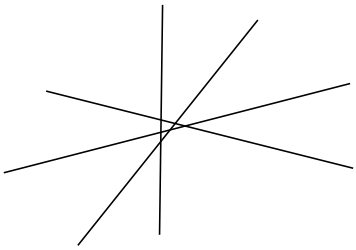
Invariants:

4 lines.



Original source details: GELFAND-P000314

Diagram and typed description.



Invariant — a number λ (one).

2 are coordinate axes.

The 3rd is diagonal.

The 4th is a graph — $y = \lambda x$.

Change the numbering: $\lambda \to 1 - \lambda$, λ^{-1}, $\lambda/(\lambda - 1)$.

Result (the proof is long).

GELFAND-P000315

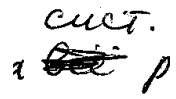
[source number: 76]



Undeciphered passage. Before “Let”, a short expression with several dense strokes and a large rounded final loop is cancelled. Its text is unrecovered.

Original source details: GELFAND-P000315

Let there be given P, E_1, E_2, E_3, E_4, then [cancelled: “syst.”?] [inserted above: syst.] decomposes as a direct sum of the following formations

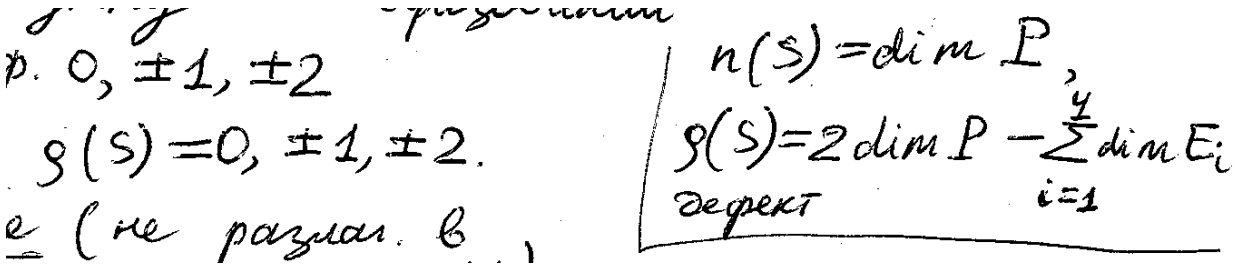


Uncertain reading. “syst.” is clearly written above the line. The earlier short expression below is densely crossed out; the candidate “syst.” remains provisional.

Original source details: GELFAND-P000315

systems with defect $0, \pm 1, \pm 2$.

$\rho(S) = 0, \pm 1, \pm 2.$



Original source details: GELFAND-P000315

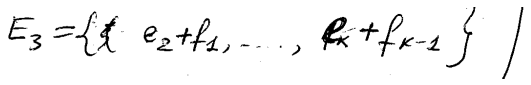
Uncertain reading. The defect symbol in the line and frame has a loop and a lower curved tail; the candidates ρ / g remain unresolved. The text provisionally uses ρ. A left and a lower rule frame the definition and “defect”, with no top or right rule.

$$n(S) = \dim P,$$

$$\rho(S) = 2\dim P - \sum_{i=1}^{4} \dim E_i$$

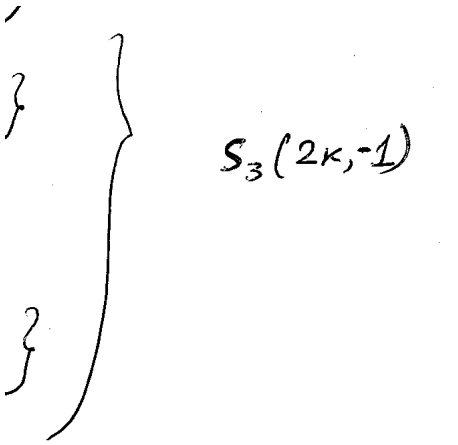
defect

Irreducible (not decomposable as a direct sum)



Original source details: GELFAND-P000315

Uncertain reading. After the opening brace of E_3, a narrow tall sign is cancelled; the candidate ξ is shown with a question mark. The last e_k is heavily overwritten; the earlier strokes are unrecovered.



Original source details: GELFAND-P000315

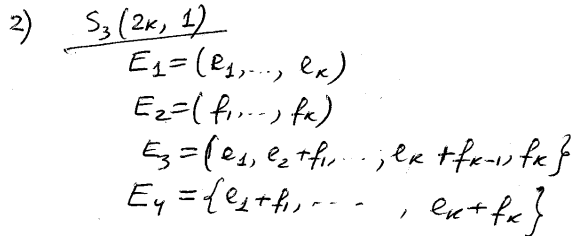
Diagram and typed description.

$$\left.\begin{array}{c} 1) \quad \text{for} \quad n(S) = 2k. \\ \text{Basis in } P \quad \{e_1 \dots e_k \; f_1 \dots f_k\}. \\ E_1 = \{(e_1, \dots, e_k)\}. \\ E_2 = \{(f_1, \dots, f_k)\}. \\ E_3 = \{[\xi?]\; e_2 + f_1, \dots, e_k + f_{k-1}\}. \\ E_4 = \{e_1 + f_1, \dots, e_k + f_k\}. \end{array}\right\} \; S_3(2k, -\underline{1})$$

---

2) $S_3(2k, 1)$.

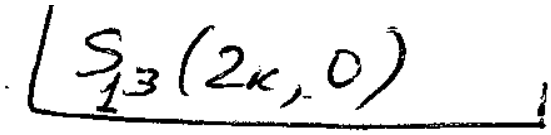
$$E_1 = (e_1, \dots, e_k).$$
$$E_2 = (f_1, \dots, f_k).$$
$$E_3 = (e_1, e_2 + f_1, \dots, e_k + f_{k-1}, f_k\}.$$
$$E_4 = \{e_1 + f_1, \dots, e_k + f_k\}.$$



Original source details: GELFAND-P000315

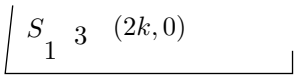
Manuscript notation. The series is labelled $S_3(2k, 1)$. The list for E_3 opens with a parenthesis and closes with a brace.

3) $\begin{matrix} E_1 \\ E_2 \end{matrix}$ — the same.

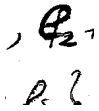


Original source details: GELFAND-P000315

Uncertain reading. The right-hand addition has 1 below S, with 3 farther right and higher, followed by $(2k, 0)$. Their positions are represented separately; the compact reading S_{13} is provisional. The frame has a left, a lower, and a short right stroke.



$$E_3 = \{e_1, e_2 + f_1, \ldots, e_k + f_{k-1}\}.$$



Original source details: GELFAND-P000315

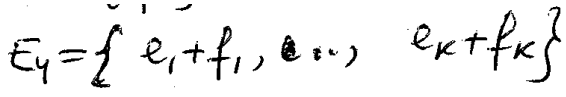
Uncertain reading. In the second position, the final e_2 is written over an earlier letter. A separate residual upright stroke resembles f, but the old letter is unrecovered.

$$E_4 = \{e_i + f_i\}.$$

GELFAND-P000316

$S(2k; 0, \lambda)$.

$$E_1 = \{e_i\},$$
$$E_2 = \{f_i\},$$
$$E_4 = \{e_1 + f_1, \ldots, e_k + f_k\},$$
$$E_3 = \{e_1 + \lambda f_1, e_2 + f_1 + \lambda f_2, \ldots, e_k + f_{k-1} + \lambda f_k\}.$$



Original source details: GELFAND-P000316

Manuscript notation. In the E_4 line, a dense e-shaped stroke precedes the ellipsis. The earlier writing is undetermined.

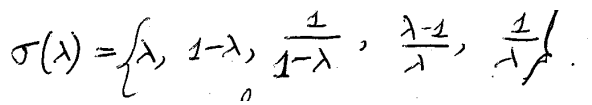
$\lambda \neq 0, \quad \lambda \neq 1.$

σ — a permutation. $\begin{matrix} 1 \longrightarrow i \\ 3 \longrightarrow k \end{matrix}$

$$\sigma(S_{13}) \xrightarrow{\text{def}} \sigma(S_{ik}).$$

$\sigma\lambda$. λ transforms as for an anharm. quadruple of points.

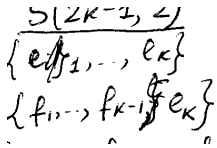
$$\sigma(\lambda) = \left\{\lambda, 1-\lambda, \frac{1}{1-\lambda}, \frac{\lambda-1}{\lambda}, \frac{1}{\lambda}\right\}.$$



Original source details: GELFAND-P000316

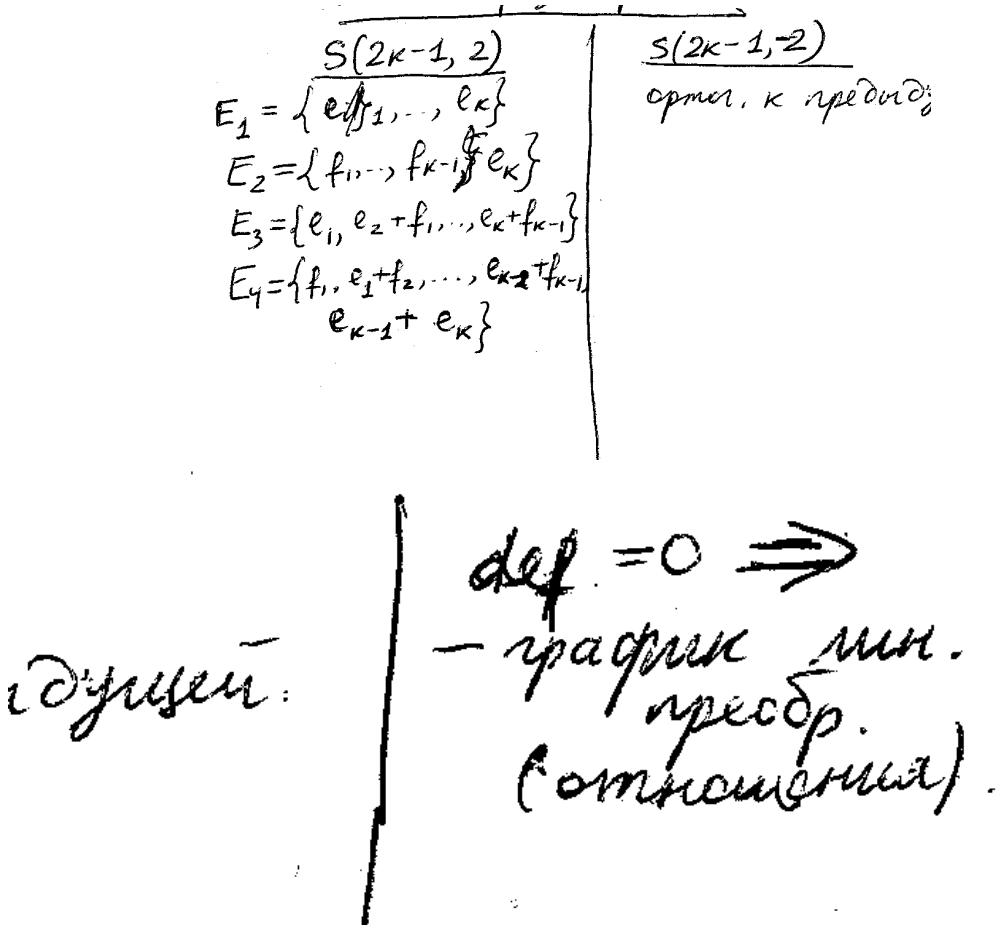
Manuscript notation. Five expressions are written inside the braces.

Odd-dimensional spaces.



Original source details: GELFAND-P000316

Uncertain reading. In the first line, near the initial subscript 1, a tall f-shaped sign is cancelled; [f?] marks it provisionally. In the second line, the old brace after f_{k-1} is heavily overwritten, with a loop above it. An added e_k stands at the right before the final brace; no separate comma is visible before it.



Original source details: GELFAND-P000316

Diagram and typed description.

$S(2k-1,2)$. $E_1 = \{e_1$ [f?]$, \dots, e_k\}$, $E_2 = \{f_1, \dots, f_{k-1}$ [overwritten] $e_k\}$, $E_3 = \{e_1, e_2 + f_1, \dots, e_k + f_{k-1}\}$, $E_4 = \{f_1, e_1 + f_2, \dots, e_{k-2} + f_{k-1}, e_{k-1} + e_k\}$.	$S(2k-1,-2)$, orthog. to the preceding one.	def. $= 0 \Rightarrow$ — the graph of a lin. transformation (relation).

def 0 does not occur in odd dimension.

def(-1) occurs for relations.

general position — without a parameter, and degenerate — with a parameter.

GELFAND-P000317

[source number: 77]

These trajectories.

Example. $S(5,2) \oplus S(1,2)$.

$S(5,2)$ is stable (i.e. deform. gives an isom. syst.).

(Indecomp. systs. without parameters are always stable).

$S(5,2)$ and $S(1,2)$ are stable, but $S(5,2) \oplus S(1,2)$ is unstable and under deform. passes into the stable syst.

$$S(3,2) \oplus S(3,2).$$

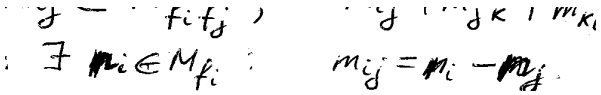
It is interesting to find a simple proof.

(The decomp. into irreducibles is unique if they are distinct, and the number is always the same)

Kazhdan.

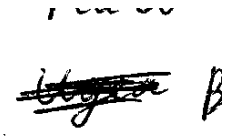
Let $m_{ij} \in M_{f_i f_j}$, $\quad m_{ij} + m_{jk} + m_{ki} = 0 \in M_{f_i f_j f_k}$.

Need to prove: $\exists m_i \in M_{f_i}$, $\quad m_{ij} = m_i - m_j$.



Original source details: GELFAND-P000317

Uncertain reading. The letters before i after $\exists$ and before i, j in the difference are heavily reinforced. The visible final reading is rendered as m_i, m_j; earlier strokes cannot be separated.



Original source details: GELFAND-P000317

Uncertain reading. Before “Choose”, a short word with loops and a lower tail is crossed out. The candidate “Будем” (we shall) remains uncertain; the complete earlier reading is undetermined.

Choose m_1, the others are determined uniquely and without contradiction.

Suppose the denominators of all elems. do not contain f_1 and all rels. hold without multiplication by f_1.

Then one must put $n_1 = 0$.

Let $m_{1i} \in M_{f_i}$,

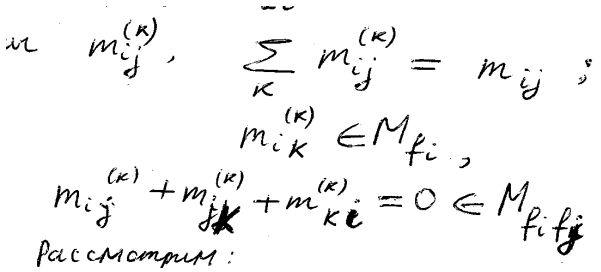
$$0 = m_{1i} + m_{ij} + m_{j1} \in M_{f_i f_j}.$$

Put $n_i = m_{1i}$.

Lemma.

$$\exists \text{ cocycles } m_{ij}^{(k)}, \qquad \sum_k m_{ij}^{(k)} = m_{ij};$$

$$m_{ik}^{(k)} \in M_{f_i}, \qquad m_{ij}^{(k)} + m_{jk}^{(k)} + m_{ki}^{(k)} = 0 \in M_{f_i f_j}.$$



Original source details: GELFAND-P000317

Uncertain reading. The k in the second subscript jk, the final i in the third term and the last j in $M_{f_i f_j}$ are reinforced. Their visible final forms are rendered provisionally; earlier strokes are unrecovered.

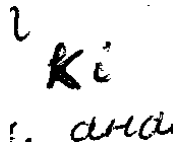
Proof. Consider:

$$m_{ij}^{(N,k)} = f_k^N m_{ij} \quad \text{— cocycles, for sufficiently large } N$$

GELFAND-P000318

satisfy all conditions.

$$\exists N: \quad m_{ki}^{(N,k)} \in M_{f_i}.$$



Original source details: GELFAND-P000318

Uncertain reading. The first letter of the lower index is heavily reinforced, with an upper loop; the visible final form is rendered as ki. Earlier strokes within k cannot be separated.

The 2nd property is proved similarly.

$$m = 0 \text{ in } M_{f_i f_j f_k} \quad \Longleftrightarrow \quad f_k^N m = 0 \text{ in } M_{f_i f_j}.$$

Choose N common to all k.

Let $\sum g_k f_k^N = 1$. Put

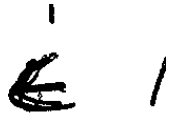
$$m_{ij}^{(k)} = g_k m_{ij}^{(N,k)};$$

they satisfy all conditions of the lemma. ■

Now it suffices to consider $m_{ij}^{(k)}$.

$$n_i^{(k)} = m_{ik}^{(k)} \in M_{f_i} \text{ by assumption.}$$

$$m_{ij}^{(k)} = n_i^{(k)} - n_j^{(k)} \in M_{f_i f_j},$$



Original source details: GELFAND-P000318

Uncertain reading. The sign before the final M has three horizontal endings and several reinforced oblique and curved strokes. The visible form is rendered as $\in$; earlier lines are unrecovered.

from the last condition of the lemma.

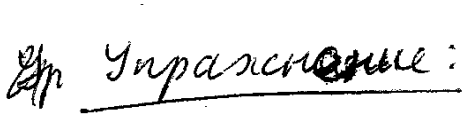
Example. The sheaf O^*, $X = \operatorname{Spec} A$.

$$\Gamma(U, O^*) = \{\gamma \in \Gamma(U, O) \mid \gamma \notin p,\ \forall p \in U\}.$$

(a subsheaf of the structure sheaf)
Consider the simplest alg. variety —
— a torus with a point removed.

$$P = y^2 - x^3 + 1; \qquad A = \mathbb{C}[x, y]/p.$$

[Reinforced before the heading: “Упр” (Ex.).]
Exercise: $H^1(\operatorname{Spec} A, O^*) \neq 0$.



Original source details: GELFAND-P000318

Uncertain reading. Before the full heading “Упражнение” (Exercise), the short “Упр” is superimposed on earlier loops; the earlier letters are undetermined. A dark reinforced loop also occurs within the full heading; the final word is rendered as “Упражнение”, without reconstructing earlier strokes.

$S = A^n - \{0\}$.

$$H^i(S, O) = 0 \text{ for } i \neq n-1 \text{ and } i \neq 0.$$

For $i = n - 1$ it is infinite-dimensional and easily described. $H^0 = A$.



Original source details: GELFAND-P000318

Uncertain reading. At the start of the final sentence, the letter before $= n - 1$ is heavily reinforced and has a separate dot above it. It is rendered as i; earlier strokes within the lower loop are undetermined.

GELFAND-P000319

[source number: 78]

78

Theor. of de Linya.
X — a complete variety,
Y — a subvariety, locally



Diagram and typed description.



Original source details: GELFAND-P000319

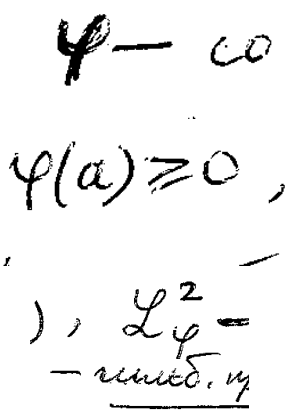
a direct prod. of transverse planes.

$f \quad = 0$ on the variety (codim. 1).
(may)

$$\left\{\frac{df}{f}\right\} \quad \text{— generating diff. forms.}$$

Resolution of sing. reduces everything to this case.
The ring does not depend on the method of resolution.

Sinai.
α — the alg. of quasilocal observ., φ — a state (functional),
$L^2_\varphi \longleftarrow$
— Hilb. space

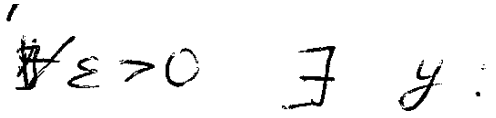


Uncertain reading. The reinforced initial letter has a right loop and a descending vertical stroke. Its visible form is rendered as φ; the positivity condition and the subscript on L^2 provide adjoining forms. Earlier strokes in the reinforced letter cannot be separated.

Original source details: GELFAND-P000319

$$\varphi(a) \geq 0, \text{ if } a \geq 0, \qquad \varphi(1) = 1.$$

M — a weakly closed ring of opers.
Powers's theorem. Let $\forall x \in \alpha, \quad \forall \varepsilon > 0, \quad \exists y$:



Uncertain reading. Before $\varepsilon > 0$, a $\forall$ is written over dark earlier strokes; a separate $\exists y$ is clearly written at the right. The earlier sign beneath the reinforced $\forall$ is unrecovered.

Original source details: GELFAND-P000319

$$|\varphi(xy) - \varphi(x)\varphi(y)| \leq \varepsilon \|y\|,$$

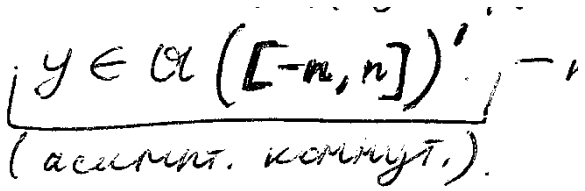


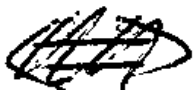
Diagram and typed description.

$\underbrace{y \in \alpha([-n,n])'}$
(asympt. commut.)

Original source details: GELFAND-P000319

— the commutant (commuting matrices).
Then M — a factor.

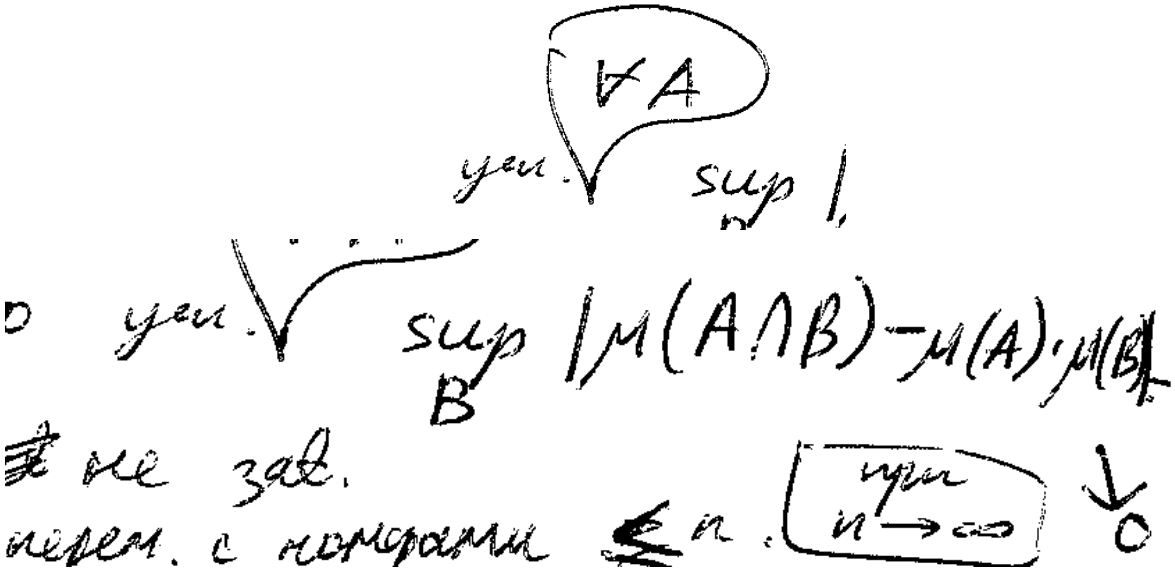
φ — of measure type



Original source details: GELFAND-P000319

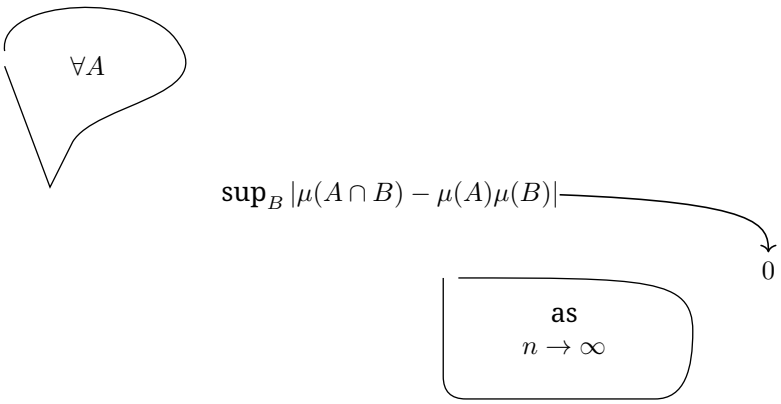
Undeciphered passage. Between “of measure type” and the dash, an entire local passage is enclosed and crossed out; its verbal or mathematical content is unrecovered.

— this is the cond.

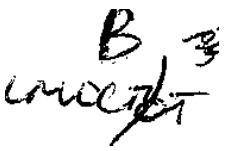


Original source details: GELFAND-P000319

Diagram and typed description.



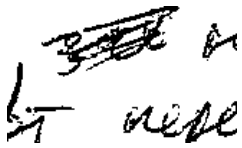
(a cond. of asympt. independence type)



Original source details: GELFAND-P000319

Uncertain reading. The ending of “независимости” (independence) has a dark overlap with a long lower stroke and an upper horizontal line. The upper-right curved stroke is provisionally rendered as a closing parenthesis; the earlier ending is unrecovered.

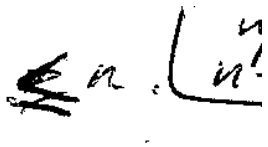
B, [a short word is cancelled; candidate “зав.” (depends)?]



Original source details: GELFAND-P000319

Uncertain reading. After B and before “не зав.” (does not depend), several horizontal and oblique strokes cross a short entry. The candidate “зав.” (depends) remains uncertain.

does not depend on variables numbered $\leq n$.



Original source details: GELFAND-P000319

Uncertain reading. The sign before n is heavily reinforced; an upper angle and two lower strokes are visible. Its final reading is rendered as $\leq$; earlier strokes are undetermined.

(the first n coords. take $\forall$ values)

Products of measures with prob. $\geq c > 0$ satisfy this cond.

State obtained from a Hamiltonian — a factor.

GELFAND-P000320



Original source details: GELFAND-P000320

Uncertain reading. Cancelled earlier date: candidate $2\ldots$/XI; the complete day is undetermined.

$16/\overline{\text{XI}}$. Kazhdan.



Original source details: GELFAND-P000320

Uncertain reading. In the left margin below the cancelled date and to the left of “Consider”, an isolated dense mark has no established verbal reading.

Consider $A^n \setminus \{0\} = X$.

$$H^i(X, O) = ?$$

$$A^n = \text{Spec}\, A[x_1, \ldots, x_n].$$

$\{0\} = (x_1, \ldots, x_n)$ and all prime ideals containing it.

(A — a field [$\leftarrow$ / —?] it is max and, hence, only itself)



Original source details: GELFAND-P000320

Uncertain reading. The short, almost double horizontal mark between “A — a field” and “it is max” has candidates $\leftarrow$ and a dash; the thickened left end does not establish an arrowhead.

Removing a closed subset — an open one remains.

Restr. of a sheaf to any open set is defined.

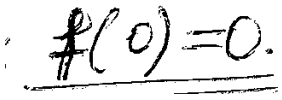
(Open sets in X — those that are open in A^n).

An open set in X — (basis)

$$U_f \cap (U_{x_1} \cup \ldots \cup U_{x_n}).$$

$(\mathbf{C}[x_1, \ldots, x_n])$.

Open sets U_f correspond to those f, for which $f(0) = 0$.



Original source details: GELFAND-P000320

Uncertain reading. The letter before $(0) = 0$ is reinforced and crossed by additional strokes; the final loop is provisionally rendered as f. Earlier strokes cannot be separated.

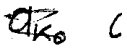
$$X = \bigcup_{i=1}^{n} X_i, \qquad X_i = U_{x_i} = \operatorname{Spec} \mathbf{C}[x_1, \ldots, x_n, x_i^{-1}].$$

$$X_{k_0,\ldots,k_m} = \operatorname{Spec} \mathbf{C}[x_1, \ldots, x_n, x_{k_1}^{-1}, \ldots, x_{k_m}^{-1}].$$

m-dim. chains — sections on $X_{k_0,\ldots,k_m}$.

$$a_{k_0,\ldots,k_m} = \frac{P(x_1, \ldots, x_n)}{(x_{k_0} \ldots x_{k_m})^N};$$

H^0:

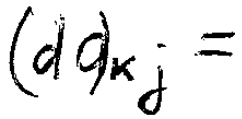


Original source details: GELFAND-P000320

Uncertain reading. After H^0, a cancelled entry resembles a_{k_0}; the lower rounded sign and the strokes beneath the cancellation do not establish the reading conclusively.

$$a_k = \frac{P_k(x)}{x_k^{e_k}}.$$

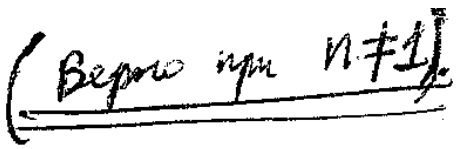
$$(da)_{kj} = \frac{P_k(x)}{x_k^{e_k}} - \frac{P_j(x)}{x_j^{e_j}} = 0 \qquad \forall k, j.$$



Original source details: GELFAND-P000320

Uncertain reading. The second sign in $(d\ldots)_{kj}$ has a loop and a right stroke close to the closing parenthesis; candidates are $(da)_{kj}$, $(dd)_{kj}$, $(dg)_{kj}$. The first candidate is provisionally shown in the formula.

Hence a_k — a polynomial for all k.



Original source details: GELFAND-P000320

Diagram and typed description.

True for $n \neq 1$!

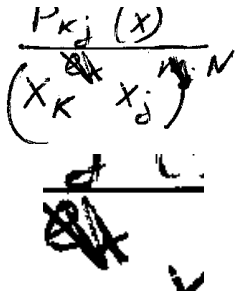
GELFAND-P000321

[source number: 79]

79

H^1:

$$a_{kj} = \frac{P_{kj}(x)}{(x_k x_j)^N}.$$



Uncertain reading. Earlier separate powers above x_k and x_j are cancelled; candidates e_k, e_j and n_k, n_j remain undetermined. The common exponent N at the right is clear.

Original source details: GELFAND-P000321

Boundary:

$$\frac{P_{kj}}{(x_k x_j)^N} + \frac{P_{js}}{(x_j x_s)^N} + \frac{P_{sk}}{(x_s x_k)^N} = 0.$$

$$\boxed{P_{kj}x_s^N + P_{js}x_k^N + P_{sk}x_j^N = 0}.$$

Bound.

$$a_{kj} = \frac{Q_k}{x_k^N} - \frac{Q_j}{x_j^N} \quad \#;$$

Need: $P_{kj} = Q_k x_j^N - Q_j x_k^N$.

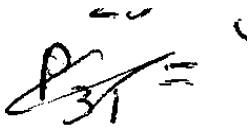
Concretely. there are 3 polyns. in 3 vars.

$$P_{12}x^N + P_{23}y^N + P_{31}z^N = 0.$$

Then $\exists Q_1, Q_2, Q_3$

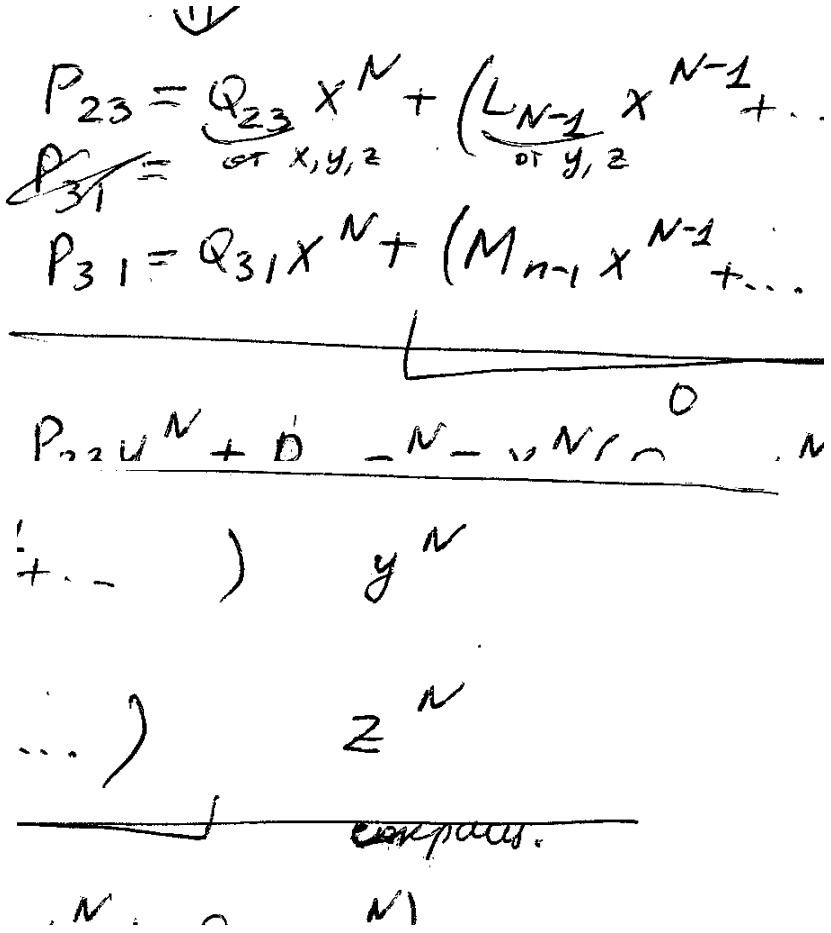
$$P_{12} = Q_1 y^N - Q_2 z^N, \quad \text{and so on.}$$

$\Downarrow$



Uncertain reading. Between the P_{23} and P_{31} rows, a short entry with subscript 31 and an equals sign is cancelled. The candidate $P_{31} =$ is rendered provisionally; long strokes cross its initial loop.

Original source details: GELFAND-P000321



Original source details: GELFAND-P000321

Diagram and typed description.

$$P_{23} = \underbrace{Q_{23}}_{\text{in } x,y,z}\, x^N + \Big(\underbrace{L_{N-1}}_{\text{in } y,z}\, x^{N-1} + \ldots \Big)\, y^N$$

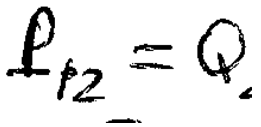
$$\cancel{P_{31} =}$$

$$P_{31} = Q_{31}x^N + (M_{n-1}x^{N-1} + \ldots)\, z^N$$

$$0 \quad \text{(cancel.)}$$

$$P_{23}y^N + P_{31}z^N = x^N(Q_{23}y^N + Q_{31}z^N).$$

$$\Downarrow$$



Original source details: GELFAND-P000321

Uncertain reading. The initial looped sign with a lower crossbar resembles f_{12} or P_{12}. In the following formula, P_{12} is shown provisionally; the letter remains uncertain.

$$P_{12} = Q_{23}y^N + Q_{31}z^N;$$

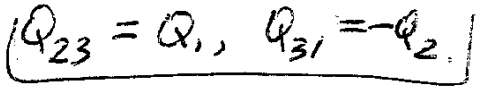


Diagram and typed description.

$Q_{23} = Q_1, \quad Q_{31} = -Q_2.$

Original source details: GELFAND-P000321

Prove $Q_{12} = 0$.

Subtract the corresponding cobound. Cocycles with 2 terms remain.

And it now suffices to consider the problem with $\boxed{P_{12} = 0}$.

GELFAND-P000322

H^1 is also easy in the general case.

H^2 is harder. (for $n > 3$).



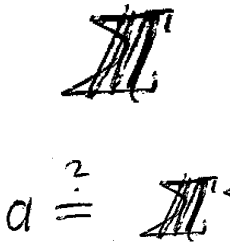
Uncertain reading. An isolated reinforced letter above H^{n-1}: candidate a; additional upper and lower loops prevent recovery of the earlier letter.

Original source details: GELFAND-P000322

H^{n-1}

$$X = \bigcup_{i=1}^{n} X_i, \qquad a = \frac{P(x)}{(x_1 \dots x_n)^N}.$$

$$a \stackrel{?}{=} \text{[cancelled sign]} \frac{\sum P_i(x) x_i^N}{\prod x_i^N}.$$



Uncertain reading. Between $a \stackrel{?}{=}$ and the fraction, a large sign with two horizontal ends is cancelled: candidates $\prod$ and $\sum$ remain undetermined. Above the equality, a separate dot below the upper curved stroke forms a question mark.

Original source details: GELFAND-P000322

$$\frac{1}{(x_1 \dots x_n)^N} \quad \text{— not representable} \quad \forall N.$$

Answer:

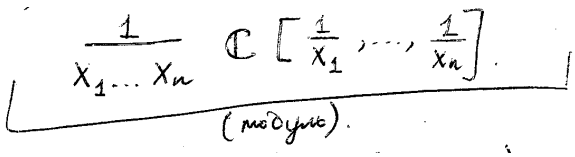


Diagram and typed description.

$\frac{1}{x_1 \dots x_n} \mathbb{C}\left[\frac{1}{x_1}, \dots, \frac{1}{x_n}\right]$

(module)

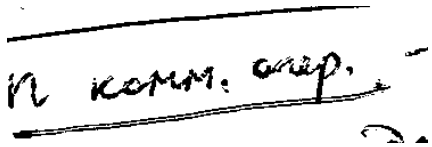
Original source details: GELFAND-P000322

A module over $\mathbb{C}[x_1, \dots, x_n]$ (over H^0).

$$a_{1\dots n} = \frac{P(x)}{(x_1 \dots x_n)^N}.$$

---

n comm. opers. — the resolv. gives nothing (as for one) —
— localiz. in 1 var. is simple, but not here.



Original source details: GELFAND-P000322

Uncertain reading. In the underlined source phrase “n комм. опр.”, the middle strokes of the final abbreviation merge; “опр.” and “опер.” cannot be distinguished confidently. The text provisionally translates the first candidate as “opers.”



Original source details: GELFAND-P000322

Uncertain reading. Above $\Longleftrightarrow$, a small enclosed sign is crossed by oblique strokes and the preceding paragraph’s line. The inner glyph is unreadable; the displayed ? is only a candidate.

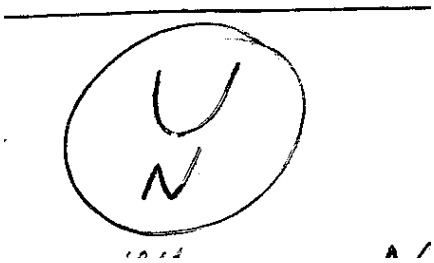
$a_{1\ldots n}$ — bound. $\overset{?}{\Longleftrightarrow}$ [unreadable sign] $P(x) = \sum P_i(x) x_i^N$.



Original source details: GELFAND-P000322

Uncertain reading. After $\Longleftrightarrow$ and before $P(x)$ is a dense oblique sign with a loop and a separate small stroke at the right. Candidates are a,“ф.”, or a cancelled letter; reinforcement cannot be distinguished from cancellation.

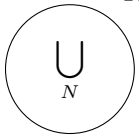
Those with degree $< N$ in every variable remain in the quotient.



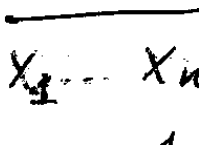
Original source details: GELFAND-P000322

Diagram and typed description.

$$\frac{C + c_i x_i + \ldots + \prod_i x_i^{N-1}}{\prod x_i^N}$$ and



There remains $\dfrac{1}{x_1 \ldots x_n}$ — the highest term. N — arbitrary.



Original source details: GELFAND-P000322

Uncertain reading. In the first lower index of the denominator $x_1 \ldots x_n$, a reinforced 1 has an additional loop and lower strokes; the final index is rendered as 1, while earlier strokes are unrecovered.

hence the answer:

$$\frac{1}{x_1 \dots x_n} \mathrm{C}\left[\frac{1}{x_1}, \dots, \frac{1}{x_n}\right].$$

GELFAND-P000323

[source number: 80]

80

A module over H^0 — mult. by a polyn. and discard integer powers;

$$\left(\text{in } \mathrm{C}\left[\frac{1}{x_1}, \dots, \frac{1}{x_n}\right]\right).$$

$$H^{n-1} = \mathrm{Hom}_{\mathrm{C}[x_1,\dots,x_n]}(\mathrm{C}[x_1, \dots, x_n], \mathrm{C}).$$

C is a $\mathrm{C}[x_1, \dots, x_n]$-module (mult. by $f(0)$).

$$\mathrm{Hom}_{H^0}(H^0, \mathrm{C}).$$

Need to construct a map

$$H^0 \times H^{n-1} \longrightarrow \mathrm{C}.$$

(product and coeff. of $\dfrac{1}{x_1 \dots x_n}$, i.e. the residue)

$(P, R) = \mathrm{Res}\, PR$ — — the constant term in what remains.

(It no longer depends on the covering)

---

i



Uncertain reading. Above the line on duality, an isolated vertical stroke with a dot is provisionally rendered as i; its role and final reading are undetermined.

Original source details: GELFAND-P000323

connection — Serre's duality theorem.

Theorem.

$$A^n = \mathrm{Spec}\, A[x_1, \dots, x_n], \qquad \{0\} = (x_1, \dots, x_n).$$

A — a module over $A[x_1, \dots, x_n]$.

$$H^0(X, O) = A[x_1, \dots, x_n],$$

$$H^{n-1}(X, O) = \text{Hom}_{H^0}(H^0, A),$$

$$H^i(X, O) = 0 \text{ for } i \neq 0, n-1.$$

Next time — project. space.
(the Kashul [Koshul?] complex).



Original source details: GELFAND-P000323

Uncertain reading. The surname after "complex" has readings "Кашуля" (Kashul) and "Кошуля" (Koshul). The joined vowel after К has an a-like shape, but remains uncertain; the first candidate is provisionally shown in the line.

GELFAND-P000324

22/XI. Plan for next semester.

1) Arnold — [cancelled: new?] Griffiths.



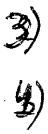
Original source details: GELFAND-P000324

Uncertain reading. Between Arnold and Griffiths, a cancelled word resembles "новый" (new); its dense ending remains undetermined.

($\int$ of rat. funcs. on various cycles — many vars.)

$$\int \frac{P(z_1, \ldots, z_n)}{Q(z_1, \ldots, z_n)} \, dz_1 \wedge \ldots \wedge dz_n$$

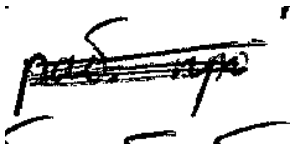
over n-dim. cycles (in $\mathbb{CP}^n$).
(Annals, $N = 1$ or 3 for 1969)
2) and Gabrielov — new Griffiths.
3) S. Gelfand — Brauer theory.
4) Sinai, Factors.



Original source details: GELFAND-P000324

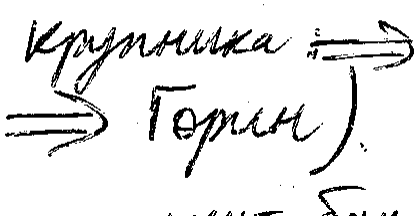
Uncertain reading. The labels of items 3 and 4 contain reinforcement and earlier intersecting strokes; the final digits are shown, while earlier forms are undetermined.

5) A new proof of Grauert's theor. (Knorr).
6) New work of Atiyah (Skew Fredholm opers. and
[cancelled: "work, opr." / "work, oper."?]



Original source details: GELFAND-P000324

Uncertain reading. Below "Skew Fredholm opers. and","раб." (work) and a continuation resembling "опр." or "опер." are cancelled; the continuation's letters remain uncertain.



Original source details: GELFAND-P000324

Diagram and typed description.

parallel ‖ works of Gohberg and Krupnik ⇒

⇒ Gorin

Gorin.

$$\alpha \xrightarrow{\varphi} \mathbf{C} \quad \text{(hom.)};$$

α — a noncommut. Ban. algebra.

$$J = \bigcap_{\varphi} \ker \varphi,$$

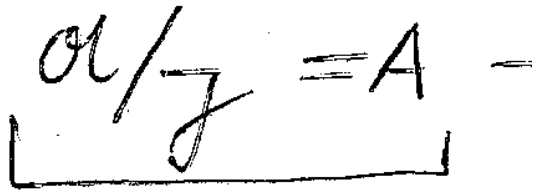


Diagram and typed description.

$\alpha/J = A$ — the algebra of symbols.

Original source details: GELFAND-P000324

Example. α — the alg. of all bounded opers. $\Rightarrow J = \alpha$ (no symbols).

$\mathcal{E}(A)$ — the gr. of invertible elements in A.

$$\alpha \xrightarrow{\psi} A, \qquad \psi^{-1}(\mathcal{E}(A)) \ni T \text{ — f-operators.}$$

$$\psi(T) \in \mathcal{E}(A), \qquad \mathcal{E}_1(A) \text{ — the conn. comp. of the identity,}$$

$$\operatorname{ind} T \in \mathcal{E}(A)/\mathcal{E}_1(A) = H^1(\mathfrak{M}_A, \mathbf{Z}).$$

(this is not a number).

GELFAND-P000325

[source number: 81]

81

On works of Gohberg and Krupnik.

Example. $|\zeta| = 1$, $|\zeta| < 1$. (the $+$ ones).

$$a \longrightarrow T_a, \qquad (T_a f) = (af)_+.$$

Consider $\alpha = \{T_a\}^?$ generated by these operators.



Original source details: GELFAND-P000325

Uncertain reading. At the upper end of the closing brace in $\{T_a\}$, an additional curl and separate dots are visible. The displayed superscript ? is only a candidate; an independent sign and earlier strokes remain undetermined.

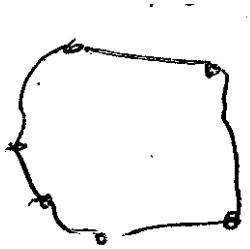
Δ — the unit disc. Opers. on $H_2(\Delta)$ (holom. inside).

$$\sum_{n=0}^{\infty} c_n z^n, \qquad \sum |c_n|^2 < \infty.$$

a — a piecewise cont. func.

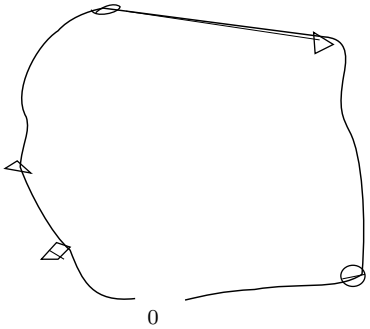
T_a is Fredholm $\iff$ $\widetilde{a}$ does not vanish at 0.

(at discontinuities, join by line segments).
The index — the number of turns.



Original source details: GELFAND-P000325

Diagram and typed description. An irregular contour with an almost straight top, curved sides and a lower gap; the round and triangular strokes are rendered by their visible shapes. Below the gap is the candidate 0. The directions of all small strokes are not established; the gap is not joined.



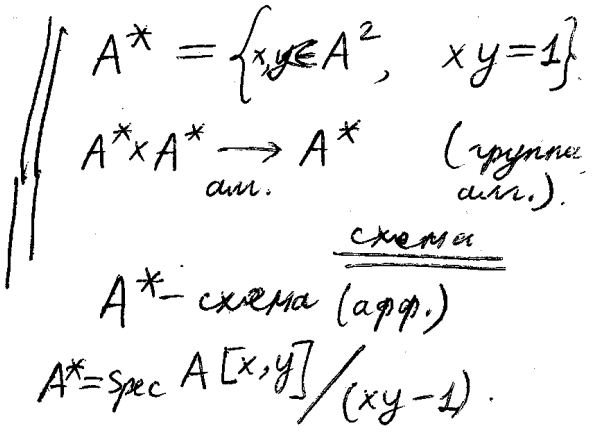
it does not pass through 0.

Arnold. On vector fields in general position.

Kazhdan. $A^{n+1} - \{0\}$.

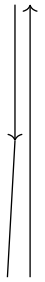
A^* acts.

$$P^n = A^{n+1} - \{0\}/A^*.$$



Original source details: GELFAND-P000325

Diagram and typed description.



$$A^* = \{x, y \in A^2,\ xy = 1\}.$$

$$A^* \times A^* \underset{\text{alg.}}{\longrightarrow} A^* \quad \text{(alg. groups)}$$

schemes

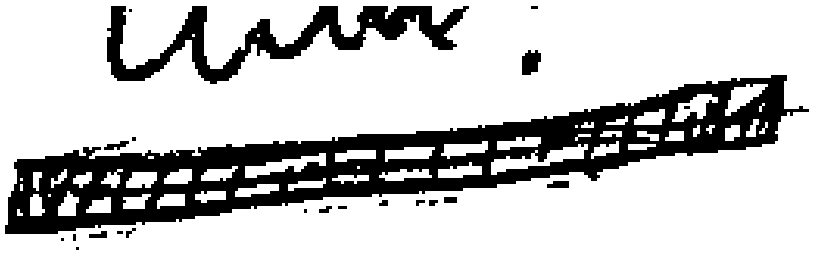
A^* — a scheme (aff.).

$$A^* = \text{Spec}\, A[x, y]/(xy - 1).$$

A precise definition of P^n.

$$A^{n+1} = \text{Spec}\, A[x_0, \ldots, x_n].$$

$P^n \in A^{n+1}$ — homogeneous prime ideals, not contained in the ideal $(x_0, \ldots, x_n)$.
Or:



Original source details: GELFAND-P000325

Uncertain reading. Under "Or" and before P^n, a long narrow mark is densely hatched. The strip and hatching are rendered; earlier content and any possible direction are unrecovered.



$$P^n \longleftarrow A^{n+1} \setminus \{0\}.$$

homog. ideals. — invariant under similarity transformations.
(the set of lines — they are points, but not closed)
A basis of open sets: $f \in A[x_0, \ldots, x_n]$ — a homog. func.,
P_f — prime homog. ideals not containing this func.:

$\Gamma(P_f, O_P) = (A[x_0, \ldots, x_n]_f)_0$ — elems. of 0 degree of homogeneity.

GELFAND-P000326

Assert:



Original source details: GELFAND-P000326

Uncertain reading. Before the final U_i, a cancelled parenthesized entry is provisionally rendered as (U_i); crossing strokes obscure the lower sign, which remains uncertain.

$$P_{x_i} \cong A^n \stackrel{\text{def}}{=} \cancel{(U_i)}\ U_i.$$

On P there are sheaves $O(k)$: ($k \in$ Z).



Original source details: GELFAND-P000326

Uncertain reading. In the first $O(k)$, the parenthesized letter is heavily reinforced; its final form is rendered as k, while earlier strokes cannot be separated.

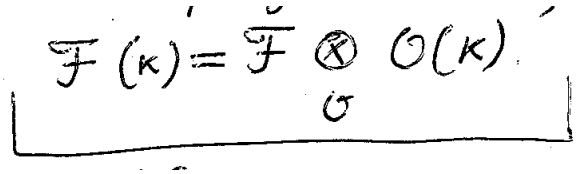
$$\Gamma(P_f, O(k)) = (A[x_0, \ldots, x_n]_f)_k \quad \text{— homog. degree } k \text{ funcs.}.$$

$O(1)$ — the Hopf sheaf; $O(0)$ — the struct. sheaf.

A coherent sheaf: loc. finitely

$$O^J \longrightarrow O^I \longrightarrow \mathcal{F} \longrightarrow 0, \qquad J, I \text{ — finite.}$$

$\mathcal{F}$ — a coherent sheaf on P^n,
Consider



Original source details: GELFAND-P000326

Diagram and typed description.

$\mathcal{F}(k) = \mathcal{F} \otimes_O O(k)$ $\qquad O = O(0).$

Serre's theorem:

1) $\forall i \quad H^i(P, \mathcal{F})$ — A-modules of finite type;

2) (obvious). $H^i(P, \mathcal{F}) = 0$ for $i > n$;

3) $\exists k_0$, such that for $k > k_0$ $H^i(P, \mathcal{F}(k)) = 0$ for [cancelled sign] $i \neq 0$.



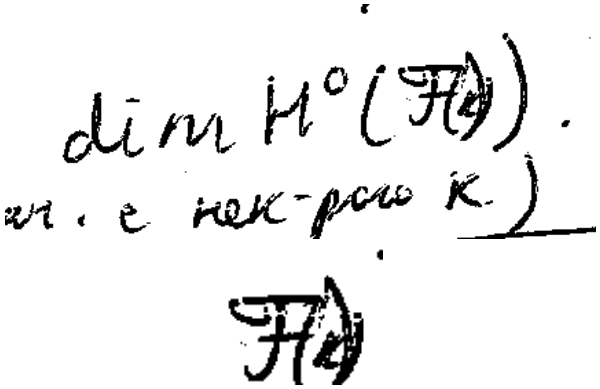
Original source details: GELFAND-P000326

Uncertain reading. Before the clear $i \neq 0$, a small sign is densely crossed by horizontal and diagonal strokes; the earlier glyph is undetermined.

Let A — a field. Define

$$\chi(F) = \sum_i (-1)^i \dim H^i(\mathcal{F}).$$

Theorem: $\forall$ coherent $\mathcal{F}$, $\chi(\mathcal{F}(k))$ — a polynomial in k.
(Serre).
(Hilbert: $\dim H^0(\mathcal{F}(k))$).

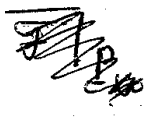


Original source details: GELFAND-P000326

Uncertain reading. Inside the parentheses after H^0 are a calligraphic $\mathcal{F}$ and a tight loop merging with the closing parenthesis. Candidates are $\mathcal{F}(k)$ and $\mathcal{F}_k$; the first is provisionally shown in the line.

(starting from certain k.)

The main lemma for Serre's theor.:
$\forall \mathcal{F}\ \exists k_0$: for $k \geq k_0$ $\mathcal{F}(k)$ is generated by its global sections
(over O).
Explanation. $P^n \supset P_{x_0}$ — aff. space.
Consider [cancelled: $\mathcal{F}|_{P_{x_0}}$?].



Original source details: GELFAND-P000326

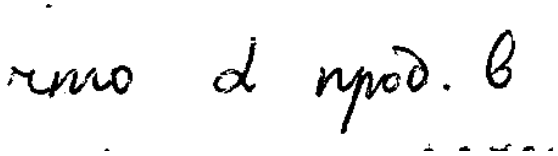
Uncertain reading. After "Consider", $\mathcal{F}$, a vertical stroke and a lower P with a dense x_0-like subscript are cancelled. The candidate $\mathcal{F}|_{P_{x_0}}$ is provisional; part of the subscript beneath the strokes cannot be separated.

$$\alpha \in \Gamma(P_{x_0}, \mathcal{F}). \qquad \exists k: \quad \alpha \in \Gamma(P_{x_0}, \mathcal{F}(k)).$$

GELFAND-P000327

[source number: 82]

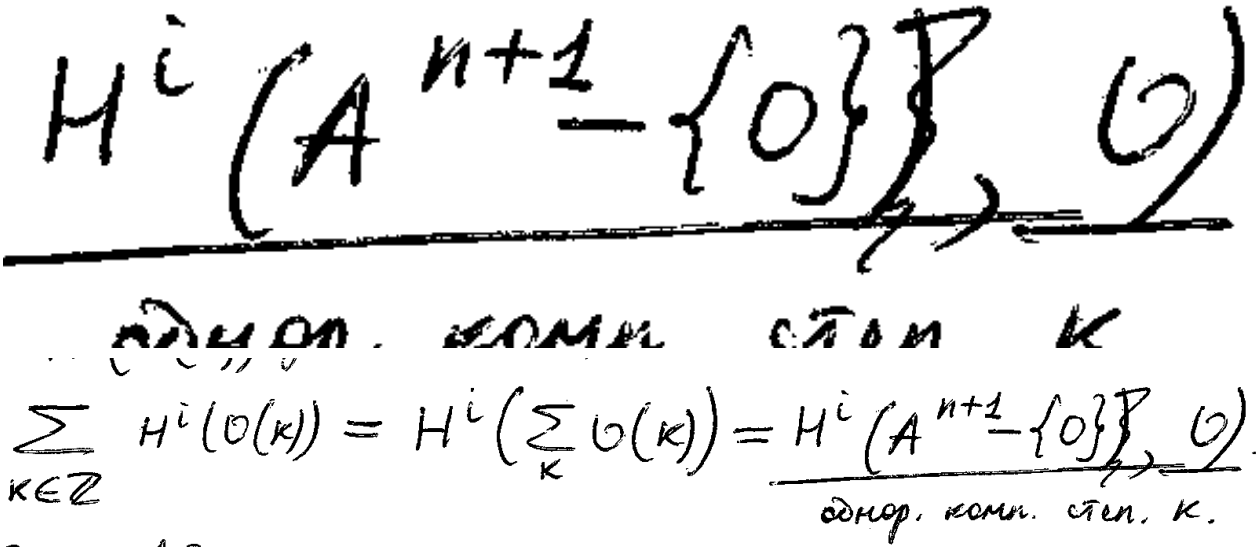
that α extends to a sect. over the whole ring.



Original source details: GELFAND-P000327

Uncertain reading. Before "extends", a loop has a tall ascending stroke. It is provisionally rendered as α; d and "2" remain among the recorded readings. The glyph is not established unambiguously.

Idea: the sections are simply related
sects. of $\mathcal{F}(k)$ — sects. of $\mathcal{F}$, with an additional pole of order k.
(mult. by x_0 — extension to global sections).
Transition to the proof of Serre's theorem.
$H^i(O(k))$ have already been computed.



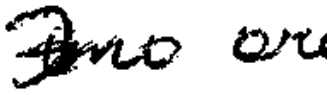
Original source details: GELFAND-P000327

Uncertain reading. After $\{0\}$, the closing mark is heavily overwritten: a tall loop and an oblique stroke cross the underline. The earlier content of these strokes is unrecovered.

$$\sum_{k\in\mathbb{Z}} H^i(O(k)) = H^i\left(\sum_k O(k)\right) = \underline{H^i(A^{n+1}-\{0\},O)}$$

homog. comp. of degree k

This is obvious.

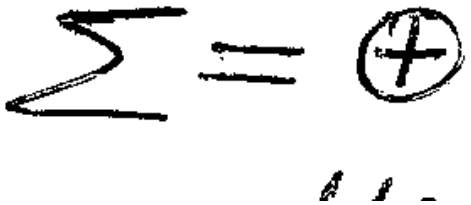


Original source details: GELFAND-P000327

Uncertain reading. The initial "Э" in "Это" (This) is heavily reinforced by transverse and looped strokes; the earlier short sign beneath them is undetermined.

Diagram and typed description.

$\sum = \oplus$.



Original source details: GELFAND-P000327

$P^n = \bigcup U_i$ and the Čech complex. One obtains homog. funcs. of all degrees.

$\mathcal{F}(k)$ is generated by global sects., i.e. $\exists$ an epimorphism



Original source details: GELFAND-P000327

Uncertain reading. A separate dot is visible below the start of the final arrow to 0; its punctuation value is undetermined. The dot and double underline are retained.

$$O^N \longrightarrow \mathcal{F}(k) \longrightarrow 0.$$

For O^N Serre's theorem is checked, and now $O^N \longrightarrow \mathcal{F}(k) \longrightarrow 0$.

From this follows Serre's theorem in the general case.

item 2 is obvious (there are no chains).

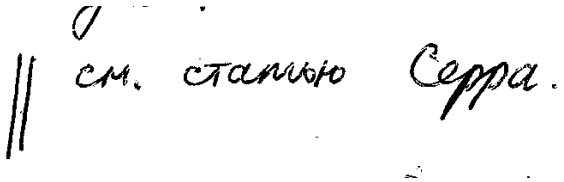


Diagram and typed description.

see Serre's article.

Original source details: GELFAND-P000327

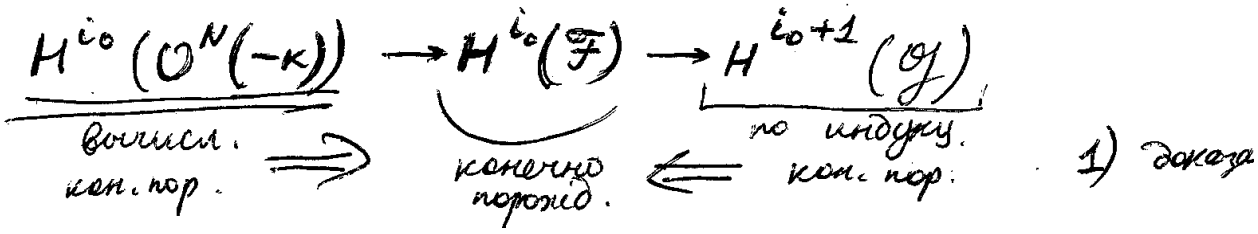
Next, induction downwards in dimensions. (simult. 1 and 3).

$i > n$ is proved.

Suppose proved for $i > i_0$.

$$0 \longrightarrow \mathcal{G} \longrightarrow O^N(-k) \longrightarrow \mathcal{F} \longrightarrow 0.$$

write the exact sequence.



Original source details: GELFAND-P000327

Diagram and typed description.

$H^{i_0}(O^N(-k)) \longrightarrow H^{i_0}(\mathcal{F}) \longrightarrow H^{i_0+1}(\mathcal{G})$

$H^{i_0}(O^N(-k))$: computed fin. gen.

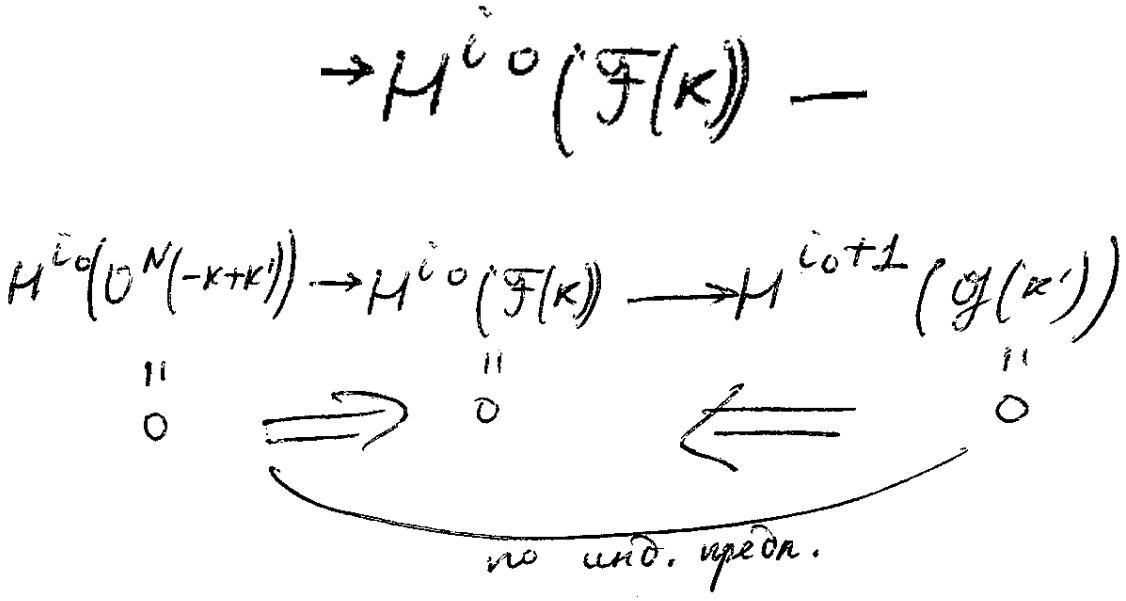
$H^{i_0}(\mathcal{F})$: finitely generated

$H^{i_0+1}(\mathcal{G})$: by induction fin. gen.

$\Longrightarrow$ $\Longleftarrow$

1) proved.

$$0 \longrightarrow \mathcal{G}(k') \longrightarrow O^N(-k+k') \longrightarrow \mathcal{F}(k') \longrightarrow 0.$$



Original source details: GELFAND-P000328

Uncertain reading. The middle term is written $\mathcal{F}(k)$; no separate prime is visible on k, unlike k' above and at the right. The reading is retained with this uncertainty.

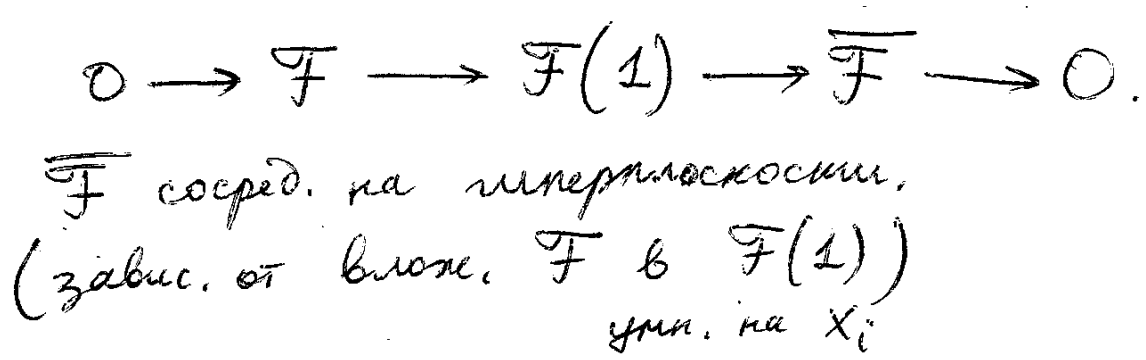
$$\begin{array}{ccccc} H^{i_0}(O^N(-k+k')) & \longrightarrow & H^{i_0}(\mathcal{F}(k)) & \longrightarrow & H^{i_0+1}(\mathcal{G}(k')) \\ \| & & \| & & \| \\ 0 & \Longrightarrow & 0 & \Longleftarrow & 0 \end{array}$$

by ind. hyp.

Corollary: ind. on dim of proj. sp.

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}(1) \longrightarrow \overline{\mathcal{F}} \longrightarrow 0.$$

$\overline{\mathcal{F}}$ concentr. on a hyperplane,



Original source details: GELFAND-P000328

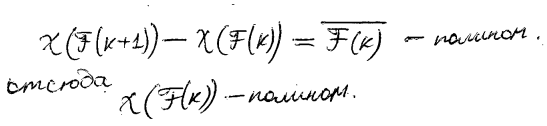
Diagram and typed description.

(depends on the emb. of $\mathcal{F}$ in $\mathcal{F}(1)$)

mult. by x_i

$$0 \longrightarrow \mathcal{F}(k) \longrightarrow \mathcal{F}(k+1) \longrightarrow \overline{\mathcal{F}}(k) \longrightarrow 0.$$

$$\chi(\mathcal{F}(k+1)) - \chi(\mathcal{F}(k)) = \overline{\mathcal{F}(k)} \quad \leftarrow \quad \text{polynomial.}$$



Original source details: GELFAND-P000328

Uncertain reading. After the equality there is no visible χ before $\mathcal{F}(k)$; a long upper stroke extends over this notation. The overlined notation and the left-pointing arrow from "polynomial" are retained.

hence $\chi(\mathcal{F}(k))$ — a polynomial.

For a point everything is trivial. (initial — 0th dim.)



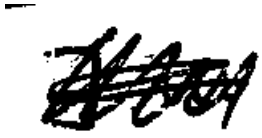
GELFAND-P000329

[source number: 83]

83

8/II-71 I. M. Gelfand.

Bott. Tang. bund. to X, integr. subbund. in TX;

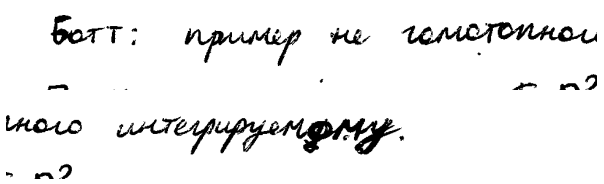


Original source details: GELFAND-P000329

Undeciphered passage. At the left, before "Suppose", a short word with tall oblique strokes is crossed out. The text of this cancelled word is unrecovered.

Suppose there is a bundle: $\exists$ an integrable foliation homotopic to it?

Bott: example of one not homotopic to an integrable one.



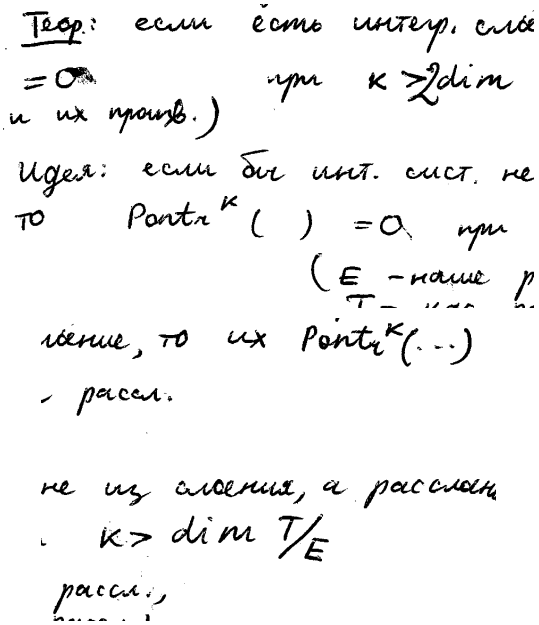
Original source details: GELFAND-P000329

Uncertain reading. In the “Bott: example” line, the provisional reading is “не гомотопного интегрируемому” (not homotopic to an integrable one). The stem “гомотопн” (homotop-) is visible, but the final word ending is heavily overwritten; earlier and latest strokes are not confidently separated. The historical alternatives “не гомотопн...” and “интегрируемых” are retained in the reading records.

There is already an example on $\mathbb{C}P^2$ — obtained using Pontr. cls.
Thm.: if there is an integr. foliation, then their

$$\text{Pontr}_i^k(\dots) = 0 \qquad \text{for } k > 2\,\text{dim bund.}$$

(and their products)



Original source details: GELFAND-P000329

Uncertain reading. The upper index is shown as k, and the small lower mark provisionally as i. At the upper right of the zero is a densely overwritten area and small strokes; the earlier writing beneath them is undetermined. The written bound $k > 2\,\text{dim}$ and abbreviation are retained.

Idea: if the integr. syst. were not from a foliation, but a bundle,
then $\text{Pontr}^k(\,) = 0$ for $k > \dim T/E$

(E — our bund.,
T — tang. bund.).



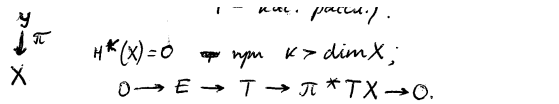
Original source details: GELFAND-P000329

Undeciphered passage. After $H^k(X) = 0$, a short looped word is crossed out. The text of this cancelled word is unrecovered.

for $k > \dim X$;

$$0 \longrightarrow E \longrightarrow T \longrightarrow \pi^* TX \longrightarrow 0.$$

and everything from naturality of Pontr. classes.



Original source details: GELFAND-P000329

Diagram and typed description.

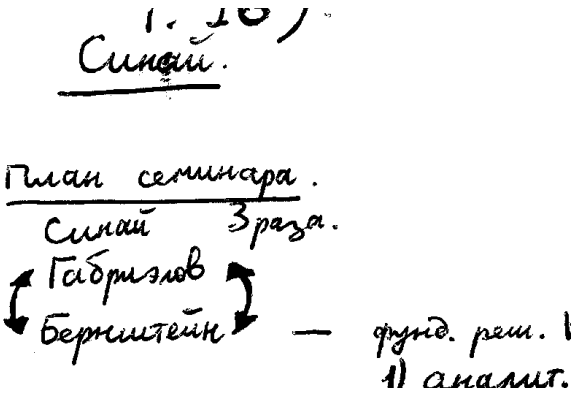
$$\begin{array}{ccc} Y & & \\ \downarrow{\scriptstyle \pi} & & H^k(X) = 0 \\ X & & \end{array}$$

(Work in the seminar on global analysis, vol. 16).

Sinai.

Seminar plan.

Sinai 3 times.
with const. coeffs.:



Original source details: GELFAND-P000329

Diagram and typed description.

Gabrielov

Bernstein

— fund. sol. ∀ eqns.

1) analytic.



Original source details: GELFAND-P000329

Uncertain reading. After “analytic.” is a crossed entry beginning with shapes resembling “во”; its remaining letters are covered by horizontal strokes. They are not reconstructed from the phrase’s meaning.

outside fin. codim.

2) analyt. cont. to the whole complex $\mathbb{C}^n$.

3) finite dimensionality. (branches — a fin.-dim. sp.)

GELFAND-P000330

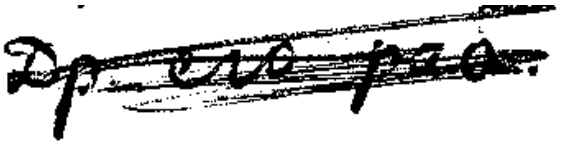


Original source details: GELFAND-P000330

Uncertain reading. The encircled, struck-out and underlined name is shown only as the candidate “Берн...штейн” (Bern...shtein). “Бернштейн” (Bernstein) and other readings of earlier strokes remain possible; historical attribution is not established.

~~Bern...shtein~~ — study analytically ord. analyt. eqns.

~~Др.... ра~~

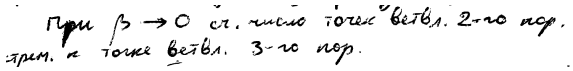


Original source details: GELFAND-P000330

Uncertain reading. A shortened entry before the equation is crossed out. The beginning “Др.” (Dr.) and ending “ра” (ra) are shown provisionally; its middle and complete reading are undetermined.

$$-y'' + x^2 y + \beta x^4 y = \lambda y.$$

As $\beta \to 0$, a ctbl. number of 2nd-order branch pts., tending to a 3rd-order branch point.



Original source details: GELFAND-P000330

Uncertain reading. Before “number” are two short letters and a dot; “сч.” is shown provisionally, with “сл.” also possible. The Russian abbreviation is not expanded; English “ctbl.” represents only the first candidate. The written orders 2 and 3 are retained.



Original source details: GELFAND-P000330

Uncertain reading. At the right, in an oval, is a 2-like mark with a separate dot below. Both are shown; their combined meaning or reading as a question mark is not established.



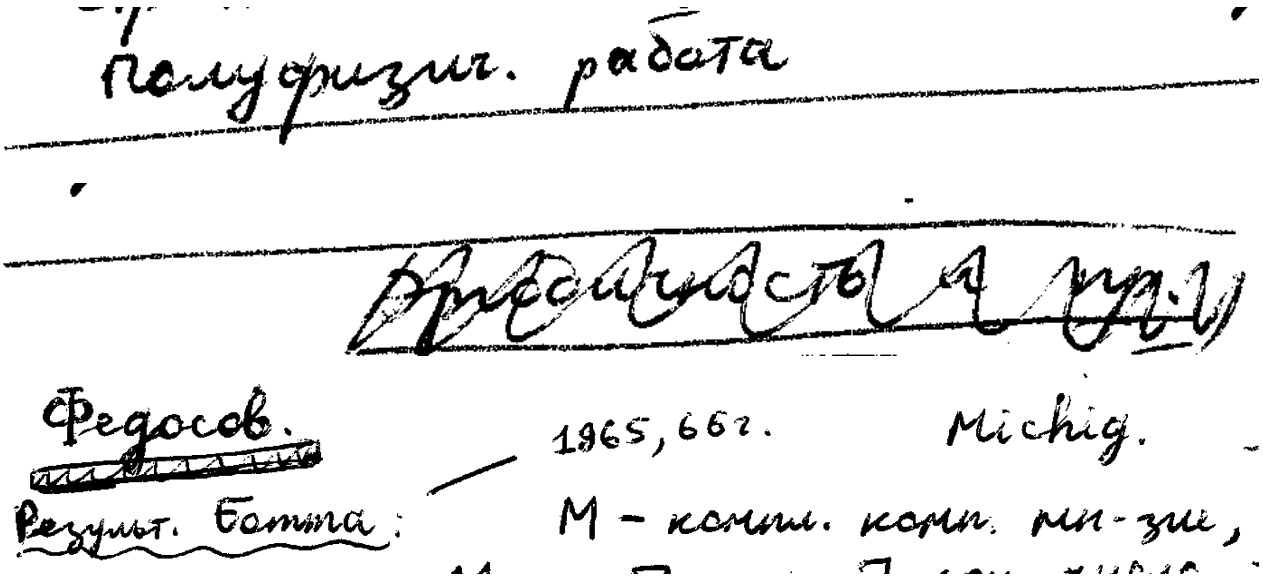
Semiphys. work.



Original source details: GELFAND-P000330

Uncertain reading. Below the rule at the right is a connected underlined entry. Only candidates are shown:“Пре...ств...”, a separate tall A/1-like sign, and an ending “ур.”/“пун...”. The complete phrase is unread.

Fedosov.

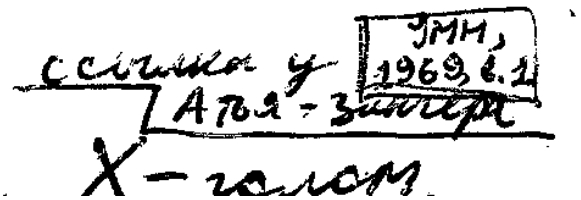


Original source details: GELFAND-P000330

Diagram and typed description.

1965, 66 Michig.

Bott's result:



Original source details: GELFAND-P000330

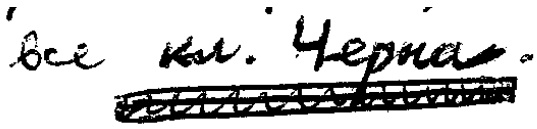
Uncertain reading. In the right margin,"reference in Atiyah–Singer" is marked by an angular frame. Above is a separate box, provisionally "UMN, 1969, 6.1"; the small figures and marks after the year are not entirely certain. The bibliographical entry is not completed.

UMN, 1969, 6.1

reference in Atiyah–Singer

M — comp. complex manif., X — holom. vect. field on M.

Let $\exists$ a fin. number of points p, where it is 0 (and nondegenerately).



Original source details: GELFAND-P000330

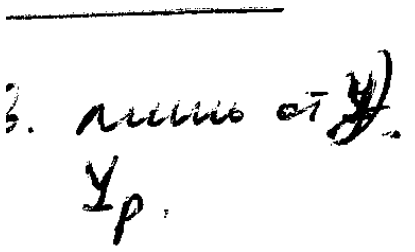
Diagram and typed description.

Using it, one can compute all Chern cls.

$$\Theta(X)Y = [X, Y].$$

Y — another vect. field.

At pt. p a lin. transformation in the fibre.



Original source details: GELFAND-P000330

Uncertain reading. After "on", the upper Y is heavily overwritten; below it is a separate Y_p. Earlier strokes at the upper letter and the value of the closing curve on the right are undetermined. The candidate Y, lower notation and curve are shown.

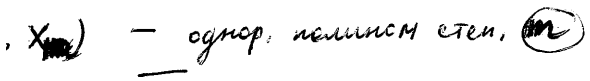
(depends only on Y)

Y_p.

Θ_p — a lin. transf. of the tang. sp.

Nondeg.: Θ_p is nondegenerate. $\forall p$.

Let dim $M = m$,



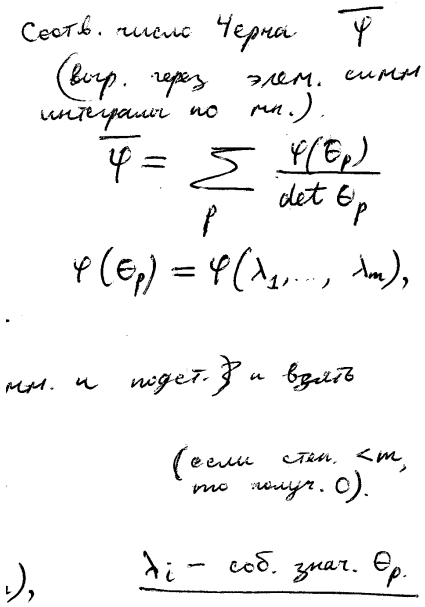
Original source details: GELFAND-P000330

Uncertain reading. The index on the final X is heavily overwritten; it is provisionally shown as m. At the right, the degree, also resembling m, is overwritten and circled. Earlier indices are unrecovered.

$\psi(X_1, \ldots, X_m)$ — a homog. polynomial of degree m

The corr. Chern number $\overline{\psi}$

(expr. through elem. symm. and subst. ξ and take integrals on "м.")



Original source details: GELFAND-P000330

Uncertain reading. After "subst." is a tall shaped sign with an upper loop and lower tail; ξ is shown provisionally, with $\mathcal{F}$ also possible. After "on" is an m-shaped sign and dot, provisionally retained as "м." without normalization to M. The full reading of these signs is undetermined.

$$\overline{\psi} = \sum_p \frac{\psi(\Theta_p)}{\det \Theta_p}.$$

(if the degree $< m$, then one gets 0).

$$\psi(\Theta_p) = \psi(\lambda_1, \ldots, \lambda_m),$$

λ_i — the eigenvalues of Θ_p.

GELFAND-P000331

[source number: 84]

This is an infinites. analogue of the Atiyah–Bott thm.

$$(e^{\text{field}} = \text{transf. of a manif. into itself})$$

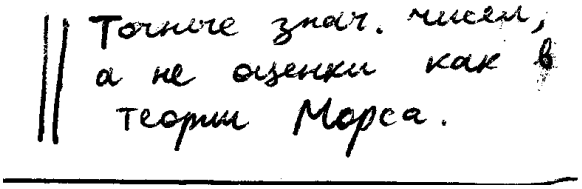
~~Ka...dan~~



Original source details: GELFAND-P000331

Uncertain reading. Before “There is”, a surname is crossed out; it is provisionally shown as “Ка...дан” (Ka...dan). “Каждан” (Kazhdan) and “Каздан” (Kazdan) remain possible; its middle and ending are hidden by dense horizontal strokes.

There is an analogue for the real case.



Original source details: GELFAND-P000331

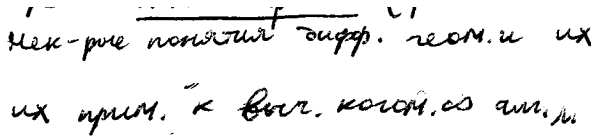
Diagram and typed description.

Exact values of the numbers, rather than estimates as in Morse theory.

Next time Sinai and the main res. of Brouwer.

15/II. I. M. Gelfand (work with Kazhdan).

Some notions of diff. geom. and their appl. to questions of



Original source details: GELFAND-P000331

Uncertain reading. The compressed line ending has the candidate “когом. со анал. м...”. Its small abbreviations and final signs at the right edge are uncertain; the full title is unrecovered.

Fedosov. References 1. Michigan Math. J., 1967, v.14, $N = 2$, 231–244.

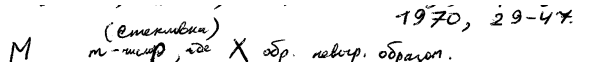
2. J. Diff. Geometry, 1967, v.1 $N = 4$, 311–330.

3. Matem., 14:2, 1970.

4. Baum, Bott, vol. dedicated to de Rham, Springer 1970, 29–47.

(Steklov)

M m-manif., where X becomes in a nondeg. way.



Original source details: GELFAND-P000331

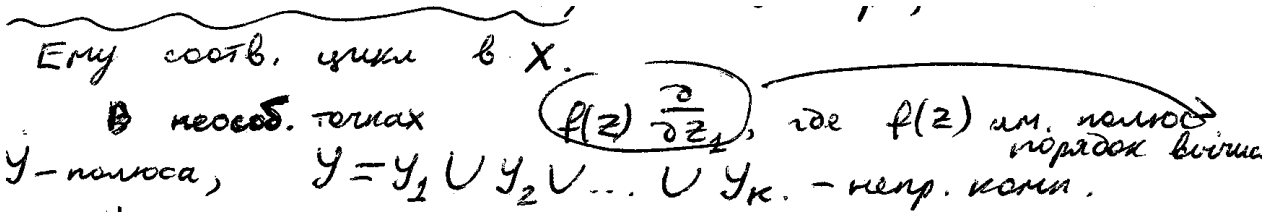
Uncertain reading. Between the references and the line about M, above “m-manif.”, the parenthesized entry reads “Стеклова” (Steklov). It is retained at this position; its attachment to the references or following line is undetermined.

X — a holom. vect. field.

Bernstein. Formula for merom. vect. fields.

V — a merom. vect. field on a comp. analyt. manif. X.

It has a fin. number of nondeg. zeros.
Poles at codim $= 1$, in general, singular.
It has a corr. cycle in X.



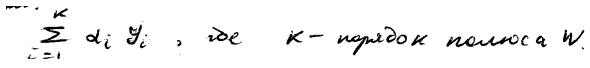
Original source details: GELFAND-P000331

Diagram and typed description.

At nonsing. pts. $f(z)\frac{\partial}{\partial z_1}$ where $f(z)$ has a pole, the order is computed.

Y — the poles, $Y = Y_1 \cup Y_2 \cup \ldots \cup Y_k$ — irred. comps.
Y_i — cycles.

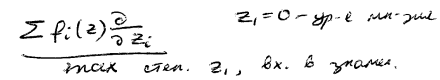
Cycle of the field: $\sum_{i=1}^{k} \alpha_i Y_i$, where k — the order of the pole of W.



Original source details: GELFAND-P000331

Uncertain reading. The final letter after "pole of" resembles W, possibly a rewritten V. The upper limit k, lower limit $i = 1$, coefficients α_i and following written "k is the order of the pole" are retained.

GELFAND-P000332



Original source details: GELFAND-P000332

Diagram and typed description.

$$\frac{\sum f_i(z)\frac{\partial}{\partial z_i}}{\text{max deg. of } z_1 \text{ occurring in the denom.}}$$

$z_1 = 0$ — eqn. of the manif.

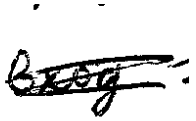
Cycles A_V.
p —



Original source details: GELFAND-P000332

Uncertain reading. The looped sign before the dash has a long descending stroke. p is shown provisionally; ρ is also possible.

~~ВХОД~~



Original source details: GELFAND-P000332

Uncertain reading. The first crossed-out word is provisionally shown as "вход" (vkhod, entry); dense strokes cross its visible letters. Its complete reading remains unsettled.



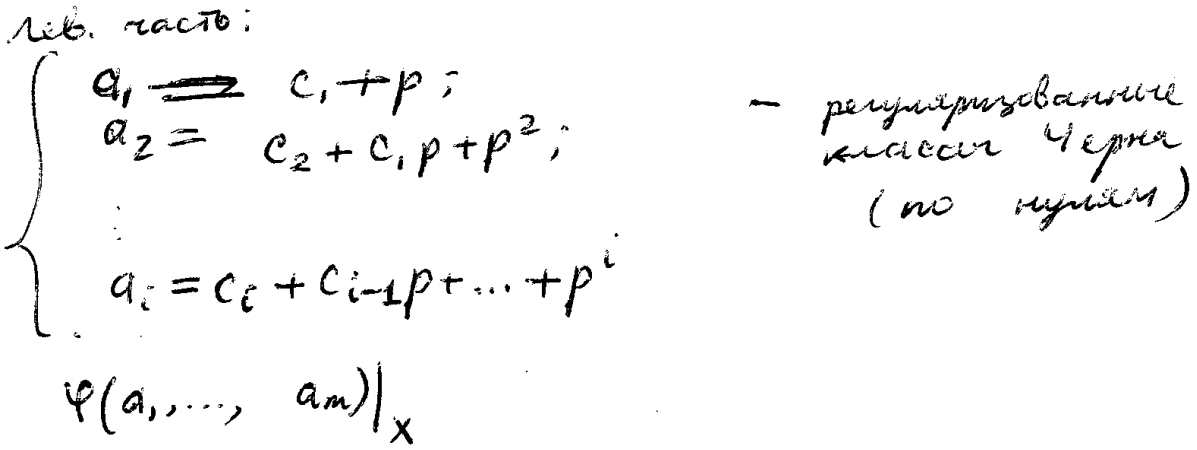
Original source details: GELFAND-P000332

Undeciphered passage. A separate second word follows, heavily crossed out; its text is unrecovered.

by Poincaré duality, 2-dim. cycles.

The right-hand side is the same.

The left-hand side:



Original source details: GELFAND-P000332

Uncertain reading. After a_1, the overwritten two-stroke sign is shown as $\Rightarrow$; a rewritten equality sign is also possible. The letter on the right is shown as p, with ρ also possible. The shared brace, all three rows and right-hand label are retained.

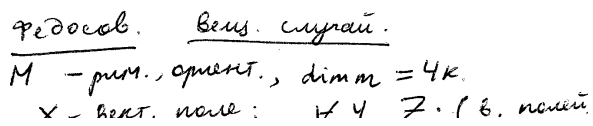
$$\begin{cases} a_1 \Rightarrow c_1 + p; \\ a_2 = c_2 + c_1 p + p^2; \\ \\ a_i = c_i + c_{i-1} p + \ldots + p^i \end{cases}$$

— regularized Chern classes (from the zeros).

$\psi(a_1, \ldots, a_m)|_X$.

Fedosov. The real case.

M — Riem., oriented, dim $m = 4k$.



Original source details: GELFAND-P000332

Uncertain reading. The initial M has a capital form; the sign after dim resembles a small m and is provisionally shown that way. The dimensional notation is retained as dim $m = 4k$.

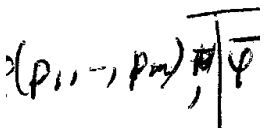
X — a vect. field; $\forall Y, Z$: (vect. fields).

$$([X,Y],Z) + (Y,[X,Z]) = X(Y,Z).$$

Displ. along traj., up to ∞ small isometries.

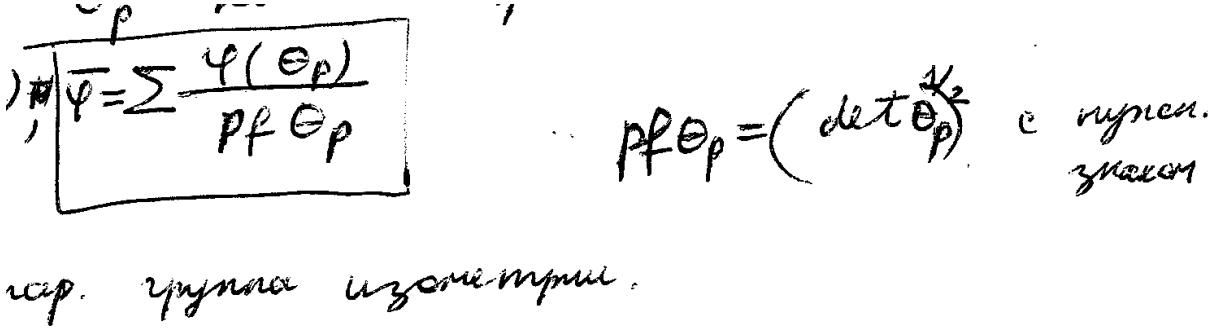
Then Θ_p is skew-symm. at sing. pts.

$\psi(p_1, \ldots, p_m)$,



Original source details: GELFAND-P000332

Uncertain reading. Between the closing parenthesis and the frame is a heavily overwritten small looped sign, resembling p. Earlier strokes are unrecovered.



Original source details: GELFAND-P000332

Diagram and typed description.

$\overline{\psi} = \sum \frac{\psi(\Theta_p)}{\operatorname{pf}\Theta_p}$

$\operatorname{pf}\Theta_p = (\det\Theta_p)^{1/2}$ with the required sign.

One-param. group of isometries.

GELFAND-P000333

[source number: 85]

22/II Sinai.

Factors — Naimark.

Next time — Gabrielov.

Arnold.

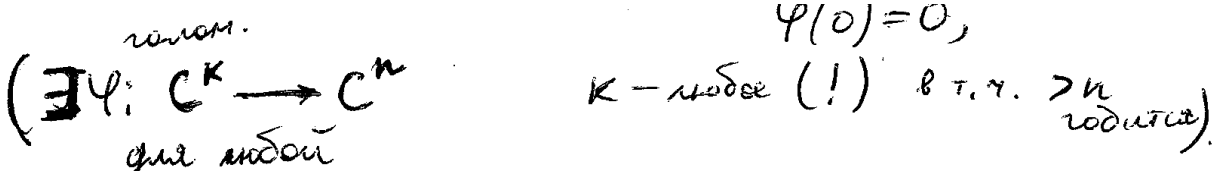
$A(\lambda)$ — a holom. family of matrices.

(or a matr. over the ring of germs of holom. fns.).

A_0 — a fixed matr., $A(\lambda)$ — its deform., $A(0) = A_0$.

Versal family: if obtained under a mapping.

$$B(\lambda) = C(\lambda)A(\varphi(\lambda))C^{-1}(\lambda); \qquad C(0) = 1, \quad \varphi(0) = 0,$$



Original source details: GELFAND-P000333

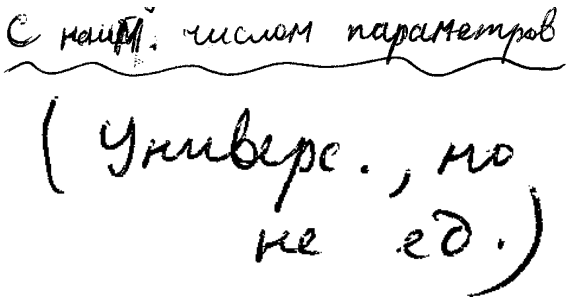
Diagram and typed description.

holom.

$(\exists\varphi : \mathbb{C}^k \longrightarrow \mathbb{C}^n$ k — any (!) incl. $> n$ works).

for any

With the least no. of parameters. (Universal, but not unique.)



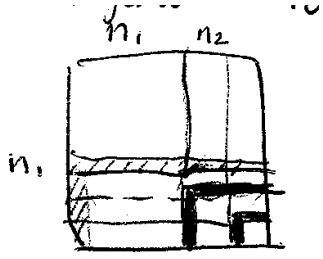
Original source details: GELFAND-P000333

Uncertain reading. The abbreviation “наим.” (least) is shown provisionally; its last letters are heavily overwritten and the earlier entry is unrecovered. The wavy underline of the whole phrase and separate right-hand parenthetical note are retained.

Answer. Versal deform. of A_0.

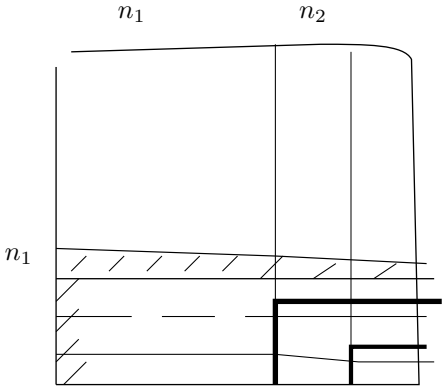
Let A_0 have 1 eigenval. (otherwise separate.)

$n_1 \geq n_2 \geq n_3 \geq \dots$



Original source details: GELFAND-P000333

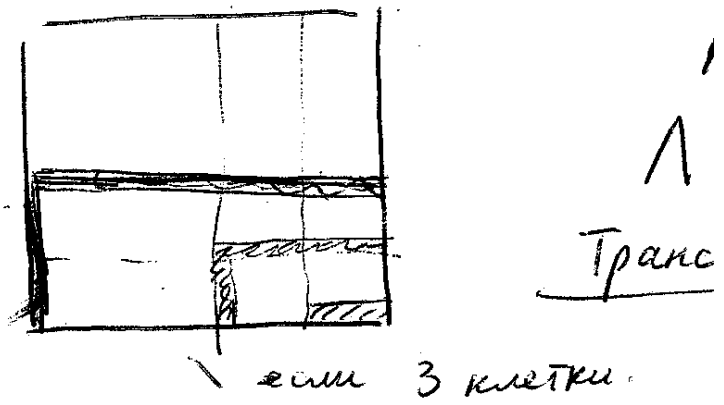
Diagram and typed description. The outer contour, two internal vertical dividers, the upper n_1, n_2 and left n_1 labels, horizontal bands, left diagonal hatching and two dense L-shaped bands at the lower right are retained. The exact cell count and earlier strokes at dense intersections are undetermined.



Insert free params. in the marked cells.

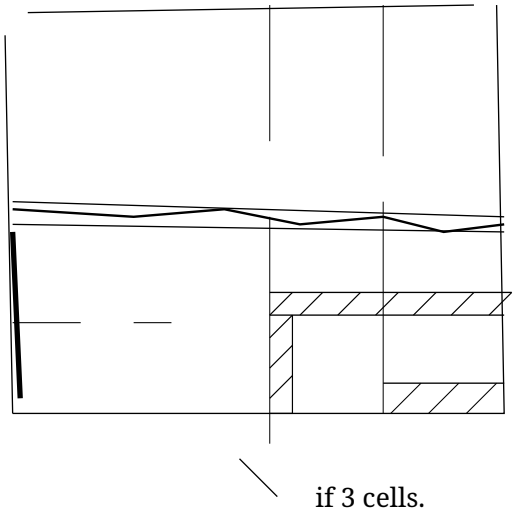
Versal deform.

Row numbered n_1, 1st column down.

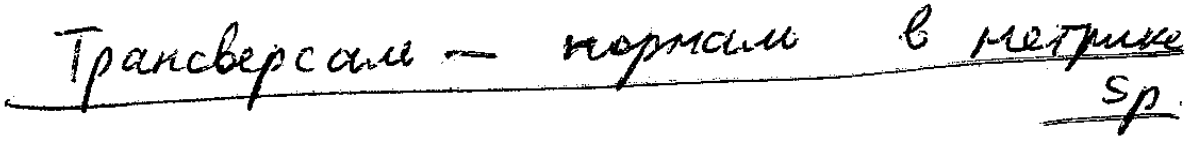


Original source details: GELFAND-P000333

Diagram and typed description. The rectangular contour, two interrupted internal vertical dividers, common dense horizontal band, two lower right hatched bands, left and internal vertical shading, slanted pointer below and label “if 3 cells” are retained. Dense intersections and the exact cell count remain unclear; no distinct arrowhead is visible on the slanted stroke.



$A_0 + \Lambda$,
Λ — free parameters.



Original source details: GELFAND-P000333

Diagram and typed description.

Transversals — normals in the matrix

Sp

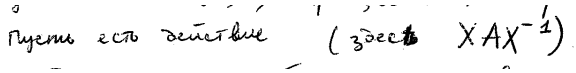
GELFAND-P000334

Sing. in a 3-param. family of matrices in general pos.

Gelfand: question:

the action of $GL(n, \mathbb{C})$ is triv., if locally — a fibration.

Let there be an action (here XAX^{-1}).



Original source details: GELFAND-P000334

Uncertain reading. The last letter in “здесь” (here) is heavily overwritten; the word is shown according to the current entry and the earlier letter is unrecovered.



Original source details: GELFAND-P000334

Uncertain reading. At the left margin, an oval encloses four slanted strokes, with a dense crossbar at the two inner strokes. The sign resembles (H) or (H); its meaning is undetermined.



cannot split — essential.

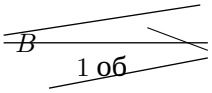
Precisely here — a versal family

describe all types of actions (tensors of rank 15, etc.).



Original source details: GELFAND-P000334

Uncertain reading. A short entry before Ad is crossed out. The left B and small 1,“об” below are shown only as candidates from visible fragments; the full reading is undetermined. The intersecting cancellation strokes are retained.



Ad — the worst action.

Hypothesis: locally there is a fibration without sing.,

in the remainder — Ad — Jordan form.

~~Near...~~

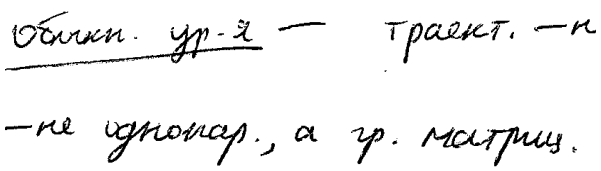


Original source details: GELFAND-P000334

Uncertain reading. After “form”, a word is crossed out; its visible beginning is “Близк...” (near...). A line crosses its middle and the ending is unrecovered. The ellipsis marks the unrecovered remainder of the word.

A nearby probl.: a generalized fn., inv. under the action of matrices.

Ordinary eqns. — traject. — not one-param., but a gr. of matrices.



Original source details: GELFAND-P000334

Uncertain reading. The middle contraction is provisionally shown as “траект.” (traject.); the historical candidate “трансв.” is also retained in the full apparatus. Its final vertical letter and join remain uncertain. The separate following contraction is “однопар.” (one-param.).